Handbook on QUARRYING



DEPARTMENT OF MINES AND ENERGY

HPRM 2018D037857

Department of State Development
Resources and Energy Division
Customer Services
Ph (08) 8463 3000

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Handbook on quarrying

5th ed.
Bibliography
ISBN 0 7308 1925 6.

1. Quarries and quarrying — Handbooks, manuals, etc. I. Armstrong, A.T. (Albert Thomas), 1906 —. Handbook on quarrying. II. South Australia. Dept. of Mines and Energy. III. Title (Series: Handbook (South Australia. Department of Mines and Energy); no.1).

622.35
ISSN 0726-1519

Edited and designed by Jon Selby
Drafting and reprographics by SADME Drafting Services Branch

Issued under the authority of The Hon. F. Blevins, MP, Minister of Mineral Resources.

Keywords: Quarrying/Manuals/Environmental management/Geology/Drilling/Exploration/Explosives/ Ground vibration/Blasting/Crushing and screening/Safety and health.

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First edition 1961
Second edition 1964
Third edition 1975
Fouth edition 1983
Fifth edition 1993
Sixth edition 2015 (electronic)

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23/03/2016 – 204571

Handbook on Quarrying

Compiled by **A.H. Dutton**

HANDBOOK No. 1
(Revised edition 1993)

DEPARTMENT OF MINES AND ENERGY
SOUTH AUSTRALIA





Stripping overburden at Wheal Hughes copper mine opencut, Moonta. (Photo 039485)

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1. Introduction

The aim of this Handbook is to give practical guidelines to quarry operators and to discuss problems connected with operations which affect the employees and the general public; it also points out obligations that arise out of legislation and the responsibilities of quarry operators to the community.

In any discussion on quarrying* three points must be recognised:

- Any stone or mineral deposit is a wasting asset — once extracted it is gone for all time.
- Deposits are fixed in position and must be worked where they occur; they should therefore be worked for maximum recovery within the limits determined by local circumstances.
- Rarely are two deposits exactly the same.

Consideration must be given to many diverse factors in the overall planning of a quarry, including:

- size, shape and type of the deposit
- rate of production and hence the life of the project
- short and long term effects of quarrying on the surrounding areas
- subsequent after-use of the land.

A few deposits can be worked with progressive rehabilitation so that the land can be restored to conform with prior or existing use. Many can be worked with a definite but different land use in mind. The aim in others must be to leave the site in an 'acceptable' condition which can be adapted to a variety of uses at some future time. The overall development plan for the area and its programmed timing will have a big bearing on the ultimate rehabilitation plan.

*When the terms 'quarry' and 'quarrying' are used in a general sense in this handbook, they are meant to cover opencut mines, gravel pits, sand pits, clay pits and other open surface excavations used to win minerals, metallic ores and extractive materials.

It is assumed in the following chapters that there are no legal difficulties which might prevent the opening up of surface workings in a particular area, and that when decisions have to be made in the exploration and planning stages, market surveys and studies of mineral and metal price fluctuations will have been taken into account.

A glossary of special terms used in mining is appended.

Before proceeding with a discussion of the practical aspects of quarrying it is appropriate to introduce aspects of the quarrying industry in South Australia.

Production of Construction Materials in South Australia

In South Australia, construction materials are classified as 'extractive minerals' under the Mining Act and Regulations. Under this Act, a levy is paid into a fund which is administered by the Department of Mines and Energy; currently this is 10c per tonne. Money from the fund is used to finance rehabilitation of areas disturbed by quarrying and to pay for research into methods of reducing the environmental impact of quarrying operations.

During the past decade, urban development has encroached upon many of Adelaide's sand and aggregate quarries; in some cases housing development has taken place immediately adjacent to existing quarries and proposed quarry extensions. This has inevitably resulted in loss of reserves and will impose constraints on operating hours and methods used by quarry operators.

Crushed rock aggregates, construction sand, brick clay and dimension stone are considered below.

Crushed Rock Aggregates

Two types of crushed rock aggregate are supplied to the metropolitan Adelaide market from metasedimentary rocks of Proterozoic age:

- carbonate rock, commonly referred to as limestone or dolomite
- quartzite (including sandstone and arkose).

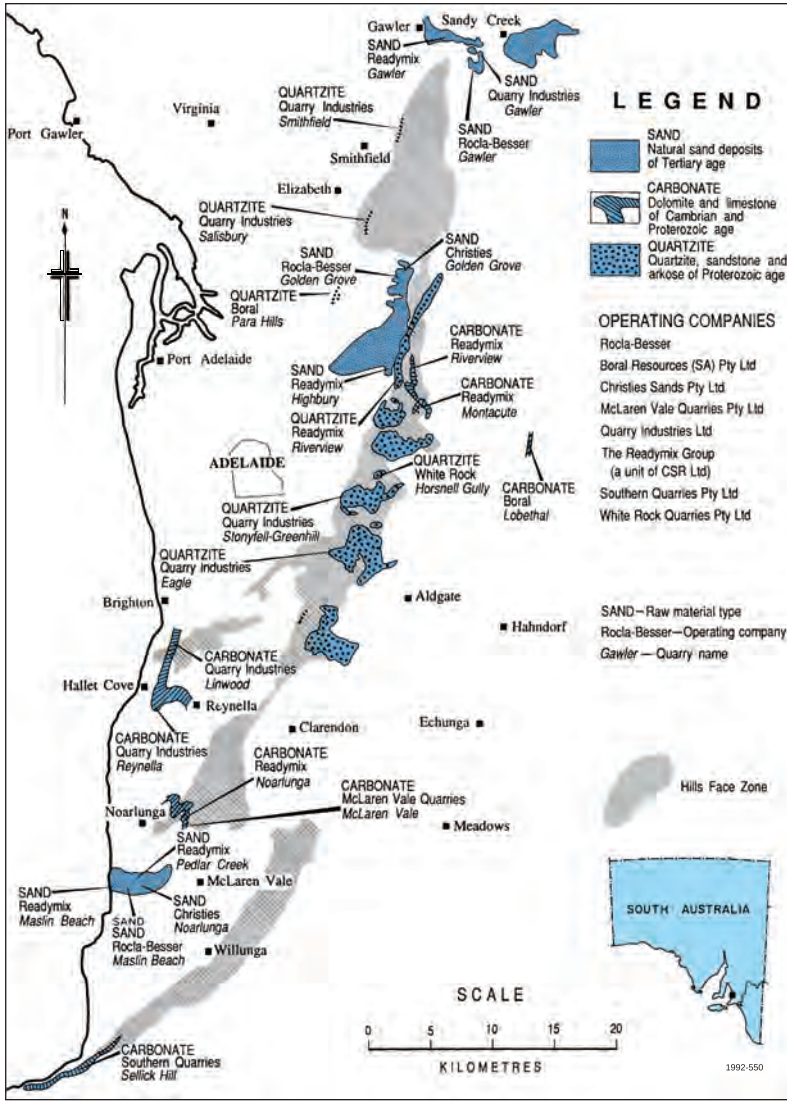


Fig. 1.1 *Principal aggregate sources for Metropolitan Adelaide.*

The distribution of these rock types and the major metropolitan quarries are shown on Figure 1.1. Total output approximates 5 million tonnes (Mt) a year of which about half is carbonate aggregate.

Although a large range of products is made from both rock types, they are not fully interchangeable. For example all of Adelaide's road sealing aggregate comes from carbonate rocks which bond more readily with bitumen than does quartzite. Quartzite quarries have a relatively high content of rubble, filling and quarry sand but only a quarter of their product is screenings (see Chapter 15 for definition). In contrast, nearly 40% of product from carbonate quarries is screenings.

Typical quarry practice is to drill 75-90 mm diameter blastholes with a spacing and burden of between 2.4 and 3.6 m with either pneumatic tracked drills, down-hole-hammer drills or hydraulic drills.

Most metropolitan quarries use bulk explosives while non-electric blast initiation systems have eliminated the use of surface detonating cord lines.

Mechanical and hydraulic rock breakers and drop balls have replaced secondary blasting in all metropolitan quarries. In these quarries trucks of from 15 to 50 t capacity are loaded at the face by 2-7 m^3 rubber tyred front end loaders. Face shovels are no longer used. Hydraulic excavators have been introduced into a number of operations and are operated in either a backhoe or shovel configuration.

Crushing plants with capacities of between 50 and 500 t/h typically comprise jaw or gyratory primary crushers and horizontal and vertical shaft impactors, with gyratory and hammer mill crushers in the secondary and tertiary stages. Throughput increases of up to 20% have been reported where hydraulic rock breakers have been installed before the primary crushing stage.

Mobile plants are usually engaged in contract crushing for government or other projects in country areas. When not engaged in contract crushing, one of the mobile plants is operated for a few months to crush sufficient material for one or more years in a number of small quarries.

Dust extraction systems have been installed in four metropolitan crushing plants. All new drill rigs are fitted with dust extraction systems while older tracked drills have cone type collector systems retrofitted.

Water trucks and delineated traffic areas minimise dust generated by vehicle movement while several quarries have installed automatic sprays to dampen access roads. Most plants have been fitted with airconditioned cabins which are pressurised with filtered air to minimise the operators' exposure to dust.

Carbonate and quartzite deposits provide some of the crushed aggregate in country areas, but other rock types are also crushed in permanent installations, including creek gravel at Port Augusta, granite at Port Lincoln, basalt near Mount Gambier, and gneiss near Murray Bridge.

Calcrete is a relatively important material in the country areas, particularly the South East, Riverland and Eyre Peninsula, where it is used extensively for sealed and unsealed road construction. Permanent installations at Ceduna and in the Riverland crush calcrete for a range of products including screenings.

Construction Sand

Almost all of metropolitan Adelaide's annual consumption of 2 Mt of natural sand comes from three Tertiary sand basins at Gawler, Golden Grove-Highbury, and Maslin Beach.

Because of high clay content, approximately 70% of the sand from the ten major sand pits shown on Figure 1.1 is treated by washing, 19% is dry screened, and the remaining 11% of untreated sand is used mainly for filling.

Tailings from sand washing are generally disposed of in deep dams in worked out pit areas. Trials have been held using shallow dams less than a metre deep which dry rapidly so that dry tailings can be used in rehabilitation, a practice which could be more widely adopted.

Sand is generally won by elevating scrapers loading from inclined pit floors to achieve blending of the horizontally bedded sands.

Small local pits supply sand to regional centres. Dry creekbeds or sand dunes are common sources throughout the State, but beach deposits of Quaternary age supply Port Lincoln and the South East areas. The Riverland is supplied from thick fluvio-lacustrine deposits of Tertiary age.

Because of the small size of most country deposits, they are generally worked by front-end loader, and sand is used directly or treated in mobile dry screening plants.

In some country centres with hardrock quarries, blending of crusher fines and dune or creek sand is practised for some purposes.

Brick Clay

Metropolitan Adelaide's annual consumption of clay for brick making is about 0.6 Mt.

Approximately 85% of this comprises weathered Proterozoic shale, the largest deposits of which are worked by elevating scrapers in the Golden Grove area.

White plastic clay is restricted to lenses within Tertiary basins. The most important deposit is at Golden Grove, where sand overburden is stripped and a bed of white clay 13 m thick is worked by elevating scraper.

During the past few years both of Adelaide's major brick producers, Nubrick and PGH Bricks South Australia, have installed computer-controlled kilns and concentrated their operations near their deposits at Golden Grove.

Clay supplies are generally stockpiled over the summer months because of difficult operating conditions during the winter.

Dimension Stone

The lack of natural stands of quality construction timbers forced the early settlers to turn to stone and brick as the main construction materials for housing and for public buildings.

Siltstone, referred to locally as 'bluestone', was initially quarried at many sites close to Adelaide, the most important being at Glen Osmond and Tapleys Hill. These quarries are now closed, but similar material is mined from quarries at Wistow and Kanmantoo.

South Australia has been Australia's foremost producer of slate from deposits which are widely distributed throughout the Mount Lofty and Flinders ranges, the principal historic sources being at Willunga and Mintaro. Quarries at Willunga produced some 20-30 million roofing slates between 1840 and 1930. Two quarries in this area have re-opened in recent years to supply a range of slate tiles, veneers, sawn building blocks and paving materials. The Mintaro quarry has been producing slate for paving, panelling, billiard table tops, floor tiles and high quality structural slate since 1856. Since the 1970s new slate quarries have been opened at Spalding and Oladdie in the Northern Flinders Ranges.

The quarries which supplied much of the sandstone for Adelaide houses and public buildings from the 1860s through to the 1950s are no longer worked. However sandstone from Basket Range, which gained popularity during the 1950s, is still available.

Mount Gambier stone, a light even textured white fossiliferous limestone, is cut from quarry floors by self-propelled circular saws with tungsten

carbide-tipped teeth. Standard ashlars are 660 x 295 x 100 mm, but other sizes and shapes are produced.

Marble has been mined in South Australia since the 1850s but today only one quarry producing Barossa white marble is in operation near Angaston.

Granite quarrying operations are concentrated on northern Eyre Peninsula and in a zone extending from the eastern Mount Lofty Ranges to the South East. Most of the Eyre Peninsula quarries produce red or pink granites, including Calca Red, Minnipa Red and an unusual pink granite marketed as Desert Lilac. A variety of gloss and textures are present in the granite suite extending from the eastern Mount Lofty Ranges to the South East. Quarries produce a green granite from near Padthaway (Tatiara Green), a blue porphyritic granite from near Kingston (Kingston Blue) and a brown granite from near Sedan (Sienna Brown). Australia's largest 'granite' quarries are at Black Hill near Mannum, and produce Imperial Black and Austral Black from an igneous rock called norite.

Production of Minerals in South Australia

Mineral resource development has been an important element in the growth of South Australia. This growth was based on copper, then iron and coal, and latterly natural gas and oil. Opal mining has continued since 1890. Production of gold, lead, zinc and other minerals was sporadic until the development recently of the huge underground copper, uranium and gold resource at Olympic Dam on Roxby Downs. Currently, the two largest open cut operations are centred on the iron ore deposits of the Middleback Ranges and the sub-bituminous coal deposits at Leigh Creek.

Middleback Ranges Iron Ore Deposits

Thirteen higher grade accumulations (iron content 60-80%) have been identified in the banded iron formations extending 60 km from Iron Knob (the earliest to be mined) in the north to Iron Duke (the latest to be developed) in the south. Total production from the area is about 2 Mt a year which is mined by open cut, crushed and loaded into trains for the Whyalla steel plant 75 km away on the coast.



Brick clay pit at Golden Grove. (Photo 038724)



Mintaro slate quarry. (Photo 036437)



Black Hill norite quarry. (Photo 040359)



Mining iron ore in the Middleback Ranges. (Photo T024065)

Leigh Creek Coalfield

At Leigh Creek, the coal is mined by an open cut method and railed 255 km to power stations at Port Augusta. The low-grade sub-bituminous coal occurs in four discrete basins over a distance of 20 km.
Mining is confined to the largest of the basins where the coal subcrops to within 10 m of the surface and dips at 10-30°.

The current mining operation involves drilling, blasting and removal of overburden in benches 12 m high until coal is exposed. Earlier, stripping was done by dragline but currently a 'terrace' or haulback system is used in which overburden is removed by shovel and truck and used as fill in areas where mining has been completed. Current annual production is 2.8 Mt of coal involving movement of 16 million m^3 of overburden.

2. Product Testing and Specifications

All stone and sand used for construction purposes should conform to specifications relevant to the product's use. Many tests have been devised to assess product suitability, most of which are beyond the scope of this handbook. A selection of the more important tests with which quarry operators should be familiar is presented below.

Sampling

Samples taken should represent the true nature of the material from which they were drawn. Results of even the most meticulous laboratory testing will be worthless if samples are not representative. This is particularly true of samples taken for sizing analysis.

Special tools have been devised to assist in sampling, including sample splitters, sampling boxes, tubes and frames. A basic guide to sampling from stockpiles, trucks and conveyors is presented in Australian Standard AS 1141-3. Large consumers such as government departments have their own sampling standards and procedures which should be followed when dealing with them.

Testing

Grading (particle size distribution) is one of the most important properties of quarry products. Products such as concrete sand or crushed rock require an even distribution of a wide range of particle sizes, but screenings for road sealing have very restricted size ranges. Grading is determined by use of sieves of specified apertures, generally with a mechanical shaker. Results are usually presented as a table showing percentage of material passing each sieve. Results may also be plotted graphically (Fig. 2.1).

Material finer than 75 microns. The silt and clay content of concrete sand is determined by washing through a 75 micron (0.075 mm) test sieve and noting loss of weight of the sample.



Sieves and splitter used in grading tests. (Photo 040470)

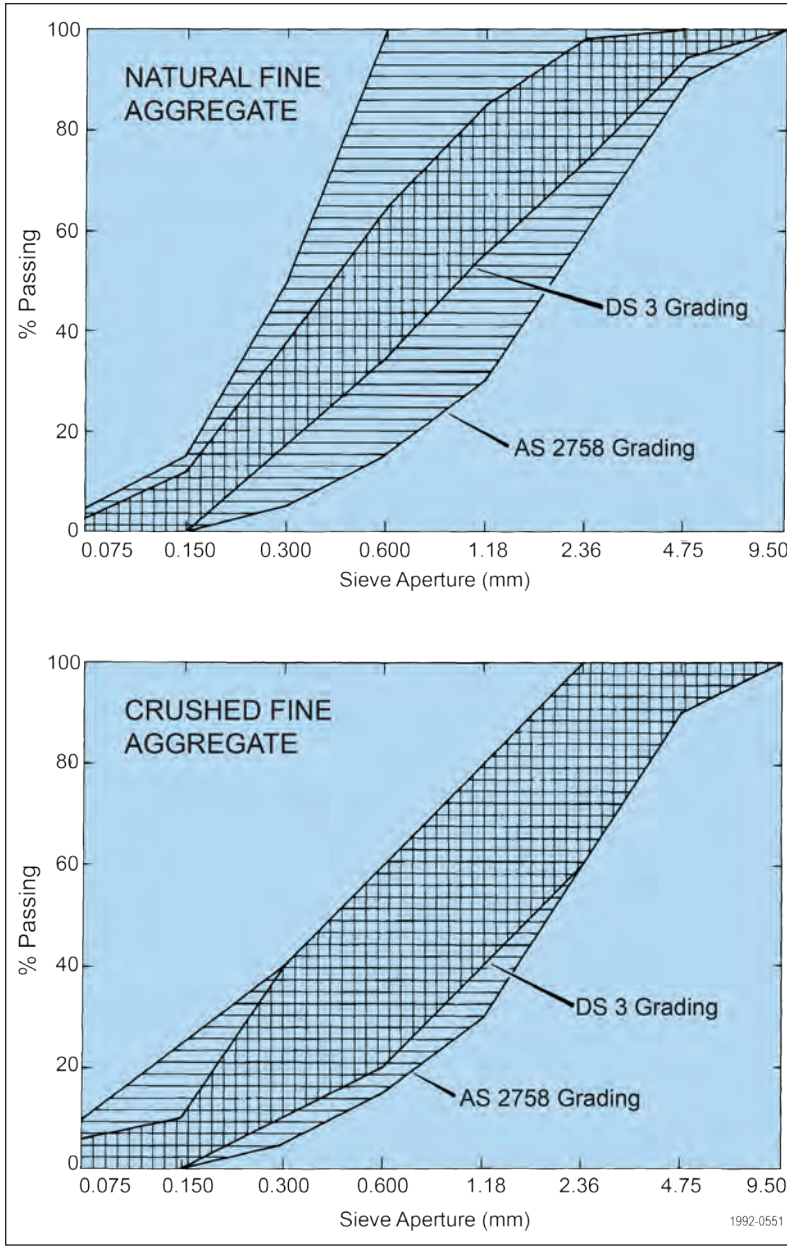


Fig. 2.1 *Aggregate grading charts.*

Los Angeles Abrasion Test. This determines the abrasion resistance of an aggregate by tumbling a sample with steel spheres in the steel drum of a Los Angeles machine. Mass and grading of test samples, speed and number of drum revolutions, and mass and number of steel spheres are specified. The result quoted is the percentage of original sample broken down to -1.70 mm during the test.

Aggregate Crushing Value. Resistance to crushing is measured by submitting a dry sample of aggregate (-13.2 mm, + 9.5 mm) contained in a prescribed metal cylinder to a specified force applied at uniform rate in a compression testing machine. The result quoted is the percentage of original sample broken down to -2.36 mm (8 mesh) during the test.

Ten Per Cent Fines Test is similar to the Aggregate Crushing Value Test, but result quoted is the force in kiloNewtons (kN) necessary to break down 10% of the sample to -2.36 mm.

Wet/Dry Strength Variation. The Ten Per Cent Fines Test is repeated on another portion of material in water saturated (surface dry) condition. Result quoted is the percentage reduction in the result when saturated as compared with the result when dry.

Sulphate Soundness Test measures resistance to cycles of immersion in saturated sodium sulphate solution followed by drying. The result, obtained by comparing the weights of the aggregate gradings before and after the test, is quoted as a percent loss. This test is used to indicate resistance of aggregate to frost or salt attack.

Bulk Density and Water Absorption. Bulk density (kg/m^3) and water absorption (per cent by weight) are determined by measuring the weight of a known volume of crushed rock before and after saturation. Water absorption may provide an indication of durability as well as potential of an aggregate to absorb water in a concrete mix. Rocks which have bulk density much lower than the density of their constituent minerals commonly show high water absorption. Note that bulk density, being the density of tightly packed crushed rock fragments, is quite different from the loose density of a product in a truck or stockpile.

Particle Shape. Flakiness and Elongation Indices are percentages by weight of flaky or elongated particles, generally determined using special gauges or proportional calipers. The test is particularly relevant to road sealing aggregates.

Washington Degradation Test is carried out on rock spalls or drill cores where the rock is suspected to contain weathering products which may lead to deterioration of aggregate in service, particularly secondary minerals in volcanic rocks such as basalts. Fines produced by agitating a sample with water in a special canister are decanted into a measuring cylinder, flocculant is added and depth of flocculate column measured after 20 minutes.

Soils Constants have many applications in engineering geology but are particularly useful for assessing handling characteristics of material of various moisture contents. Tests are carried out on the fine fraction of a graded aggregate such as crushed rock or rubble to determine:

- Plastic Limit: The per cent moisture content at which the material begins to exhibit plastic behaviour, i.e. the lowest moisture content at which an air-dried soil is plastic.
- Liquid Limit: The per cent moisture content at which the material begins to exhibit liquid behaviour, i.e. the moisture content at which a soil passes from the plastic to the liquid state.
- Plasticity Index: The range of moisture content through which the material exhibits plastic behaviour (Liquid Limit minus Plastic Limit).
- Linear Shrinkage: The per cent shrinkage on drying material from its Plastic Limit.

Alkali Reactivity. Some rocks, particularly silcretes, acid volcanics and some limestones and dolomites, react with alkalies in cement to cause deterioration of concrete. Reactivity can be determined chemically or by measurement of dimensional changes during storage of a mortar bar made from the aggregate.

Deleterious Materials. Tests are sometimes required for deleterious materials such as friable particles, organic impurities and sugar in concrete aggregates.

Specifications

The Standards Association of Australia publishes standards which nominate basic requirements for various materials, and these are often used as guides for preparation of individual works specifications. For convenience, aggregates are often designated by Nominal Size, which is usually expressed as a whole number above the sieve size through which nearly all of the aggregate passes. Coarse aggregate is defined as having a nominal size of 5 mm or more. Fine aggregate has a nominal size of less than 5 mm and includes such materials as concrete and mortar sand.

Fine Concrete Aggregate

Grading requirements and limits of deviation specified by AS 2758.1 are shown in Table 2.1 and presented graphically in Figure 2.1. Different gradings apply to uncrushed (natural) and crushed sand. Note that AS 2758.1 plots are relatively broad envelopes. Individual works specifications often require gradings with narrower limits, such as the Engineering and Water Supply Department DS3 specifications for fine concrete aggregate shown in Figure 2.1.

Table 2.1 *Fine concrete aggregate grading requirements.*

Uncrushed fine aggregate			**Crushed fine aggregate**	
Sieve aperture (mm)	**Mass of sample passing** (%)	**Maximum deviation** (±%)	**Mass of sample passing** (%)	**Maximum deviation** (±%)
9.50	100	–	100	–
4.75	90-100	5	90-100	5
2.36	60-100	5	60-100	10
1.18	30-100	10	30-80	15
0.600	15-100	15	15-60	15
0.300	5-50	10	5-40	10
0.150	0-15	5	0-25	5
0.075	0-5	5	0-10	5

Coarse Concrete Aggregate

Graded or single sized coarse aggregates may be used in concrete; examples of grading specifications for both types are presented in Table 2.2 adapted from AS 2758.1.

Other characteristics of coarse concrete aggregate are usually specified, including:

- Los Angeles Abrasion Test: Upper limits generally in range 30-40%
- Aggregate Crushing Value: Upper limits generally in range 30-40%
- Sulphate Soundness Test: Upper limits generally in range 6-12%
- Wet/Dry Strength Variation: Upper limits generally in range 25-45%.

Table 2.2 *Coarse concrete aggregate grading requirements.*

	Mass passing (%)						
Sieve size (AS 1152) (mm)	**Nominal size of graded aggregate** (mm)			**Nominal size of single sized aggregate** (mm)			
	40	20	14	40	20	14	10
75.00	100	–	–	100	–	–	–
53.00	–	–	–	–	–	–	–
37.50	85-100	–	–	85-100	–	–	–
26.50	–	100	–	–	100	–	–
19.00	30-70	85-100	100	0-20	85-100	100	
13.20	–	–	85-100	–	–	85-100	100
9.50	10-35	25-55	–	0-5	0-20	–	85-100
6.70	–	–	25-55	–	–	0-20	–
4.75	0-5	0-10	–	–	0-5	–	0-20
2.36	–	0-5	0-10	–	–	0-5	0-5
0.075	0-2	0-2	0-2	0-2	0-2	0-2	0-2

Road Sealing Aggregate

A summary of selected specifications for screening is presented in Table 2.3.

Table 2.3 *Screenings specifications, Department of Road Transport, South Australia.*

Sieve size (mm)	**Screenings 16-10 mm** (% passing)	**Screenings 14-10 mm** (% passing)	**Screenings 10-7 mm** (% passing)
19.00	100	–	–
16.00	95-100	100	
13.20	40-70	90-100	100
9.50	0-10	0-15	90-10
6.70	0-2	0-2	0-10
4.75	–	–	0-2
1.18	0-1	0-1	0-1
Los Angeles Value (maximum)	25	25	25
Flakiness Index (maximum)	30	30	30

Crushed Rock and Quarry Rubble

A summary of selected specifications is presented in Table 2.4.

Table 2.4 *Crushed rock and quarry rubble specifications, Department of Road Transport, South Australia.*

Sieve size (mm)	**Crushed rock**		**Quarry rubble**		**Quarry waste**	
	20 mm (% passing)	**40 mm** (% passing)	**20 mm** (% passing)	**40 mm** (% passing)	**75 mm** (% passing)	**40 mm** (% passing)
106.00	–	–	–	–	100	–
75.00	–	–	–	–	90-100	–
53.00	–	100	–	100	75-97	100
37.50	–	95-100	–	90-100		90-100
26.50	100	79-91	100	74-96	51-75	–
19.00	95-100	65-83	90-100	62-86	–	60-80
13.20	77-93	–	74-96	–	34-58	–
9.50	63-83	44-64	61-85	42-66	–	–
4.75	44-64	29-49	42-66	28-50	19-40	25-55
2.36	29-49	20-38	28-50	20-39	13-30	–
0.425	13-23	8-18	11-27	8-21	5-16	–
0.075	5-11	3-9	4-14	3-11	2-9	–
Liquid Limit (maximum)	25	25	28	28	28	32
Plasticity Index (maximum)	1-6	1-6	1-8	1-8	1-8	12
Linear Shrinkage (maximum)	3	3	3	4	4	4
Los Angeles Value (maximum)	30	30	45	45	50	–

3. Environmental Management

Quarrying, and to a lesser extent the investigations carried out to prove a quarry site, have the potential to cause environmental damage.

However, many of these potential impacts can be prevented and others reduced through a simple process of considering the consequences of an operation before it begins and planning activities in a way which causes minimal damage.

Some impact is usually unavoidable during the operational phase but permanent effects may be removed by rehabilitation of the site when the operation is finished.

While it is important that emphasis be placed on the detrimental effects on the environment of a development, thought should be given to the benefits to the local community and State as a whole. Among these benefits are:

- Employment opportunities for local and imported workers.
- Increase in population with consequent increases in requirement for general stores, local trades and tradespeople.
- Quarry materials produced locally will usually be cheaper than imports.
- Export earnings.
- Rates and taxes and royalties paid to Local and State Governments for local and state-wide use.
- Availability of infrastructure — roads, railways, schools, parks and other public utilities — to the local community.

To be able to take advantage of these benefits and allow the development to proceed, attention must be paid to possible effects on the environment.

Environmental management has three main elements:

1. Defining the site and the operation proposed in order to identify potential impacts.
2. Planning the operation to prevent or minimise the potential impacts.
3. Progressively rehabilitating the site; this process commences once sufficient area has been mined with the final rehabilitation removing those impacts which were unavoidable during the life of the operation.

A fourth element — monitoring — must be applied to these activities to ensure that the planning and rehabilitation phases are effective.

What is the Environment?

The environment is that which surrounds us including its effects on life.

The environment of a quarry site includes:

- Physical features — landform, lifeforms, soils, surface and groundwater, sites and structures of heritage value to Aborigines and Europeans and other natural and man-made features.
- Flora and Fauna — the vegetation and the animals it supports.
- Human features — the people who live in the vicinity, those who travel past the site and those who use the site or adjacent land for income, recreation or pleasure.

Each quarry site has its own unique environment and the impacts generated by the same scale operation at different sites will vary considerably. The effects of a large limestone quarry in desolate flat, arid land may be minimal while the impact of a small scale operation on an exposed hill or within a wetland area, close to residential land, is likely to be considerable without careful environmental management.

Why Protect the Environment?

Land and the resources it supports are community assets which will be needed by future generations as much as the present.

As temporary users of land, which often belongs to another party, the quarrying industry has a responsibility to leave the sites of its operations in the same or a better state than when it first occupied the land.

However there are more compelling reasons for planning a quarry operation to reduce environmental impacts.

A well planned proposal, which demonstrates that environmental factors have been considered and no unnecessary impacts are generated, has more chance of being approved quickly without onerous conditions being attached to the approval.

Furthermore a good environmental performance at one site will ensure that the operator will have a better chance of gaining access to the next; a bad environmental performance at one site will usually require a more costly application for the next site and the possibility of refusal because of the poor record.

Thus sound environmental management is both a responsibility to the community and the key to future access to land.

Range of Environmental Impacts

An impact may be something which interferes with or reduces the quality of the environment or the amenity of an area.

All or any of the following impacts could be generated by a new quarrying operation:

- Visual impairment — pleasant natural landscapes can be destroyed by quarry workings, waste dumps, product stockpiles, workshops, crushing or washing plants, haul roads and other elements of a producing quarry.
- Nuisances — noise and vibration generated by blasting, treatment plants, operating machinery and workshops during working hours, and dust blown from the site, can spoil the environment of those living in or using the adjacent area.
- Water pollution — water quality in surface streams and other bodies of water can be adversely affected by silt, leachates, lubricating oils and other materials washed from a quarry site.
- Habitat loss — removal of vegetation, as well as creating visual impairment, may destroy the habitats of birds and animals and cause soil erosion.

The extent to which these impacts are generated depends upon the environment at each particular site and the type and size of the operation proposed.

Impact Identification

Successful environmental management begins with identification of the potential impacts which could be generated by a quarrying operation at that particular site. It is therefore necessary to have a good description of the original site and the elements of the proposed operation.

This can best be done by using enlarged scale colour aerial photographs obtained from the State's Department of Lands. These show vegetation, drainage channels, houses, roads and other features of relevance, but should be checked for changes since the photos were prepared. Contour plans are also available; these will show hills and valleys and undulating country and are invaluable for constructing line-of-sight diagrams.

On tracing paper laid over these photographs and plans, the outline of the mineral deposit and other geological features as well as mineral lease boundaries, and details of the proposed workings can be shown.

The aerial photos and tracings can be taken into the field and viewed from various vantage points when the actual impacts of all aspects of the operation can be assessed. The layout may then be modified to reduce these impacts and a programme for environment control prepared.

Models presenting the current and future appearance of the operation and ancillary works are very useful in the early planning stages.

Visual Impact

Any surface mining operation involves disturbance of a land surface. With existing operations it is quite probable that the only control available lies in screening by earth banks or trees, and in keeping the area in a neat, tidy condition. With a new operation it may well be possible, using contour maps, to locate all excavations, plant, dumps, etc., behind natural ground depressions or to make them blend into the natural background. It may also be possible to select a starting point that is invisible, whilst simultaneously embarking on a heavy tree planting programme so that ultimately the whole area is screened. Early in the operation dumps or 'bunds' can be constructed with the outer face sloped, covered with topsoil and planted with grass. It is well to remember that the use of paint on buildings can have a pleasing visual effect and that boundary fencing needs to be kept in a neat condition. In all these matters, which are visual, it is good policy to seek advice from such experts as landscape architects and trained horticulturalists (Figs 3.1 to 3.4).

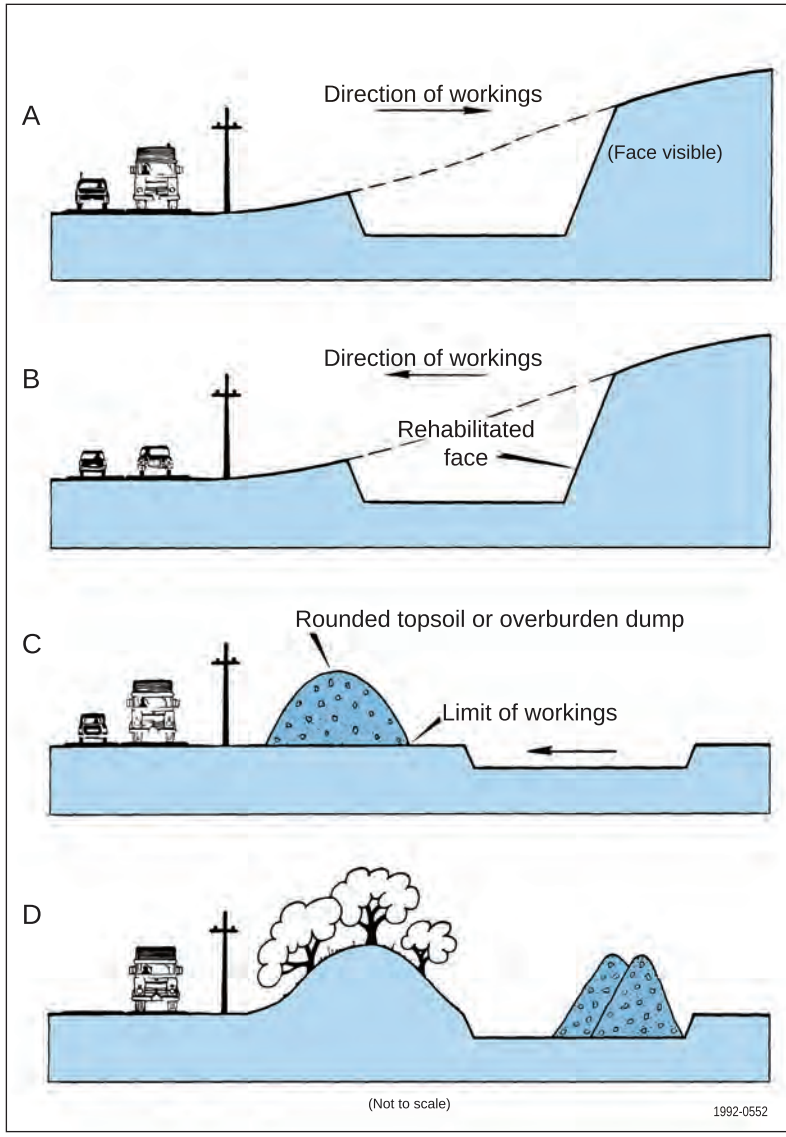


Fig. 3.1 *Direction of working influences visual impact.*

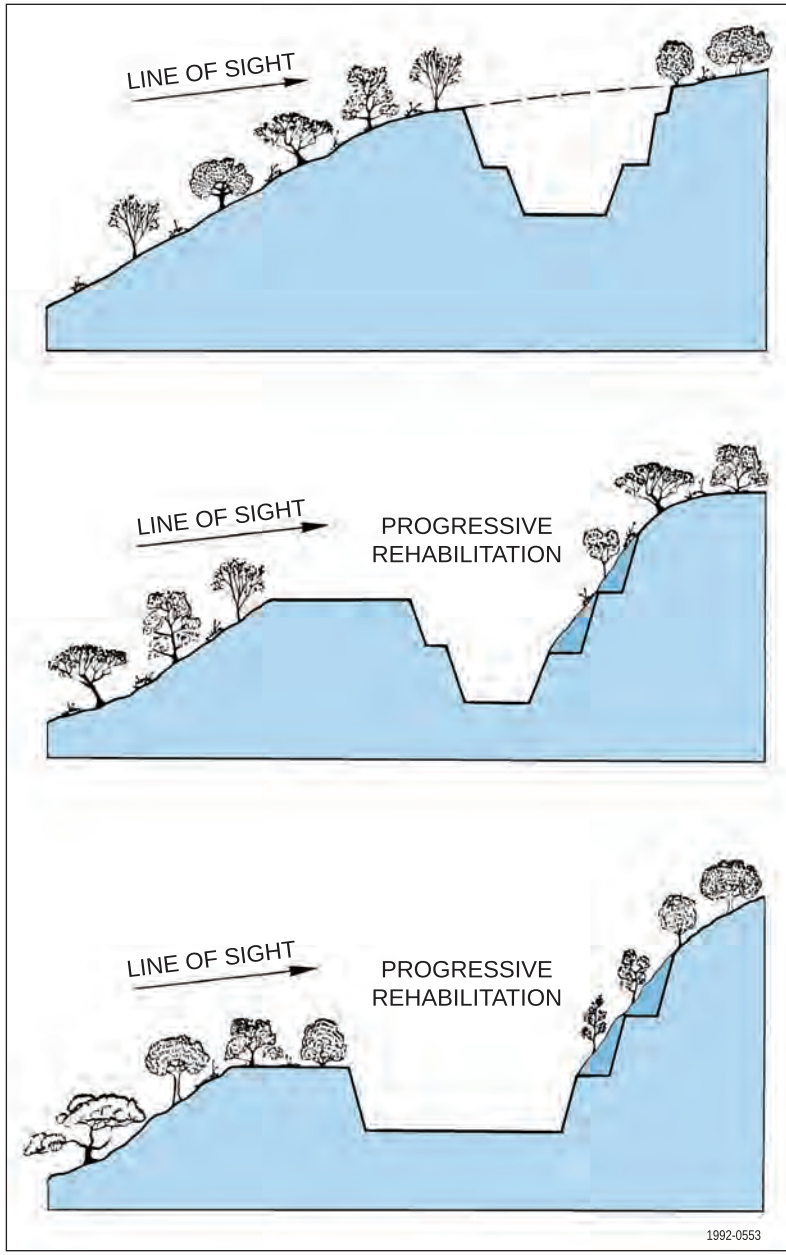


Fig. 3.2 *Screening of operations using vegetation and natural features.*

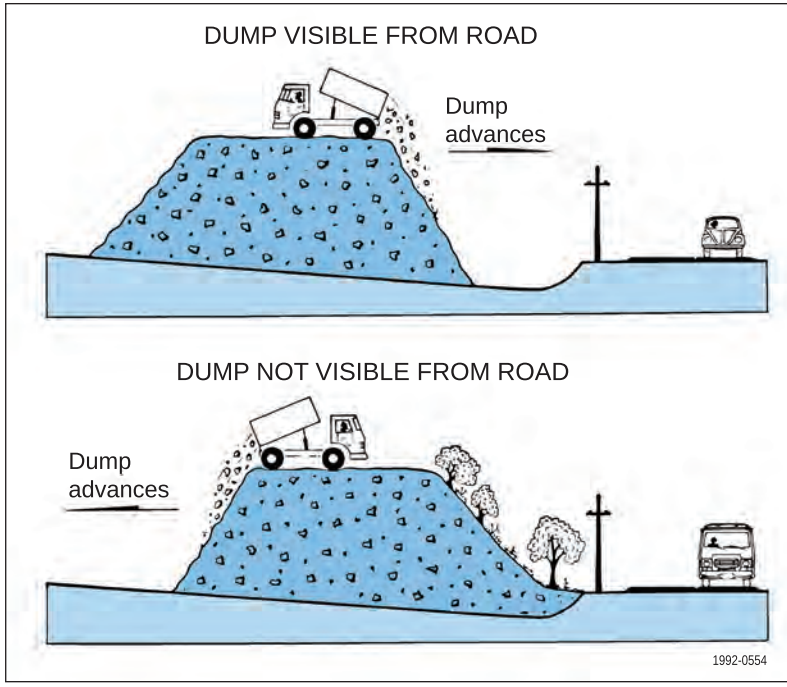


Fig. 3.3 *Reducing visual impact of an overburden dump.*

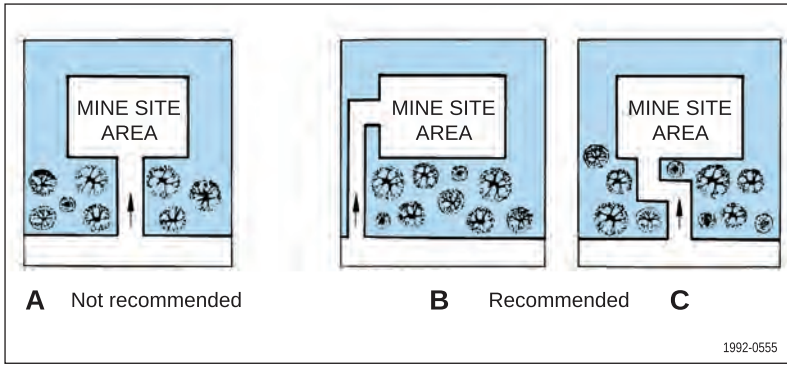


Fig. 3.4 *Use of trees and shrubs for screening.*

Noise (see also Chapter 17)

As an environmental factor, noise can be regarded as a nuisance. Quarry noises are normally associated with blasting and machinery. Blasting noises are usually attributable to poor practice and lack of management control. Machinery noises include rock drills, trucks and crushing plants. Control in these cases, as well as including the use of sound-proofing enclosures and earth mounds to deflect noise, might well have to include a restriction upon hours of work. If the operation is unreasonably close to housing, work hours may have to be tailored to fit in with the habits of the local people — nobody appreciates being wakened on a Saturday morning at six o'clock by the sound of a jack-hammer. If drilling has to be done on a Saturday, it is very likely that there is not enough drilling equipment on hand to meet production needs. Control of excessive noise often requires the attention of experts.

Dust (see also Chapter 17)

As an environmental factor it is the dust that affects persons living nearby that is of concern. Such dust is usually associated with road traffic and stockpiles. It is good management to keep stockpiles as far away from housing as possible. The use of water sprays can be beneficial both on stockpiles and on roads. Strict attention to vehicular speed may be an effective control. All vehicles leaving the property should have their loads wetted down. It is well to bear in mind that if a situation becomes very severe due to prevailing winds, then the final solution may be to cease operations until the weather improves. A drastic step perhaps, but if it is the only solution then it should be taken.

Dust from blasting is another area which may become troublesome. It may well be that the explosive being used is too strong for the rock being broken. Again it should be borne in mind that when there is an adverse wind blowing, blasting should not be undertaken. This necessitates the manager having his operations under control so that there is always sufficient broken rock lying on the floor — or in stockpiles — in order that the plant can continue working.

Dereliction of Land

Dereliction of land results from poor planning and poor housekeeping. Overburden dumps scattered needlessly about the property, heaps of unwanted rubbish, rags, metals and worn out engines and plant, and rusting sheds and buildings all give a derelict look to any operation. Dumps should be deliberately constructed to blend into the general topography. Never build a dump without first stripping and carefully storing all the top soil and vegetative material that would otherwise be buried beneath the dump: this material can be used at a later stage to give vegetative cover to the dump. Remember that trees and grasses will not grow on bare rock dumps.

Associated Activities

The most common activity associated with any quarry operation is the increase of traffic on adjacent roads. In addition it has become accepted practice to have premix concrete plants adjacent to a quarry. This activity, as well as increasing traffic density, usually results in an overall spread of working hours. As operations expand, more and more business people begin calling at the operation. All these factors increase the background noise levels. It should be remembered that much of this traffic needs parking space on the property — trucks must not be parked in the street obstructing the normal traffic flow. Driver education is important: courteous, careful drivers will bring the goodwill of your neighbours.

Pollution

Pollution is taken to include the deleterious effect on water supplies, stagnant water, or water to be used for recreational purposes, by acid or alkaline solutions or oily materials as well as offensive smells or gases, excessive dust and noise.

Not many stone quarries have overburden materials or waste rocks which are in themselves, or are likely to become, health hazards. However, the siting of a dump over a water course without provision for drainage could result in stagnant water accumulating in parts of the workings or the general area. Besides the provision of a suitable drain, a trap or settling pond is also required, otherwise the run-off from the dump could cause gravel or slime to flow onto adjoining properties or choke water courses lower down.

Dump materials in which sulphide minerals are present or which contain readily soluble minerals can cause sterilisation of land. Streams in the area can also be affected. This is particularly noticeable when the flow in the streams diminishes in the summertime.

Management needs to be aware that groundwater is a precious commodity. It cannot be seen and therefore is easily overlooked. Most Departments of Mines have hydrogeology sections which can advise on a given locality and it is wise to consult them. A groundwater resource once fouled may be lost forever.

Tailings dams should be sited at the head of catchments to prevent through-flow, and they should be drained and eventually covered.

Common causes of complaint near some quarries are the presence of oil, kerosene or grease in creeks or water courses, the smell from accumulations and the effect on vegetation when water from these sources is used on

gardens. Grease and oil traps should be built in drains near workshops and places where vehicles or machines are likely to cause spillage. They should be regularly cleaned out and the waste oils collected and removed from the property. Some local government authorities may demand complete containment of water within the site.

Rehabilitation

In some areas the existing use of land can be preserved. In other areas a new use can readily be prescribed for the property, taking into consideration its ultimate changed form. In a third category are those workings where no future use is apparent at the planning stage.

The 'amenity' of an area according to the Shorter Oxford English Dictionary means 'The quality of being pleasant or agreeable'. The amenity conditions imposed on quarry operations will vary greatly, since it is obvious that the amenity of a seaside suburb cannot readily be compared with that of a scenic mountain area or an area in a remote locality.

A few quarry sites can be restored almost to the original condition. Many lend themselves to a different after-use which may be of more value to the owner and the community than the original. In some instances where there was sparse growth of vegetation, a more luxurious appearance can be given to the landscape. Obvious questions can be posed. What is the nature of the planned district landscape? What will the site look like at the end of the quarry life? Would it be out of step with the surroundings? Would there be a better utilisation of the land?

There must always be an initial break in the harmony of an area with the first excavation for a quarry, just as there would be for a highway, a dam, or even a building. The more quickly measures are taken to remove or mask any scars the better. Even camouflage may be acceptable initially. But whatever preparatory scheme is adopted, it should, where possible, be related to the ultimate plan for rehabilitation and after-use of the land. Where an after-use cannot be laid down, the short and long term rehabilitation, while aiming to keep the site in harmony with its surroundings, must have flexibility to enable a variety of uses to be ultimately considered. It should make the site capable of some redevelopment.

When to start

Rehabilitation should start as soon as practicable, in some cases before actual site stripping or excavations. Due regard must be given to the season of the year as it may affect the best time for planting, but shrubs or trees must not be planted too close to blasting areas or dusty places.

In the case of dumps, rehabilitation can commence as soon as sufficient area has consolidated. Only enough room should be left on the dump for a road to the tipping edge, with space to manoeuvre for tipping. Dust on the road must be kept down by spraying the surface with oil emulsion and reducing the speed of trucks.

In the case of slime dams it is necessary to select special trees, shrubs and grasses that will flourish in damp and possibly acid conditions. If this vegetation is tested at an early stage it may, if successful, not only camouflage the dams but also help to consolidate them.

Where to start

As a general rule work should start at the top and progress downwards. Scars on higher ground are very noticeable from the plains and should be kept to a minimum. Dump slopes on sections which are in their final form can be vegetated. Vegetation is useful for covering up but it can also bind a surface and prevent erosion. Planting can be made at an early stage on ground that will not be disturbed by the actual workings and so could mask excavations planned to be made later.

What to plant

Preference for planting should be given to trees, shrubs and grasses indigenous to the area and encouragement should be given to the voluntary, so-called ‘colonising’, plants which are introduced to the site naturally, the seed having been brought in by birds or blown in by the wind.

After-Uses

Ultimate after-uses for a worked out quarry site include:

- outdoor recreation area (sport, picnic, or scenic)
- agricultural land
- pasture
- timber
- housing
- light industrial
- outdoor recreation area (swimming pool, boating lake)
- water supply, dam.



Stonyfell Quarry: working face in 1980. (Photo T022560)



Rehabilitated face in 1991. (Photo 040386)



Purple Downs Quarry, Roxby Downs: spreading topsoil on worked out quarry floor. (Photo 039289)



Topsoil after contour furrowing to assist revegetation. (Photo 039290)

Preliminary uses and work for the first six after-uses listed would be preparing the ground by:

- using the quarry as a tip for suitable material
- levelling the land using dumped material
- landscaping using appropriate dumped material.

The last two after-uses would require landscaping above the water line and possible sealing with clay if above the ground water level.

Some South Australian examples of rehabilitation:

Tea Tree Gully sand pits -recreation
Torrensville (Brompton clay pits) — light industry
Fort Glanville sand pit — caravan park
Tea Tree Gully Quarry — recreation
Maslin's Beach sand pit — recreation
Normanville sand pit — recreation and building sites
Glen Osmond Quarry — residential.

In considering the material that may be used for filling excavations, attention should be paid to its possible effect on any water supply and whether its use will cause a health hazard or pollution. Checks should be made of the ability of the material to pack down solidly or be compacted. If the material won't take loads, problems with foundations could result and the rehabilitated workings may have to be restricted to 'open space' or recreational areas in what might otherwise have been a suitable housing development scheme.

These overall considerations of the final position have some influence on the planning of the quarrying operation, and could affect bench interval, ultimate face slopes, size and positioning of dumps, and slime or tailings dams. The timing of programs for the re-vegetation of some worked-out areas, dumps and dams is also relevant.



Sewage treatment works built in an abandoned sand pit at Christies Beach. (Photo T024627)



McElligotts Quarry rehabilitated as a picnic area. (Photo 039129)

4. Geology

Useful rock and mineral deposits have formed in the earth's crust as a result of past geological events and processes. Rarely are two deposits alike because the features which control the quality and quantity of material are numerous and variable. Although the location and evaluation of mineral deposits usually require the specialised techniques of the professional geologist, a working knowledge of geological features and processes can help the quarry manager in day-to-day operations.

Rocks and Minerals

Rocks are composed of aggregates of minerals. Some rocks, such as sandstone, limestone and iron ore consist predominantly of only one mineral, but most rocks comprise mixtures of minerals. For example, granite is made up of quartz, feldspar and mica with smaller quantities of other (accessory) minerals.

Although over 2 500 different minerals are known, less than 20 are commonly found in rocks. The most frequent of these are quartz, feldspar, mica, amphibole, pyroxene, calcite, dolomite and various iron and clay minerals. An elementary book on mineralogy will help quarry workers to become familiar with these minerals.

Classification of Rocks

Rocks are classified according to their physical and chemical characteristics (lithology) and their age of formation. Both of these systems are used on modern geological maps.

Lithological Classification

The type of minerals present, their relative abundance, grain size and arrangement (texture) form the basis of the lithological classification. For example a rock composed of quartz, potash feldspar and mica is classed as a granite, but where soda feldspar predominates, the name granodiorite is used. These rocks can be further described as fine or coarse grained. If the same minerals are arranged in distinct bands the rock is called a granite gneiss.

Limestone consists predominantly of fine particles of calcite but if the calcite is coarse grained and crystalline the rock is classed as marble.

Using the lithological classification three groups of rocks can be defined:

Igneous rocks are emplaced in a molten state as 'magma' either at depth in the Earth's crust or on the surface through volcanoes. The depth of emplacement controls the rate of cooling and thus the grain size. The slow cooling rate of deep-seated magmas allows the formation of large well developed crystals and a coarse grained texture results (e.g. granite). Magmas which flow onto the Earth's surface from volcanoes cool rapidly and are fine grained (e.g. basalt and rhyolite).

Igneous rocks are generally hard and, when not excessively coarse grained, are suitable for concrete and other aggregates. Volcanic rocks require careful testing because they often contain secondary minerals which may have deleterious effects on concrete.

Examples of igneous rocks quarried for aggregate in South Australia are the basalts at Mount Schank and Mount McIntyre in the South East, granite at Port Lincoln, and rhyolite which has been quarried for rail ballast near Tarcoola and Keith.

Sedimentary rocks are formed through the accumulation of mineral particles or sediments. These may be derived from an eroding land mass and transported by water or wind to be deposited in layers; they may result from the accumulation of shells or other organic material; or they may be composed of minerals precipitated by chemical reactions.

As successive layers of sediment accumulate, the buried layers are compacted and lose water. The mineral grains are often cemented together by circulating fluids.

Many sedimentary rocks are deposited under the sea (marine deposits) but accumulations in fresh water lakes, along major river systems and on the land surface are common. Later earth movements, such as mountain building processes, can elevate marine sediments to thousands of metres above sea level.

Sedimentary rocks show great variation in physical properties, even within individual deposits, and very careful testing is necessary before quality can be assured.

Clays, shales and limestones are examples of sedimentary rocks and the dolomites and quartzites which supply metropolitan Adelaide with aggregate are examples of sedimentary rocks which have undergone low level metamorphism.

Metamorphic rocks form when existing rocks are changed by heat and/or pressure to develop new minerals and/or textures.

Although sedimentary rocks undergo minor changes during compaction, the original features are clearly preserved. However, if sedimentary or igneous rocks become deeply buried or are involved in major earth movements where conditions of temperature and pressure are drastically changed, new minerals are formed and other features such as banding are imprinted on them. In other instances, such as the conversion of limestone to marble, no new minerals are formed but the grain size of the original calcite is considerably changed.

Many metamorphic rocks are hard and even grained (e.g. quartzite), making them suitable for aggregates. In others, the arrangement of mica minerals causes the rock to break into thin platey fragments unsuitable for aggregate (e.g. schist and slate).

In South Australia gneiss is quarried at Rocky Gully near Murray Bridge. Similar rock has been used extensively for road construction on eastern Eyre Peninsula.

Classification by Age

Since the earth's formation about 4 500 million years ago, its crust has undergone dramatic changes.

Systematic study of the fossil record and, more recently, measurements of radio-active decay in certain minerals, has permitted the geological time scale (Fig. 4.1) to be compiled. All of the rocks in the earth's crust can be assigned to the particular period of geological time in which they were formed.

Rocks of similar mineralogy can differ considerably in age and this has important consequences in the search for new deposits. For example in South Australia, fossiliferous limestones of the Cambrian Period are hard and suitable for aggregate but Tertiary fossiliferous limestones, although chemically similar, are soft enough to be sawn into building blocks. This is because the older rocks have undergone metamorphism.

Rock units of similar lithology which have wide regional extent or possess other characteristic geological features, are commonly given a

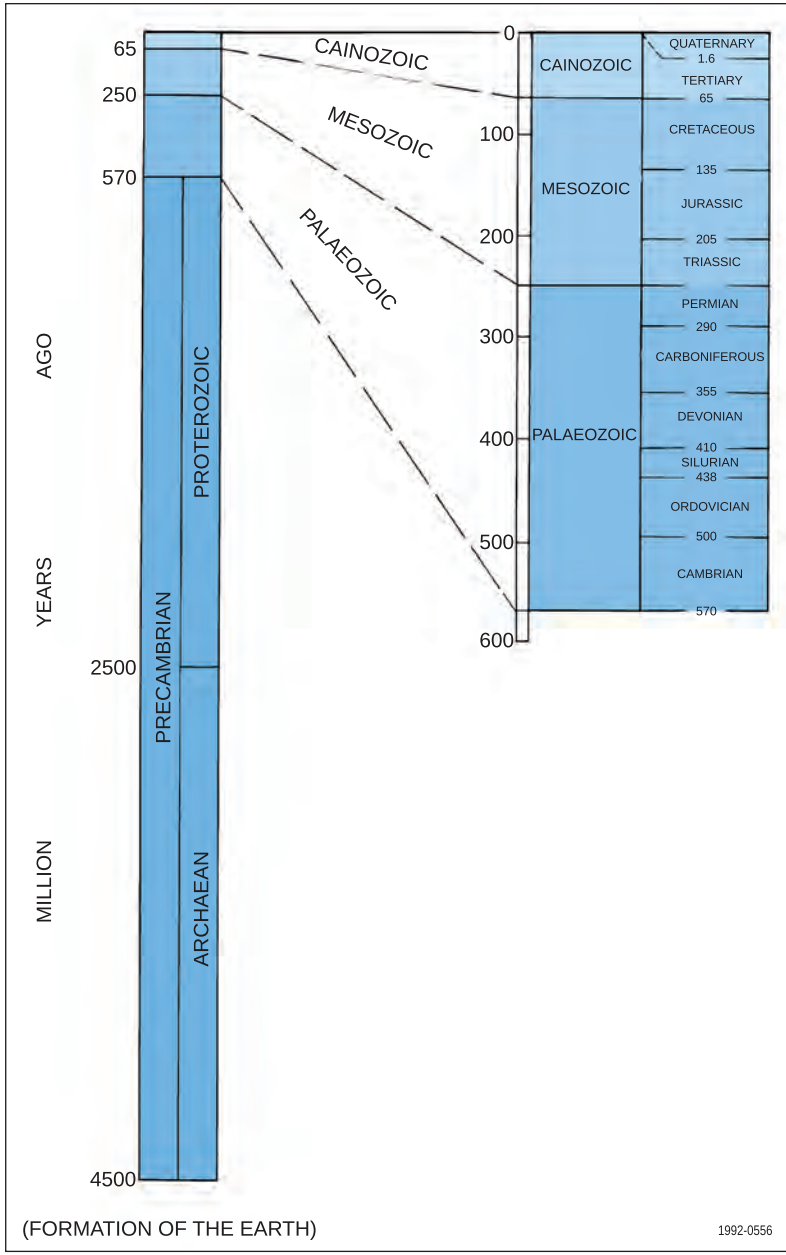


Fig. 4.1 *Geological time scale.*

formal (stratigraphic) name which is used on geological maps. The name incorporates a lithological description coupled with the geographic locality in which the unit is well exposed. Thus we have: Brighton Limestone; Palmer Granite; Hindmarsh Clay; Leigh Creek Coal Measures. In some cases a significant unit has persistent but variable lithology and names such as 'Radium Hill Metamorphics', 'Renmark Beds', 'Marree Formation' are used. The lithological range is then described on the legend of the geological map.

Rock Structures

The earth's crust is not stable but is subject to continual change by the agents of erosion and from deformation associated with the relief of internal stresses.

Stress relief processes are permanently expressed in the rocks by structures such as folds, faults and joints. This section is concerned only with the rock structures which affect the distribution of useful deposits, the properties of the materials in the deposits and the safety of quarry faces.

Crustal stresses vary greatly, ranging from those on a continental scale which were responsible for building high mountain ranges, to those of more local extent such as earthquakes and the opening of joints along the walls of deeply-incised river valleys.

Rocks differ in physical properties such as strength and melting point while individual lithological types react differently to the same set of conditions. For example, under extreme pressure a hard brittle rock such as quartzite will fracture or snap whereas a limestone will adjust by flowing into areas of lower stress.

Folds

Sedimentary sequences laid down in a horizontal position are frequently compressed by lateral forces into folds, much as the pages of a book crumple when pushed from the side.

An anticline is a fold where the beds arch upwards while in a syncline the beds are bent downwards into a trough. The axial plane of a fold is the plane which divides the fold as symmetrically as possible (Fig. 4.2).

In some folds the axial plane is vertical but more commonly it is inclined (Fig. 4.3). Where the axial plane is inclined to such a degree that the limbs (flanks) dip in the same direction, the fold is said to be overturned. A fold is said to plunge when the axis (the trace of the axial plane on a particular bed) is not horizontal (Figs 4.4 and 4.5).

Individual folds range from many kilometres across to only a few millimetres. In large scale folds, the anticlinal axial region is often eroded and only the limbs are seen.

Folding can affect the reserves of material available for quarrying in a deposit (Fig. 4.6).

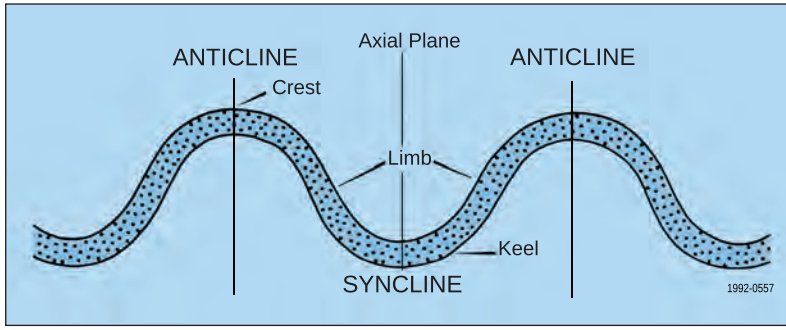


Fig. 4.2 *Elements and types of folds.*

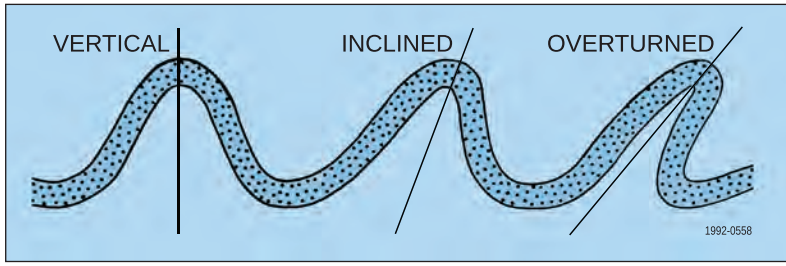


Fig. 4.3 *Attitude of axial plane.*

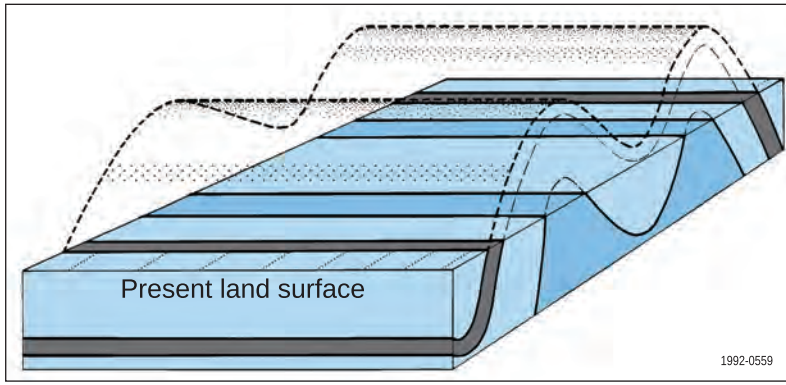


Fig. 4.4 *Non-plunging folds.*

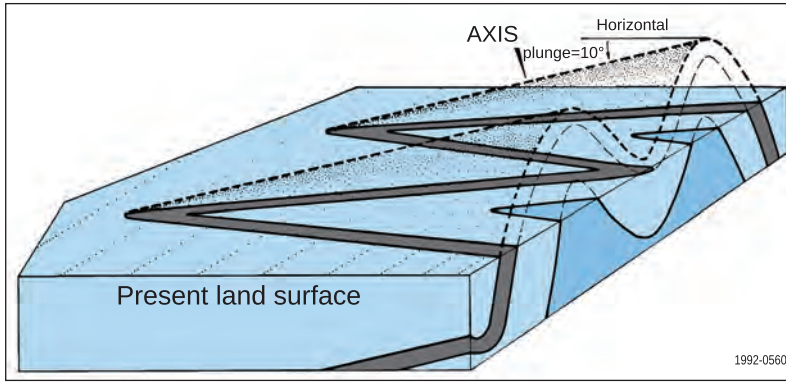


Fig. 4.5 *Plunging folds.*

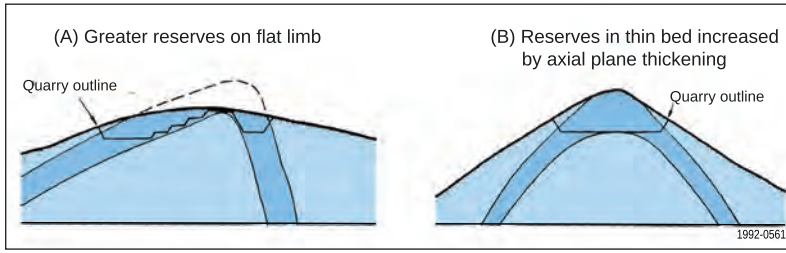


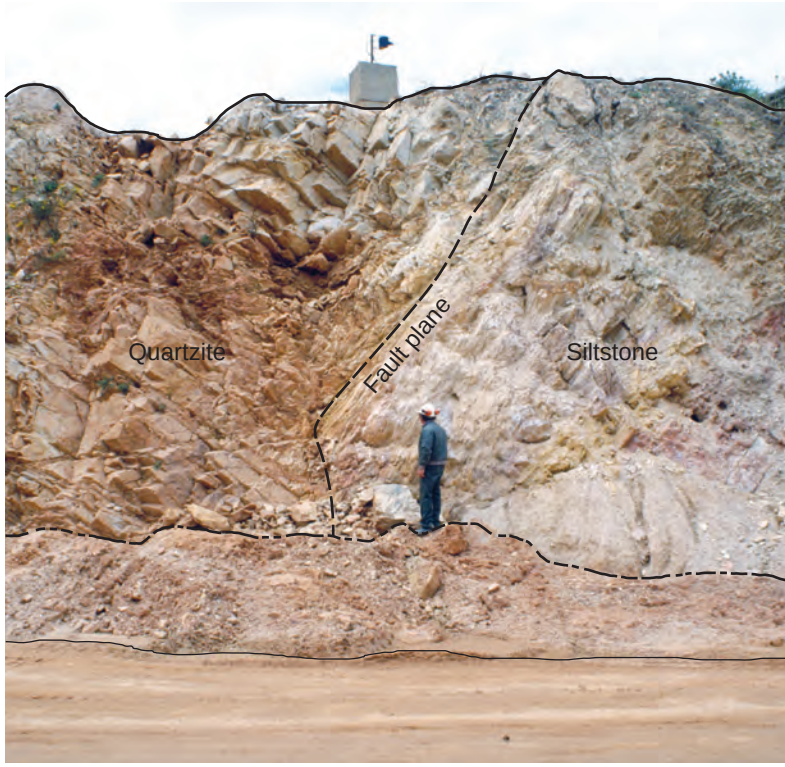
Fig. 4.6 *Effect of folding on available reserves.*



Syncline at Linwood Quarry. (Photo 040432)



Complex anticline at Riverview Quarry. (Photo 040433)



Fault bringing quartzite into contact with siltstone at Para Hills Quarry. (Photo 040434)

Faults

A fault results when deformation is so intense that disruption or breakage of a rock mass occurs. Although more easily seen in sedimentary rocks, faults are also common in igneous and metamorphic rocks.

Movement along a fault plane may be either horizontal (strike faults) or vertical (dip faults). In many cases the movement is oblique, giving both a horizontal and vertical displacement.

Faults vary from major dislocations where movement continues intermittently over long periods of time and where overall horizontal displacement may measure many tens of kilometres or vertical displacement many hundreds of metres, to smaller movements of only a few centimetres. In the former, the fault zone may be several hundred metres wide and be made up of many individual fault planes.

Where the movement is in a nearly vertical plane, faults are classified as normal or reverse (Fig. 4.7).

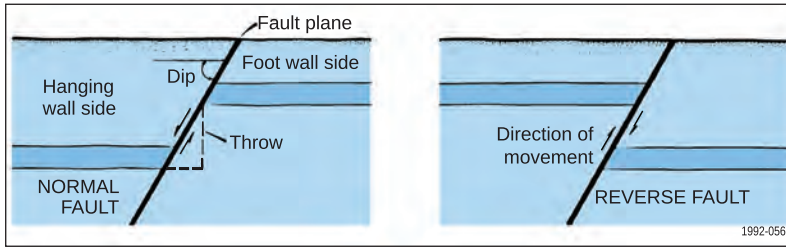


Fig. 4.7 *Fault types and elements.*

The rock adjacent to the fault plane is often broken (brecciated) and frequently the fault zone is filled with pulverised rock (gouge). These features may make the rock unsuitable for use. Depending on its dip, a fault can make quarry faces extremely dangerous (Fig. 4.8). This is particularly so when water, entering from above, mixes with the clay gouge and may soften it but more importantly will increase the hydrostatic pressure between the fault surfaces tending to separate them. Faults may have a significant effect on available reserves and if undetected can have a disastrous consequence on the economic life of a quarry (Fig. 4.8).

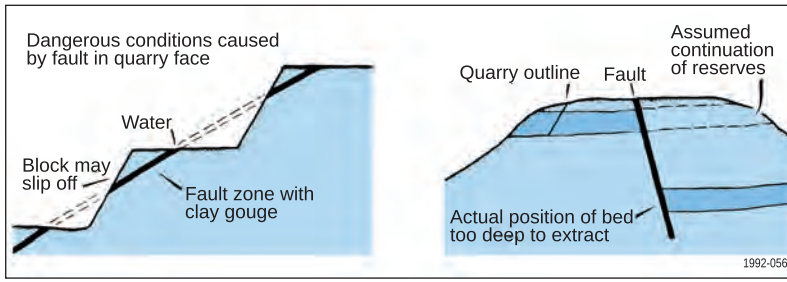


Fig. 4.8 *Effect of faults on safety and reserves.*

Bedding

Sedimentary rocks are made up of layers of particles usually laid down horizontally under water. This layering (bedding) can be clearly seen if the size of particles in neighbouring layers changes abruptly. When there is a pause in deposition so that the upper surface hardens, or when there is a change in the type of particles, the two beds are sufficiently different with the result that, on subsequent exposure in a quarry or cliff face, they will break along the bedding plane. In some cases breakage only occurs along the bedding plane after the rock has been exposed to the atmosphere.

As in the case of faults, the dip of bedding planes relative to quarry or cliff faces can have a major effect on safety: dip away from the face may not give rise to problems but dip towards the face can be very dangerous.

The size to which sedimentary rocks break after blasting depends on the spacing of bedding plane weaknesses, so careful testing of rocks with closely spaced bedding planes is necessary to ensure their suitability for the coarser ranges of aggregate.

Rocks with widely spaced bedding planes are referred to as ‘massive’; those where more closely spaced bedding plane weaknesses cause large slabs of rock to break off are described as ‘flaggy’.

Cleavage and Schistosity

Fine grained rocks of sedimentary origin frequently display closely spaced parallel planes of weakness lying at an angle to the bedding planes. This feature is referred to as cleavage and is due to the alignment of platy minerals such as mica into parallel orientation at right angles to compressional forces. Cleavage may be developed locally in the limbs of folds where the orientation

of mineral grains and cleavage planes approximates that of the fold axis, or it may be a product of strong regional compressive forces developed during later geological events. In the latter case the cleavage direction is constant over a large area and may not necessarily be parallel to the axis of local folds. In cases where cleavage planes are the dominant rock weakness, bedding planes are often visible only after close scrutiny.

Rocks in which platy metamorphic minerals such as mica and chlorite are well developed with parallel orientation to the extent that they impart a cleavable texture to the rock are known as schists.

Joints

Most rock faces are broken by smooth faced fractures which in sedimentary rocks are not parallel to bedding and along which there is no evidence of movement. These are referred to as joints. They originate in a number of ways, for example as a result of decreases in volume during the cooling of igneous rocks, during folding or faulting, or from the relief of lateral stress developed after the removal of overlying rock by weathering and erosion.

Joints are generally straight (planar) but may be curved. They may extend for only a few metres or may be up to several tens of metres long.

Joints always occur in parallel groups or sets and the spacing between individual joints may range from a few centimetres to several tens of metres. Joint sets may dip at any angle and frequently two sets, often at right angles, are present.

Where the spacing is not too close, joints are of great benefit in the breaking of rock faces during quarrying. However, like faults and bedding, they can create dangerous conditions in quarry faces.

Joint planes are usually more open near the surface and may be filled with clay or coated with deleterious iron minerals. Below the water table, joints provide storage space for groundwater and good supplies can often be obtained by drilling at sites where groundwater can enter outcrops of hard closely jointed rocks (Fig. 4.9).



Bedding in quartzite, Para Hills Quarry. (Photo 040435)



Joints controlling final shape of quarry face, Stonyfell Quarry. (Photo 040436)

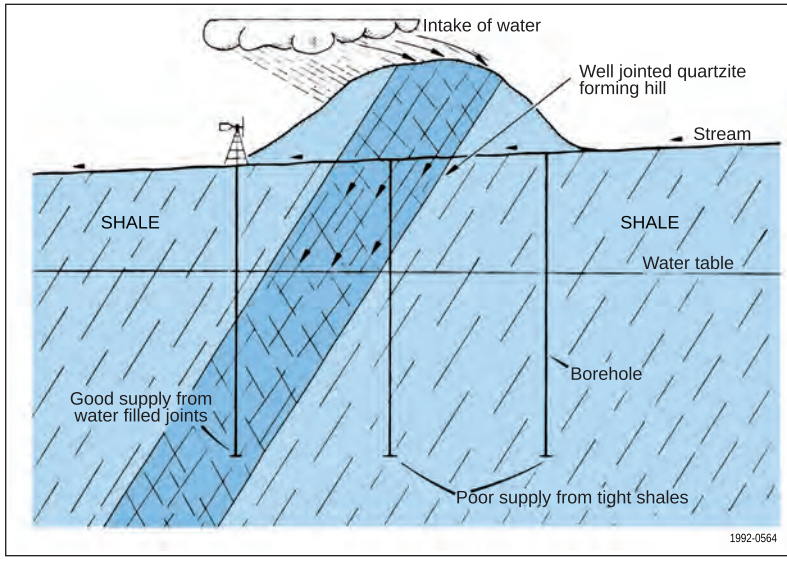


Fig. 4.9 *Effects of jointing and rock type on water supply.*

Strike and Dip

The attitude of all planar and near-planar rock structures such as bedding, faults and joints may be recorded by two simple measurements (Fig. 4.10).

The strike of a plane is the direction of a horizontal line drawn along the plane, and in perfectly flat terrain is the direction of the trace the structure makes on the land surface. It is measured with a compass and expressed in degrees clockwise from magnetic north.

The dip is the angle which a line drawn down the plane, at right angles to the strike, makes with the horizontal. It is expressed in degrees. Note that a horizontal bed has a dip of zero degrees and no strike measurement.

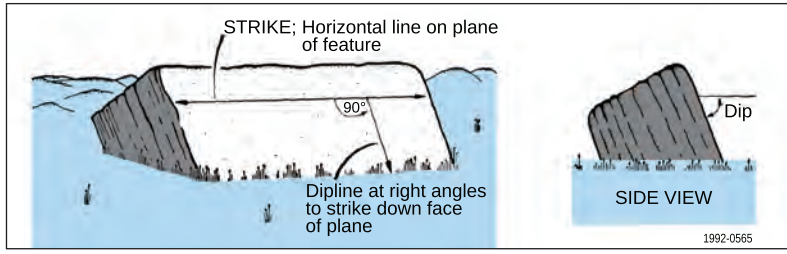


Fig. 4.10 *Strike and dip.*

Weathering and Erosion

Throughout geological time the surface zone of the land has been subjected to continual attack from the agents of weathering and erosion. In some cases useful materials result: for example, the alteration of slates to clay shale suitable for brickmaking, or the accumulation in stream channels of sand and gravel for use in concrete. However hard rocks may be softened and altered by weathering so that they become unsuitable for aggregates. Useful deposits may be concealed by a covering of erosion products.

A knowledge of these processes and their effect on the surface is necessary when evaluating mineral and rock deposits. Figure 4.11 shows how a completely erroneous estimate of reserves could be made from the surface inspection of a thin quartzite bed covered by the products of its own erosion.

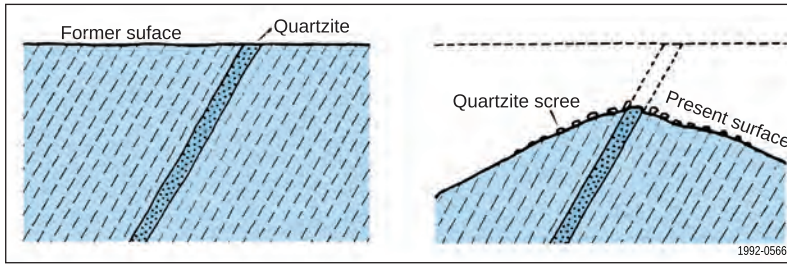


Fig. 4.11 *Scree slopes developed from thin quartzite, giving erroneous surface indication of volume of reserves.*

Application of Geology

Rock and mineral deposits are non-renewable resources and new supplies will be required to provide for the needs of future generations.

In the settled areas the more obvious deposits have already been opened and new deposits in these areas are harder to find and subject to environmental constraints. Major development projects in more remote areas require large quantities of materials and usually there are no existing sources of supply. Modern mining and planning legislation requires the submission of detailed quarry plans for development work and rehabilitation, and companies obliged to do this find that the forward planning contributes significantly to efficiency and economy.

The State Geological Surveys provide a wide range of services, from the publication of geological maps showing the regional distribution of potentially useful deposits, to detailed reports on individual properties. These data are available free or for only a nominal charge and the quarry operator is urged to use these services wherever possible. Larger companies with many holdings employ their own geological staff to handle these problems.

5. Drills and Drilling

The main types of drills used for drilling blast holes and for sampling rock formations in exploration and production operations are hammer (percussion) drills, rotary drills and diamond drills. Jet piercing is used to a limited extent overseas for blast holes, originally in extremely hard iron ores. In South Australia at the norite workings at Black Hill and the granite quarry at Calca, jet burners are used in conjunction with pneumatic hammers to cut out blocks of dimension stone. Auger drills are used in soft formations for sampling and occasionally for blasting. Cable tool (churn drill) plants are occasionally used for constructing water wells and for sampling in soft to medium hard formations but are rarely used for blasting. Reverse circulation drilling is now an important technique used in exploration of materials.

Hammer Drilling

This is the most direct descendant of the original hammer-and-tap method of blast hole drilling and merely replaces the hand operations with mechanical hammering and rotation powered by various means.

Until recently, the most common source of power was compressed air, but units now range from electric, hydraulic and small petrol internal combustion powered hammers to huge diesel powered hammers used primarily for pile driving. Drills used in mining are almost invariably of pneumatic or hydraulic type.

Pneumatic percussion tools used in quarrying and mining range from small ‘jack-hammer’ machines to ‘long-hole’ drifters mounted on a wide range of carriers, and ‘down-hole hammers’.

Surface Hammers

In all drills fluid pressure is used to force a piston back and forth to strike the drill steel, piston reciprocation being achieved by means of a self-acting or automatic valve. The valve action is set by the designer and built into the machine.

Hydraulic top-hammer drills use hydraulic oil instead of compressed air as the energy transmission medium. The higher pressure obtainable with hydraulics results in greater penetration rates, reduced energy costs and increased efficiency (Fig. 5.1).

While in smaller machines the steel is rotated by the piston actuating a rifle bar and ratchet, in the large drifter type drills rotation is achieved by an independent rotation motor which is usually built onto the back end of the machine.

Rotation must be provided so that successive blows of the bit are moved around the face of the drill hole in such a way as to allow the rock to break (spall) into chips of the maximum size that can be flushed from the hole. Rotation must not be so fast that the spacing between impacts is too great to create chips that will break off. Generally, the more abrasive the rock being drilled, and therefore the greater gauge wear on the bits, the slower should be the rotation speed.

The smoothness of rotation is dependent on the number of pawls and teeth on the ratchet ring and the fastest, most even rotation is achieved with four pawls and an uneven number of teeth. Even wear results and back-lash is reduced to a minimum because the bar needs to rotate only one-quarter of a tooth to engage the new tooth. Rotation speeds of 30-80 rpm are normal (Fig. 5.2).

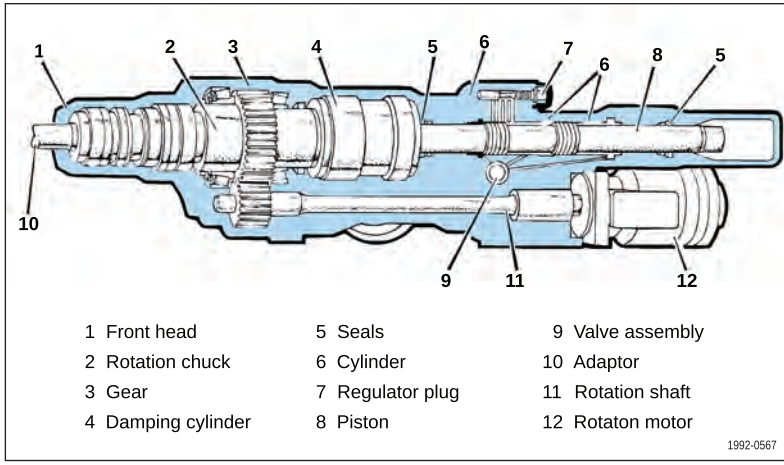


Fig. 5.l *Section through hydraulic rock drill (Atlas Copco).*

The effect of thrust on the rotation speed and the performance of conventionally rotated rock drills is very important. If the thrust is too great and the bit is jammed on the rock face, or if the hole is being drilled through broken ground and the drill rod becomes jammed, the drill can stall. If insufficient thrust is applied, the bit may lose contact with the rock face during the rebound (return of piston) and the drill steel can spin freely and rotate far more that it should. Rebounding also causes loosening and cracking of the carbide inserts. Thrusts of 9-90 kg per 25 mm of bit diameter are used with percussion drills.

Lubrication is an important feature of rock drill operation and a good quality rock drill oil must be used to lubricate moving parts in the usually cold, moist environment associated with the use of compressed air and flushing water. It is strongly recommended that air line lubricators be placed not more than 3 m from the drill and that each drill be provided with an individual lubricator.

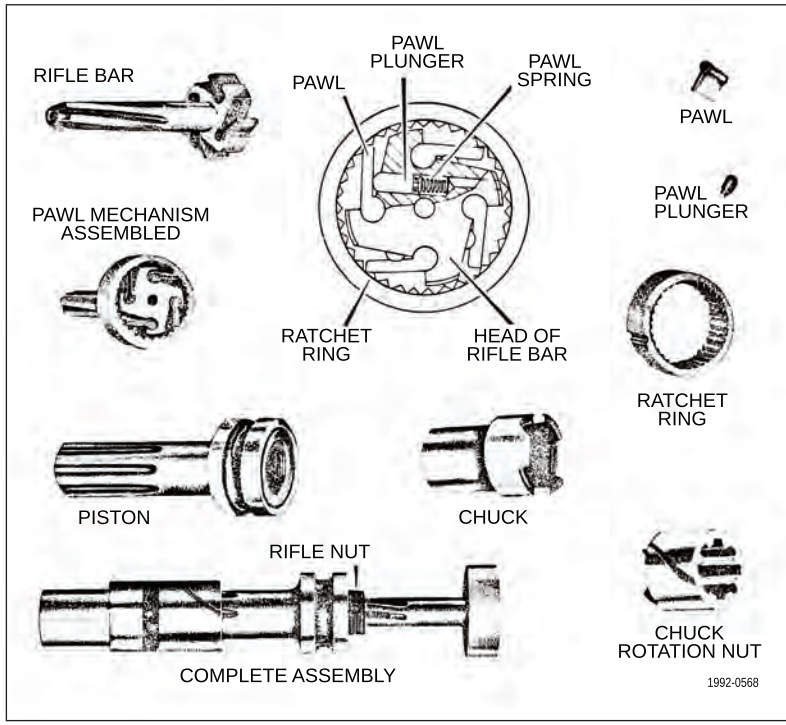


Fig. 5.2 *Details of rotation mechanism in a pneumatic rock drill.*

Cuttings are removed by water and/or air delivered through the hollow drill steel. The amount of flushing medium required depends on hole angle, size, depth, drill steel dimensions and the formation being drilled. It should be noted that a surface hammer drill will cease to operate if the backpressure of water encountered in the hole is greater than flushing air pressure.

Drill steels are available in a wide range of sizes and with various thread connectors. Thread forms used are of a long pitch with minimum notching, such as rope threads, ISO buttress threads and Hi-leed threads.

Tungsten carbide-tipped drilling bits are either integral with the rod for comparatively small diameter holes or, more commonly, detachable. The main types of detachable bits are: pigtail (for light extension drilling equipment), bottom drive (the steel butts against the strong head of the bit), shoulder drive (the steel is upset to provide a shoulder), and the taper drive (ideally used with a copper thimble and primarily in mining operations often with air leg machines).

Both cross and button bits are available in most types of bits. Button bits are appropriate for hard ground and cross bits for softer formations. Generally the button bits give a smoother drilling operation and a more nearly circular hole. Also, because of the rotation and blow rate imposed by the machine, button bits perform better in large diameter holes.

Down-Hole Hammers

In general, more efficient use of compressed air power results from the use of down-hole hammers when drilling deeper holes, as the piston blows are transmitted directly to the bit without losing percussive energy through the drill steels. The drill string (or drill rod) can be made considerably lighter as it merely imparts rotation to the bit and acts as an air passage to the hammer. A comparatively large diameter, thin wall, internal flush drill pipe allows use of lighter hoisting equipment, reduces frictional losses in the pipe and lessens the annular area between drill pipe and wall of the hole. More effective hole cleaning (and less regrinding of cuttings) becomes possible with greater up-hole air velocities and more efficient use of compressed air.



Jet burning at Black Hill norite quarry. (Photo 040437)



Down-hole hammer drilling. (Photo 040438)

Whereas surface hammers are usually described by the piston diameter, the down-hole hammer is designated by its overall diameter. Thus a 108 mm hammer has a piston approximately 95 mm in diameter and will drill a 120-127 mm hole using, say, 90 mm drill pipe. A hammer of this size requires approximately 7 m^3/min of air with a minimum 7.0 bar pressure drop across the hammer. This imposes two limitations:

- If water is intersected in the hole so that back pressure is placed on the hammer exhausts, then the pressure drop across the hammer and penetration rates are reduced; with depth and more water a stage is reached where the hammer will not operate.
- The hole and drill pipe sizes must be such that the air velocity is sufficient for effective hole cleaning to avoid regrinding of rock cuttings. An accepted minimum up-hole velocity is 14 m/s but with heavy cuttings such as iron ore, 20-25 m/s is required, and most operators will seek 20 m/s in average conditions.

Figure 5.3 shows details of the down-hole hammer operating cycle. When the bit is lifted from the hole bottom the full air volume is allowed to pass without operating the hammer. This allows accelerated initial and periodic hole cleaning. The hammer commences operation as soon as the exhaust valve is closed by running the bit to bottom.

A choke to regulate air through the hammer and a non-return valve to eliminate invasion of the hammer by water and cuttings are generally provided in the back-end of the hammer.

Independent top rotation motors powered either pneumatically or hydraulically are usually used on drilling rigs designed for down-hole hammer work, but kelly drive, rotary table machines can be used so long as rotary speeds are adequate. (Kelly drives are best suited for tall masts and deep holes where it is desirable to reduce top weight.)

Slow rotation (10-25 rpm) is required for most hammers to minimise abrasive wear on carbide inserts and rapid loss of gauge.

A down-hole hammer, like a surface hammer, requires a moderate amount of down pressure to effect maximum penetration rates. Typical total weight range on the bit for a 150 mm bit would be from 450 to 1000 kg.

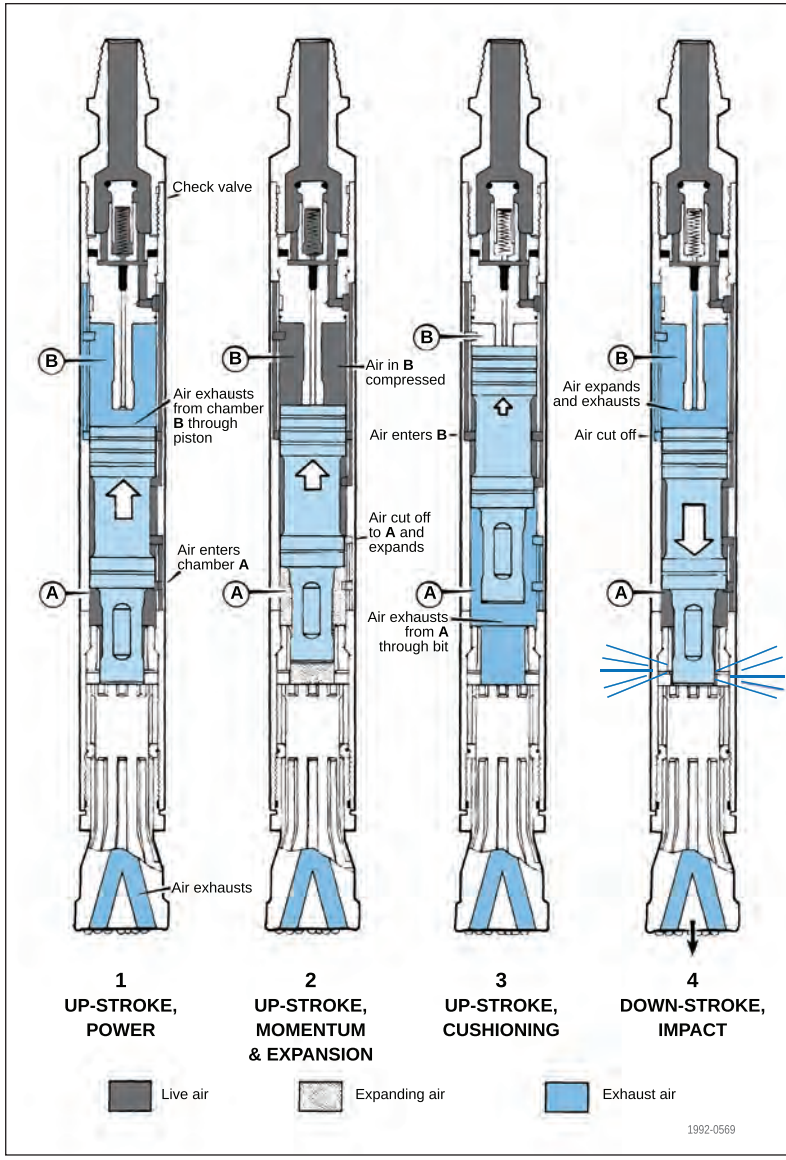


Fig. 5.3 *Section through a down-hole hammer (Ingersoll Rand DHD-260).*



Down hole hammer. (Photo 040439)



Kelly drive for rotary rig. (Photo 040440)

Water may be injected into air lines to prevent the hole from 'gumming up' when drilling through strata containing small quantities of water. From 5 to 10 litres/min is usually sufficient.

Other air additives may be used to create a mist or foam to assist in hole stabilisation. It is essential, however, to allow no foreign solid matter into the air stream and therefore the hammer. Drill pipe must be kept clean and dirt-free inside and each joint should be blown clean before being added to the drill string.

Machine Mounting

Jack-hammers, weighing usually 15-30 kg, are often hand-held. For near-horizontal holes pneumatic 'air-legs', and for up-holes 'stopers' provide both support and thrust for these hammers.

Wagon drills are the simplest form of mobile mounting and these light (often three-wheel) trailers are used for both surface and down-hole hammers. The carrier incorporates an adjustable mast, or slide, which allows the feed mechanism to advance the surface hammer or the down-hole hammer rotation motor with a set feed pressure. Wagon drills are usually not self-propelling, although some have small pneumatic motors fitted to the front (steering) wheel to negotiate short distances between holes over level country. Most wagon drills are of comparatively light construction to carry 75-100 mm rock drills weighing up to 90 kg and operating with 75-100 mm bits and 25-40 mm drill steels. Alternatively 60-105 mm down-hole hammers can be used with 40-90 mm drill pipe. Both units operate with an auxiliary compressed air supply.

A separate feed and hoist motor operated by air or hydraulic power and semi-automatic controls provides hoisting and variable feed by way of a chain. Thrust is limited by the weight of the machine. This is often less of a limitation with hammer drilling than with rotary drilling.

Air tracks are heavier machines. These tracked carriers are suitable for either surface drills or down-hole hammers and are invariably self propelling with sufficient power to tow the necessary compressor on reasonably flat ground. Drive motors, rotation motors, feed mechanism and auxiliary operations can be either pneumatic or hydraulic. Modern air tracks incorporate an hydraulically positioned drill boom that is capable of drilling holes in a wide area in front of the tracks and at virtually any angle from horizontal to several degrees behind vertical. Heavy air tracks can carry one or, in some cases, two rock drills of up to 150 mm (piston diameter) or equivalent down-hole hammer equipment.

Wagon drills and air tracks are primarily designed for one-person operation.

Truck or crawler mounted drills are unitised self-powered and self-propelled rigs which are available in a very wide range of sizes and configurations. Generally, these rigs are designed for down-hole hammer or conventional rotary-air-blast drilling.

The majority of blast hole drills used in quarry and open cut mining operations are track-mounted, while exploration or water well drills are usually mounted on appropriately sized trucks, using their power sources, but larger units may have one or more auxiliary motors to power some or all of the drill components.

Hydraulic power is replacing air power for drilling component motors — rotation, hoisting, feed, pumps, injection pumps, dust extraction systems etc. The hydraulic pump powering these motors is driven by the rig motor as is the main compressor which provides air for the down-hole hammer and/or flushing.

Large blast-hole machines now incorporate a number of refinements designed for more efficient drilling and greater operator comfort; these features include:

- Hydraulic levelling jacks.
- Centralised controls now often placed in an enclosed, noise proofed, air conditioned cabin.
- Semi-automatic pipe handling, allowing up to 45 m of drill pipe to be held in the mast and added to the main drill string as required without any manual handling.
- Auxiliary hoisting gear for adding drill pipe to the automated pipe handling assembly.
- Effective dust collection and sampling equipment to virtually eliminate the hazard of dust to operators and machinery.
- Separate hoisting winch.
- Hydraulically powered breakout and make-up of drill pipe connections (tool joints).
- Hydraulic mast-raising cylinders with the ability to drill angle holes, and angle measuring devices; this latter feature is generally limited to about 30° off vertical.

Surface Versus Down-Hole Hammers

The selective use of surface hammers is generally limited to shallow drilling (to about 30 m) and holes less than 90 mm in diameter.

Advantages and disadvantages of **surface hammers** are:

Advantages

- low unit cost
- rugged construction
- heavy percussion blow available with large piston to bit diameter ratio
- all expensive components are on the surface
- lower cost bits
- amenable to hand operation
- air and/or water flushing possible.

Disadvantages

- depth limited due to percussive power loss in drill rod and couplings
- hole size is limited owing to power loss and small drill rod inside diameter for flushing medium
- noisy
- hole sizes generally not sufficient for slurry explosives
- large machines and associated mountings are very heavy
- hole direction (straightness) is difficult to control.

The **down-hole hammer** has the following features:

Advantages

- percussive blow is applied direct to the bit, irrespective of depth
- depth capabilities dependant chiefly on air pressure
- hole flushing is more efficient as a result of large inside
- diameter of drill rod and low pressure loss
- more economical use of air as hammer exhaust is used for hole flushing
- quiet operation, particularly with hydraulically powered surface components
- lighter construction of drill unit possible
- lighter drill rods
- greater hole sizes possible — suitable for slurry explosives
- hole direction generally easier to control
- can utilise high pressure air (20-30 bar)
- availability of misting and foaming agents to maintain hole stability.

Disadvantages

- minimum hole size of 100 mm
- lower piston diameter/bit diameter ratio
- expensive hammer down hole
- more expensive bits
- need to maintain operational air pressure drop across the hammer limits depth capacities below water. (This can also apply to surface hammers drilling deep holes).

Rotary Drilling

Auger Drills

Auger drilling is restricted to soft formations such as sand, clay, coal and the softer shales and sandstones; its main use is for sampling. Bucket or short flight augers require the cutting head to be withdrawn from the hole to raise and so remove or spin off the drill cuttings. Continuous flight augers have flights which can be extended in the same way as drill steel extensions and the cuttings are brought to the surface continuously.

Hole diameters range from 50 to 1500 mm and depths to 90 m.

Rotary speeds are slow but torque requirements are high — generally necessitating heavy, high power machines. Cutting shoes are generally replaceable steel teeth or blades, hard faced by electric welding techniques or with the addition of tungsten carbide.

Augers are also used to drill horizontal holes up to 3 m in diameter and 120 m deep to extract coal from the final limits of open cut coal mines.

The larger sizes such as the Calweld (bucket type) drill are not used in quarrying but have a widespread use on opal fields for exploration and for shaft construction.

Reverse Circulation Drills

Where hole sidewall failure is a problem, either because of difficult hole progress or contamination of drill cutting samples, reverse circulation drilling can be used. In this method the circulating fluid does not come in contact with the drillhole wall (Fig. 5.4).



Auger drill. (Photo 035777)



Calweld (bucket type) drill. (Photo 035776)

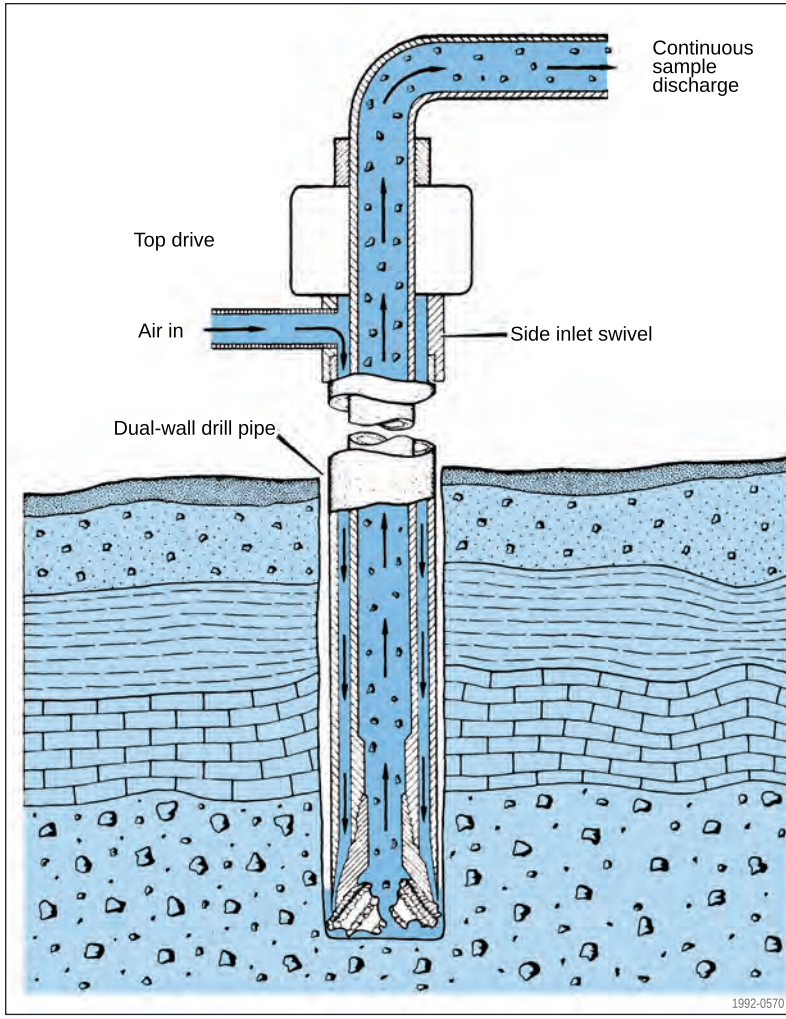


Fig. 5.4 *Reverse circulation drill.*

Rotary Drills

Rotary drilling with air, water or mud circulation involves four basic functions:

- rotation of the drill bit attached to the drill rods
- provision for applying adequate weight on the drill bit
- ability to hoist the drill rods from the hole and add additional rods as necessary
- pumps to force a suitable fluid around the bit and so carry the formation cuttings to the surface.

Each of these functions can be achieved in a number of ways, depending on the nature of the operation.

The depth and size of hole required, the purpose for which the hole is constructed, the type of formation being drilled, the terrain and the availability of services, all are factors which decide the overall type of machine and the drilling method required to achieve economical performance.

The general use of ammonium nitrate-based explosives has necessitated larger diameter blast holes for most open pit mining and quarrying operations, and both down-hole hammer and air rotary drilling have been introduced to meet this demand.

Circulatory Fluids

Air circulation flushing enables rapid drilling, minimises the time involved in setting up and moving, and eliminates the need to cart water or introduce other circulation fluids. The compressor is often an integral part of the drilling unit.

Low pressure air from 5.5 to 7.0 bar is sufficient for holes up to 36 m deep and an annular velocity of from 15 to 25 m/s is sought for effective hole cleaning. This velocity is achieved by using appropriately sized air compressors and drill pipe for the hole size required.

Bad standing country and/or a high water table can preclude the use of air as a circulatory medium — particulary in exploration drilling — and in these cases water or mud liquids are used to stabilise the hole, cool the bit and remove the cuttings. A positive displacement pump unit is required in such cases to circulate the liquid at appropriate volume and pressure. An annular velocity exceeding 1.0 m/s for water and 0.67 m/s for mud is generally required to clean the hole effectively and to prevent regrinding of drill cuttings.

Operational Features

Practically all of the larger machines used in quarry or open pit mine operations are capable of drilling angle holes to 20° off vertical and are mounted on tracks which are powered by either hydraulic or electric motors. The largest rigs (bit diameter over 250 mm) normally drill only vertical holes.

The smaller units are often truck mounted as are the more mobile exploration and development drilling machines. The latter drills are generally equipped with both air compressors and mud pumps to cater for a wide range of rotary drilling applications. These units are often restricted to vertical drilling and may be fitted with an additional hoisting winch and mast working platform or 'monkey-board' to enable drill pipe to be racked vertically and so eliminate the need to lay down drill pipe when changing the bit on deep holes.

The more conventional rotation of rotary table and kelly bar has given way to either a telescopic kelly, to enable the required depth to be drilled without adding additional pipe, or the provision of hydraulic top rotation drive and carousel, to enable one person to add successive lengths of drill pipe safely and efficiently.

Rotary drilling requires considerable weight on the bit and heavy-wall pipe and adequate pull down. The machine itself must therefore be heavy enough to counter this pull down.

Dust suppression is an important feature of all blast hole drilling, particularly where the country being drilled is siliceous or contains fibrous material such as asbestos. If water is not used, cyclones or other extraction devices must be incorporated on the surface to collect the cuttings and clean the return air.

Injection systems may be used to add water, detergents, oil or foaming agents to the air stream to facilitate hole stability and air cleaning.

Bit types vary according to the physical nature, hardness and abrasiveness of the formations being drilled. Soft materials such as coal, clays and some shales can be drilled with blade or finger bits which usually have tungsten carbide hard facing or cutting edges. Harder formations, such as limestones, marbles and schists can be economically drilled with roller rock bits with either hard faced steel teeth or tungsten carbide insert teeth on the usually three cone configuration of the bit.



Rotary rig drilling inclined blast holes at Penrice Quarry. (Photo 040441)

In some hard rock areas (iron ore, for example) where hole sizes up to 400 mm are used and big tonnages are produced, rotary drilling is proving economical and penetration rates are becoming better than those achieved with down-hole hammers.

Many operators have demonstrated that rotary drilling with air flushing can provide a rapid and economical means of drilling blast holes in average ground, and even in very hard rock they may show performance advantages over down-hole hammer units when large diameter holes are required.

The following advantages and disadvantages apply to **rotary drilling**:

Advantages

- generally lower cost than for down-hole hammer drilling
- no expensive percussive tools
- lower pressure air
- less expensive compressor unit
- heavy-wall drill pipe and sturdy connections minimise downhole losses, particularly with large diameter drilling
- wide range of bits available
- down-hole hammers can be used instead of rotary bits.

Disadvantages

- need to provide adequate pull down usually demands a heavier machine than the down-hole hammer tools
- slower penetration rates in hard formation on holes of smaller diameter (say less than 150 mm)
- heavy duty sealed bearing roller cone button bits are generally restricted to commonly used oil field sizes.

Diamond Drilling

The diamond drill has been used in underground mining for the accurate drilling of long blast holes and ring patterns and in surface mining for drilling very hard rock; but nowadays its main application is in exploration work for securing of core.

The requirements of a diamond drill are similar in all respects to that of rotary drilling. The conventional bevel gear drive, which imparts rotation to the drill rod string through the limited travel of an hydraulic feed head and a manual chuck, is replaced on modern machines by hydraulically powered top rotation motors. This eliminates the need for re-chucking after each 600-900 mm of feed and enables longer passes (up to 6 m) to be continuously drilled with controlled weight on the bit.

Wire line coring equipment, whereby the inner tube of the core barrel and therefore the core can be recovered on a wire line and overshot (a 'hook' being used to 'catch' the core barrel), also enables more continuous drilling. With this equipment drill rods require pulling from the hole only to replace the bit and this, as well as the considerable saving in time, minimises hole damage and possible bit damage when commencing each core run.

The use of wire line equipment necessitates a second winch to carry the light (4.7 mm) cable which must be capable of running the overshot inside the rods to the bottom of the hole to pick up the inner tube of the core barrel.

Diamond drills must be capable of angle drilling and, with the exception of the pump unit, all functions are normally powered by the one power source. Most surface drills are petrol or diesel powered while those used underground may be driven by air or electricity.

Water is the common circulating fluid used, although considerable success is now being achieved with air circulation in shallower holes. The use of wetting agents (primarily soluble oils) in small (1-3%) concentrations also enhances diamond bit life and generally assists rod lubrication so achieving smooth running.

Diametral clearances between bits, barrels, rods and the walls of the hole are very close in most diamond drilling and, until the development of low solid polymer mud systems, this precluded the use of mud fluids with standard equipment. Standard hole sizes range from 29.8 mm diameter (core 18.3 mm) to 75.0 mm diameter (core 43.7 mm).

Exploration drills vary in size and capacity from small air leg or bar mounted machines capable of drilling 30-150 m deep, to large diesel powered skid mounted units with a capacity in excess of 3 000 m. Common drills are diesel powered, often trailer mounted with integral mast and are capable of drilling to 600-1 200 m, depending on size of hole and equipment being used.

Choice of Drill

Among the factors to be considered in making a choice of a drill for blast hole work in quarries are the following:

- characteristics of the rock formation: thickness, dip, jointing, cleavage, and bedding planes
- characteristics of the rock itself: texture, hardness, toughness, abrasiveness, grain size, uniformity
- locality
- height of faces in the proposed quarry

- blasting practice including
  - limitations on the amount of explosive fired per delay
  - angle of holes to be drilled
  - diameter of holes to be drilled.

The overall cost of winning rock should include drilling, blasting, loading, hauling, clean-up, and primary crushing. In these circumstances it is not necessarily the drill with the cheapest cost per metre which is the most economic; it may well be the one that can put down more holes per shift or one that can drill a larger hole in a certain time. The blasting practice has to be considered with the drilling operation. A little more money spent on drilling and explosives to obtain good fragmentation could result in much improved loading, hauling and crushing costs, and produce the final product at a lower cost per tonne.

The drillers' logs and the sludge, chip and core samples of the drill holes put down in the preliminary exploration of the deposit are of great help in assessing the drillability of the rocks. Nevertheless, the choice of drills for a particular quarry is largely determined by on-site tests with various drills, including the blasting of charges in the test holes. For these tests as much weathered rock as possible should be removed from the drill site.

Compressors

The choice of drilling and other air operated equipment will influence the size and type of compressor to be installed. The use of portable compressors has become universal, superceding a central compressor house with its long pipelines and consequent friction losses.

Types of Compressor

While the reciprocating (piston) type of compressor operating at up to 800 rpm is still in use, it is being replaced by the rotary screw compressor which has the following advantages:

- compactness
- a small number of moving parts, generally six (four bearings and two rotors)
- no metal to metal contact (approximately 0.05 mm clearance between rotors)
- cooler air produced through isothermic compression
- quieter in operation than a reciprocating compressor.



Diamond drilling. (Photo 040442)



Typical drill bits. From left to right: (top row) crossbit for surface hammer. crossbit for downhole hammer, tungsten carbide (TC) button bit for downhole hammer, dragbit with TC inserts, replaceable blades for dragbit; (bottom row) tricone roller bit with TC buttons, milled tooth tricone roller bit, TC chip core bit, TC insert calyx core bit, diamond core bit. (Photo 040443)

General Requirements for Air Reticulation

Proper lubrication is essential and, where possible, air-line lubricators should be used.

The compressed-air supply must be clean, and provision must be made to drain receivers and air-lines of condensate and oil.

Air-lines should be sufficiently large to reduce friction and provide a reservoir of air close to the tools. Sharp bends should be avoided and flexibility of connection is most desirable.

The minimum diameter of the air main depends upon the volume of air required at the tools and the loss of pressure through friction in the distribution system.

To determine the air requirements of the total number of rock drills, what is termed a 'diversity factor' is used. This takes into account that, under normal operating conditions, all available drills and other air operated equipment are not working at the one time as various operations, such as steel changing, must be carried out while the air is cut off. The altitude above sea-level also must be considered. Table 5.1 gives the multipliers to be used in this regard:

As previously mentioned the trend is towards portable rather than stationary compressors, but in either case skilled and efficient servicing is essential as a compressor out of action means loss of hole production. Such loss is accentuated if only one large compressor is installed. Thus, it is better to install two or three smaller units than one large one.

Compressed air is costly to produce and constant checking of pipes, hoses and couplings for leaks is essential for economic operation.

Table 5.1 *Multipliers to determine air consumption of rock drills at various altitudes.*

Altitude (m)	**Number of drills**					
	1	2	3	4	5	6
0	1.00	0.90	0.90	0.85	0.82	0.80
300	1.00	0.95	0.93	0.87	0.84	0.83
600	1.10	0.97	0.95	0.92	0.88	0.86
900	1.10	1.00	1.00	0.95	0.92	0.90
1200	1.10	1.05	1.03	0.97	0.94	0.93
1500	1.20	1.10	1.07	1.02	0.98	0.96

6. Exploration

This chapter deals mainly with methods of outlining and reporting on possible aggregate sources, and gives only passing attention to commercial, environmental and planning matters which need to be addressed during the assessment.

The choice of an area in which to search for a deposit is determined by market requirements and economic cartage distance. For example, requirements for a deposit to supply a single crushing contract for road sealing aggregate or rail ballast will differ from those for a permanent quarry near one of the major population centres.

Investigation generally follows five distinct stages:

Stage 1 Office Studies

As much relevant information as possible should be gathered on the search area, including the following:

Geological maps

The most detailed maps available should be used, preferably at a scale of 1:50 000 or better. Old 1:63 360 (1 inch to 1 mile) maps contain useful data; some are out of print but may be available in libraries. 1:250 000 scale maps are widely available but detail which can be presented at this scale is limited.

Geological Reports

Mineral resource reports, or reports of previous quarry site investigations may be available from the various State Geological Surveys.

Bore Records

Borehole records are maintained by the State Geological Surveys. Boreholes with full geological logs are most valuable and drillers' logs may contain useful data.

Aerial Photography

Coloured aerial photographs with stereoscopic coverage are useful for locating outcrops. In South Australia these are available from the Department of Lands.

The main aim of the office study is to select geological units and sites to visit during the field reconnaissance. Geologists at the Department of Mines and Energy can offer advice at this stage.

Selection of target geological units depends on the type of material being sought. For example:

- If sand is required for concrete manufacture, coarse creek sands will be sought in preference to fine dune sands, although dune sands may be adequate for filling or packing purposes.
- If screenings are required for road sealing, bitumen affinity is important so dolomites, basalts or rhyolites are likely to be sought in preference to quartzites. Quartzite screenings however are adequate for use in concrete.
- The presence of some shaly interbands in a quartzite unit may be advantageous in the production of road base or sub-base where minimum plasticity of the fines is specified. Such interbands may be deleterious if a deposit is to be used solely for the production of concrete aggregate.

Environmental Factors

Environmentally sensitive areas such as native vegetation, heritage sites, geological monuments and water courses should be identified during office studies.

Legislation

Before commencing a field programme, advice on legislative requirements should be sought from the Department of Mines and Energy.

Stage 2 Field Reconnaissance and Site Selection

During the field reconnaissance, one or more sites are selected for preliminary site investigation. This selection is made by examining the best possible exposures of the selected geological units to confirm that they are likely to be suitable. Exposures in road and rail cuttings and creekbeds give an idea of depth of weathering and the nature of the fresh rock. Spoil from dams and post holes may also provide useful data.

When the most suitable unit for the purpose has been selected, sites within it which may be amenable to quarrying are chosen for further investigation.

Topography plays an important role in site selection, not only because of practical quarrying considerations such as haul road and plant location, but also through its effect on quarry visibility and the extent to which further screening can be achieved by mounding and tree planting. Environmental factors such as native vegetation, heritage sites, proximity to housing and impact of trucking routes should also be taken into consideration.

Stage 3 Preliminary Site Investigation

This phase of the investigation should provide an indication of the quality of the material and a preliminary reserve estimate as well as identifying environmentally sensitive areas, so that a decision can be made on whether a detailed site investigation is warranted.

For a sand deposit this generally entails a broadly spaced program of machine auger or hand auger holes.

The first stage in assessment of a rock quarry site is geological mapping to determine the limit of the deposit, the distribution of the various geological units which comprise it, and their structural relationships. Such mapping is done either with a theodolite and staff or an electronic surveying instrument or on an enlarged aerial photograph. The Department of Lands publishes orthophoto maps which superimpose topographic contours on an aerial photo at 1:10 000 or 1:2 500 scales. These are particularly useful for geological mapping.

Backhoe trenching in areas of poor outcrop is often a cheap and effective way of locating geological boundaries or of detecting the presence of poorly outcropping inferior material such as shaly interbands within a quartzite or dolomite deposit. Backhoe trenches may provide samples for initial testing. Results of testing of surface spalls should be treated with caution as outcropping rock is often much harder and more erosion resistant than other parts of the deposit.

Petrographic examinations of rock samples should be made to confirm suitability and to detect deleterious constituents such as secondary minerals which are common in some basalt deposits.

Preliminary reserve estimates are made from geological cross sections.

Stage 4 Detailed Site Investigation

This phase of the investigation confirms reserves and provides sufficient data to prepare a development plan. It should not be hampered by lack of funds: there is little point in investing millions of dollars in plant and equipment if the deposit contains a large proportion of unsaleable material, or if, for some geological reason, reserves are substantially less than expected.

In sand deposits this stage involves infill auger drilling to supplement the preliminary work. Contamination is often a problem with open flight augers, so some reverse circulation drilling is advisable, particularly for testing deposits deeper than about 8-10 m (Fig. 5.4).

Investigation of a rock quarry site should include a diamond drilling program. Small diameter cores may be adequate for visual logging and petrographic examination, but core for mechanical and physical tests should be at least 50 mm in diameter. The diamond drilling program aims to:

- confirm geological structure and distribution of the different rock types
- assess degree of weathering and alteration
- assess nature and spacing of jointing and other rock defects which may have an effect on blasting procedures or stability of quarry faces
- provide samples for petrological, physical and mechanical testing.

Diamond drill holes should be carefully sited to provide maximum useful information (Fig. 6.1).

Percussion drilling is of limited use in evaluating rock deposits, because of up-hole contamination and the fineness of samples. Penetration rates and colour of dust or sludge may be useful indicators, but generally percussion drilling results should be used with extreme caution as they are often misleading. Chip samples from reverse circulation drilling are more reliable, but for quarry site investigations there is no substitute for a carefully planned diamond drilling program.

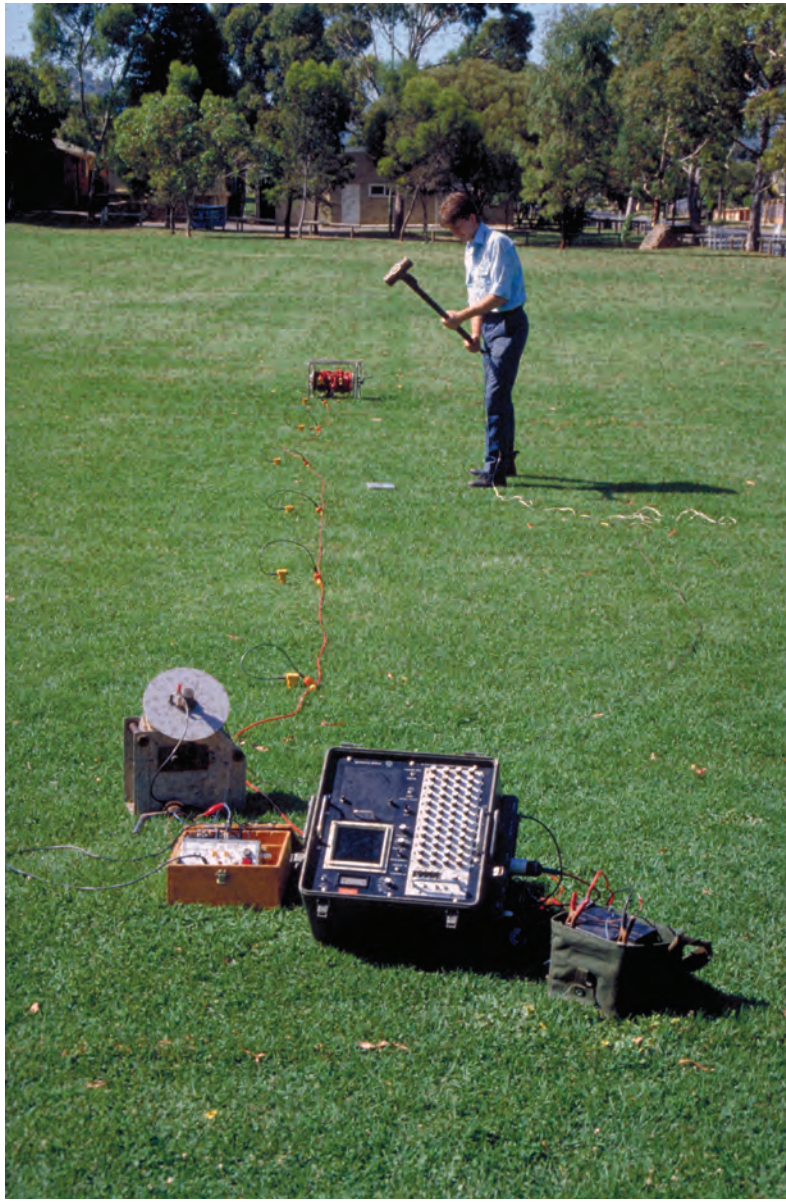
Geophysical techniques may be of use in some circumstances; for example, seismic techniques can assess rippability of surficial materials such as calcrete.

The presence of groundwater has a major impact on mining operations. If mining near or below the water table is contemplated professional advice should be sought to design a programme to investigate de-watering and mining techniques.

A geological plan and cross-sections are compiled and reserves calculated.



Diamond drilling to prove reserves at Williamstown kaolinite - sillimanite opencut. (Photo 036420)



Hammer seismograph used in assessing rippability. (Photo 040444)

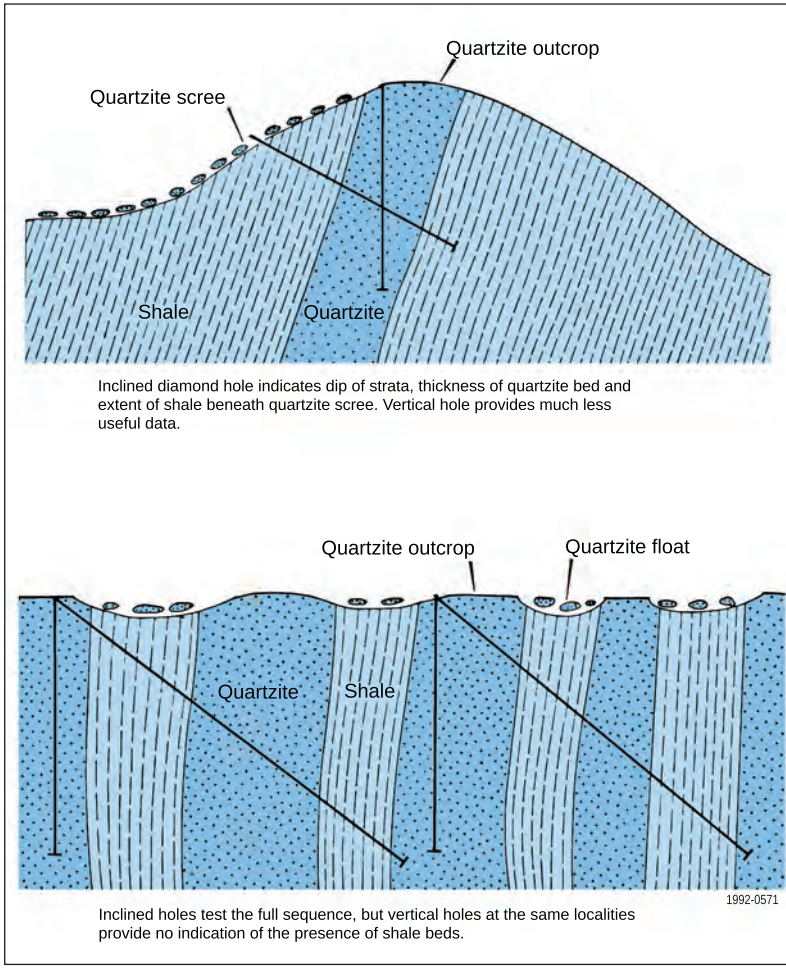


Fig. 6.1 *Use of inclined drillholes for proving reserves.*

Stage 5 Preparing a Development Plan

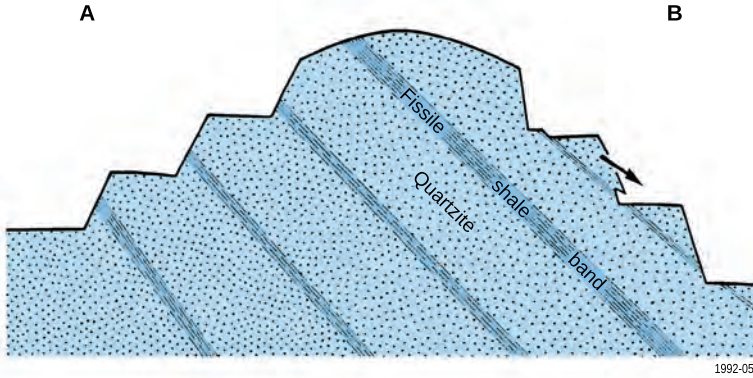
Information gathered during the detailed site investigation, including data affecting quarry face stability (Fig. 6.2), is used to compile a development plan (see Chapter 13).

A development plan should include:

- geological plan and sections showing extent of the deposit and reserves
- ultimate limit of proposed workings
- stages of development and progressive rehabilitation
- location of overburden dumps, crushing plant, buildings, haul roads, tailings dams, screening mounds, tree planting etc.
- rehabilitation plans.

A carefully prepared development plan assists the operator with long term planning and prevents costly mistakes such as siting plant or overburden dumps on future reserves. Information compiled during preparation of a development plan can also be used in obtaining the necessary environmental approvals and as documentation for obtaining finance.

The Department of Mines and Energy has published Guidelines for the preparation of a development programme which lists matters which should be considered and the form in which the programme should be presented.



Quarry opened in hillside at **B** will be more susceptible to instability than one opened at **A**. Similar instability can be caused by strongly developed joints or intersecting joint sets.

Fig. 6.2 *Geology and quarry stability.*

7. Explosives

Explosives are solid, liquid or gaseous substances or a mixture of substances which, on application of a suitable stimulus to a small portion of their mass, are converted quickly into other, more stable substances, largely or entirely gaseous, with the development of heat and pressure.

In all explosive reactions at least part of the products must be gaseous so that work can be performed by their expansion. When blasting powder 'explodes' it produces 500 times its own volume of gases when measured at ordinary temperatures, but it also produces heat which expands these gases to 4 000 times the volume of the original blasting powder. Similarly, the gases formed by the explosion of one volume of nitro-glycerine (liquid) would occupy 1 200 volumes at ordinary temperatures, but at the temperature of the explosion (about 3 000°C) these would expand to 10 000 volumes if unconfined.

The explosives should be buried in the rock to be broken, that is, not only set in a hole but also properly stemmed, so that as much of the expanding forces and shock waves as possible are used to fracture and displace the rock. 'Stemming' is aggregate, cuttings or other inert material which is run into the hole to confine the explosives. When charging flatly inclined holes, it may be necessary to press (tamp) the explosive and stemming. In general the harder and stronger the rock, the faster and more intense the explosive action should be (see also Chapters 8 and 9).

The initiation of an explosion is accomplished by flame in the case of blasting powder, but 'high' explosives require a severe shock or detonation wave, while blasting agents like ANFO require an even greater shock.

Velocity of Detonation (VOD)

Velocity of detonation is the velocity at which a column of an explosive detonates, that is the velocity at which the chemical reaction and hence the shock wave are propagated through the explosive. The VOD of any explosive usually varies with the degree of confinement and the diameter of the column;

it also varies with explosive particle size and density. Manufacturers generally state the conditions under which their measurements are carried out.

The greater the VOD the greater the shattering effect of an explosive. For hard rocks and metal cutting, an explosive with a high VOD is recommended but, since density and strength of the explosive also play a part in the shattering effect, all three must be considered in the selection of the explosive.

Types of Explosives used in Quarrying

Commercial explosives generally fall into one of the classes below as defined in the Regulations under the Explosives Act:

- propellant powders (classification 1.1C)
- blasting explosives (classification 1.1D)
- detonating (initiating) explosives (classification 1.1B).

'Low' Explosives

Classification 1.1D explosives include gunpowder (black powder, blasting powder) which is the oldest explosive known and is composed of 10% sulphur, 15% charcoal and 75% saltpetre (KNO_3). It has been called a 'low' explosive because it does not need a shock wave for initiation (although it may be detonated by concussion) and it has a heaving rather than a shattering effect. It is commonly ignited by the spit of a safety fuse and it burns at an extremely rapid rate referred to as 'deflagration'.

'High' Explosives

High explosives detonate when properly initiated with a detonator of sufficient energy to cause stable propagation of the shock wave through the explosive. Propagation of the shock wave is accompanied by a rapid chemical reaction that provides the energy necessary to maintain the wave. Velocity of detonation (VOD) ranges from 2 200 to 7 500 m/s.

Gelignite AN60 (based on nitroglycerine), once the most common explosive used in general blasting work, for priming and for wet holes, is no longer manufactured in Australia. It is being replaced by watergels and emulsion type explosives which are less sensitive to accidental initiation from shock or mishandling. Emulsions may be purchased as packaged explosives or in bulk. Packaged emulsions have a short storage life of approximately 12 months and hence extra care should be taken to ensure rotation of explosive stocks — first purchased should be first used. The life of bulk emulsions is even shorter.

Watergel explosives contain no ingredients which are explosive in themselves. They consist of a supersaturated aqueous solution of ammonium nitrate (AN) and sodium nitrate mixed with a carbonaceous fuel sensitiser such as ground coal, wheat flour or molasses. They also contain gums and crosslinking agents. Examples are Riogel and Tovex 800.

Emulsion explosives consist of microdroplets of supersaturated AN solution within an oil matrix. The introduction of air bubbles (voids) increases sensitivity. Voids may be introduced by physical methods (glass microballoons) or chemical gassing. Examples are Powergel 3151 and Emulite 130.

It should be noted that Classification LID also covers non detonator-sensitive explosives which require a primer for proper initiation of the explosion such as AN-fuel oil mixtures (ANFO); these latter are sometimes called 'blasting agents'.

Detonating (Initiating) Explosives

Detonators can be set off by flame, heat or shock. They have a very high VOD producing a detonation wave which can initiate an adjacent high explosive or high explosive priming charge.

Primers and Boosters

A 'primer' is a plug of high explosive used for transmitting and intensifying the detonation wave from the initiator to the explosive column. The plug of AN60 gelignite with detonator inserted, which was formerly the main device for priming, has largely been replaced by packaged watergels and emulsions or by cast primers. Examples are 'Pentolite' which is a 50:50 mixture of TNT (tri-nitro-toluene) and PETN (penta-erythritol-tetranitrate), 'Anzomex' and 'HDP' (high density primer). Primer sizes range from 26 g to 2.8 kg to suit various drillhole diameters and explosives columns. They have a very high VOD (7 500 m/s) and are waterproof. A 'booster' has the function of maintaining propagation of the shock wave and of allowing the explosives column to detonate at maximum VOD.

Ammonium Nitrate (AN) Mixtures

AN by itself is not an explosive but is an oxidising agent. It becomes an explosive when mixed with a prescribed amount of combustible material such as oil (commonly diesel), coal dust, molasses etc. Hence it must be stored at least 7 m clear of any fuel. The mixture requires a primer to initiate it into the self-propagating rapid reaction of a detonation. AN readily dissolves in water and may absorb water from the air when the humidity is greater than 60%. It must be stored in a clean dry location, preferably sheltered.

AN in the form of prills is mixed with fuel oil (FO) in the proportions of 16:1 by volume or 5.7% FO by weight (commonly 3 litres of FO to 40 kg of AN) to produce ANFO. This is oxygen balanced to produce the minimum amount of noxious fumes. It will also release the optimum amount of energy when detonated. Only fuel oil with a flash point greater than 61°C must be used; the use of other fuels will introduce dangers of flammability.

Uniform mixing produces better blasting results and is best ensured by using 'Waxoline', a red dye which is mixed with the fuel oil to indicate when an even mix has been achieved.

ANFO is an explosive with a VOD ranging from 2 000 to 4 000 m/s in holes from 40 to 250 mm in diameter.

Moisture rapidly reduces the sensitivity of ANFO to detonation and may cause loose prills of AN to cake together making it difficult to mix and handle. If not used promptly, it must be stored in an explosives magazine licensed for this purpose and packaged and labelled appropriately.

ANFO is the cheapest and most convenient explosive in use for quarrying but is of little value in wet blast holes.

Mixing methods fall into two categories:

- batch methods, by hand or small concrete mixer
- continuous methods, by auger-mixers with a metered FO spray or by introducing spray-metered FO into a stream of AN as it is poured into a blast hole. These methods must be approved by the Department of Labour.

Bulk deliveries of non-explosive ingredients to blasting sites for mixing as required are becoming more common. Among the advantages are:

- The raw ingredients are non-explosive (until mixed) and so can be stored and transported as non-explosive materials; however they are classified as dangerous substances.
- All of the ingredients to make a wide range of explosives arrive together and so multi-product options are available on site; this allows for changes in ground conditions (and water in holes) to be accommodated as they occur.
- The required explosive is mechanically mixed exactly to specifications.
- Large blasts can be quickly prepared.
- Labour costs are reduced.



Hand-operated auger mixer for ANFO. (Photo 033153)

ALANFO is made by the addition of 10-15% aluminium powder to ANFO. Weight strength is increased by up to 45%. Use of ALANFO has diminished due to the advent of the lower cost emulsions and watergels together with the hazards associated with aluminium powder.

Heavy ANFO is the product of mixing ANFO with an emulsion or watergel matrix whereby the density is increased to 1.00-1.35 and the bulk strength by up to 80%. The greater the amount of emulsion, the higher the density, the VOD and the water resistance.

Low density ANFO has been designed to reduce the density of ANFO and thus obtain a cheap low strength explosive for special purpose use. Polystyrene in the form of expanded beads is mixed with ANFO in proportions up to 50:50. It is used to minimise overbreak, reduce cracking and give safer walls or backs (in the case of tunnelling). Great care must be taken to avoid segregation during charging. The oil used in this product is not straight diesel but has to be modified to prevent deterioration of the PS beads. For example SANFOLD has a 5% emulsion content to make the product 'tacky'. Examples are AMEXLD, ISANOL and ANFOPS.



Range of commercial explosives available (Dyno Wesfarmers Ltd).



Bulk delivery and onsite mixing of explosives at Linwood Quarry. (Photo 040445)

The mixing of ANFO is regarded as the manufacture of an explosive and hence a licence to mix is required from the appropriate authority (in South Australia the Dangerous Substances Branch of the Department of Labour).

Properties of Explosives

Sensitivity

The sensitivity of an explosive is a measure of its ability to detonate under a given stimulus. The explosive must be sufficiently insensitive to shock and heat so that it is safe to manufacture, handle and place into position. However, it must not be so insensitive that it will fail to explode on initiation in a blast hole. (Note here that wet or freezing conditions affect some explosives adversely.)

To improve safety when handling explosives there is a trend towards the manufacture of less sensitive bulk explosives; these require primers to provide the energy necessary for their detonation. Most of the new emulsion and watergel explosives can be made with a wide range of properties and sensitivities in order to meet particular requirements.

Relative Weight Strength

Relative weight strength is the amount of energy released by a given weight of an explosive relative to the amount of energy released by an equal weight of standard ANFO (0.8 g/cm^3).

The relative weight strength of an explosive can be determined by:

- Measurement techniques which include the underwater test — the intensity of the shock pressure and the time of bubble oscillations when a known weight of the explosive is detonated under water; these results are compared with results achieved in a similar test using standard ANFO.
- Calculation techniques which include use of proprietary thermodynamic computer programs which predict the rate of energy release from a particular explosive given its ingredients.

When comparing products from different manufacturers care must be taken that the method of determining the quoted relative strength is understood.

Relative Bulk Strength

Bulk strength is the energy released from a given volume of explosive compared with the energy released from an equal volume of ANFO of density 0.8 g/cm^3. It is often important in quarrying because the volume of explosive that can be used is determined by the diameter and length of the drill hole.

As an example, an explosive with a bulk strength of say 160 has 60% more energy potential than an identical hole filled with ANFO to the same depth.

It will be noted that using cartridges will have the effect of reducing charging density.

$$\text{Relative Bulk strength (\%)} = \frac{\text{Relative weight strength (expl.)} \times \text{Density (expl.)}}{\text{Density (ANFO)}}$$

(expl. = explosive under consideration)

Water Resistance

The water resistance of an explosive is its ability to detonate after being exposed to water. This characteristic is most important when it is necessary to blast wet holes. ANFO, the cheapest explosive, has no water resistance. A water resistant explosive such as a watergel or an emulsion should be used in holes containing water.

Recent developments have led to the use of ammonium nitrate emulsion blends which can be mixed in varying proportions depending on the degree of water resistance required. Examples of these products are Energan (low resistance), Detamax 001, ANFO Extra (sometimes known as Heavy ANFO) and Powergel 2500 or Detapower 400 (high resistance).

Fumes

Fumes produced by a commercial explosive after detonation include carbon dioxide, nitrogen and water vapour (steam) which are not noxious, and carbon monoxide and oxides of nitrogen which are. The proportions of these gases vary with explosive type and with the conditions under which it is exploded. Fumes do not usually constitute a problem in surface blasting but can be a hazard underground if ventilation is not adequate.

Detonating Explosives

Plain Detonators (used with safety fuse)

These consist of an aluminium tube, closed at one end, about 42 mm long by 6 mm in diameter. At the closed end is a base charge of PETN on top of which is a smaller charge of an initiator composed of a mixture of lead azide, lead styphnate and aluminium powder (ASA). The latter mixture explodes when initiated by the spit of a safety fuse and in turn detonates the base charge (Fig. 7.1).

Detonators must be removed from their cartons carefully. The shell should be tapped to rid it of any dust. The fuse must be cut square at the end and inserted into the open end of the detonator to the full depth, without twisting, and the metal detonator crimped around the fuse. Crimping must be done on a special crimping machine, which makes a water-proof joint, or with non-sparking hand crimpers (Fig. 7.2).

Instantaneous Electric Detonators

These contain the same explosive compounds as plain detonators. They are made of aluminium, and have a 'fuse head' incorporated in them. A wire bridge is built into this fuse head, and an electric current flowing through heats the wire, ignites the fuse head and the detonator is exploded. In 'protected' detonators the fuse head is surrounded by a protective insulating sheath. The lead wires, which are generally yellow PVC insulated, enter through a neoprene rubber plug.

Delay Electric Detonators

These are similar to instantaneous electric detonators except that a delay element or fuse is inserted between the fuse head and the detonating charge (Fig. 7.3). These detonators are used to fire explosives in sequence with the advantages of better fragmentation, reduced ground vibration and reduced overbreak. Sequence firing with delay detonators enables the firing of a large blast in one operation with the delays allowing time for some ground movement between blast holes.

Delay detonators are of two types: long delays (half second intervals) and short delays (25-30 millisecond intervals). Long delay detonators are mainly used in shaft sinking and tunnel driving where more time is required for the ground to clear between blast holes.

Each detonator is marked with a numbered tag or stamp on the base which indicates where in the sequence it will detonate.

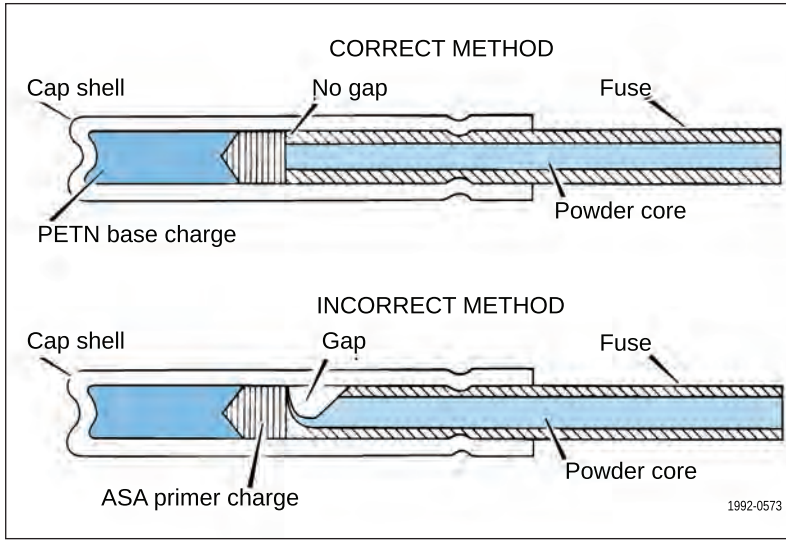
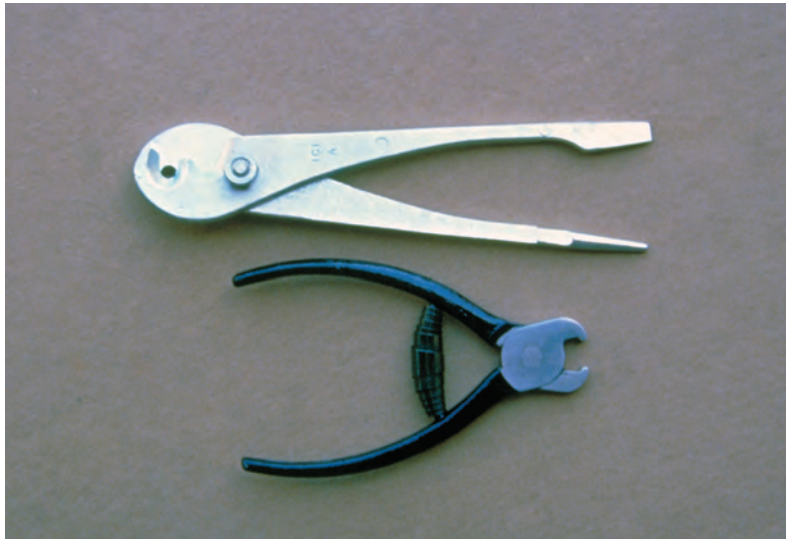


Fig. 7.1 *Fuse capping with No. 8 detonator.*



Hand crimpers for detonators (ICI Explosives). (Photo 040446)

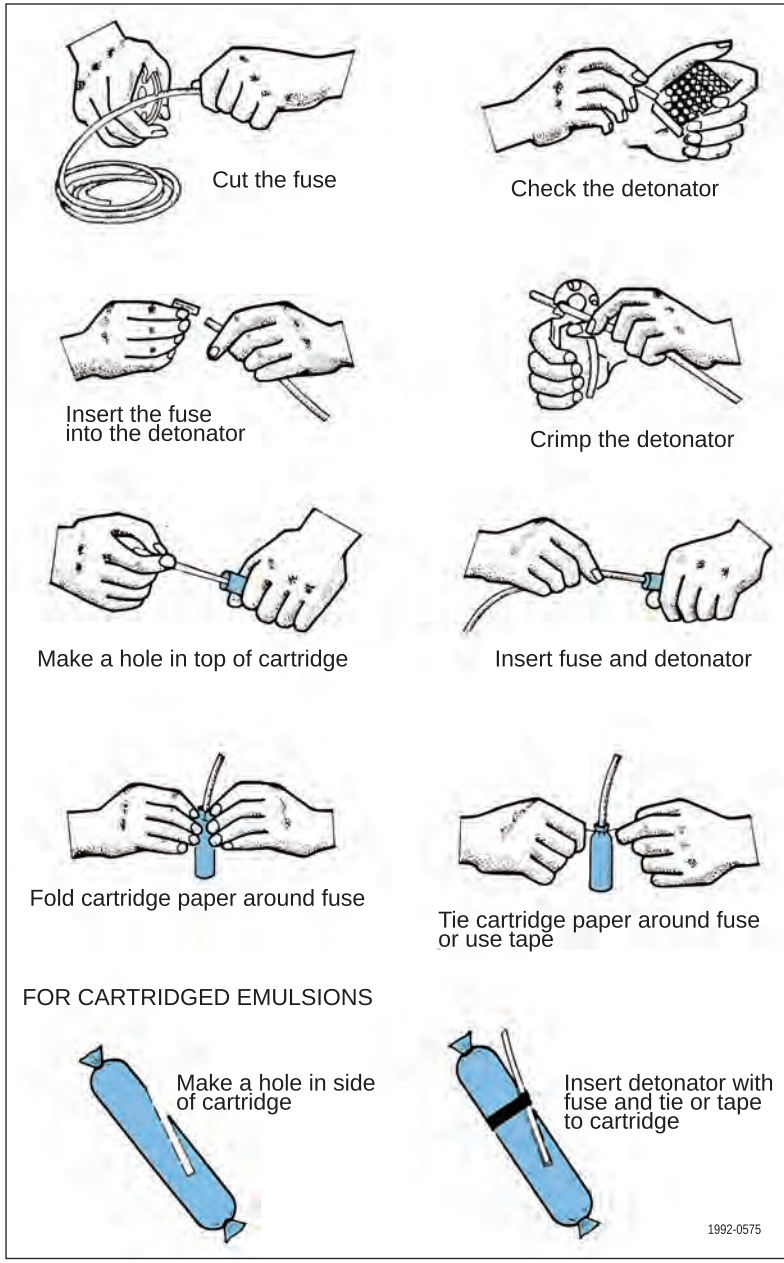


Fig. 7.2 *Preparation of primer.*

It is imperative that detonators from different batches or manufacturers are not used in the same shot. To do so could lead to a serious misfire.

Signal Tube Detonators

Signal tube (non-electric) detonators differ from electric detonators only in that they are ignited by a shock wave of a velocity of 2 000 m/s which passes to the detonator along a factory-connected hollow plastic tube. The tube is 3.0-3.7 mm OD and 1.0-1.5 mm ID; its inside surface is coated with a 0.1 mm film of explosive (HMX plus aluminium, 18 mg/m nominally). Signal tube detonators are available in a series of delay numbers similar to short delay electric detonators (Fig. 7.3).

The use of common in-hole delays and surface sequencing is allowing greater control over blast timing and hence improvement in blast performance and increasing blast size.

Increasing industrial usage suggests that the non-electric initiation system will eventually supersede electric firing. Examples are Primadet with Trunk Line Delays, Riodet, Nonel with UB Connectors.

Detonating Cord

Detonating cord consists of a core of PETN enclosed in layers of nylon textiles and/or an outer plastic sheath. Cords are classified according to the PETN core load which ranges from 0.5 g/m ('Detaline') to 70 g/m ('Shearcord').

A plain, electric or signal tube detonator taped onto the side of the cord must be used to start the shock wave running at approximately 7000 m/s; this may be interrupted by detonating relay connectors to achieve the desired delay pattern.

Detonating cords with a core loading of 5 g/m PETN, or greater, can be used to side initiate other cords. Cords with less than 5 g/m require special detonating connectors which are initiating class 1.1B explosives. Those containing less than 10 g/m may not reliably side initiate a column of high explosive and need to have detonators attached to provide force to initiate a primer. Even 10 g/m cord may not initiate an explosion in a 100 mm column of ANFO without a primer. The heavier cords can be used to transmit the shock wave from an initiation point along the cord to other cords knotted and joined to it which ultimately initiate the main explosive charges.

Detonating cords are water resistant and safe to handle as they cannot be initiated by friction, electrical currents or ordinary shock. However, they are

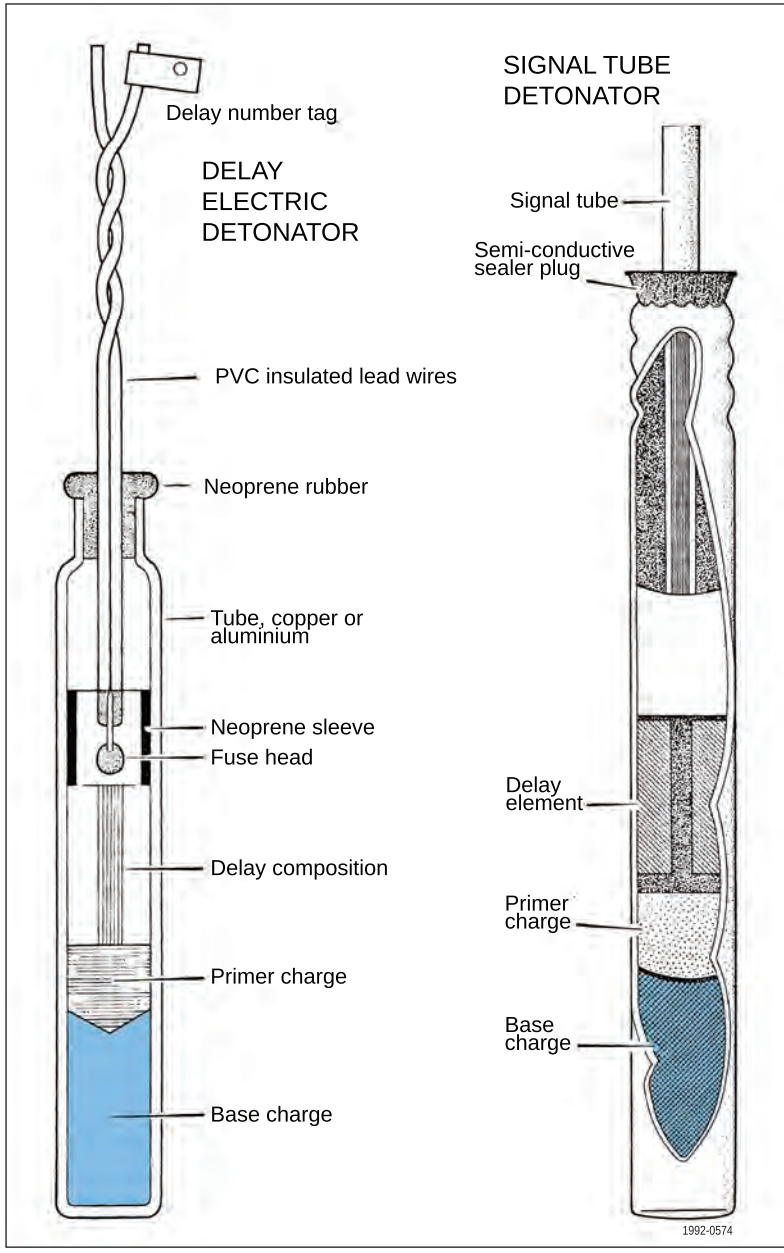


Fig. 7.3 *Delay detonators.*

subject to water migration into cut ends and can be initiated by severe impact or friction. Detonating cords are not used in metropolitan locations because of noise disadvantages.

Initiation of Explosive Charges

NON-ELECTRIC INITIATION

Safety Fuse Initiation

Safety fuse is manufactured by forming a train of black powder within several layers of textiles and water proofing material (e.g. bitumen). The outer covering also prevents the burning fuse from side spitting and igniting other fuses.

Safety fuse is designed to burn at 100 seconds a metre at sea level (plus or minus 10 seconds), and this burning rate should be tested when starting a new roll and before using it for every blast by cutting off a metre and timing the burning with a stop watch.

Fuse should be stored in cool, dry conditions, and should never be kinked. Fuse that has become damp should not be dried and used, but should be destroyed.

Holes may be fired in the correct order by varying the length of the fuses and lighting the shortest first; no fuse to be lit should be shorter than the legal requirement which in South Australia is two metres.

To prepare a primary charge for fuse firing, a hole is made in the end or side of the cartridge with a non-ferrous skewer; the detonator is inserted to point along the cartridge and the fuse tied tightly to the cartridge with string or tape (Fig. 7.2).

The main force of a detonator explosion is in the direction in which the detonator is pointing. When using safety fuse, the priming charge should normally be the last one inserted in the hole followed by the stemming; this ensures that the fuse is not kinked.

It is permissible to light one fuse with a match, the usual method being to hold the match head against the powder core and strike the box across the match head, the subsequent flare igniting the fuse.

To light more than one fuse, up to the legal maximum of 15 fuses, a two-minute fuse lighter should be used. Fuse lighters should be kept in a dry store room and never stored with other explosives.

Where a large number of charges are to be fired by the safety fuse method, multiple fuse-igniter cartridges or plastic igniter cord should be used. A fuse-igniter cartridge is about the same size as a shotgun cartridge into which eight fuse tails can be inserted. A master fuse is lit which ignites the compound on the bottom of the cartridge and thence the other fuses.

When a large number of shots are too widely separated, plastic igniter cord is used with special bean hole connectors attached to the safety fuses to light the fuses. Igniter cord burns with an irregular intense flame which can ignite other fuses (or may start unwanted bushfires). Fast burning igniter cord burns at the rate of 3.3 seconds a metre.

Signal Tube Initiation

The shock wave which travels through the plastic connecting tube may be started by a special non-electric starter, a plain or electric detonator, or detonating cord with a core loading greater than 5 g/m PETN. Blasting patterns can be arranged similar to those using electric delay detonators. Trunk lines may be interrupted by trunk line delays to achieve correctly sequenced blasting (Fig. 7.4).

Common in-hole delays may be sequenced by surface trunk line delays which allow unlimited delay timing.

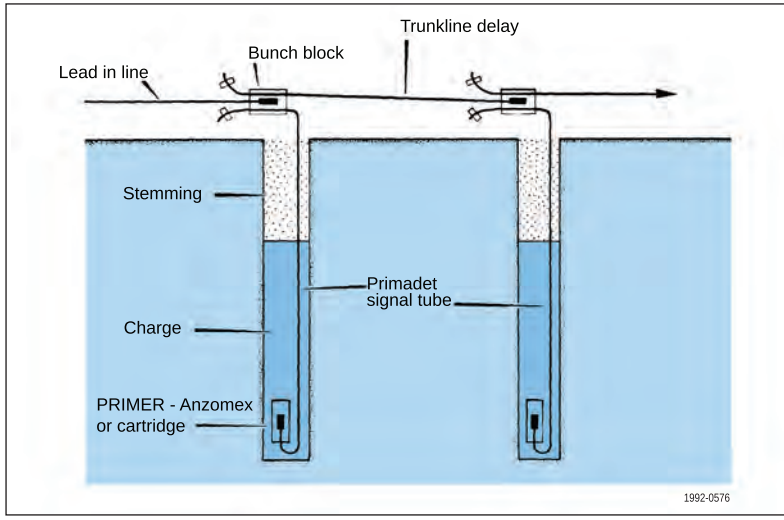


Fig. 7.4 *Primadet and trunkline delay.*



Non-electric detonation. (Photo 040447)

The advantages of the signal tube initiation system are that it is not affected by electrical hazards, it does not side-initiate other explosives, it is quick and simple to hook up (saving time and reducing the possibility of misfires) and there is little noise; it is also safe and easy to handle. Other advantages include firing large shots with environmental control, timing flexibility, sequencing the blast after loading and reducing stock inventory.

Detonating Cord Initiation

Various joints used in connecting detonating cords together are shown in Fig. 7.5. In firing a series of charges with high core loading detonating cord, only one detonator is required for initiation. This improves safety by elimination of detonators from the charged holes. The charging of holes is made easier, quicker and safer, and the danger of incomplete detonation and burning explosives is minimised; also the ever-present hazard when unexploded detonators and explosives remain in a hole after a round has been fired is reduced. Detonators need not be present on the job while the holes are being charged, and are not attached to the detonating fuse until the last minute when they are bound to the detonating cord with insulating tape, the detonators pointing in the desired direction of propagation (Fig. 7.5). However chance of a misfire from a cut-off is greater if no in-hole delay is used.

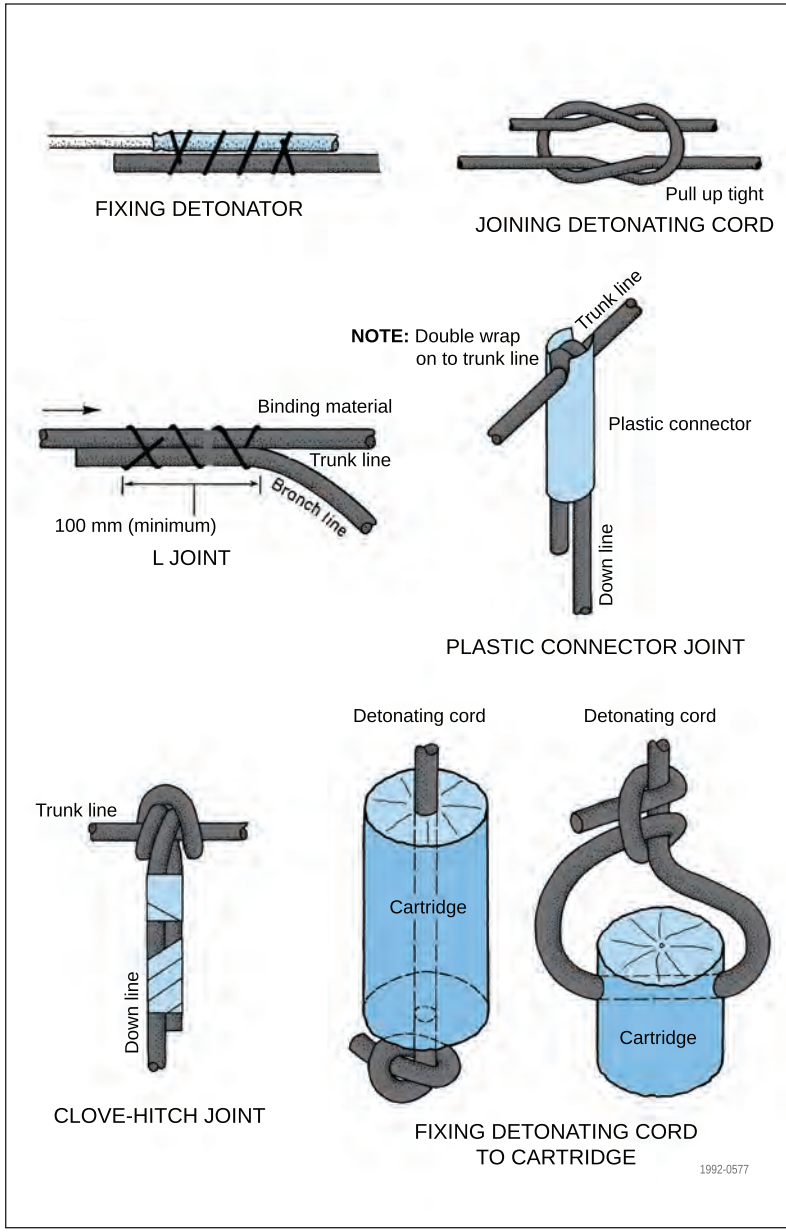


Fig. 7.5 *Detonating cord connections.*

Disadvantages of high core loading (10 g/m or more) detonating cord include disruption of stemming by detonation of the down line, side initiation of the explosive column, possible desensitising of slurry/emulsion/watergels and problems of noise resulting from unconfined cord on the surface. Detonating cord lines on the surface should wherever practicable be covered with 200 mm of sand to muffle the noise and reduce the probability of complaints from residents in surrounding areas.

The trend in detonating cords is towards lighter core loadings, from 10 g/m to 1 g/m. Cords with core loadings of less than 5 g/m will not side initiate other cords, or ANFO columns.

Detonating cords may be interrupted by millisecond detonating relay connectors (DRCs) to achieve a correctly sequenced blast which will result in better fragmentation and reduced ground vibration (Fig. 7.6). It should be noted that DRCs must not be used with cord loading less than 5 g/m; for this application millisecond (MS) connectors are appropriate. MS connectors are now more commonplace than DRC.

Delay connectors contain two detonators and hence must be treated and stored as detonators.

ELECTRICAL INITIATION

For electrical firing the detonator should be embedded in the primer cartridge and the lead wires hitched around the cartridge. In bottom priming the detonator must be pointing upwards.

Electrical circuits may be either series or parallel or a combination, but in quarry blasting with exploders the series system which requires low amperage and high voltage is most commonly used. The parallel system requiring high amperage and relatively low voltage is usually applied to large underground blasts using mains power. Further, checking the continuity of a series circuit is much simpler than for parallel or mixed circuits.

To fire electrically, connecting wire, firing cable, an approved blasting circuit tester and an exploder with sufficient capacity to meet the requirements, as well as the electric detonators, are necessary (Fig. 7.7).

Connecting wire (bell wire) in a series circuit is used to connect the two free ends of the detonator lead wires circuit to the firing cable or direct to the exploder. The wire is usually 25 SWG copper wire with PVC insulation and is considered expendable as it is usually damaged in the blast. Blasting cables are a heavier gauge wire of lower resistance, used to extend the firing

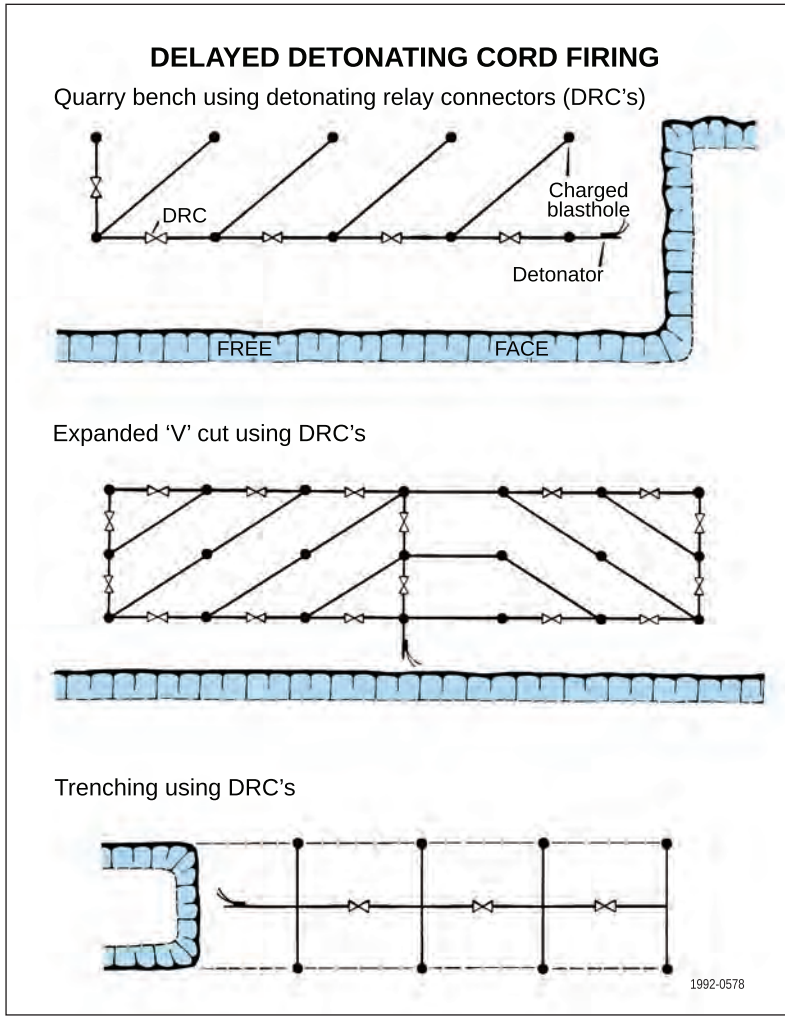


Fig. 7.6 *Blasting layout using detonating cord for delayed firing.*

position to a safe place. Blasting cables are tested for continuity with the ends connected together, and for short circuiting with the ends open, using a blasting circuit tester prior to firing.

Blasting Circuit Tester

Only an approved blasting circuit tester, for example a 'Detameter', may be used to test a firing circuit. An electrician's circuit tester must never be used because it may pass a sufficiently large current through the circuit to initiate a detonator.

Exploders

Exploders are available in size ranges from a single shot machine (used in coal mines) to machines capable of firing upwards of 100 detonators. A check should be made that the exploder is permitted in quarry operations.

All exploders used must be fitted with a locking device or have a removable handle which will render the exploder safe when the handle or key is removed.

Exploders should be tested periodically to determine whether they are operating effectively by using a blasting rheostat in accordance with the instrument directions (Fig. 7.8).

Examples of exploders are:

Army Mk VII Rack-bar Exploder. This is a powerful dynamo-type exploder with a vertical rack bar, which on the downward movement operates an armature. At the end of the stroke an internal short circuiting device opens, and the current generated by the rapidly revolving armature passes into the firing circuit. The exploder is rated as a 15-shot machine.

Nobels Rack-bar. This is similar in operation to the Mk VII, but more powerful, and is rated as a 30-shot machine.

Nissan F-3 30-Shot. This is a capacitor type exploder which is charged up by a replaceable D cell battery. The exploder is small, easy to carry, and features a removable firing key, a charge indicating light, and a safety firing button.

Beethoven Dynamo Condenser-Exploder. This machine has been designed for the firing of 100 detonators connected in series or 200 detonators in a carefully balanced series-in-parallel circuit. The exploder is built on a metal chassis and contained in a moulded bakelite moisture-proof case.

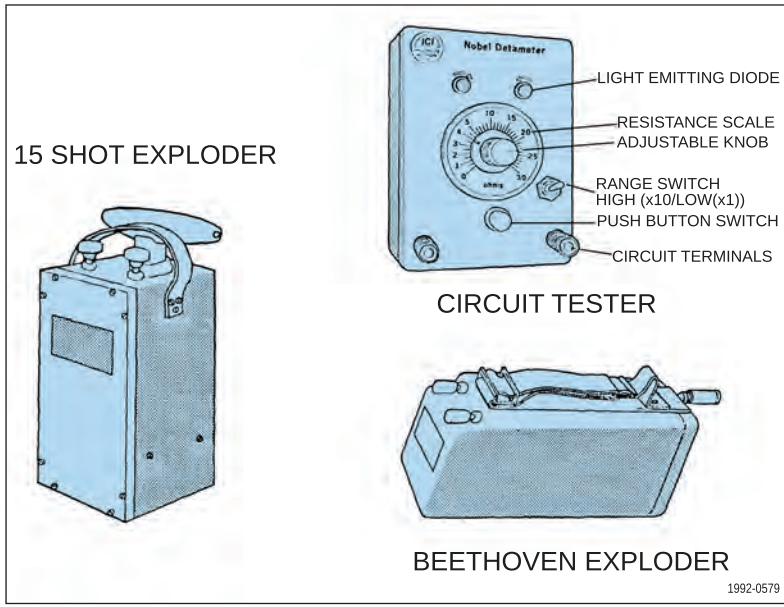


Fig. 7.7 *Exploders and circuit tester.*

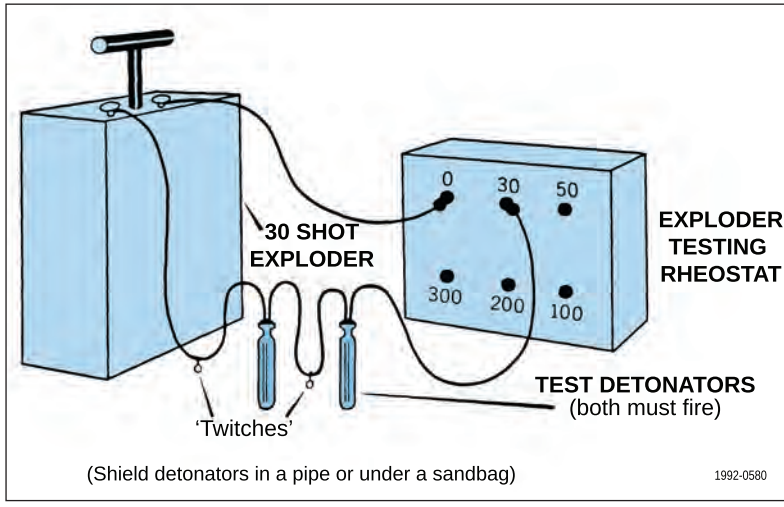


Fig. 7.8 *Testing exploder with rheostat.*

A leather carrying container with strap is provided and the whole unit weighs about 4.5 kg.

A hand-driven revolving generator produces a voltage which charges a 6 microfarad condenser to a potential not less than 1 200 V. A neon lamp indicates when the condenser is charged, and a button switch applies the voltage from the condenser to the output terminals. Removal of the charging handle discharges the condenser; this is a safety provision to avoid human contact with the high voltage terminals.

The exploder can be tested by use of an iridio-platinum wire tester provided. The tester is mounted on the output terminals and the fuse wire, carried on a reel on the tester, is drawn between spring loaded clips. The exploder is then prepared for firing in the usual way, and on pressing the button the voltage passing through the iridio-platinum wire should fuse it.

Sequential Exploder. The BM175-10ST sequential exploder is a capacitor discharge exploder capable of firing 10 separate circuits, one after the other, at selected time intervals. This enables the firing of large delayed blasts to comply with ground vibration limits when the number of shots exceeds the range of delay detonators available. The operation of this exploder is complex and good operator training is required.

Other Exploders. There are various other exploders for use with special devices such as bridgewire detonators or Magnadet detonators but these are not now used in quarrying.

Firing From Power Mains

This method of firing is used mainly in tunnels and underground workings. As a general rule the reticulation of mains power to the quarry faces is not practicable.

Firing From Batteries

Firing direct from batteries is prohibited in South Australia.

Electric Circuits

The connection of detonators to form an electrical circuit can be either series, parallel or a combination. Parallel firing is not used in quarry blasting because it requires a high current initiation system which can usually only be provided from a mains power firing system.

Series Circuits. These are made by connecting one wire from each electric detonator to one wire from the next, and so on, thus forming a single continuous circuit (Fig. 7.9). The free wire at each end of the circuit is joined to connecting wire which in turn joins to the firing cable. The current must pass through each detonator, one after the other, to complete the circuit. The advantage of the series method is that the complete circuit can be tested for continuity with a Detameter.

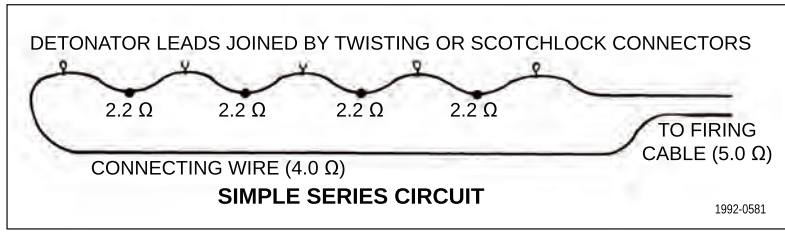


Fig. 7.9 *Simple series circuit.*

The total resistance of the circuit is equal to the combined resistance of the electric detonators, plus the resistance of the connecting wire and cable in the circuit. For example, 15 electric detonators with 3.1 m lead wires are connected in series with 45 m of No. 25 gauge connecting wire (bell wire) and 90 m of firing cable. The resistance of an electric detonator with 3.1 m leg-wires equals 2.2 Ω. The total resistance of the circuit is then:

Detonators	15 x 2.2 = 33 Ω
Bell wire	45 m=4 Ω
Firing cable	90 m =5 Ω
Total resistance	42 Ω

Thus when the circuit is tested with a Detameter a reading of approximately 42 on the low scale would be expected. If the reading is much higher, lower or nil, a fault would be suspected and recheck of the wires and connections must be made.

With a series circuit the current supplied should be at least 1.5 amps (A). The voltage required for a circuit is equal to the total resistance of the circuit in ohms multiplied by 1.5 A. Thus for the above series circuit the voltage required should be greater than 42 x 1.5 = 63 V.

Parallel Circuits. These are made by connecting one leg wire of each detonator to a common point or wire and the other leg wire to a second common point or wire (Fig. 7.10). Each detonator in a connection of this kind

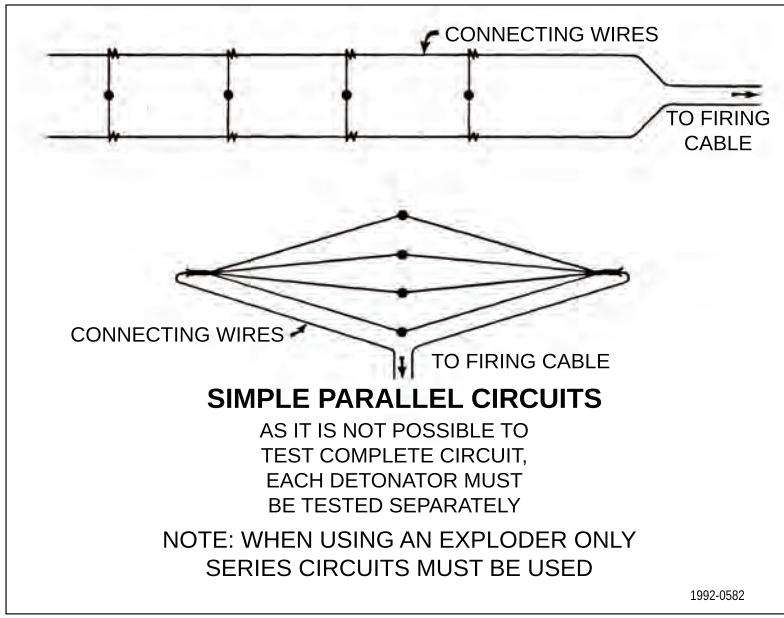


Fig. 7.10 *Simple parallel circuits.*

offers a path for the electric current independent of all the other detonators in the circuit. An important disadvantage is that it is not possible to test the circuit as a whole (i.e. that each detonator is connected properly) using a circuit tester. Parallel circuits have overall low resistance and require high current with moderate voltage to initiate them successfully. They are more susceptible to misfires and should be avoided in quarrying.

Testing Circuits

To avoid misfires it is necessary to test individual detonators and complete circuits. This is done by means of a special blasting circuit tester (e.g. Detameter). For detonator testing, the detonator is placed in a block of wood or pipe away from the person testing, and the ends of the lead wires are placed one on each terminal of the tester. Failure to obtain a change in indicator lights, at the expected resistance value, or failure to obtain a deflection of the needle on the scale, means that the detonator is defective and it should be destroyed.

The shot-firing cable should be tested both in open end and closed circuits. The completed firing circuit must be tested before attempting to fire, and this testing must always be done at the firing end of the cable well away from the charged holes.

Hazards with Electric Firing

Lightning. It is necessary to stop all blasting work and clear personnel from work areas, including quarry, tunnels and shafts, when an electrical storm comes within 8 km of the site.

If such storms are a regular occurrence, it may be best to use signal tube detonators or detonating cord instead of electric detonators. Even then, personnel should always be removed to a safe distance on the approach of a storm.

Warning units are available which indicate the approach of a thunderstorm. These can be extremely useful in areas prone to 'dry' lightning strikes.

Static Electricity. Static electricity is generated by friction, for example dust storms, dry snow storms, moving conveyor belts, escape of compressed air, and pneumatic loading of explosives. Blasting should not proceed during severe dust storms or periods of high temperature and low humidity.

In areas of low humidity care should be taken to uncoil lead wires along the ground. Lead wires should not be dropped onto the ground whilst holding the detonator.

When pneumatic ANFO loaders are used they must be fitted with a semi-conducting loading hose and the loader must be earthed. The operator should wear antistatic boots.

Stray Currents and Power Lines. There can be a danger of stray currents from electrical equipment or faults occurring in power transmission lines. Care should be taken to ensure that blasting cables or other parts of the firing circuits do not come in contact with electric cables, power tools, generators, or steel pipe or rail which may carry stray currents.

When blasting within 100 m of a power line carrying more than 440 V, electrical detonator circuits should be wired in twisted or closely paired conductors which are insulated from the earth, and all joints of the firing circuit should be taped.

Recommended safe distances for electric blasting near high voltage transmission lines, i.e. greater than 33 000 V, varies from 100m to 1 320 m depending on the operating voltage and the length of the detonator lead wires.

Special care should be taken to ensure that the firing cables are not thrown up over power lines by the blast.

Radio and Electromagnetic Transmitters. No hazard exists from transmitters for electric detonators when they are transported bunched and electrically shorted out as supplied by the manufacturer.

However, the wires of a blasting circuit can act as an aerial to pick up energy from various types of electromagnetic transmitters sufficient to cause a premature explosion. The recommended safe distance depends on the frequency and power of the transmission. Car and CB hand-held (walkie-talkie) transmitters should not be taken within 50 m of an electric blasting circuit.

The following recommended minimum safe distances from radio and TV transmitters should be observed:

Transmitter Power Range (W)	Minimum Safe Distance (m)
5-25	50
25-50	70
50-100	100
100-500	220
500-1000	300
1000-5 000	700
5 000-10 000	1000
10 000-50 000	2 200
50 000-100 000	3 000
The power output of some local transmitters is:	
ABC FM	10 000W
5AN	80 000W
TV	100 000W
5CL	50 000W

Explosives Regulations

Each State has its own explosives regulations and persons using explosives must check with the appropriate State authority.

In South Australia there are two Acts of Parliament governing explosives. The Mines and Works Inspection Act, 1920-78, covers the use of explosives and the issue of Blaster's Permits. The Explosives Act, 1936-82, covers the manufacture, storage, transport, and purchase of explosives.

Every person charging or firing explosives in a quarry must be either the holder of a Blaster's Permit or a trainee working under supervision of the holder of a Blaster's Permit.

Candidates for a Blaster's Permit must first undergo an initial training period, before taking a practical test and oral examination covering the usage of explosives and the requirements under the Explosives Regulations.

Magazines

Explosives magazines must be licensed under the Explosives Act — refer to the Department of Labour. They must be substantially constructed to provide good security against theft and must be located at a specified safety distance from other buildings and structures. A mound of earth to the height of the eaves around the magazine is preferred, to deflect the blast upwards from any accidental explosion of the magazine. Dry grass and other combustible material should be cleared for a distance of at least 7 m all around the magazine.

Detonators must not be stored with other explosives. A separate detonator magazine is required. Fuse lighters must not be stored with explosives. The magazine must not have any iron or steel in the interior; copper nails or brass screws are used for internal fixtures.

The words 'Explosives' or 'Explosives, Detonators' must be painted in a conspicuous place on or near the magazine.

Guidance notes for the construction and operation of explosives magazines are available from the Dangerous Substances Branch of the Department of Labour.



Magazines for explosives. (Photo 040448)

Transport

In South Australia licences are issued by the Department of Labour for the transport of explosives in motor vehicles — up to 60 kg, up to 265 kg and over 265 kg. Vehicles must display explosives signs, carry a fire extinguisher and meet other requirements according to the quantity carried. Up to 3 kg of explosives can be transported without a licence in a vehicle displaying explosives signs, carrying a fire extinguisher, and which has a suitable means of securing the explosives.

Destruction of Defective Explosives

Nitroglycerine Compounds

Nitroglycerine explosives, for example gelignite, if stored for excessive periods or in extreme heat, can exude nitroglycerine which puts them into a highly dangerous state in which friction or shock could cause an explosion.

Nitroglycerine is an oily substance and if cartridges are exuding oil they should not be used in blasting operations, but destroyed. Cartridges of gelatinous explosive should be laid in a furrow of sawdust or newspaper in a 'train' consisting of overlapping cartridges. No more than 50 kg of explosive should be laid in any one train, but several trains may be burnt simultaneously. Lighting kerosene or distillate is poured over the train; it is ignited by a few metres of binder twine soaked in kerosene and should be burnt against the wind.



Preparing defective explosives for destruction by burning. (Photo 033152)

The area used should be free of loose stones, and operators should retire to a place from which the burning can be observed at a safe distance; explosions have been known to occur during these operations. Fire bans and other restrictions must be observed.

Watergel and Emulsion Explosives

Watergel and emulsion explosives have a limited shelf life, usually from 6 to 12 months depending on storage conditions. Care should be taken to ensure that old stock in the magazine is used first.

These explosives may be destroyed by burning in a similar manner to that used for gelatinous explosives. They are much more difficult to burn due to their water content and require more fuel, diesel or kerosene for example, on the train to maintain sufficient heat for complete combustion. It is usually best to destroy them by placing them in a blasthole with good explosives.

ANFO Mixture

A small quantity (up to 2 kg) can be scattered over 20 to 50 m^2 of bare ground where there is no combustible material. Larger quantities should be dispersed in water, where the water will not drain directly into rivers or dams, or preferably destroyed by burning.

Detonating Cord

May be detonated in quarry blast holes or alternatively laid out in parallel lines on sawdust or newspaper and burnt similarly to gelatinous explosives.

Detonators

Small numbers (ten or so) of detonators may be destroyed by exploding them with a good detonator in sand or in a large diameter steel pipe.

Detonators may be bundled with the closed ends together and placed in a strong bag which is buried in sand or in a large diameter steel pipe. A good detonator, preferably with a small amount of explosive, should be used to detonate them. Exploding detonators throw shrapnel and may scatter unexploded detonators if they are not bundled together and buried. Search the area afterwards for any unexploded scattered detonators.

Other methods of destroying detonators are:

- wrap red cord (10 g/m detonating cord) around twenty or so detonators and blast
- push them into a plug of explosive and blast
- place them with explosive down a blasthole to be blasted.

If in doubt about the destruction of explosives, an Inspector of Mines or Inspector of Explosives should be consulted.

8. Ground Vibration and Air Overpressure

Explosives detonating in a blasthole release energy by the almost instantaneous formation of gases contained within the blasthole under high pressure and temperature, and by the generation of a shock wave that radiates in all directions from the confined blasthole (Fig. 8.1A).

The shock wave is essentially a pressure or compression wave in which the rock particles vibrate backward and forward in the direction of propagation. It induces tensile stresses which generate radial fractures in the rock. Compare striking a stone with a hammer — it splits in the direction of the blow, i.e. radially (Fig. 8.1B).

When the compression wave reaches a free surface it is reflected as a tension wave and the rock slabs off backwards from the surface. The fractures are opened up and extended by the gases. Meanwhile release of the gases as the rock is broken loose produces compression waves in the air which cause airborne audible and subaudible vibrations known as air overpressure (see later).

The radiating shock waves create vibrations within and at the surface of the rock called surface waves in which the vibrations take place transversely (i.e. at right angles to the direction of propagation) and also waves in which the particles move in a circle as in ocean waves. Compare rippling of the surface of a pond — when a stone is dropped in, a cork will bob up and down (Fig. 8.1C)

Failure to control these air and ground vibrations can create a nuisance for neighbours and may damage property.

An efficiently designed blast hole will yield the maximum volume of well fragmented material with controlled rock movement and a minimum of air and ground vibration.

Ground Vibration

Vibration Terminology (Fig. 8.1D)

Amplitude (A)	The maximum displacement from the position of rest measured in millimetres (mm).
Frequency (f)	The number of oscillations per second that a particle undergoes under the influence of the vibration wave expressed in cycles per second (c/s) or Hertz (Hz).
Particle Velocity (v)	The rate of change of amplitude expressed in millimetres per second (mm/s). This is the velocity of movement of a particle in the ground brought about by the vibratory motion, and is not to be confused with velocity of propagation of the wave through the rock mass. Since the speed is continually changing from zero to a maximum and then reversing direction, the peak particle velocity (PPV) is a useful value for reporting.
Acceleration (G)	The rate of change of particle velocity expressed in mm/s^2.

For the common sine wave form of vibration the following relationships apply:

$$v = 2\pi fA$$

$$G = 2\pi fv$$

$$G = 4\pi^2 f^2 A$$

Measuring Ground Vibration

A ground vibration monitoring instrument (vibrograph or seismograph) has a sensor (geophone) which is firmly placed on or in the ground.

The sensor contains three mutually perpendicular transducers (magnets suspended inside coils) which generate small electric currents the voltages of which are proportional to the velocity of ground particle movement. The readings can be converted to particle acceleration or amplitude as required in the three directions: to and from, left to right, and up and down.

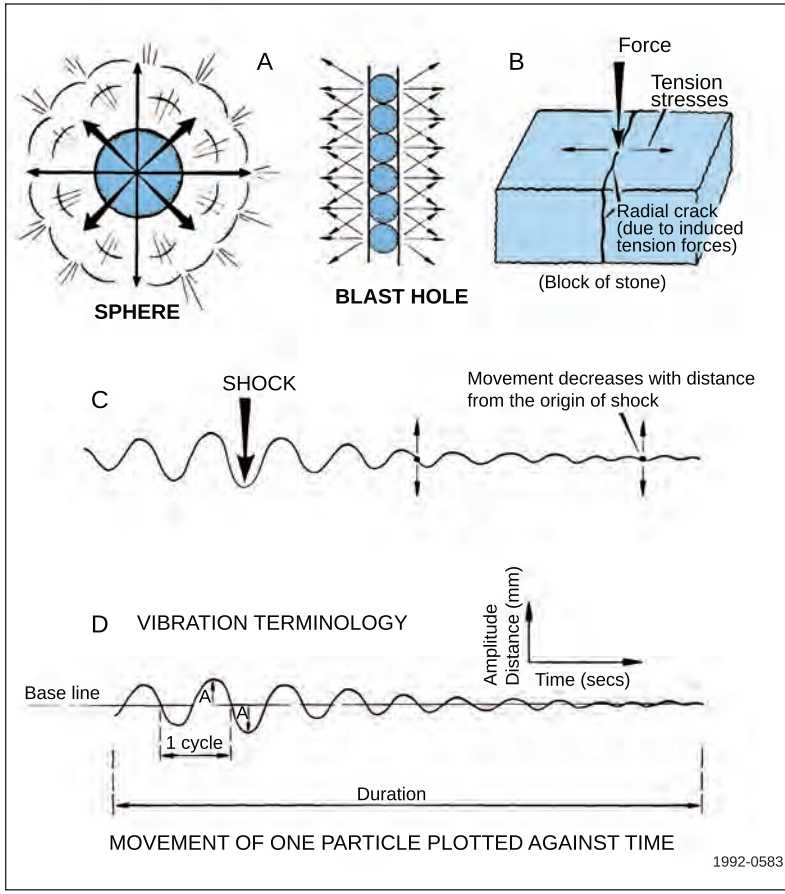


Fig. 8.1 *Blasting and ground vibration theory.*

The movements in each direction are combined to give a resultant, the maximum value or peak of which is known as the PPV.

In more technical terms, if V_X, V_Y and V, are the three mutually perpendicular velocity components taken at the same instant then

$$V_P = (V_X^2 + V_Y^2 + V_Z^2)^{½}$$

The maximum value of Vp is the PPV of the vibration.

These data can be processed to obtain either graphical representation or the maximum value of each of the components and the peak vector sum of the ground vibration. Alternatively the data can be stored digitally on a tape or chip for downloading on to a computer for analysis. Besides producing maximum values and graphical representations, frequency distributions, accelerations and displacements can also be obtained.

Ground vibration monitoring equipment should cover a range of at least 2 to 100Hz.

Damage

The response of buildings and other structures to ground vibration is complex and depends on many factors. Chief among these are magnitude and duration of motion, nature of the ground, interaction between ground and structure, and design and natural frequency of vibration of the structure.

Permanent ground distortion created by gas pressure in cracks developed during blasting will affect only structures very close to the blast.

Settlement of foundations during or after blasting due to the vibration compaction of the underlying soils or sands, particularly if they are cohesionless or saturated (compare with reported occurrences during some earthquakes), is extremely rare but the possibility should be checked at sites where the appropriate conditions prevail.

Cracking in structures due to blast vibrations occurs as a result of differential vibratory movement between particles in a structural member or between a member and a surface, or between structural members and joints. Each of the members and components has a natural frequency and if these natural or resonant frequencies coincide with the frequency of the ground vibration wave the effect is magnified. Because these natural frequencies are low it follows that ground vibrations of low frequency, say from 4 to 14 Hz, are likely to lead to an increase in damage for a given PPV.

It should be noted that changes in environmental conditions, such as temperature and humidity, and operations like slamming a door or driving a nail, can induce strains in a structure comparable with or greater than those caused by vibrations from quarry blasting.

Examples of causes of structural failure **other** than blasting are:

- foundation failures:
  - use of poor materials
  - settlement, due to the weight of the structure, particularly if on clays
  - water-logging of soil
  - differential settling, due to building partly on fill and partly on undisturbed ground
- movement of soil or material in which the footings are set due to alternate shrinkage and compaction caused by drying and wetting of the soil
- faulty design
- faulty construction.

Cracks in plaster walls and ceilings may also be caused by shrinkage of the paster, cement, or timber framing.

Vibration cracks in walls usually consist of two sets at about 45 degrees to the horizontal in vertical walls, while cracks in ceilings follow the same pattern. Many people, after experiencing ground vibrations or air blast, find in their houses shrinkage and other cracks that they had not seen before, and are ready at once to blame the blasting. Where pre-blast inspections are made of houses and cracks pointed out to householders, it is quite common to find that people living in a house had no idea of the presence of some cracks, even noticeable ones. It follows that an obvious precaution to avoid complaints is to survey all houses in the vicinity before blasting commences making sure that owners sign the survey report.

Human Response

Human response to vibration is extremely variable. People can feel ground vibration levels as low as 0.5 mm/s and many are distressed by vibrations which are considerably less severe than those which can cause damage to buildings. The psychological impact is increased if the vibration is associated with substantial noise such as a heavy truck being driven on a nearby road. The rattling of windows due to air overpressure is often confused with ground vibration.

Humans are affected by the amplitude of the ground vibration and also by its frequency. If the vibration is of high frequency people can tolerate less in the way of movement. A person should never be subjected to a PPV greater than 25 mm/s.

Australian Standard

It is now generally recognised that vibration levels with PPV in excess of 50 mm/s will cause damage to residential structures, although the onset of cosmetic damage can be associated with levels as low as 12.5 mm/s for particularly sensitive structures.

The frequency of vibration is also an important factor. Ground vibrations with frequencies of from 4 to 12 Hz will tend to be amplified by residential structures, increasing the likelihood of damage.

Table 8.1 gives the present Australian Standard (AS) recommendations for limiting PPV, as calculated under 'Measuring Ground Vibrations'.

Table 8.1 *AS 2187, Part 2-1983 for limiting peak particle velocities*

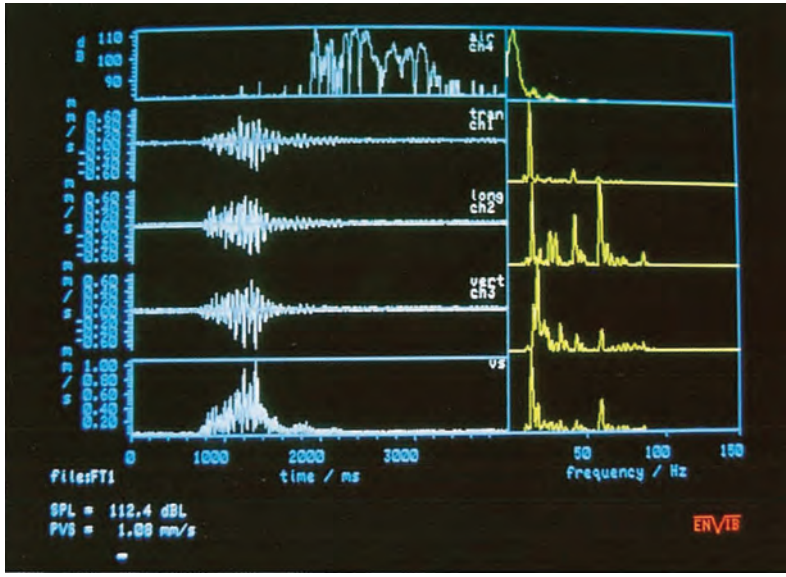
Type of structure	**PPV** (mm/s)
Historical buildings or monuments and buildings of special value or significance.	2
Houses and low-rise residential buildings; commercial buildings not included in category below.	10
Commercial and industrial buildings or structures of reinforced concrete or steel construction.	25

Note: *For environmental reasons acceptable limits outside normal working hours for residential areas are considerably less than the above figures.*

The PPVs in the table have been selected to minimise building damage and take no account of human discomfort and the effect of sensitive equipment within the building. PPV currently being proposed by various State Regulatory Authorities to consider human discomfort and to control complaints is 5.0 mm/s.



Equipment used to measure both ground vibration and air overpressure. (Photo 040449)



Typical vibrograph readout.

Controlling Ground Vibration

The amount of vibration is affected mainly by the amount of explosive used per delay, the distance from the blast, the nature of the intervening ground and the blasting pattern used. The presence of presplitting cracks (see Chapter 9) between the blast and nearby residences would appear to make no difference to the vibration.

The nature and condition of the rock or material in which a shot is fired affect both the frequency and the amplitude of the ground vibration but do not greatly affect the particle velocity. This is because PPV varies with the product of frequency and amplitude and so the high frequency and low amplitude in dense material tend to balance each other as do the low frequency and high amplitude in soft material. However, this does not apply to saturated ground in which the frequency and hence PPV are increased.

Several formulae have been proposed for calculating particle velocities using a range of test results and assumptions dependent on local conditions. However, the most reliable method is onsite testing using trial blasts monitored by a seismic instrument from which PPVs can be determined.

As a rough guide the following formula based on 'scaled distances' can be used:

$$PPV = 1140 (R/\sqrt{W})^{-1.6}$$

where R = distance (m)
W = explosive per delay (kg)

A selection of W values using this formula to give a PPV of 8.7 mm/s are:

Distance from nearest residence (m)	Maximum explosive per delay (kg)
20	0.9
40	3.6
60	8.2
80	14.5
100	22.7
150	51.0
200	90.7
250	141.7
300	204.0

Predicting Ground Vibration

An accurate knowledge of the actual vibration levels likely to be generated at nominated places by blasting operations can be obtained only by conducting a seismic survey. This consists of monitoring a number of small blasts at varying distances and plotting the results. Most of the factors which may affect the type and magnitude of the vibration are then automatically included in the analysis.

The formula given previously shows that PPV is a function of $R/\sqrt{W}$ where R is the distance of the point under consideration from the blasthole and W is the weight of the maximum instantaneous charge. The PPVs for various vibrograph records are taken at various distances (R) for a given charge and the values are plotted (using log-log paper) against the ratio $R/\sqrt{W}$ called the 'scaled distance' (Fig. 8.2).

This can be done for a range of directions radially from the blast site to cover possible difference in rock and ground conditions.

From the graph the value of scaled distance can be selected for any specified PPV; then, knowing the R for the site, the charge weight can be calculated.

As a rough guide the chart in Figure 8.5 gives values that can be used for a preliminary or trial blast.

When using a generalised chart like this it must be remembered that in some rock types such as basalt the actual PPV may be as low as 20% of the calculated value, and in some sandstones as high as three times the value, as derived from the scaled distance formula.

Air Overpressure

When an explosion takes place in the confined space of a well stemmed blasthole the enormous pressure of the gases ('bubble energy') forces the fractured rock outward. Airblast and noise are produced when part of the energy escapes to the atmosphere causing a pressure or compression wave in the air. In such a wave, particle vibration takes place in the direction of propagation causing successive compression and rarefaction of the air. The pressure associated with this wave is in addition to normal atmospheric pressure (approximately 100 kPa) and is called 'overpressure'.

These pressure waves cover a wide range of frequencies, some of which are audible and are known as sound or noise waves but most of which are at frequencies below the audible range (less than 20 Hz) and these are generally known as ‘air blast’ or ‘concussion’.

Measuring Air Overpressure

Air overpressure is commonly recorded in decibels (dB) which measure all frequency components such that

$$dB = 20 \log_{10} (P/P_o)$$

where P = measured overpressure in kPa

P_o = reference pressure (2×10^{-8} kPa)

The decibel scale is logarithmic so a difference of 20 dB from, say, 100 to 120 dB represents a ten times increase in pressure.

Sound level meters (sensitive pressure gauges) usually have three scales (dBA, dBC and dBlin) in the first two of which emphasis or weight is given to particular frequencies in the audible range, but to measure true overpressure covering all frequencies it is necessary to use the **peak-linear-hold setting**. The reading can be made in dBlin or kPa. The symbol dB is frequently used for observations in all scales and hence it is important to be aware of the context in which it is being used.

Air blast overpressure monitoring equipment should cover a range of at least 2 to 250 Hz.

Air blast measuring devices frequently consist of a sophisticated microphone linked to the seismograph so that the three ground vibration vectors and the associated air blast wave can be recorded together.

Damage

Residential buildings as a whole have a natural frequency of vibration of from 4 to 20 Hz and therefore air blast vibrations at these frequencies will be amplified by the structure, and increase the risk of damage and complaint, if amplitude and PPV of the vibrations are sufficiently high.

It has been found that poorly mounted, pre-stressed windows might crack at 150 dBlin with most cracking occurring at 170 dBlin. Structural damage would not be expected below 180 dBlin. Rattling of windows takes place at levels above 120 dBlin (Fig. 8.3).

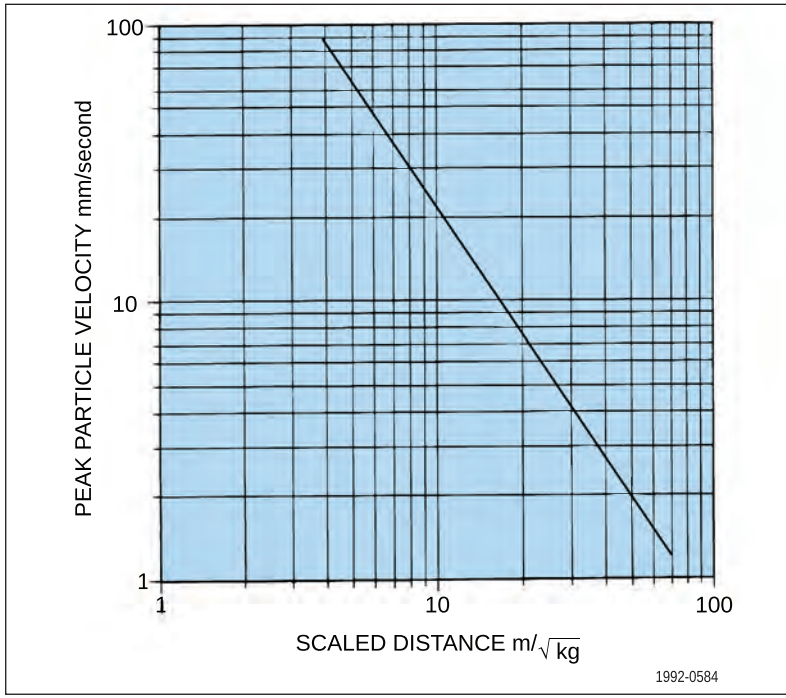


Fig. 8.2 *PPV plotted against scaled distance.*

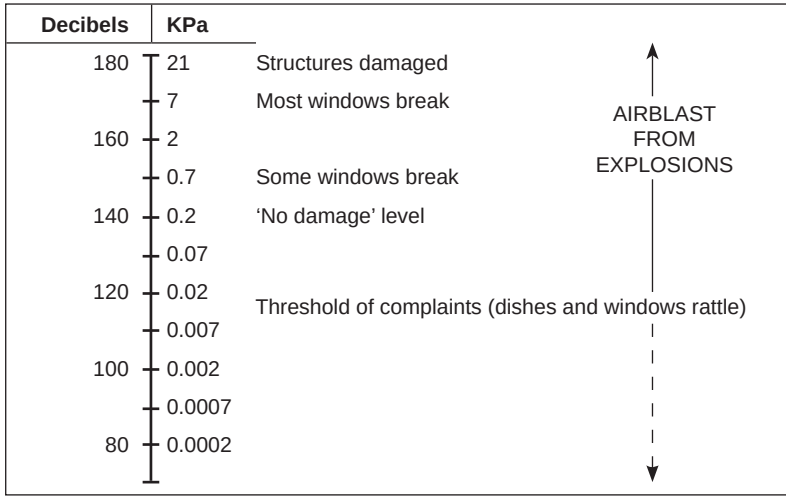


Fig. 8.3 *Human and structural response to sound pressure level.*

Human Response

The human body has a resonance in the range 3-12 Hz and resonance associated with incoming ground or air vibrations at these frequencies will lower a person's threshold of disturbance and discomfort. Also the susceptibility of individuals will vary depending on their activity at the time and other factors such as age, mental attitude and previous exposure. Response levels can be significantly altered by good public relations.

Approximate reactions for different air vibration levels are (Fig. 8.3):

Peak Linear Pressure		**Effect**
(dBlin)	(kPa)	
100	0.002	not noticed
110	0.006	common and acceptable
120	0.020	people may become annoyed

Note: These are instantaneous peak linear pressures and are not to be confused with the more sustained and selective dBA scale used in measuring noise (see Chapter 17)

Standards

Although it is currently accepted by Australian authorities that damage will not occur below 128 dBlin (0.050 kPa) some regulatory bodies have defined limits to ensure that no more than 5% of blasts exceed 115 dBlin and that noise does not exceed 120 dBlin.

Proposed Australian Standard

Table 8.2 gives the draft Australian Standards proposed for maximum levels of airblast to avoid human discomfort and structural damage.

Controlling Air Overpressure

The magnitude of air overpressure at a particular point, and hence possible damage, depends on many factors including:

- type of explosive and its VOD
- distance from blast
- amount of explosive per delay
- burden and spacing
- stemming quality and amount
- direction of initiation
- initiation system

- topography
- atmospheric conditions
- rock type and jointing

Some of these factors can be controlled by the blaster while others must be taken into account when designing the blast and timing the firing. For example, it may be desirable not to fire at times of heavy cloud cover or temperature inversion (where air temperature is warmer at higher altitudes) to avoid reflection of noise back to ground. Firing when the wind is blowing away from residential areas may help to reduce complaints.

Secondary blasting, when detonating cord and very little or no stemming is used, is a common source of complaints. The solution is obvious — cover the cord or use a different form of initiation, use a rock breaker or a primary blasting system that leads to better fragmentation.

Greater care in designing the blast should be exercised when the section being blasted is open to residences and not shielded by other faces or topography.

Table 8.2 *Draft AS DR 91095:R for limiting levels of airblast.*

Time of explosion	**Maximum airblast level to avoid human discomfort** (dBlin)	**Maximum airblast level to avoid structural damage to residential and industrial buildings** (dBlin)
Monday-Saturday 9.00 am - 5.00 pm	120-125*	133
Monday-Saturday 5.00 pm - 9.00 am	115-120+	133
Sunday and Public Holidays	115-120+	133

*Ten percent of explosions may exceed 120 dB
+Ten percent of explosions may exceed 115 dB

The blast should be designed so that gases are not vented out through cracks, inadequate burden, weak ground or poor stemming, before they have been able to create the amount of breakage of which they are capable.

Predicting Air Overpressure

A rough estimate of the decrease in air overpressure with distance can be made by means of 'cube root scaling' using the formula below:

$$K = R/\sqrt[3]{W}$$

where K = cube root scaling factor

R = radial distance from the explosion (m)

W = weight of the explosive per delay (kg)

and substituting for K in the following formula:

$P(Pa) = 3.3K^{-1.2}$ and $dB = 20 \log_{10}(P/P_o) = 165 - 24\log_{10}K$

where $P_o = 10^{-8}$ kPa

It should be stressed that these formulae relate to properly confined charges. For unconfined charges, e.g. a 'plaster', a 'cratered' explosion or a poorly stemmed hole, air overpressure may be up to 30 times higher and the decibel level 30 dBlin higher than for a correctly charged blasthole.

The charts in Figures 8.4 and 8.5, based on the above formulae, may be used as a first approximation.

For particular locations, charts of overpressure plotted against cube root scaling factor can be prepared as described earlier under 'Predicting Sound Vibration'. These estimates must always be reviewed and modified if necessary in the light of the factors discussed under 'Controlling Air Overpressure'. To obtain a rough preliminary indication of air overpressure and ground vibration, the chart in Figure 8.5 may be used.

Service to Industry

For many years the SA Department of Mines and Energy has conducted tests and monitored blasts. The instruments used for measuring ground vibration and air overpressure are:

- Micro Innovations PVMA 5 and PVMA 6 vibration monitors
- Blastronics QMX quarry monitor.

A guideline document *Control of ground vibration airblast overpressure from quarry blasting* is available from the Department.

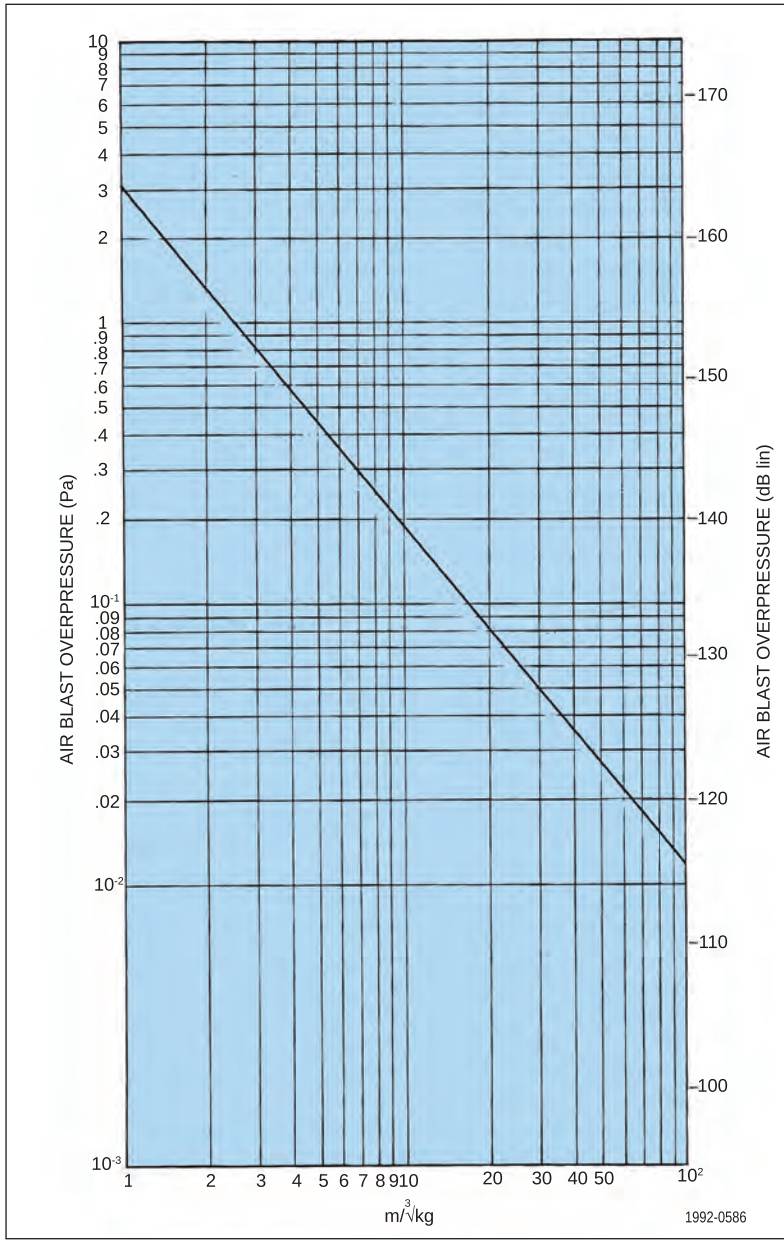


Fig. 8.4 *Air blast overpressure plotted against cubed root scaled factor (K).*

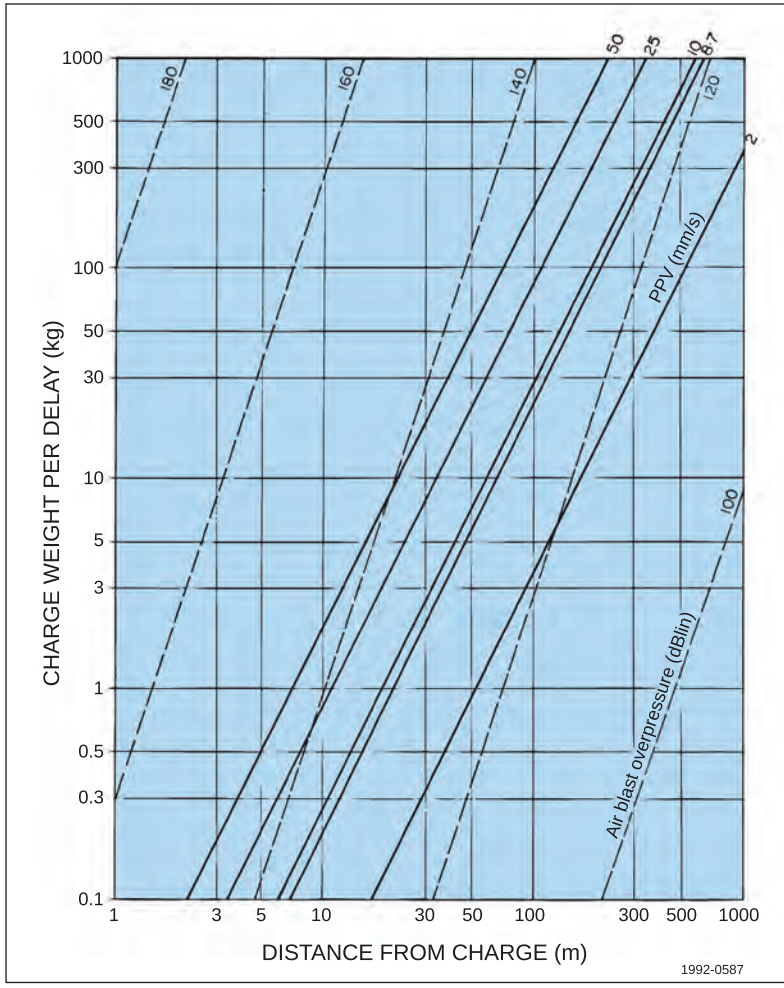


Fig. 8.5 *Ground vibration and air blast overpressure plotted against charge weight per delay and distance from charge.*

9. Quarry Blasting

General Considerations

Purpose of Blasting

The total cost of operating a quarry to deliver a required range of products comprises several components each of which affects the others. For example: a saving in drilling costs may lead to an increase in blasting costs; a reduction in drilling and blasting costs may produce large stones which will increase the cost of crushing; use of a small crusher may increase drilling and blasting costs, and so on. The aim in blasting should be the production of rock from the quarry in a size and form that will facilitate subsequent operations and lead to minimum overall costs.

The size and/or physical shape of the broken rock affect in no small way loading, transporting and crushing, and influence the choice of equipment for these tasks. Improved fragmentation can improve plant capacity without costly additions or modifications. Quarrying techniques are determined by two main factors — safety and economics, and there is a necessity at each quarry to consider the optimum bench height, the correct borehole pattern, a suitable borehole diameter, burden and spacing.

Blasting practice must be considered in conjunction with the equipment used in the various quarry operations for:

- **Loading:** Shovel excavator, front end loader, bulldozer, wheeled scraper, dragline.
- **Transporting:** Truck, railway, skip, dumper, conveyor.
- **Crushing:** Jaw, gyratory and other crushers.
- **Winning of dimension stone:** Crane, cableway, wire saw.

Physical Characteristics of Rocks

For the rocks themselves, hardness, softness, toughness, brittleness and plasticity are important factors in blasting. For the rocks in beds or massive formations the bedding planes, schistosity, joints and faults, and the many combinations of these factors determine drilling and blasting details including the choice of explosives.

Where rock is soft, fractured (broken by joints), weathered or laminated, breaking and loosening for subsequent loading may sometimes be achieved by ripping with or without the help of explosives. This topic is covered in more detail under 'Ripping' in Chapter 11.

Amenity Considerations

Amenity has become more important in recent years and this can affect bench heights, direction of faces, drilling equipment, transporting methods, positioning of plant and roadway location, as well as blasting practice.

Primary Blasting

Blasting Theory

When an explosion takes place, the chemical energy of the explosive is liberated and the compact explosive becomes transformed more or less immediately into a glowing gas mixture with an enormous pressure.

The high pressure shatters the area adjacent to the drill hole and exposes the space beyond that to very great tangential stresses. The diameter of the hole is enlarged and a system of radiating cracks issues from its perimeter.

The shock created by the explosion produces compression waves; if there is a free face in front of them the waves reflect from this surface and give rise to tensile stresses in the rock which may cause part of the rock on the free face to 'slab off'. Rock is stronger in compression than tension. The larger the charge the greater the thickness of layer torn off. With a sufficiently large charge, a crater reaching back to the charge will result. The angle of breakage is approximately 90°. In general the smaller the rock resistance, the greater will be the outbreak angle. The direction of the quarry face relative to the schistosity or cleavage in the rock will also affect the size of the outbreak angle.

A vertical hole in a horizontal surface, where that surface is the nearest free face , will break out a crater as in Figure 9.1 if the relationship between charge and depth is right.

A vertical hole or steeply inclined hole adjacent to a free face will break out a wedge-shaped piece of rock as in Figure 9.2.

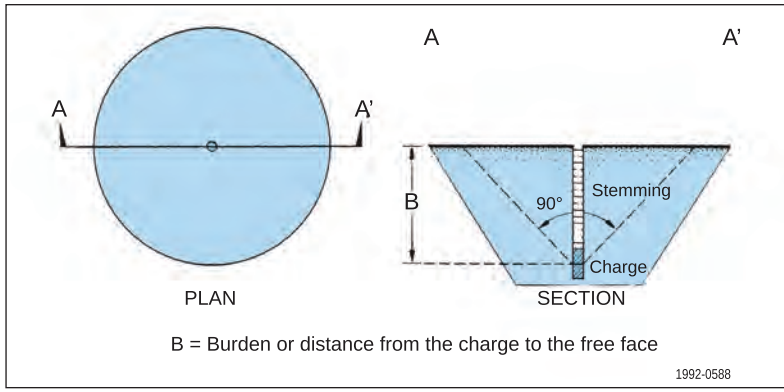


Fig. 9.1 *Result of firing vertical hole in a flat surface.*

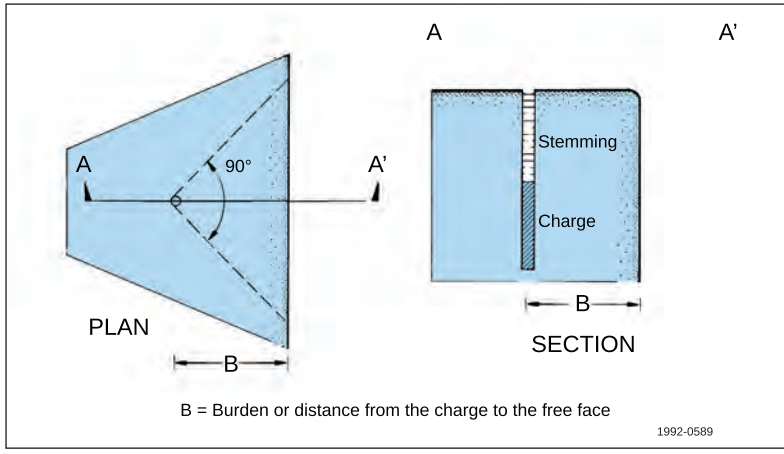


Fig. 9.2 *Result of firing vertical hole near a vertical face.*

Burden

The charge in a borehole, in addition to breaking rock and displacing a mass of rock, must have sufficient force to shear off the rock at floor level to ensure smooth working of that bench. The maximum amount of work in breaking out a rock face therefore has to be done at the bottom of the hole and this is the reason for the burden usually being taken as the distance from the bottom of the hole to the free face measured at right angles to the axis of the hole. It must be remembered that the amount of explosive required at a point in the hole depends on the condition of the rock forming the burden at that point.

Figure 9.3 shows an extreme case where the distance to a vertical face exceeds the depth of the hole; the result of firing an adequate charge will be the conical shaped hole YXZ.

Where the distance to a vertical face equals the depth of the hole, the result may be as shown in Figure 9.4. In this case the break would be along the line Y to Z, if the charge were strong enough; however, in practice the charge would tend to blow a crater Y^1 XW^1. Note that in Figures 9.1, 9.3 and 9.4 point charges are assumed. For plugs or short columns of explosive take the centre of the charge as the point of explosion (X), i.e. the hole would need to be deepened to allow for this or the depth of the crater would be reduced.

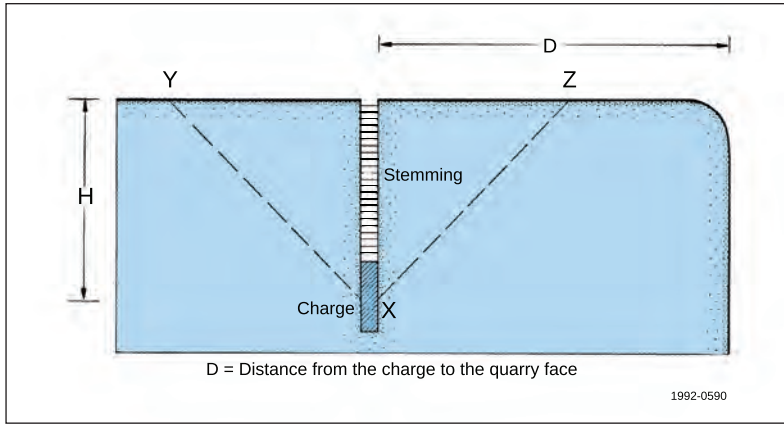


Fig. 9.3 *Section showing distance to quarry face greater than depth of charge.*

With a hole bottoming at the quarry floor, the indicated result would be shown in Figure 9.5. In practice a toe of rock would result as shown. This failure to bring out the toe is explained by the fact that only a portion of the compression wave is reflected from the free face, as shown in Figure 9.6. This can be overcome by increasing the size of the hole at floor level by 'bulling' or deepening the hole (sub-grade drilling) or using a stronger explosive at the bottom of the hole; the most common technique is to bore holes 'below grade', i.e. below the level of the bench floor.

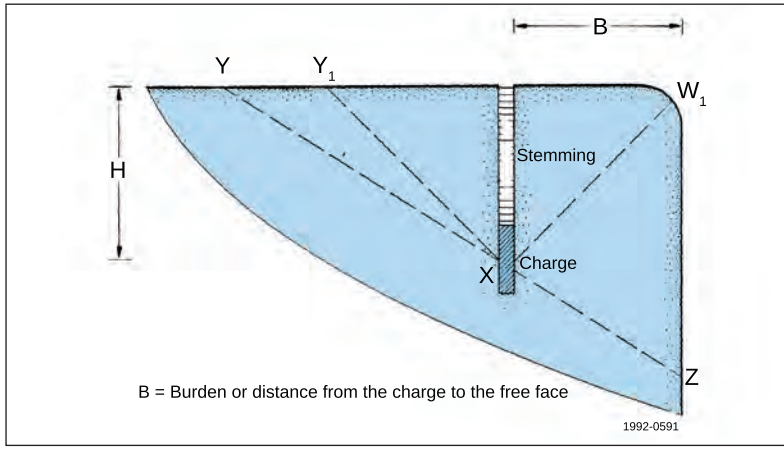


Fig. 9.4 *Section where burden equals depth of charge.*

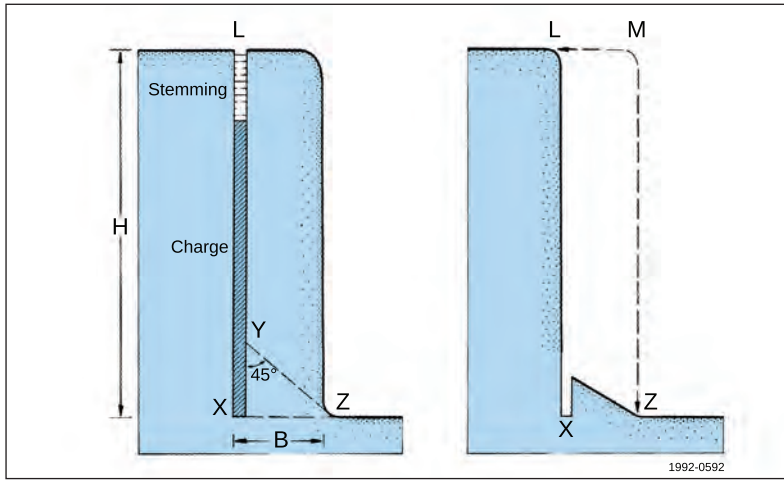


Fig. 9.5 *Section showing hole bottoming at quarry floor.* ***Left:*** *Angle of break 45°.* ***Right:*** *Actual break.*

With inclined holes, owing to the fact that more of the compression wave is reflected back as tension waves, the amount of residual toe is decreased (Fig. 9.6). Refer later under 'Angle Drilling'.

Spacing

Jointing of the rock is one of the factors in determining spacing between holes in a row; the outbreak angle is another. If simultaneous firing is used with holes spaced so that the spacing is equal to twice the burden in a rock with an outbreak angle of 90°, the result should be as shown in Fig. 9.7. If the spacing is equal to the burden, as illustrated in Fig. 9.8, the outbreak angles would intersect at P, leaving the rock in the triangle MNP outside the theoretical outbreak angles. In practice, however, the area broken will be LMNO. The incipient radial cracks resulting from the compression wave will combine with pre-existing joints to split the rock between the holes.

With delay blasting on the other hand, a new free face is developed after each hole is fired. For the situation in Figure 9.8, after M explodes N would tend to burst towards the nearest free face at P, and a bump in the face would develop between N and the next hole in the row. These theoretical conditions are important in the design of quarry blasts.

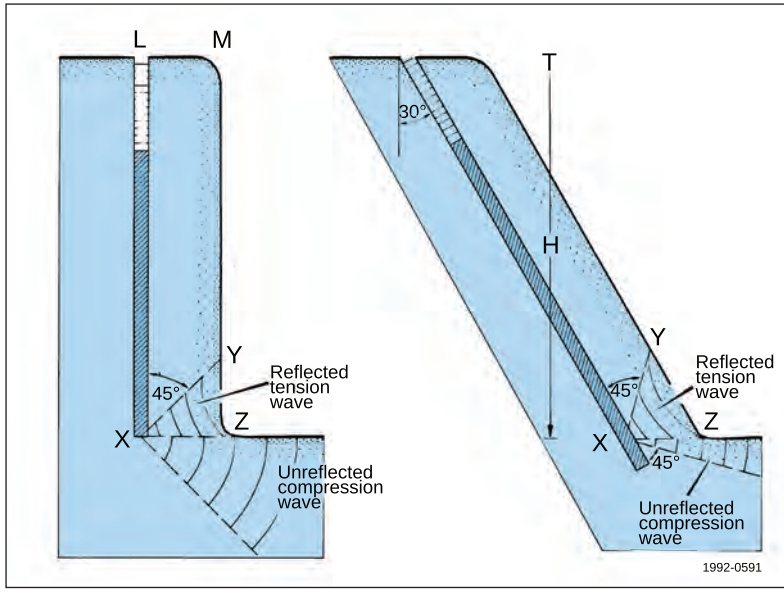


Fig. 9.6 *Section showing shock wave reflection.* ***Left:*** *Vertical hole.* ***Right:*** *Inclined hole.*

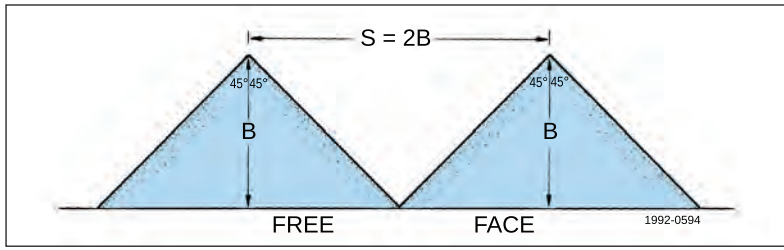


Fig. 9.7 *Plan showing result when spacing equals twice the burden.*

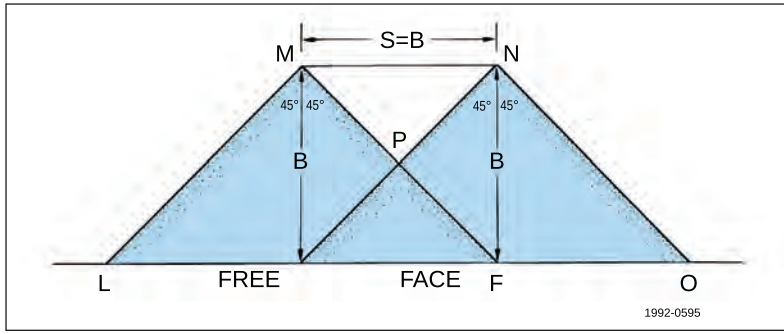


Fig. 9.8 *Plan showing result when spacing equals the burden.*

Experience shows that increases in the burden result in poorer fragmentation, tend to leave an unbroken toe and decrease the general quality of breakage much more quickly than changes in the width of spacing. Therefore, if an increase in the tonnage per metre of hole is desired, it is better to extend the spacing between holes than to increase the burden.

Fragmentation

Good fragmentation of rock in the primary blast is of utmost importance in obtaining the lowest overall cost of quarry product.

Generally, fragmentation is dependent on the amount of explosive used in relation to the volume of rock to be broken.

However, it can be improved by:

- better distribution of explosive charge over the length of hole
- shorter spacing distance
- shorter burden distance
- use of staggered patterns
- use of short delay initiation
- use of inclined holes
- optimising stemming.

Selection of Burden, Spacing and Hole Diameter

The burden, spacing and hole diameter can be adjusted within limits to meet requirements of the loading equipment and plant and the overall cost of the operation. A few empirical rules are given below for guidance in determining these blasting parameters. Usually the higher the bench the greater the burden selected, but remember that the hole diameter must then also be increased.

Rules for guidance:

1. Burden — between 20 and 40 times the borehole diameter; the lower figure applies when using low density explosives in a dense formation
   — between 20 and 40% of height of face; the higher the face the lower the percentage.
2. Diameter of borehole — between 0.5 and 2% of height of face
   e.g. for height 20 m, diameter 100-400 mm
   for height 10 m, diameter 50-200 mm
   for height 6 m, diameter 30-120 mm.
   (Note: Small hole diameters and correspondingly smaller burdens, give better fragmentation and less ground vibration, and allow the use of lighter drilling equipment and small rounds. These are points to be noted in difficult quarrying situations near housing).
3. Drill below grade (below quarry floor) a distance of 1/3 burden.
4. Spacing may be equal to, or up to 2.5 times greater than, the burden depending on the outbreak angle and the drilling and firing pattern. In many situations the best ratio has been found to be spacing = 1.15 times burden (as in a staggered equilateral pattern).
5. Stemming length should be not less than burden distance or approximately 25 times hole diameter and the stemming material should be a clean 7-10 mm aggregate product rather than drill chippings and dust.

The following formulae may be used for the calculation of borehole charges, burden and spacing:

a. Amount of rock broken by each hole = B S H G (t)
where B = burden (m)
S = spacing (m)
H = height of face (m)
G = density factor (t/m^3 in solid condition)

b. Amount of explosive required per hole = G B S H F (kg)
where F = explosives ratio (kilograms of explosive per tonne of rock)

In practice it will be found that explosives ratios vary considerably, depending on the rock being blasted. Ratios of from ½ kg/t to 1/20 kg/t are common.

Trial Blasts

Determination of boring patterns and explosive charges depends not only on the factors outlined above — particularly the value of explosive ratio to be used — but also by the effect of local rock structures such as bedding planes, faults and joints. Primer position and appropriate decking in the blast hole are also important (Fig. 9.9). Consequently, in any new situation or where results are unsatisfactory, trial blasts should be made to obtain the best burden, spacing, depth below grade, hole diameter, loading and so on, for the tonnage and type of product required from a blast.

Among the points to be noted during and after each trial blast (or for that matter any blast) are noise, ground vibration, air concussion, rock pile profile, scatter, fragmentation, backbreak, and toe.

These observations will usually lead to alterations being made to the design of the next blast. Care must be taken not to alter too many of the parameters at once, otherwise the results of the next blast may not be positively correlated.



Drilling quarry blast holes. Note enclosed cabin. (Photo 036414)



Blasting iron ore in the Middleback Ranges. (Photo 037739)

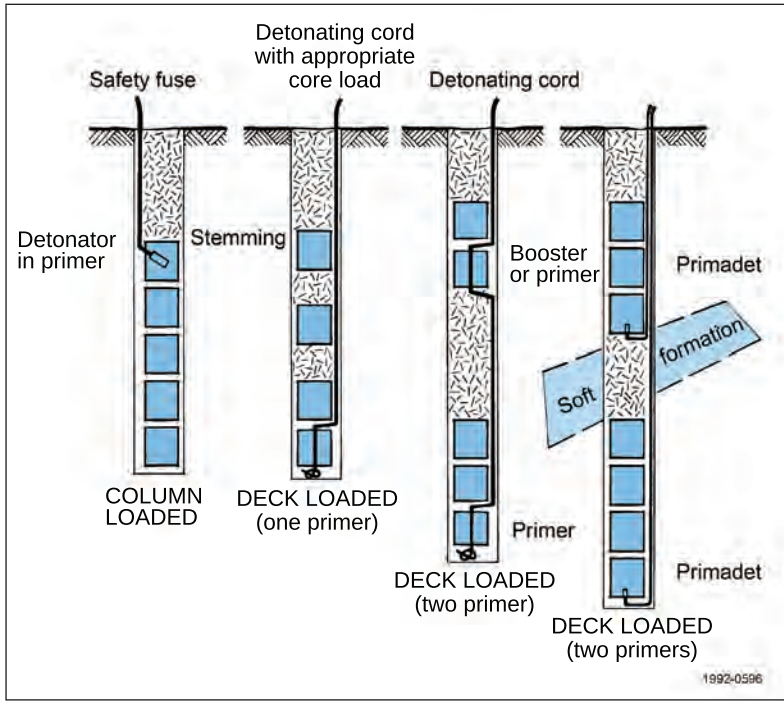


Fig. 9.9 *Types of loading for cartridges. Note position of primer.*

Profiling Blast Holes

Figure 9.10 is a section through a typical quarry face showing a blasthole which illustrates how the burden can vary from point to point down the hole. Clearly for optimum blasting results the charging in the hole should be determined not only by the rock and its structure but also by the burden at each point in the hole. In order to achieve the most efficient combination of drilling and charging, profiling the face is desirable.

Profiling is best done using a laser instrument which measures distance and angular position of points in the face from the instrument base; from these measurements an associated computer produces a series of cross-sections of which a typical one is shown in Figure 9.10. The position of the drill hole can be determined in the same way by sighting a rod placed in the hole.

It is said that the reduction in flyrock and blasting costs will go some way towards paying for the system. It is worth noting that new legislation in the United Kingdom requires that quarry rock faces be surveyed prior to blasting.

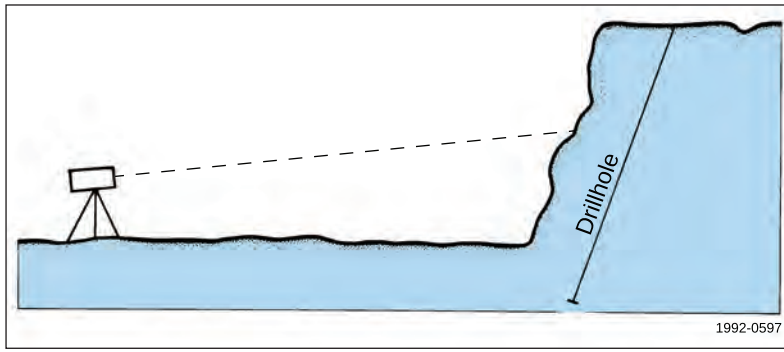


Fig. 9.10 *Profiling a working face.*

Blasting in Single Holes

In some low-production quarries, with limited capital and small equipment, the rock may be blasted down by means of single holes. These holes may be bored to a general pattern across and down a face, but in reality they are random holes, in that they are located and directed to take advantage of any joints and bedding planes in the rock, or of fractures caused by previous firings.

Single holes may also be fired in very jointed rock, where the ground just needs a shake up for digging and loading. Sometimes the proximity of housing may limit blasting to single holes. In dimension stone quarries single holes, or two holes bored to join and form a V, are used to split large stones in place. Single holes may also be used to clean out a zone of unwanted broken material.

Blasting in Rows

In practically all surface mining and quarry blasting a number of charged holes are fired in each blast, and in most of the regular production blasts the holes are arranged in rows. Each charge in a blast sets up stresses in the rock and generates pressures with the gases produced. The firing of adjoining charges in a blast puts extra stresses on the rock between holes, and this coupled with rock collisions makes for better fragmentation.

Single row firing is practised at many quarries, but where circumstances permit, multi-row firing is used. As a general rule three rows would appear to give best fragmentation and heave characteristics. In both single row and multirow firings, delay blasting contributes to the fragmentation effect by

providing a series of new free faces. Delay blasting also enables the total amount of charge to be increased many times without increasing ground vibration.

A few examples of single row and multirow patterns are given in Figure 9.11.

Using non-electric systems such as Primadet Trunkline Delays and MS Connectors an unlimited number of delays can be incorporated in one blast (Figs. 9.12 and 9.13).

Choice of Explosive

Factors to be considered in selecting an appropriate explosive include:

- Tenacity and other physical characteristics of the rock.
- Presence of bedding planes, joint planes and similar planes of weakness.
- Whether the hole is 'dry' or 'wet'.
- Diameter of the boreholes: large diameter holes (and small diameter holes that are 'bulled' or 'chambered' — see later) call for use of low-strength, low-density explosives. In harder tenacious formations small diameter holes are more economical to drill with percussive drills but higher strength explosives are required.

Quantity

In determining the type, strength, quantity and cartridge diameter of explosive to be used for any particular job the considerations are:

- Applicability to the particular circumstances.
- Safety of life and property.
- Efficiency (measured as tonnes of rock per kilogram of explosive and varying from 2 to 20 depending on rock type and conditions, e.g. free face or no free face).
- Economy (expressed as total cost of drilling, charging and firing and covering cost of explosives, accessories, stores, labour, capital charges and including secondary breaking per tonne of rock broken).

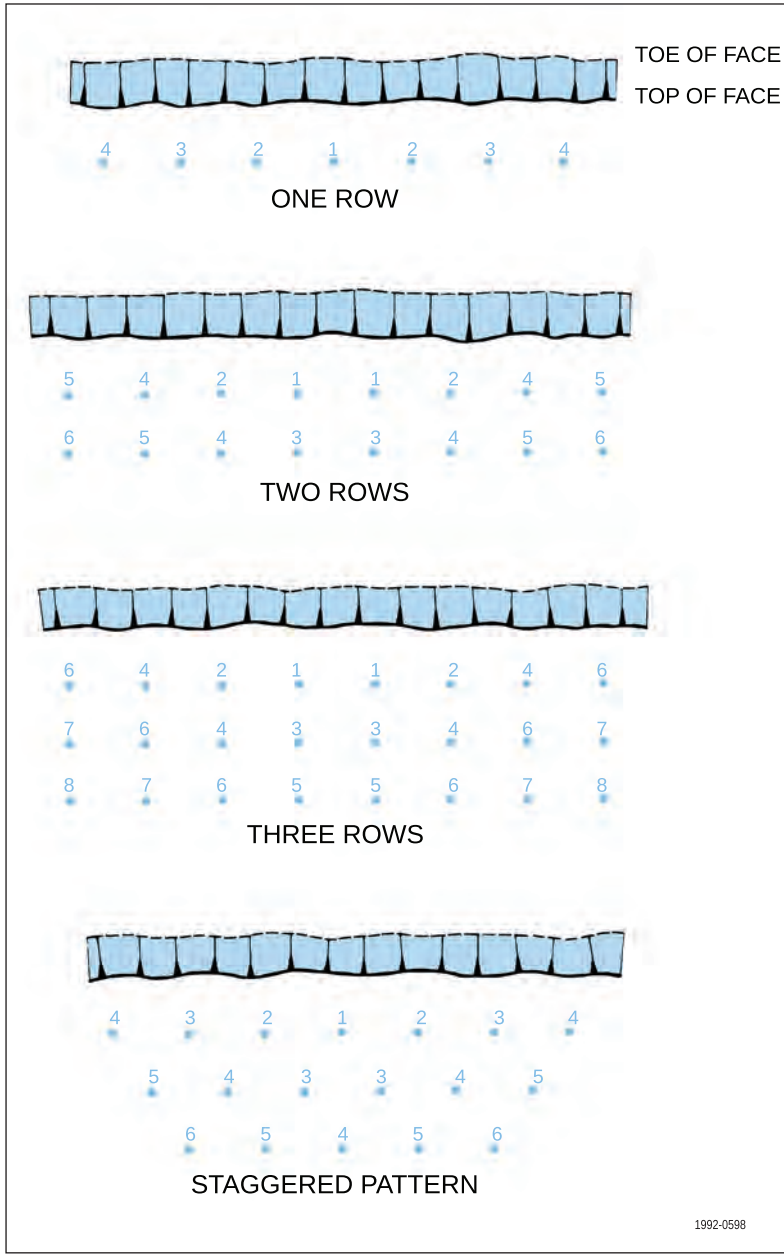


Fig. 9.11 *Firing patterns to give centre piles.*

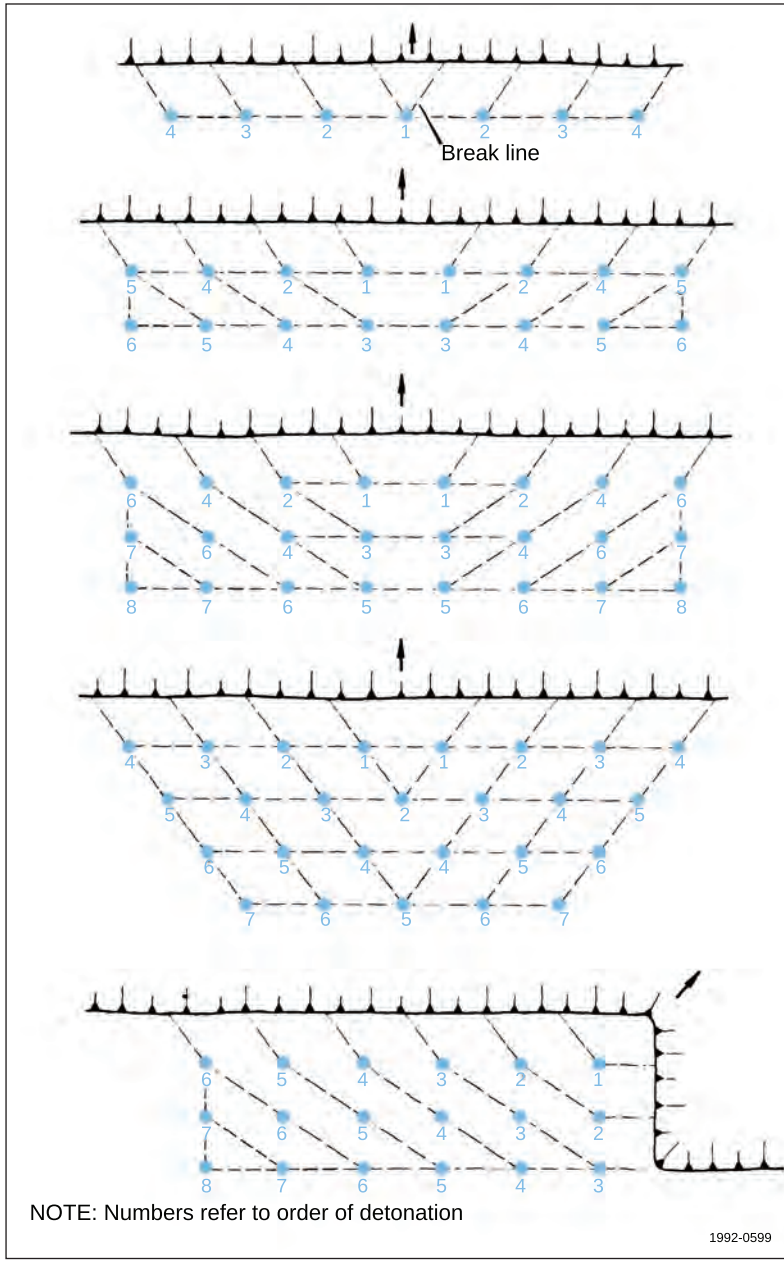
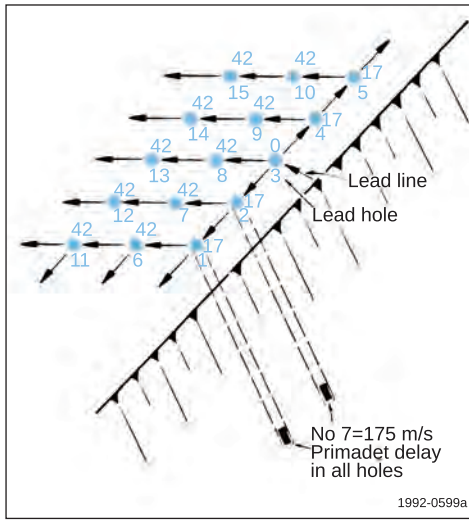


Fig. 9.12 *Firing patterns using Primadets.*



Hole number	Firing time from start (m/s)		Order of hole exploding
	Surface	Primer	
1	34	209	3
2	17	192	2
3	0	175	1
4	17	192	2
5	34	209	3
6	76	251	6
7	59	234	5
8	42	217	4
9	59	234	5
10	76	251	6
11	118	293	9
12	101	276	8
13	84	259	7
14	101	276	8
15	118	293	9

Fig. 9.13 *Layout using signal tube delays.*

After the type, strength and density of explosive have been determined for the particular conditions, the quantity to be used depends on:

- hole diameter
- spacing
- hole depth
- burden.

Angle Drilling

Drilling of inclined holes (up to 30° from vertical) is now common practice in quarries, a typical angle being 15°. The angle depends on the dip of the strata, the cleavage and jointing in the rock, and the angle which the quarry face makes with the strike direction of these features.

The advantages claimed for angle drilling are:

- The collar of the hole is a greater distance from edge of bench, there is less overhang, and the face is cleaner leading to a safer operation for both men and equipment.
- Improved fragmentation and greatly reduced secondary blasting cost.
- Reduced toe problems.
- Back break difficulties practically eliminated.
- Increased muckpile throw — better for front end loaders.

- Greater uniformity of burden throughout the length of hole.
- Less sub-grade drilling leading to less floor damage and better drilling through that floor later.
- Progressive reduction in explosives consumption with increased inclination from the vertical.
- Less vibration.
- Reduction in blasting, loading, and crushing costs.

Among the disadvantages are:

- More difficult layout and drill set-up required to ensure correct burden, drilling angle and direction.
- Drilling is more likely to be inaccurate resulting in too much or too little burden near toe.
- There can be more fly rock.
- Deviations can cause environmental problems.
- Difficulties increase with hole diameter due to rod handling problems.

Smooth Blasting and Pre-Splitting

Smooth blasting and pre-splitting techniques involve the firing of lightly-charged, closely-spaced holes around the perimeter of a section of rock.

If firing is done with or after the main blast it is called 'smooth' blasting. The spacing of the holes should be about 15 times their diameter. The ratio of spacing to burden should be close to 0.5 to 1, but can be as high as 0.8 to 1.

If firing is done before the main round blast, it is called 'presplitting'. The spacing of the holes should be about 10 times their diameter. Use 180-280 g of explosive per m^2 of sheared area if Powergel 3131 or Powergel Perimeter is used. All holes should be fired simultaneously.

Smooth blasting or pre-splitting have the following benefits:

- For surface excavations (quarry faces, spillways, road cuttings and so on): safer, smoother faces, less back break, less erosion or ravelling in the long term.
- For trenches: less excess material handling, less vibration from main charge, more fragmentation, and safer sides to the trench.
- For tunnels: less overbreak, less excess material handling, less concrete to line tunnels.



Presplit quarry face. (Photo 040451)

Tunnel ('Coyote') Blasting

Where very large quantities of rock are to be blasted and the conditions are suitable, large quantities of explosives can be placed in excavated chambers or tunnels. After the position and quantity of the explosives have been determined the tunnels are excavated to the required size. The explosives are packed into the chambers in their original cases and the connecting passageways back-filled. Charges are usually fired electrically with detonating fuse as a complementary detonant.

Suitable conditions could be:

- Quarries and large sidehill cuts where conventional drilling methods are impracticable because of high costs.
- Rock types where the desired degree of fragmentation can be obtained simply by dislodging and heaving out a large mass of material.
- The production of large rock pieces used for breakwaters.

As a general rule coyote blasting cannot be considered near residential areas because of the large quantity of explosives detonated on a single delay.

'Bulling' or 'Chambering' Holes

The bottom of a small diameter borehole (such as made by a diamond drill or wagon drill) may be enlarged by exploding in it a small charge of a strong explosive, preferably by means of an electric detonator. No stemming other than water should be used. The effect is to pulverise the rock around the charge and to expel it from the mouth of the hole as dust. The bottom of the hole is thus enlarged in the form of a bulb-shaped chamber which will accommodate a larger quantity of explosive. Care must be taken to allow the bottom of the hole to cool down before adding further explosives.

The procedure is repeated until the chamber is large enough to hold the calculated amount of explosives for the main blast, but it should be noted that there is no easy way to measure the size of the cavity and it is easy to cause damage further up the hole which might necessitate redrilling. The method has been used chiefly in quarry work in bench holes. The holes can be drilled from small benches on high sloping faces or to break vertical faces.

A free-running explosive in the bulled holes is more effective than cartridged explosives as the former type fills the entire cavity. Bulling has an application where drilling is slow and costly or where large hole equipment is not available. Fragmentation is usually not good with the bulled hole method, and a ragged face usually results.

Quarrying using Small Diameter Holes

Some small and medium sized quarries which are equipped with relatively small primary crushers use small diameter holes drilled the full depth of the quarry face to ensure improved fragmentation and greater output. Others, without the facility to drill deep holes, are using methods based on short small diameter holes drilled by means of jackhammers, air-leg machines, and wagon drills.

Holes are usually limited to depths of from 5 to 6 m because of increasing difficulties of handling drill steel and removal of cuttings.

Faces higher than 6 m can be broken by using a combination of vertical, inclined and horizontal holes.

Quarrying using Large Diameter Holes

Where a quarry is set up to handle large daily tonnages, loading arrangements on the quarry floor and the size of primary crusher usually permit rock to be broken to a larger size in the primary blast than would be desirable in a quarry with small size equipment. This allows use of large diameter blastholes.

For drilling large diameter holes in quarrying operations, percussion drilling and rotary drilling machines are in use: percussion drilling for the harder rocks and rotary for the softer formations. For very soft material auger type drills are used.

Most rock quarries use blast holes 75-120 mm in diameter with patterns 1.8 x 2.1 m to 4.0 x 4.0 m, although holes up to 450 mm in diameter with correspondingly larger patterns are used in the large metal and coal open cut operations.

Hole depths are determined by face heights which are commonly 8-15 m but range up to the limit set by regulations (20 m in South Australia with provision for greater heights by special permission).

Holes are drilled from 0.75 m to several metres below the quarry floor level to assist in breaking out the toe.

Hole patterns can be square (or rectangular) or staggered; it should be noted that holes drilled in a square can be fired to a V pattern giving effective spacing greater than effective burden with better fragmentation.

Secondary Breaking

To obtain good fragmentation a good blasting design should be used for the primary charges. Secondary breaking, whether by blasting or mechanical means, to enable oversize rocks to be reduced so that they can be handled by the loading equipment and the primary crusher is very expensive.

Apart from manual spalling with sledge hammers or use of a plug and feathers in a short drill hole, there are several methods by which this secondary size reduction can be accomplished:

Popping

A ‘pop’ is a hole of small diameter (30-40 mm) bored approximately to the centre of the boulder, charged with explosives and fired by a detonator and safety fuse, or by other means. An advantage of using this method is the small quantity of explosives required — the aim being to break the rock to the size required without undue scattering; 75-100 g of explosive per m^3 of stone is commonly used, but the charge can vary widely in different types of rock. Provided a suitable initiating system is employed, for example plastic igniter cord, a large number of pops can be fired at once. Disadvantages are fly rock scatter on the floor of the quarry, air blast, and the labour, compressor and drill required to bore the pops; it is also necessary to move equipment to a safer position.

Plastering ('Mud-capping' or 'Blistering')

A charge is placed on top of the boulder — if possible in a depression — and covered with mud or clay and detonated as above. Advantages with this method are no boring required, little fly rock and a job quickly done. Disadvantages are that it requires about four times as much explosives as are used in popping and the noise and airblast constitute a nuisance. It is not tolerated in residential areas and is not recommended.

Drop Ball

In this method a cast steel weight on a hoist rope is dropped from a crane on to the oversize stone. It is the experience in South Australia that there are substantial savings in the use of the drop ball for secondary breaking, not only directly in drilling and explosives costs but indirectly, in that there is no need to clear equipment out of the quarry prior to a blast and no bulldozing of the scattered rock on the floor of the quarry — in short there are no interruptions to the loading operations. Drop balls weigh from 1.5 to 2 t. Often shovels or draglines which are no longer suitable for front line production are converted to drop ball machines considerably reducing the capital required if a new machine were needed.

However, the process is slow, requires the stones to be laid out on firm ground and some degree of operator skill is necessary.

Rock Breakers

Pneumatic and hydraulic rock breakers mounted on hydraulic excavators are now widely used in quarries to break oversize stones left after primary blasting. The latest designs are able to automatically vary blow energy and frequency according to rock hardness and ground conditions.

They are also used to break oversize stones bridging or blocking primary crushers and are invaluable in applications such as excavating, trenching, demolition or concrete breaking, particularly near occupied buildings.

Much of this work can be done by contractors who move from site to site as required.

The Rocjack and Darda machines have found a special application in boulder breaking in places where explosives cannot be used. A hole is bored into the boulder, a wedge device is inserted and a hydraulic pressure exerted on the wedge to break the rock.



Drop ball. (Photo 040452)



Hydraulic rock breaker. (Photo 040453)

Other Methods

Electric energy and ultrasonic waves have also been used to fracture boulders or rock in place. Neither of these methods is generally applicable at this stage to boulder breaking in quarrying.

Summary

Quarry blasting practices at some South Australian operations are shown in Table 9.1, while Table 9.2 shows further details of the drilling, blasting, loading and transport equipment as well as information about the production from each quarry.

Table 9.1 *Blasthole loading practice at some South Australian quarries.*

Quarry	**Rock**	**Face height** (m)	**Burden** (m)	**Spacing** (m)	**Diameter of hole** (mm)	**Explosives**	**Explosives ratio** (kg/t)
Ardrossan	Dolomite	9.0	5.0 3.0	5.0 3.0	159 76	ANFO	0.16
Iron Duke	Iron ore	8.0	5.0	6.0	250	ANFO	0.24
Iron Knob	Iron Ore Waste	10.0 10.0	5.0 6.0	6.0 6.0	250 250	ANFO in dry holes; 20 kg Tovex/Riogel cartridges in wet holes	0.20
Klein Point	Limestone	4.0-13.0	3.0	3.0	90	Powergel, ANFO	0.28
Leigh Creek	Overburden	12.0	9.0 11.0	9.0 11.0	200 225	ANFO	0.18 (kg/m^3)
Linwood	Dolomite	12.0-16.0	3.0 3.3	3.3 3.6	75 89	ANFO ANFO	0.15 0.17
Mount Schank	Basalt	8.0	3.0	3.0	93	ANFO	0.09
Para Hills	Quartzite	15.0	3.0	3.0	75	ANFO	0.08
Penrice	Limestone (marble)	14.0	3.5	4.0	115	ANFO, Energan, Powergel	0.13
Rapid Bay	Limestone	10.0	4.4	5.1	140	Powergel with ANFO	0.16
Reynella	Dolomite	15.0-18.0	3.0	3.3	89	ANFO	0.22
Riverview	Dolomite	14.0	2.1	2.4	76	ANFO, Riogel	0.20
Salisbury	Quartzite	12.0-18.0	3.0	3.0	89	ANFO	0.24
Sellicks Hill	Dolomite	14.0	3.5	4.2	114	ANFO	0.11
Stoneyfell	Quartzite	12.0-20.0	3.0	3.0	89	ANFO	0.30
White Rock	Quartzite	10.0	2.7	2.7	75	ANFO, Riogel	0.08

Table 9.2 *Some South Australian quarry practices.*

	Ardrossan Quarry	Black Hill Quarry	Golden Grove	Iron Duke Quarry	Iron Knob Quarry	Klein Point Quarry
Stone or ore	Dolomite	Diorite dimension stone	Sand and gravel	Iron ore (haematite)	Iron ore (haematite)	Limestone
Face height (m)	9	2.5-4.0	3-15	8	10	4-13
Minimum width working bench (m)	30	3	6-30	10	10 (Haul roads approx. 25 m wide)	27
Face slope (degrees)	60	Vertical	60-70	60	45-63	22-24
Overall quarry slope (degrees)	46	—	45	45	33-45	—
Driling equipment	2 x GD RCD 930 1 x IR Crawlair LM300	Drifters Diamond wire	—	BE 45R	BE 45R	1 x Drifter
Hole diameter (mm)	GD 159 IR 76	30-35	—	250	250	90
Depth below grade (m)	1.5 1	—	—	2	2	—
Hole pattern (m) (burden x spacing)	5 x 5 3 x 3	Closely spaced (12-20 cm)	—	5 x 6	Ore 5 x 6 Waste 6 x 6	Slab holes in hard patches
Explosives	ANFO	K-pipe ANFO	—	ANFO	Dry holes: ANFO using ST bowl mixers Wet holes: cartridged Tovex (20 kg)	ANFO
Priming	225 g booster	—	—	HDP 450	HDP 450 0.5 kg Austin primer	Powergel (Magnum)
Initiation	Safety fuse/Detaline	Safety fuses with detonator	—	Nonel U & UB series with 3.5 g/m cord	Safety fuses, None1 U & UB series, and 3.6 g/m detonating cord	Safety Fuse with 5 g detonating cord
Stemming material	Drill cuttings	Drill cuttings	—	Drill cuttings	Drill cuttings	Drill cuttings

Table 9.2 *Cont.*

	Ardrossan Quarry	Black Hill Quarry	Golden Grove	Iron Duke Quarry	Iron Knob Quarry	Klein Point Quarry
Depth of stemming (m)	3.5 minimum	—	—	5	4-5	1
Explosives ratio (kg/t)	0.16	—	—	0.24	0.20	—
Loading equipment	2 x 1.9 m^3 RB shovels 1 x 2.5 m^3 RB shovel 1 x Cat 980 loader 1 x Komatsu W A 450 1 x Komatsu WR 500	FEL	2 x 996 Cat FEL 2 x 623 wheeled scrapers (20 m^2) 1 x Cat D9 dozer	Cat 992 FEL	2 x Cat 992c FELs	1 X lO^3 FEL 1 X 5.4^3 FEL
Transport equipment	6 x 16 t trucks (highway) 3 x 35 t trucks (off highway)	Crane	do.	Komatsu HD 785/3	4 x 785/3 Komatsu dump trucks (85 t)	2 x 85 t Cat 777 dump trucks
Stripping method	—	Excavator FEL	do.	FEL and dump truck	Drill, blast, load and haul	Bulldozer 1 x Cat D9G 1 x Cat D10
Production	Ore - 900 000 tpa* Waste - 900 000 tpa	1000 m^3 pa	150 t/h	—	Ore: 1.01 x 10^6 tpa Waste: 2.23 x 10^6 tpa	1.6 x 10^6 tpa limestone 250000 tpa waste

* tpa = tonnes per annum

Table 9.2 *Cont.*

	Leigh Creek Coal Mine (Open Pit)	**Linwood Quarry**	**Mount Schank Quarry**	**Para Hills Quarry**	**Penrice Quarry**
Stone or ore	Coal (sub-bituminous) with limestone overburden	Dolomite Limestone	Basalt	Quartzite	Limestone (marble)
Face height (m)	12 (overburden)	12-17	8	15	14
Minimum width working bench (m)	40	20	20	40	15
Face slope (degrees)	70	65-75	Vertical	27	70
Overall quarry slope (degrees)	40-50	45-55	—	—	65
Driling equipment	IR DM 35 IR DM 40	GD Hydratrack IR LM 400c	Crawlair IR DXL 800 cfm ECM 350	Contract	Atlas Copco DHH down hole hammer IR (DM 25 SPH), single pass DHH or rotary
Hole diameter (mm)	200 225	75-89	93	75	115
Depth below grade (m)	Nil	1	—	1	1
Hole pattern (m) (burden x spacing)	9 x 9 11 x 11	3 x 3.3 3.3 x 3.6	3 rows of 3 x 3	3 x 3	4 x 3.5
Explosives	Dry: ANFO Wet: Powergel emulsion	ANFO Energan Powergel	ANFO Powergel (Magnum)	ANFO	ANFO Energan Powergel gold emulsion
Priming	454 g single booster	Amex boosters	Anzomex	Powergel	‘P’ Boosters
Initiation	None1 3.6 g down line	Primadets trunk line delays	None1	Electric	None1
Stemming material	Drill cuttings	7 mm aggregate sand	10 mm quarry rubble	7 mm aggregate	6 mm screenings or drill cuttings
Depth of stemming (m)	5-6	2.5	2	2.5	4.5
Explosives ratio (kg/t)	0.16	0.17	0.09	0.08	0.13

Table 9.2 *Cont.*

	Leigh Creek Coal Mine (Open pit)	Linwood Quarry	Mount Schank Quarry	Para Hills Quarry	Penrice Quarry
Loading equipment	2 x PH 2800 electric shovels (26 m^3)	Cat 988B Liebherv 40 t excavator	988B wheel loader	1 x FEL	1 x WA 600 FEL 1 x O&K face shovel
Transport equipment	19 x Titan 33-15B haulers (155 t)	Cat 769c dump trucks (35 t)	1 x 35t-769 dump truck	1 x 50 t Euclid	3 x 50 t and 2 x 35 t dump trucks
Stripping method	Truck and shovel	Excavator	Bulldozer	FEL	—
Production	2.8 x 10^6 tpa coal 16.0 x 10^6 m^3 tpa overburden	300 t/h 600 000 tpa	120 tpa	300 t/h	1 x 10^6 tpa 250 000 tpa overburden

Table 9.2 *Cont.*

	Rapid Bay Quarry	**Riverview Quarry**	**Sellicks Hill Quarry**	**Stoneyfell Quarry**	**White Rock Quarry**
Stone or ore	Limestone	Dolomite	Dolomite	Quartzite	Quartzite
Face height (m)	9 and 16	14	14	12-22	10
Minimum width working bench (m)	40	20	15	20	10
Face slope (degrees)	70	68-70	75	75	80
Overall quarry slope (degrees)	45	–	55	45-55	—
Driling equipment	GD airtrack Halco 400c	Atlas Copco hydraulic rig	Halco 650 DHH GD airtrack	Atlas Copco 812 hydraulic rig	Crawlair VL 140 ECM 350
Hole diameter (mm)	Airtrack 90 Halco 140	76	105-114 76	89	75
Depth below grade (m)	1.5	1.2	1 0.75	1	1.2
Hole pattern (m) (burden x spacing)	4.4 x 5	2.1 x 2.4	3.5 x 4.2 2.2 x 2.2	3 x 3	2.7 x 2.7
Explosives	ANFO	ANFO, Riogel, watergel	ANFO, Energan, ANFO	ANFO, Energan, Powergel	ANFO and AN60
Priming	Powergel Magnum (10 kg front row 5 kg back row)	Riogel	Powergel Magnum + Anzomex Powergel	Amex booster Powergel plugs	AN60
Initiation	Full None1	Detaline	Electric MS delay detonators (Primadet)	Primadet trunk line delays	Primadet MS delays
Stemming material	Drill cuttings	20 mm screenings	14 mm screenings Drill cuttings	7 mm aggregate	Drill cuttings
Depth of stemming (m)	3	1.8	3	3	4
Explosives ratio (kg/t)	0.06	0.14	2.5	0.30	0.80

Table 9.2 *Cont.*

	Rapid Bay Quarry	Riverview Quarry	Sellicks Hill Quarry	Stoneyfell Quarry	White Rock Quarry
Loading equipment	1 x 2.7 m^3 electric shovel 2 x 5.4 m^3 FEL	2 x Cat 980 FEL shovel	Cat 988b Cat 980c	988c Cat loader	2 x Cat 980c FELs
Transport equipment	1 x 35 t Cat 769 1 x 35 t Euclid R35	3 x 35 t Komatsu 2 x 980 FL Cat	2 x 25 t Euclid dump trucks	50 t dump trucks	1 x 35 t Cat 769B 2 x 30 t Wabco 1 x 35 t Euclid
Stripping method	Nil	Bulldozer FEL 30T trucks	Cat 225 excavator	Drill and blast where necessary	Bulldozer
Production	40 000 tpa	760 000 tpa waste 380 tpa dolomite	180 t/h	700 000 tpa	240 t/h

10. Loading and Transport

Earthmoving involves two principal activities: loading ore or waste from the working face onto a transport system, and transport of the material to its final destination. Different equipment is commonly used for loading and hauling, but some types of equipment can perform both tasks. The removal of waste is sometimes referred to as stripping — a term also applied to the removal and stockpiling of topsoil. Occasionally one will see a reference to stripping equipment as being different from mining equipment, but this usually refers to stratiform deposits such as coal. In quarrying no such generalised distinction occurs.

The range of loading and transport equipment is endless. There are many manufacturers each making many types of machines. Most of this equipment can be grouped into a few categories, each with a particular and limited application. In this chapter the mechanics of the more commonly used equipment and the applications to which each is best suited will be described.

Integrated Loading and Hauling Systems

Integrated loading and hauling systems are combinations of equipment which carry out both the loading and hauling functions. Some individual pieces of equipment can carry out both functions.

Dozers

A bulldozer (‘dozer’) is a tractor equipped with a front pusher blade which can be raised or lowered by hydraulic control and the combination is used for digging and pushing broken material. Angle dozers have blades which can be set at angles to cast dirt on either side. The motor is mounted on crawler tracks or four rubber tyred wheels. The distance the material is pushed is usually not more than 100 m, preferably downhill. The output of a dozer depends on the traction system (crawler or wheel), power, grade, distance, blade size and shape, and the material being pushed.

Examples of tasks for which crawler dozers may be used are:

- Pushing waste direct to a final or interim disposal site.
- Pushing material over the edge of dumps.
- Pushing ore or waste into a receiving hopper to feed a crusher/ conveyor system or to fill trucks.
- Pushing material to a front-end-loader or, rarely, a power shovel.
- Cleaning up a bench or around a shovel.
- Ripping soft or medium hard jointed rock. For this task one, two or more ripper teeth are attached to a frame mounted at the rear of the dozer; the ripped broken material can then be pushed into piles or loaded by a bowl scraper.
- Push-assisting a bowl scraper.
- Levelling ground in reclamation work.
- Performing construction work, e.g. roads, drains, pipe laying and conveyor shifting.

Scrapers (‘Bowl scrapers’ or ‘Wheeled scrapers’)

The scraper is an articulated rubber tyred vehicle: the front section consists of engine, transmission, two driving wheels and the operator’s cab; the rear section consists of the bowl, elevator mechanisms (if fitted) and the rear wheels. Twin power scrapers have the second engine, transmission and drive axle in the rear section. There are two main classes of scraper: open bowl and elevator; the latter use either an elevating chain conveyor or a pair of augers to move the dirt back into the bowl.

In operation the scraper approaches the dig site in low gear as fast as possible slowly lowering the bowl with its cutting edge into the dirt. If necessary a dozer is positioned behind the scraper to push it forward. The operator controls the dig depth by controlling the height of the cutting edge. When the bowl is full, the operator raises the bowl, lowers the apron to retain the dirt and drives to the dump site. To dump — which is done on the move — the operator raises the apron and activates the push plate to force the dirt forward and out of the bowl.

Bowl capacities range up to 30 m^3. Loading is very fast — 20 to 40 seconds for a 20 m^3 scraper -and wherever possible should be carried out downhill. Scrapers are best suited to dirt or soft rippable material which breaks up readily into small lumps although open bowl scrapers can handle lump materials up to 600 mm sizing.

In quarrying applications, scrapers are most commonly used for topsoil removal, but are also used for bulk earth movement, particularly in clay and sandpits. Because of the extra weight involved in the bowl and (where fitted) the self-loading mechanisms, haulage by scraper is more expensive than haulage by truck, and scrapers are generally used only where the one way haul is less than a kilometre, i.e. when the lower loading costs offset the higher haulage costs.

Draglines

The dragline is a crane-like machine consisting of a machinery house to which is attached a long boom; the machinery house is mounted on a rotating frame on an undercarriage. From the end of the boom a bucket is suspended on a hoist wire rope. The bucket can be raised or lowered with the hoist rope and pulled towards the machinery house by the drag rope; both ropes are driven by hoists in the machinery house. The bucket is filled by dropping it into and pulling it through the muckpile; it is then lifted and swung round to the dump and emptied by releasing the tension on the dragrope. Smaller draglines (up to 20 m^3 bucket capacity) are mounted on crawler tracks, larger machines (5-100 m^3) move by means of a unique walking mechanism — hence the term 'walking dragline'. Draglines are among the biggest of earthmoving machinery and can weigh up to 6 000 t. In the right conditions they are the most economical means of moving earth up to a distance of twice the boom length. Draglines are best suited to the following conditions:

- digging below grade (i.e. below the base of the machine)
- depth less than 60 m
- digging below water
- soft or well blasted material
- movement distance no more than twice the boom length
- minimum moves between digs.

The cost of moving material by dragline is so cheap that it is usual for the range of the dragline to be further extended by rehandling half or more of the material. Careful planning is required to optimise the geometry of the dig to minimise rehandling.

Draglines are most commonly used in coalmines to remove waste from above flat lying coal seams, but they are also widely used in the construction industry and in mineral and sand mining. They are not ideally suited for loading trucks, but can be used for this purpose in special cases particularly if a hopper is provided. Draglines usually operate independently of other equipment, but it is common for a dozer to accompany the larger machines to level the working area. The smaller draglines are diesel powered (usually



Scraper excavating construction sand at Golden Grove. (Photo 040454)



Dragline shifting overburden. (Photo 040455)

diesel-electric) while machines above 5 m^3 bucket capacity are invariably electrically powered.

Typically, a dragline will dig one bucketload every minute. In a year, on a 7-day 24-hour roster, a large dragline will move from 250 000 to 300 000 bucketloads.

Because of its deep digging capacity and its ability to work from outside the excavation, the dragline is frequently used in wet pit operations, e.g. sand and gravel pits, and in special applications such as the gypsum workings at Stenhouse Bay and Lake Macdonnell where a 0.6 m^3 machine digs 125 m^3 per hour. Some performances in sand and gravel are:

0.8 m^3 bucket — 150 m^3 per hour
1.5 m^3 bucket — 250 m^3 per hour
1.9 m^3 bucket — 290 m^3 per hour.

Loading Systems

Loading equipment is used for transferring material from the face to the transport medium. Separate equipment is required for transport.

Rope Shovels (‘Power Shovels’ and ‘Face Shovels’)

The rope shovel is an electric or diesel powered loading machine, mounted on crawler tracks, and fitted with a rotating upper works and boom and bucket. The rope shovel is designed to dig and load at or above the level of its tracks and is the workhorse of the surface mining industry. Since the first steam driven machines were built in the 1880s, virtually every major open pit mine has relied on shovels as the prime loading tools. Shovels can be used in hard rock applications. They are approximately double the weight of an hydraulic excavator of the same capacity and have a very long working life, commonly in excess of 30 years.

The shovel usually digs a face slightly lower than the boom point sheave, to enable the dipper to reach all parts of the face and prevent dangerous overhangs from forming. The average face height range for large shovels is 10-15 m. Dipper capacities range up to 45 m^3.

In operation the dipper is lowered to the ground by letting the hoist rope run out, and pulled back close to the track by racking back on the crowd. At this point the front of the dipper rests on the ground; the hoist rope is then wound in and at the same time the dipper stick is crowded out. This causes the dipper to move first forward into the muckpile and then upwards. In good conditions the dipper should be full half-way up the face. The upper works

are then swung to position the dipper over the truck, and the door which forms the bottom of the dipper is opened to release the load. The dipper is then swung back to the face and returned to its original position flat on the ground. During this motion the door is closed by gravity and secured by a spring loaded latchpin. A typical cycle is 30 seconds or less.

The rope shovel is a very heavy, relatively immobile piece of equipment which needs a considerable amount of assistance to work efficiently. Although digging forces are high, the shovel operates best in well blasted material which rills to the bottom of the face. A rubber tyred or crawler dozer is needed to clean up spillage. Floors must be kept flat and smooth and hard ridges must be removed or point pressures can cause severe track damage. Shovels are not at their best when working on slopes, although a slope of 1 in 20 is allowable.

Three standard loading techniques are used: single side loading, double side loading and drive past. The aim is to maximise the time that the shovel is actually digging, and to minimise the time lost by waiting for a truck to be positioned. In single side loading, the trucks are positioned on one side of the shovel only, usually the driver's side. This results in a break in production while the loaded truck departs and an empty truck is positioned ready for loading.

In double side loading, the empty truck is positioned on one side of the shovel while a truck is being loaded on the other side. The shovel can thus complete loading one truck and immediately commence filling the next, without losing time for truck positioning.

Drive past loading requires a long bench, and usually a wide turning area. Trucks do not reverse for loading, but are driven past the shovel pausing to be loaded. Drive past loading, when properly set up, is probably the most efficient of all. It is also safer and preferred by truck drivers as no reversing is needed.

Regardless of the method of loading, the shovel dipper must never pass over the truck driver's cabin or the trailing cable (Fig. 12.1).

A variant of the face shovel is the stripping shovel which is usually a much larger machine with a very long boom and stick. In operation it stands on the floor, picks up waste from the face, rotates 90-180° and places material directly on the spoilpile.

Hydraulic Excavators

The hydraulic excavator is a diesel-hydraulic loading machine mounted on crawler tracks (except some smaller machines which are mounted on rubber tyres) fitted with rotating upper works and an articulated boom and bucket system. The bucket may be designed for digging above grade (front shovel) or below grade (backhoe). Capacities vary up to 30 m^3.

Nearly all hydraulic excavators are diesel driven but electric power is optional in the larger machines. Since the diesel or electric motors serve no other function than to power the hydraulic pumps, there is essentially no difference in practice between the two as far as performance is concerned, but costs and reliability do differ.

Hydraulic excavators are essentially stationary loaders, which do not move from their position while loading. They have very high digging forces, and can be used in all hard rock applications as well as for specialist jobs such as trenching.

The front shovel hydraulic excavator is fitted with a front end designed for digging and loading above the level of its tracks, although most can dig a considerable distance below. In operation, the bucket is crowded into the face, and filled by rolling it back. The bucket is then hoisted and at the same time the upper works swing to position the bucket over the hauler. The bucket then opens releasing the load. Front shovel excavators can fill the bucket at any point in the face — bottom, middle or top — whereas a rope shovel can load only by raking the face from bottom to top. The front shovel excavator can be used to pick a large boulder out of the top of the face or, more importantly, to dig ore from the top of waste or vice versa. The geometry also permits the front shovel excavator to follow a rising contact. Most front shovel excavators are fitted with bullclam buckets which empty much faster than front end loader type buckets.

The backhoe excavator is fitted with a front end designed to dig below grade but most have a considerable reach above grade as well. The bucket is reversed on the backhoe, facing towards instead of away from the operator. In the ideal setup, the backhoe sits on a bench several metres above the floor removing material below and in front of its tracks and placing it in a hauler positioned on the lower level. This considerably reduces cycle times and increases productivity for the excavator — but truck costs are increased. Backhoes can also be used to load trucks on the same level as themselves.



Mining with a rope shovel at Leigh Creek Coalfield. (Photo 040456)



Hydraulic excavator loading limestone at Angaston. (Photo 040457)

Hydraulic excavators are designed for use in the most severe digging conditions although the front shovel excavator is more suited to severe conditions than the backhoe. They serve the same function as a rope shovel as a primary production loader, but are considerably more mobile. In the smaller sizes, the capital cost of an excavator is considerably lower than that of an equivalent rope shovel.

Front-End Loaders

The front-end loader (FEL) is a four-wheeled diesel driven machine with a loading bucket on the front end. Bucket capacities range up to 20 m^3 and power up to 900 kW. Most large FELs are articulated, with the engine in the rear segment and the operator's cabin and bucket mechanism in the front segment. Steering is achieved by means of hydraulic cylinders acting about a pivot between the two segments. Smaller machines are rigid bodied and fitted with four-wheel steering. Very small machines are rigid bodied with skid steer.

Some smaller machines are mounted on crawler tracks. These machines are less mobile, but are more suitable for loading in boggy conditions and are capable of digging tighter dirt because of their higher crowd capability. A tight turning circle can be obtained by counter-rotating the tracks.

Cycle times vary from 20 seconds for the smaller wheeled machines up to 40 seconds for the very large machines. Most wheel loaders are four-wheel drive, with the power being transmitted from the engine to the wheels through torque converter, transmission, driveshafts, differentials and planetary drives. The very large machines are diesel electric, with an electric motor in each drive wheel.

In operation, the FEL is driven into the muckpile with the bottom of the bucket flat on the ground. The crowd force is applied by the wheels. The bucket is filled by raising the boom at the same time crowding the face and rolling the bucket back. When the bucket is full, the FEL is reversed away from the face and at the same time the bucket is hoisted. The loader is then turned about 45°, driven forward again turning about 45° to finish up about 90° from its original direction. Bucket hoist continues during travelling. When the bucket is fully hoisted the loader is driven forward so that the bucket is over the truck, and the bucket emptied by rolling it forward. The FEL then returns to the face and the procedure is repeated. When operating in good conditions, the FEL will move backwards and forward no more than 1.5 wheel revolutions.

The much less used ‘overloader’ is not suitable for rough digging, but since these do not need to turn to tip, they have at times been found useful in some restricted sand pits, gravel pits and shell grit workings. Both rubber tyred and track machines have been used. They operate with a forward thrust from the tracks or wheels, then the full bucket is lifted, the machine reversed and the bucket lifted over the machine to tip into a truck stationed behind.

Bucketwheel Excavators

Bucketwheel excavators (BWEs) dig by means of a rotating wheel of buckets at the end of a long boom (Fig. 10.1). Material from the bucketwheel is carried by conveyor belt to the discharge boom, where it is discharged either directly into the dump, on to a conveyor or into trucks. The bucketwheel boom can be raised and lowered through an angle of about 30°, and slewed through an angle of about 90°. The discharge boom remains stationary or can be slewed independently of the bucketwheel boom. The entire structure is usually mounted on crawler tracks.

The largest BWEs are capable of moving up to 300 000 m^3 per day. Machines of this size are invariably integrated with a conveyor system to transport the material to its final destination. These conveyors are over 3 m wide and travel at speeds of up to 10 m/s. In such a configuration, the bucketwheel, receiving hopper, face conveyor, distribution centre, trunk conveyor, dump conveyor, tripper and spreader together constitute a single integrated loading and transport system.

Smaller BWEs can be used to load trucks, but this usually results in a reduction in output.

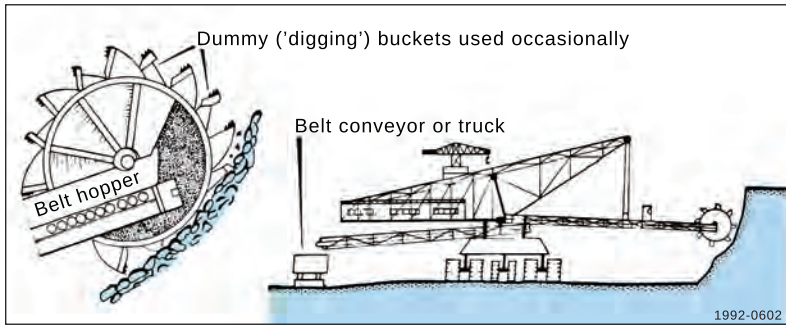


Fig. 10.1 *Principles of the bucket wheel excavator.*

Bucketwheel excavators are most suitable for soft free-flowing easy digging material. They are generally unsuited for handling blasted material because of the small size of the individual buckets.

The larger BWEs are all electric powered, while the smaller machines are diesel-electric or diesel-hydraulic. Where digging conditions are suitable, the use of a BWE will result in lower loading costs than a shovel or FEL of similar capacity.

Also categorised with BWEs are the so-called continuous surface miners which consist of a rotating tooth drum mounted on a crawler mounted tractor. These machines are designed to remove thin horizontal slices, and are ideal for the selective mining of thin coal seams. They are usually used to load trucks but can be also utilised with a conveyor system.

Bucket Chain Excavators

Bucket chain excavators have a series of buckets fixed to a chain which is mounted on a long boom (Fig. 10.2). The excavators can dig either downwards or rarely upwards. They cannot dig hard ground and they cannot be used to mine selectively. They are capable of high production (in excess of 1 500 t/h) in suitable ground. Their main application is for the bottom cut in a wet lignite pit.

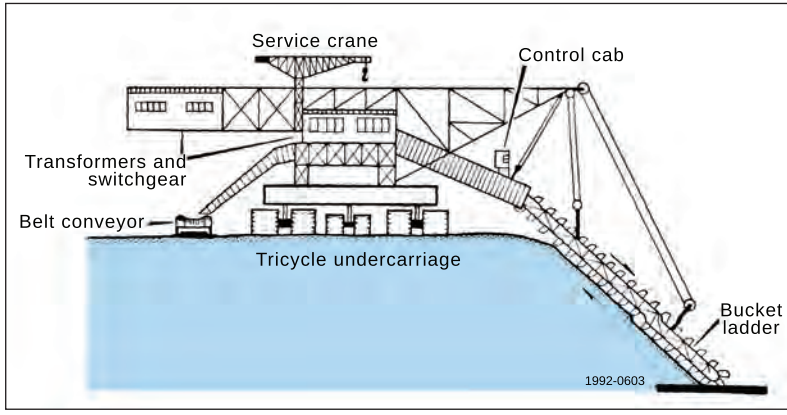


Fig. 10.2 *Principles of the bucket chain excavator.*



Bucketwheel excavator operating in an opencut coal mine. (Photo 040458)

Cranes

Mobile cranes or stiff-legged derrick cranes are used for lifting and loading dimension stone such as slate, marble, granite.

Mobile cranes or floating cranes, with grabs, can be used for sand and gravel.

Transport Systems

Trucks

The most common form of transport in mines and quarries is the diesel engine powered, rigid bodied or articulated, rear dump or bottom dump haul truck. Haul trucks range in size from 10 t up to 220 t capacity.

A wide range of trucks exists, and the quarry operator has a choice between:

- articulated, rigid bodied, or unitised
- diesel-mechanical or diesel-electric drive
- one, two or three axle drive
- rear dump or bottom dump (rarely side dump)
- steel or aluminium bodies.

Rigid bodied haulers are nearly always two axle rigs, with single tyres on the front steering axle, and duals on the rear drivers; they are invariably rear dump.

Articulated trucks are basically of two types: the prime mover/trailer truck similar to the familiar semi-trailer where the front section is a tractor and the rear section a bottom dump or rear dump body; and the all wheel drive rear dump truck manufactured specifically for use in boggy conditions. The former use a turntable mounting, the latter a universal articulation joint.

The unitised truck does not have a separate frame. The body itself is used as a structural member, and the axles and power plant are mounted on the body. Unitised trucks are usually bottom dump with rear engine and are extremely manoeuvrable, of large capacity (up to 150 t) but are generally suited only for flat or shallow hauls.

Most haul trucks are diesel powered mechanical drive, but trucks above 120 t capacity are commonly diesel-electric, with electric wheel or traction motor drive. Mechanical drive trucks are available in capacities up to 170 t and electric drive trucks in capacities up to 220 t.

Belt Conveyors

Belt conveyors are widely used in mines and quarries for transporting material from the face to the treatment plant or to the waste dump. Typical conveyor applications in mining include:

- integrated conveyor-tripper-spreader systems which operate in conjunction with BWEs
- conveyors which transport material from an in-pit crusher or hopper to the surface
- modular conveyors, in lengths of up to 50 m, which are used to transport material from the face to a main trunk conveyor.

The main limitation with conveyors for the transport of primary material from the face is that size reduction to less than 1/3 to 1/4 of the belt width is usually required before material is placed on the conveyor. If size reduction is required in a subsequent part of the process, then no additional expense is involved in this step; however, if no further size reduction is required, which is commonly the case with waste materials, then the cost of this stage has a serious effect on the economics of belt conveying.

Ancillary Equipment

Graders

Graders are specifically designed for construction and maintenance of road surfaces. They remove spillage from the road to prevent tyre damage and also cut and reshape the road formation to maintain level, grade, crowning and smoothness. This applies to both expensive permanent roads and the temporary roads on benches and dumps.

Graders have a long wheelbase with the blade mounted between the single steerers and tandem drivers. This arrangement maximises control over the blade. Most recent machines are articulated to improve manoeuvrability. The blade can be moved to almost any position. Graders of between 12 and 25 t are commonly used on mine sites.

Water Carts

Water carts are needed to:

- maintain the optimum moisture level in the road formation to prevent unravelling and to minimise maintenance
- suppress dust which is environmentally troublesome, can be a health hazard and can reduce visibility to a dangerous degree.

Lighting Plants

Some of the larger quarries and open pits continue working after dark. Even though the shovel or other loading equipment may be fitted with floodlights and the haulers with head lights, it wise to illuminate the general face working area and the dump tipping area so that all workers are visible to operators.

11. Quarrying Methods

In South Australia the term ‘quarrying’ is taken to include all surface methods of obtaining metals, minerals, coal and construction materials.

The method of quarrying depends on many factors including:

- location of the deposit
- its size, shape and depth
- type of material/mineralisation
- nature of the material and surrounding rock
- amount and type of overburden
- equipment available
- product specification (size, shape, grade, tonnage etc).

Earlier methods such as hand loading, rail, cableway and inclined haulage in quarries, and panning and sluicing of sands will not be covered in this text; for information on these reference should be made to mining text books.

Stripping

Stripping is the process of removing overburden to expose material for quarrying.

The nature and amount of overburden on a mineral deposit, the rate at which the overburden has to be moved, the distance it has to be transported and the topography largely determine the methods by which it can be stripped. In some cases the overburden is consolidated and must be blasted before removal. Some sedimentary rocks, although consolidated, contain numerous planes of weakness such as joints, shear and cleavage planes and can be ‘ripped’. Other materials are loose and easily moved. Lastly, there is a variety of materials to be moved from wet ground or from under water. The final choice of stripping method and machinery to be used is one of economics — capital and operating costs.

The equipment can be either:

- The same as that used for the subsequent quarrying operations, e.g. shovel and truck removal of overlying waste rock after drilling and blasting in a hard rock quarry, or scrapers removing sand overlying a clay deposit.
- Of a different size and type, e.g. a dragline removing overburden to allow hydraulic shovels to load coal; a bulldozer ripping and pushing weathered rock off a deposit being worked by drilling and blasting before removal by a FEL and trucks; or sluicing for sand.

Major options for stripping include in-pit against out-of-pit dumping, or truck haulage versus in-pit crushing and conveying.

Ripping

Ripping is an alternative to drilling and blasting.

Modern ripping equipment usually consists of up to three single teeth or 'tynes' mounted on the rear of a crawler tractor or bulldozer. Because rocks are extremely variable in physical characteristics the question of whether, in a given situation, large scale ripping is justifiable or even possible has required extensive and costly onsite testing. In 1958 the Caterpillar Company developed the refraction seismograph as an aid to answering this question. The principle involved is that seismic (shock) waves travel through rock at speeds which depend on the consolidation or hardness of the material. The equipment measures the time of arrival at the instrument of seismic waves induced in the rock, and from this it is possible to determine the seismic velocity and thickness of the various layers of material beneath the surface to depths of 10 m or more. Given this information, whether or not the rock can be ripped will depend on the number and shape of the tynes, the weight and power of the tractor and the down pressure on the tynes. Tables and charts assist the interpretation of results, but it must be remembered that this process is only a guide as to which situations will warrant full scale testing.

There are two main types of tyne mounting — hinged and parallelogram. It is desirable also to have a facility to alter the angle of entry of the tip to suit local conditions. Replacable shoes will increase the life of the tip. For occasional tough going a second tractor can be used to push the first tractor-ripper, while some blasting may be needed to break up hard patches of rock.

Caterpillar have released a vibratory ripper in which blows are repeatedly delivered to the ripper tyne (and hence to the cutting tip) by a hydraulic hammer built into the ripper mechanism. This considerably increases productivity in very hard ground.

Example. Calculate the cost per cubic metre for ripping weathered shale using a large dozer equipped with a single tyne ripper. Assume that the dozer operates for 8 hours, 220 days per year, that the average depth of ground ripped per pass is 80% of maximum penetration, and that the overall operator-machine efficiency is 75%. For this example neglect turning time. Given also that:

	Depth of penetration of tyne	=	0.6 m
	Distance between passes	=	l.0 m
	Time for 100 m pass, one way	=	5 min
	Annual fixed cost of owning the plant	=	$88 000
	Hourly cost:		
	fuel	=	$10.00
	lubrication oil	=	$1.00
	repairs and maintenance	=	$30.00
	tyne tip	=	$6.00
	labour	=	$20.00
Calculation:			
	Depth of cut 0.6 m x 80%	=	0.48 m
	Width of cut	=	1.0 m
	Volume per 100 m	=	48 m^3
	Time for 100 m run	=	5 min
	Overall volume per hour = (48/5) x 60 x 75%	=	432 m^3/h
	Operating cost = $10 + 1 + 30 + 6 + 20	=	$67/h
	Annual cost per hour = $88 000/(8 x 220)	=	$50/h
	Total cost per hour = $67 + 50	=	$117/h
	Total cost per cubic metre = 117/432	=	$0.27

Moderate Scale Stripping

Mobile earth moving equipment is commonly used in stripping operations, the principal machines being the FEL, the scraper, the bulldozer and the bulldozer with ripper. Generally they are used in soils and the less consolidated rocks, but even in harder rocks the availability of cheaper explosives has enabled them to compete with the more conventional stripping equipment — the shovel and the dragline.



Scraper assisted by bulldozer shifting overburden at Mintabie opal field. Note ripping tyne on bulldozer used to loosen the sandstone host rock. (Photo 039412)

For many stripping operations on relatively flat ground where the lead (movement of material) is about 150 m, the bulldozer will be the choice. It can be used with a ripper and it can be used in ground requiring a 'shaking up' with explosives. It also finds many applications in cleanup and other work in quarries when not on stripping.

The FEL has application in loose sand or gravel or well-fragmented material either to load into trucks or to tip over a bank.

The scraper is ideal for moving soft or ripped material distances of up to one or two kilometres. Simple seismic geophysics can be used to identify and delineate the strata where bulldozing, ripping and scraper loading and transporting can be applied.

For general all round work in hard digging conditions where the material is semi-consolidated, or is not well-fragmented in blasting, the crawler mounted shovel with dump trucks is the most satisfactory combination for shifting stripped material.

The dragline is particularly useful where it can dump off the mineral deposit into a worked out area. It can also be used to load into hoppers for conveyor belt haulage, or into dumpers, provided the material is loose or well fragmented. It is at its best in those situations where it is necessary to work from a position out of the pit, and especially one filled with water. In some wet pits floating grab cranes, dredges and the like can be used.

In suitable topography, where water is available and the overburden is soft, there is a special application for water power in the form of hydraulic jets to remove overburden, the resultant slurry being drained or pumped away.

Large Scale Stripping

For large scale operations, the most common items of equipment are stripping shovels and draglines, and bucket wheel and bucket chain excavators. The first two can dig in hard rock that has been blasted and suitably fragmented. Bucket sizes being equal, the stripping shovel will shift more rock than the dragline but cannot move it as far. It is, however, a more expensive machine, and it requires more maintenance in hot, dry conditions. It must stand on the deposit itself to dig overburden, whereas the dragline can stand out of the excavation.

The bucket wheel and bucket chain excavators are best suited to soft formations but the bucket wheel excavator is now being adapted to harder formations. It is a most effective machine for loading large quantities of unconsolidated ground in broadacre operations.

Note: The term ‘strip mining’ refers to the quarrying method most commonly used for coal and is described later.

Quarries or Open Pits

Hillside

This is the most common form of hardrock quarrying in South Australia. The quarry is best developed in a series of benches beginning at the top of the deposit and progressing downwards with particular attention being paid to the environment and to overburden stripping and dumping (Fig. 11.1).

The access road is usually constructed outside the quarry and in order to maintain a safe and suitable gradient it is often constructed in a series of zig-zags.

Working this type of quarry from the top bench downwards is the most common method and has the following **advantages**:

- Each bench can be almost fully worked out before there is a need to develop the next one down. This spreads out evenly over the life of the quarry the periods of low productivity which occur when developing new benches. In contrast, commencing the bottom bench first results in each successive working face being advanced a short distance until the increasing face height necessitates the creation of the next bench up the hill. This can result in the high cost of developing the quarry occurring over the first few years of operation rather than being spread over the life of the project.
- Progressive rehabilitation can be carried out on the completed upper benches as the workings proceed down the hill side thus assisting in minimising the visual impact. Developing from the bottom results in a situation where all faces could be visible early in the life of the quarry.
- In some cases overburden and waste rock can be progressively spread on completed benches as part of quarry rehabilitation and thus minimise out-of-pit dumping.

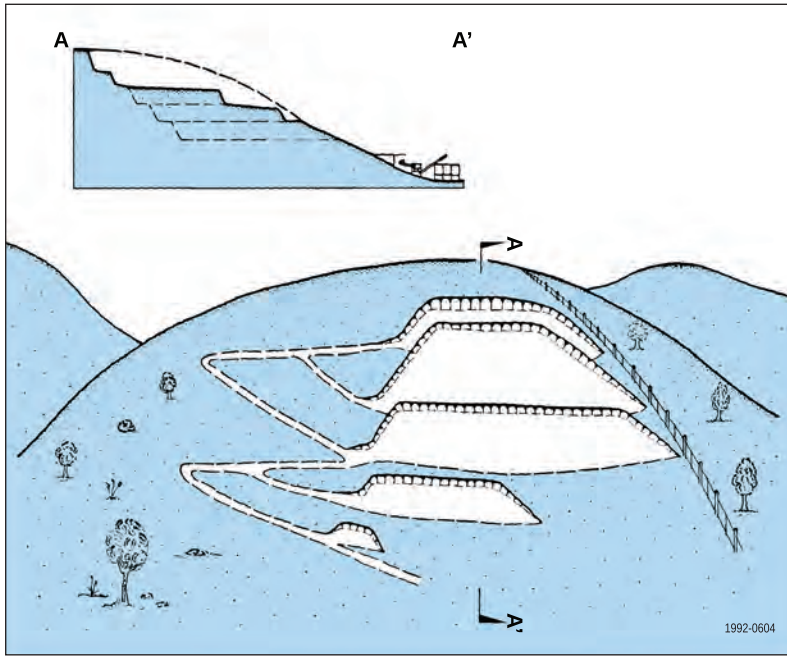


Fig. 11.1 *Hillside quarry.*

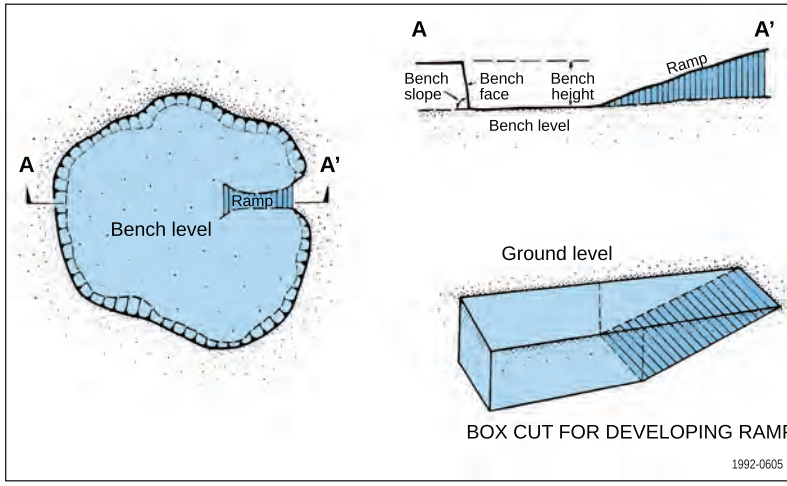


Fig. 11.2 *Single bench quarry (access to single bench).*

However, there are several **disadvantages** which must be considered when commencing at the top of the quarry:

- More road development is required for access to the top bench before quarrying can commence.
- Blending of stone from the top benches with that from the bottom to meet specifications would not be possible without large stockpiles.

Face heights are typically 10-15 m and the gradient of the access road is usually in the order of 1 in 10 (10%).

In most cases the plant is located on or below the level of the lowest bench.

Single Bench

Single bench quarries are used for shallow, flat or near flat deposits such as sheet limestone, sand and gravel pits and coal. The depth of quarrying is dependent upon:

- depth of the deposit
- regulatory requirements
- maximum stable face heights and slopes.

In South Australia single bench pits are typically less than 15 m deep and ramp slopes are usually 1 in 7 or flatter.

Access to single bench quarries is either by an external or internal ramp (Fig. 11.2).

Production is limited only by the capacity of the equipment that can be employed and the number of areas along the face that can be excavated simultaneously.

This method is amenable to progressive rehabilitation, using the placement of overburden inside worked out sections of the quarry.

Multibench

This method is used when the deposit is to be excavated to a depth greater than the height of the working face.

Where the width and length of the deposit are similar, excavation usually takes place in horizontal slices extending over the area of the deposit. Thus the overall direction of working is vertically downwards. Haul roads are

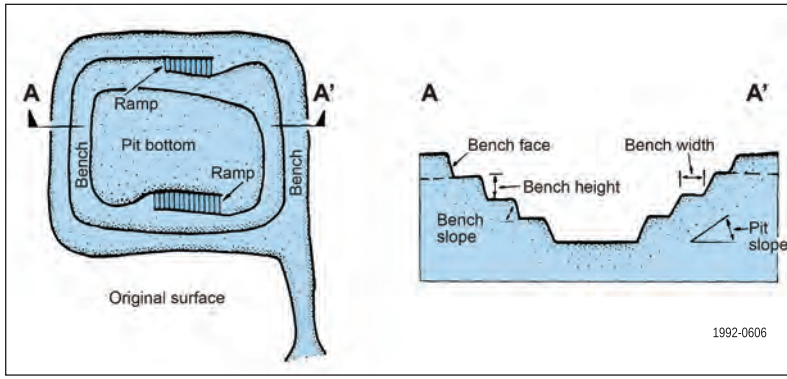


Fig. 11.3 *Multiple-bench open pit mine (spiral).*

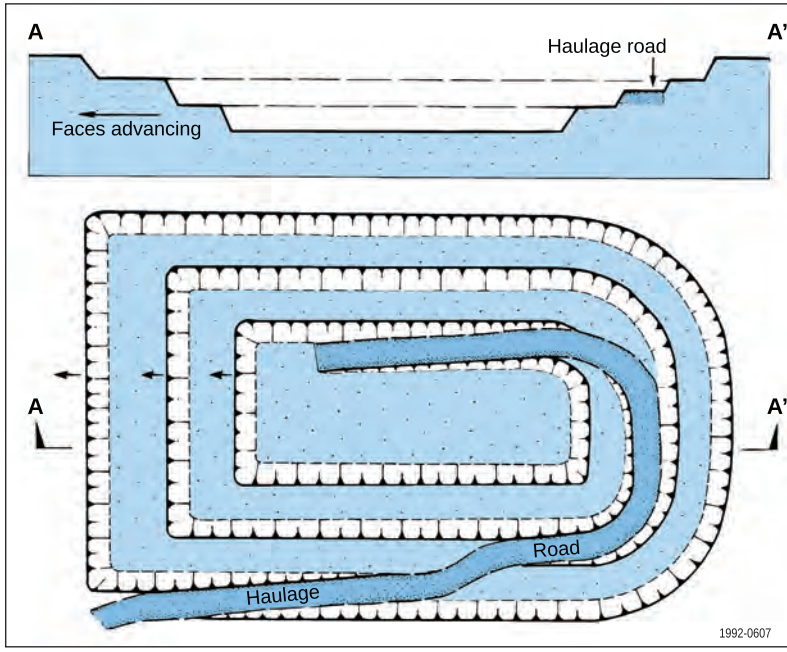
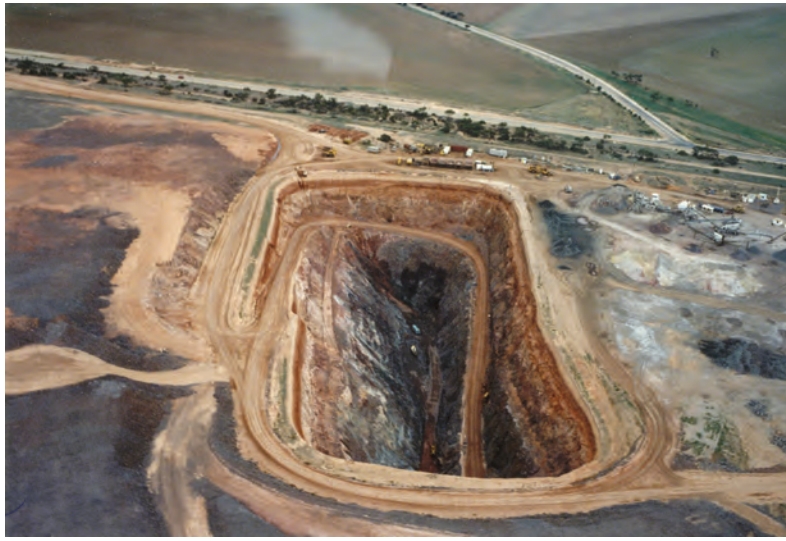


Fig. 11.4 *Multiple-bench open pit mine (elongated).*



Greenhill Quarry: A typical hillside quarry. (Photo 038723)



Poona copper mine: A multiple-bench open pit. (Photo 039558)

constructed on the (usually final) pitwalls and if truck haulage is used the grade is approximately 1 in 10. As the pit extends downward the roadway spirals down the pit walls to the working bench (Fig. 11.3). All waste rock must be hauled outside the pit as this method is not generally suitable for in-pit dumping.

In the case of long, narrow, flatly dipping deposits the initial phase is to excavate to the final depth of the pit with the haulroad located in a side pit wall (Fig. 11.4). When the deposit has been opened up to full depth, excavation proceeds horizontally along strike on each of the working benches. Initially overburden must be dumped outside the pit but, as the excavation extends in the direction of strike, it is often feasible to progressively backfill and rehabilitate.

Truck and shovel or FELs are the most commonly used equipment for these types of operation but in some very large operations other combinations including bucketwheel excavators, in-pit crushers and conveyors are employed.

Variants of multiple bench quarrying include the use of either a haulage shaft with drifts (tunnels) to each bench or a horizontal or inclined adit to remove rock from the pit (Figs 11.5 and 11.6). These types of operation are more particularly applied to mining metalliferous ores and, as extensive underground development including the provision of a crushing station is often required, they are rarely used for quarrying stone.

Strip Mining

Strip mining is the term used to describe the method whereby overburden is continually excavated in strips along strike to expose the material to be mined, with the overburden being placed in the previously mined-out strip (Fig. 11.7).

Mining of coal is the most common application of this technique but it is suited to other near surface shallow dipping deposits which may or may not require blasting. Typically, coal thickness varies from 1 to 10 m up to 50 m with strip ratios as high as 30:1 in favourable conditions. Economically feasible strip ratios depend upon the value of the product compared with the cost of removing overburden.

The most common types of equipment used for removing the overburden are draglines, bucket wheel excavators, and in some cases truck and shovel combinations.

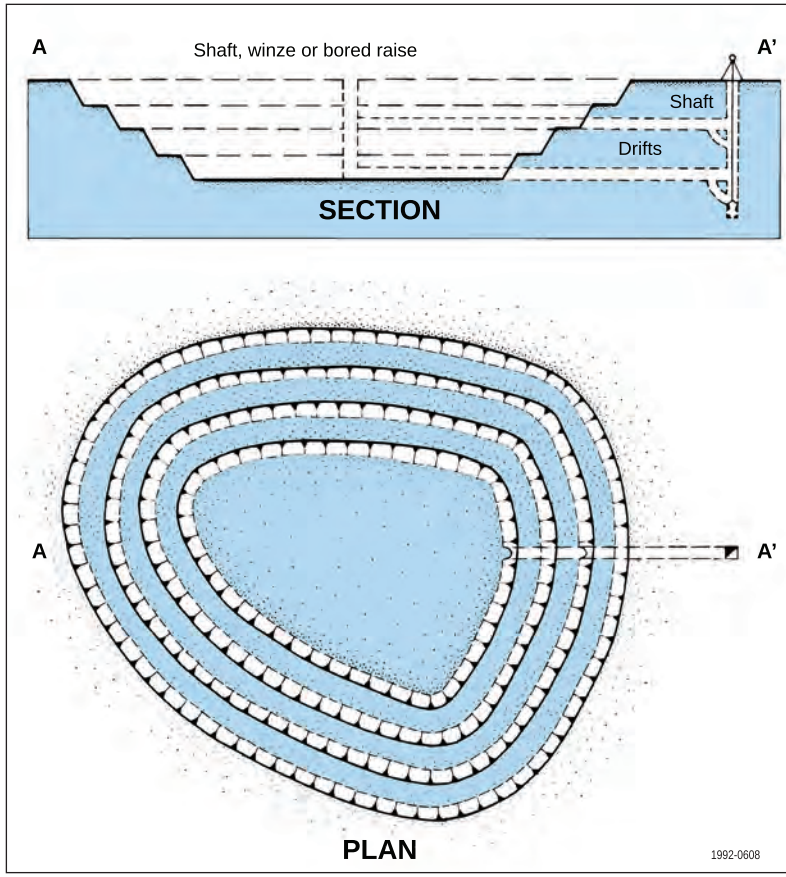


Fig. 11.5 *Open pit, external shaft haulage.*

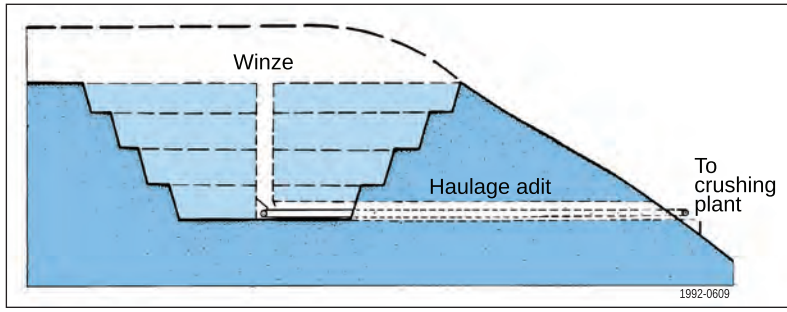


Fig. 11.6 *Open pit, adit haulage.*

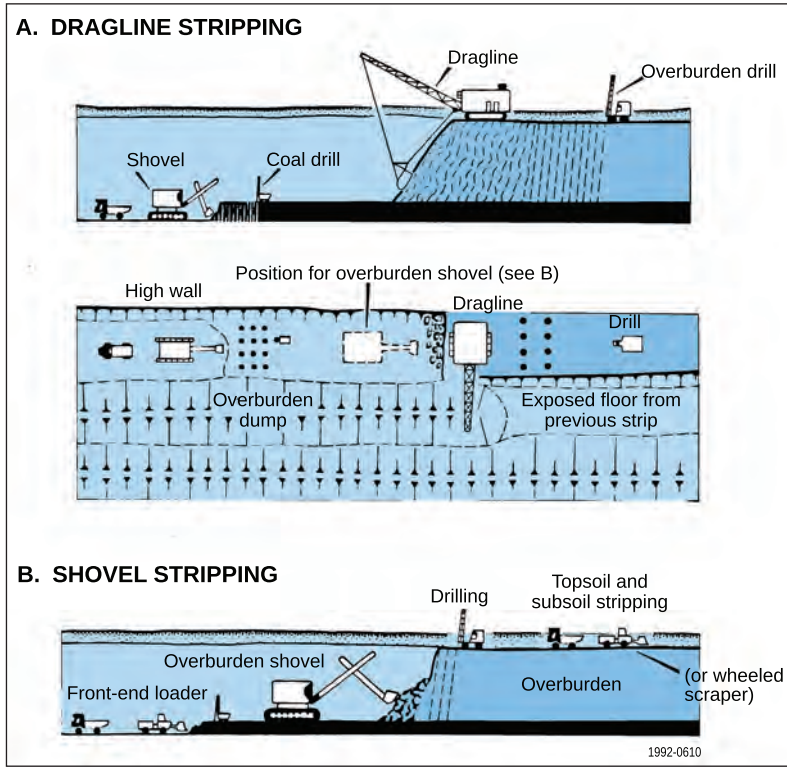


Fig. 11.7 *Coal strip mining.*



Coal strip (terrace) mining at Leigh Creek (ETSA). (Photo 039735)



Dimension Stone

Dimension stone is rock which is removed in blocks for use in special purposes such as building blocks, grave headstones, paving or sculpture.

Where the dimension stone is uniform and the deposit is more or less homogeneous, the pit is taken down in a series of steps or benches. The depth and width of these benches may be determined merely by the size of the block which can be cut with the boring equipment or which can be handled by the crane or lifting equipment available. On the other hand they may be determined by the presence of faults or planes of weakness in the rock.

Where the desired material occurs as boulders (as with some deposits of marble) or in separate layers of a main bed (as with slates) it may be necessary to base the plan of working the pit on the lifting apparatus (e.g. a crane) that will eventually be needed to give access to the whole deposit, and work out details of the method of separating the useful material as work proceeds and its structure is exposed.

The method of breaking or splitting the stone depends on its hardness and cleavage or fracture planes. For example, slate is split using wedges and the sheets broken off to manageable size using short jackhammer holes without explosives. Marble may be broken into blocks using closely spaced drill holes and split by wedges or light charges of blasting powder.

Harder diorite is broken to size in a manner similar to marble or by using the technique of jet piercing — a flame produced at the end of an iron pipe by burning kerosene or diesolene in oxygen or air. A water jet is sometimes incorporated in the ‘bit’ to assist spalling of the minerals in the heated rock.

Limestone and sandstone may be cut using a circular saw usually tipped or impregnated with an abrasive-resistant material such as tungsten carbide.

Soft Rock

Shale or hard clay may be obtained by drilling, blasting and loading with an FEL into trucks; in some cases it can be directly excavated and loaded by hydraulic excavator. In other pits it is common to first use a bulldozer-ripper to loosen the material and then load it using a bulldozer to push it into heaps with subsequent handling by an FEL. Alternatively, the loosened material may be loaded and transported using a wheeled scraper.

Plastic clay is mainly excavated during the drier months of the year.



Cutting limestone blocks at Mount Gambier. (Photo 040360)



Mechanical salt harvesting at Dry Creek. (Photo T013520)

Sand and Gravel

Depending on the depth (thickness) of the deposit a single or multiple bench system may be adopted. In general the face slopes need to be much flatter than for hard rock.

Digging and transport operations vary as follows:

- FEL and motor truck.
- Dragline and motor truck.
- Bulldozer pushing into heaps or into a chute for truck or conveyor transport.
- Wheeled or tractor scraper.
- Hydraulic monitor — rather like a large fire hose nozzle which is mounted on a fixed stand with a swivelling head.
- Dredge with a cutter head and suction hose for harder material or buckets on an endless chain for softer material. (The dredge floats on a pond of water contained in the excavation.)

Most of the sand is washed, but a proportion is suitable 'as mined' or requires dry screening only.

Salt

In Australia all salt is produced by solar evaporation of saline water, usually sea water. In solar salt operations the saline water is passed through a series of ponds, progressively increasing in salt concentration as the water evaporates. The final stage of evaporation, when the valuable sodium chloride is precipitated, occurs in the crystallisers. Great care is taken to ensure that any gypsum (calcium sulphate) precipitates out before the liquid enters the crystallisers and that the liquid is removed from the crystallisers before magnesium chloride (bitterns) comes out of solution. The salt is deposited in the crystallisers in flat layers up to 300 mm thick and is 'harvested' by machines specially designed for the job.

The salt is stockpiled and allowed to drain prior to further processing. At Dry Creek in South Australia the salt is re-dissolved in water and pumped to the processing works, but it is more common to handle the salt as a solid.

12. Quarry Unit Operations

Sequence of Quarrying Operations

Once the necessary amount of stripping has been completed the quarry is ready for production.

The sequence for any one blast may include:

- preparation for drilling, including clearing the area and possible removal of irregularities in the face or floor
- drilling
- charging holes with explosives
- blasting
- barring down or otherwise making the face safe
- loading and transport of rock
- cleaning up of floor.

With a long bench, or with more than one bench or floor in a quarry, there may be several blasts in the course of preparation or clean up. The number would depend on the output required from the quarry and the type of plant — drilling, loading, transporting and so on — available as well as the production faces available. It is necessary in the timing of the various operations to make sure that they are conducted safely and do not interfere with one another.

Drilling

In laying out a new face or re-establishing an old one, special attention must be given to 'lining up' the face. Excessive toe bumps, backbreak, incorrect slope of the face and other irregularities must be eliminated by minor trimming operations. Once the face is aligned, care is required to ensure that drilling and blasting keep the face moving in the correct direction on the selected slope.

With pattern drilling, holes must be parallel to one another and be at the correct inclination. In some quarries it is regular practice to make a profile sketch of the face opposite the hole position using information obtained

in simple surveying with rods, plumbline and tape. In others, a line of paint spots or stones is made back from the previous row of holes. In some circumstances the services of surveyor may be required to accurately position the holes in relation to the face and geological structure so that the blast will not result in undesirable flyrock. In quarries, particularly those for mineral production, it is common to collect, examine and assay the cuttings from a hole before charging it.

Drills can be aligned from the collar by using markings set out by a surveyor, or they can be positioned at right angles to the row of holes as marked above. The inclination or slope of the drill boom can usually be fixed with a clinometer. But even if the drill and holes are set out correctly, it requires attention on the driller's part to ensure that the holes go where intended.

The drill must be set firmly. Check for worn chucks and rods, or rods too small in section when using percussion drills. Watch for excessive thrust on the rods and bit. Watch for fissures and for changes in rock hardness passing from one bed to another by observing the drilling rate. These fissures and hardness changes can cause marked hole deviation. With angle drilling there is a tendency for the drill rods to rest on the underside of the drill hole, causing the bit to upturn and to deviate from the intended angle of drilling. If possible, it is good practice to observe the tracks of drillholes in old blasts to see if there is a regular trend.

Drilling dust. In dry drilling considerable dust is produced. The dust must be dealt with in such a way that there is no danger to the health of any person and that it does not create an unacceptable visible dust cloud.

A collector, fitted with flexible rubber top through which the drill stem passes, can be placed around the collar of the hole. A venturi (air ejector), run by a small amount of air from the compressor, can be used to entrain the dust and pass it to a cyclone mounted on the drill rig. Some cyclones have extractor fans to draw the dust-laden air through them.

Wet drilling. This often causes problems with recovery of cuttings where assays are required. Where it is possible to drill wet, a detergent added to the water helps the wetting action. Vapour drilling, that is drilling with a fine mist spray, may be effective against dust. However, care must be exercised to see that the drill steel is moved up and down the hole at intervals to ensure that there is no excessive build up of cuttings on the side of the hole which may trap the bit.

Blasting

It is important that there should be a record of each hole bored, with notes on depth, direction and dip, and on the strata passed through, and a sketch to illustrate important features. From these details a blast pattern is determined. The sketch of the layout of the blast itself should be accompanied by notes on the explosives used, method of charging, initiating and so on.

Finally, after each blast the results should be carefully assessed and recorded: noise, dust, ground vibration and air concussion, explosive factor, fragmentation, back-break, toe bumps, disposition of rock pile, scatter or fly rock, climatic conditions and anything else that could be relevant under the local conditions.

Loading

Loading by power shovels and large hydraulic excavators is mostly confined to the bigger operations such as the Leigh Creek coal field and the Middleback Ranges iron ore operations. Shovels are now rarely used in the extractive minerals quarries in South Australia, the smaller and more mobile rubber tyred FEL having taken over.

Positioning of trucks for loading by a rope shovel has been covered in Chapter 10. Figure 12.1 shows three methods commonly used.

Most quarries adopt a system of signals using a horn or hooter when a truck is being backed, spotted and then moved away after loading. The shovel driver should determine when the truck can be moved into the loading position. He should sound the signal which should be answered by the truck driver when backing; it is best if the drivers can see each other as well as exchanging signals. The space into which the truck is backing must always be visible to at least one of the drivers in case anyone walks into the danger zone. Method C in Figure 12.1 is thus the safest spotting method — it is also efficient.

Positioning of trucks for loading by FELs is also covered in Chapter 10. The truck should be backed into a position alongside the loader but far enough away so that the loader can reverse out and turn to discharge its load. Note that the smaller the angle through which the loader has to turn, the quicker the cycle time.

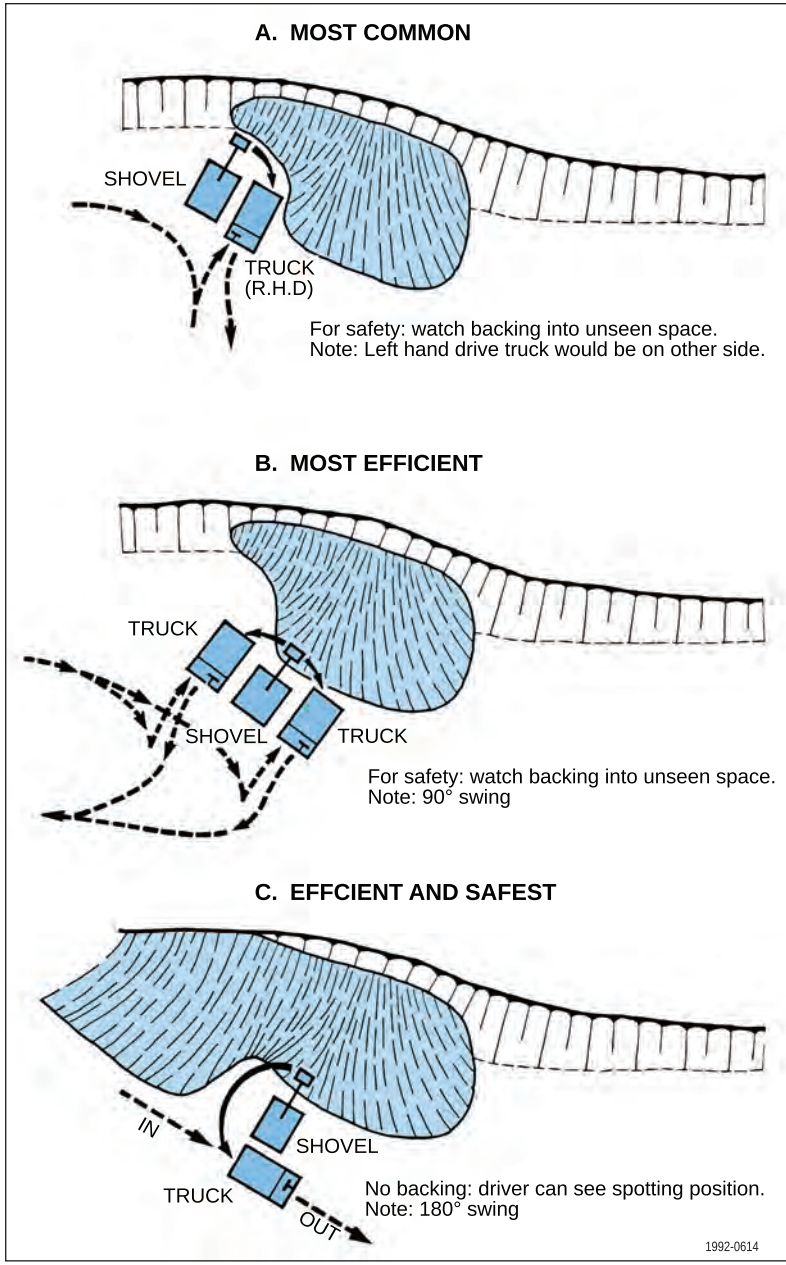


Fig. 12.2 *Positioning of trucks for loading.*

Transport

Each day, before taking out a vehicle, drivers should check tyres, fuel, oil and water. They should run the engine to ensure its smooth operation and move off, testing steering and brakes before proceeding to the job. This is in addition to the regular routine servicing.

Roads

Roads should be well made to take the heavy vehicle loads. They should be of suitable grades and widths (approximately 3.5 x truck width for two way roads) and finished with a smooth surface contoured for proper drainage (Fig. 12.2). This will ensure less tyre and road surface wear, a shorter cycle time, and it will be less tiring for the driver. Further, it should result in less maintenance on vehicles.

Where practicable, road gradients should be not greater than 1 in 9, with no sharp transitions. It is quite common to see empty trucks taking the outside of a road, while the loaded truck takes the inner side, firstly to keep the loaded truck from breaking the road edge and secondly to give it safer running in case of brake failure. However, the determination of whether to drive on the left or right side will depend largely on the layout of the haulage roads in the quarry or pit and the connecting links with the crushing plant. Nearly all big trucks are LHD and hence these are the most commonly used in quarries unless there is a very good reason for right hand drive.

Roads need to be cleaned down regularly, graded and well watered to maintain optimum moisture content in the surface and keep down dust.

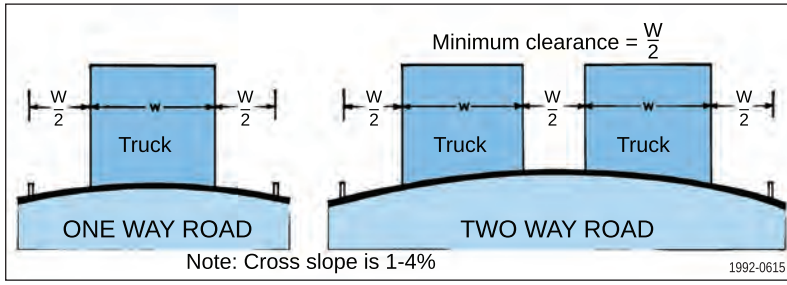


Fig. 12.2 *Typical road sections.*

Reclamation

As soon as possible after completion, each bench or suitable section of the quarry should be reclaimed or treated in accordance with the overall reclamation program (see Chapter 3).

Dumps and Slime Dams

All dumps, silt, slimes and water storage dams should be constructed so as to blend in with the surrounding countryside. Moreover, they must be properly designed and rigorously controlled during tipping or filling operations.

Dumps

Factors to be taken into consideration in selecting the site and building the dump include:

- environment (proximity to housing and visual impact)
- topography
- foundations
- drainage
- construction.

In some cases advantage can be taken of natural features or existing excavations to place dumps in positions where they are not generally visible from housing or main roads. In others the dumps themselves, suitably landscaped and planted with vegetation, can act as screens for the workings. They may also serve as noise barriers and as wind breaks or dust barriers. Where the dumps serve no particular useful purpose in this regard, they should be constructed to conform with local topography.

For designing the dump an estimate of the quantity of material to go into it is first required. A study of the geology of the site and tests of the rock characteristics may allow a reasonable estimate of the waste disposal requirements to be made. In dolomitic siltstone quarries in South Australia as much as 25% of the production may be waste, and in mineral operations over 90% may be waste.

Consideration of the groundwater and physical features of the area, including drainage patterns should be made. It is essential that the dump is built on a solid formation capable of withstanding the weight of material placed on it; any topsoil should be stripped from the dump foundation. Provision must be made for suitable drains underneath the dump. This applies whether the dump is on level ground or part way up a hill. Dump

drainage may be connected into the natural drainage system of the area and, if necessary, provided with a sediment settling pond to catch silt that may be eroded from the exposed surfaces of the dump or dam.

Dumping. The safest way to tip on a dump, but unfortunately the most expensive, is to back the truck towards the edge, dumping the waste at least 3 m back from the edge, and then pushing the material over the edge with a bulldozer (Fig. 12.3). Next best is to use a well secured stop block, or to maintain a 'windrow' near the edge which will stop the truck (Fig. 12.4).

Under no circumstances should drivers back to the dump edge where there is no stop block or windrow unless they have the help of a person to act as 'spotter', or there are spotting posts to guide them. They should not use the rear vision mirror to back. The driver of a dozer operating on the dump normally acts as a spotter. The truck driver should not back onto unconsolidated material; a safety belt should always be worn, particularly while hauling and backing up to dumps.

The dump should be built up in lifts consistent with maximum safety so that the material will be consolidated by passage of machines backward and forward. At suitable height intervals during construction the outside face should be stepped back to provide greater stability. In the final stages of construction in non-arid regions, faces can be smoothed over with a bulldozer

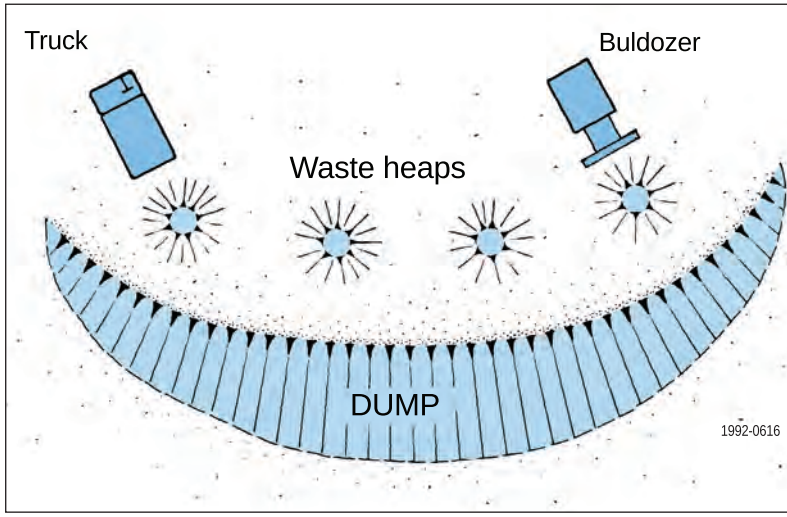


Fig. 12.3 *Trucks tipping overburden on a dump.*

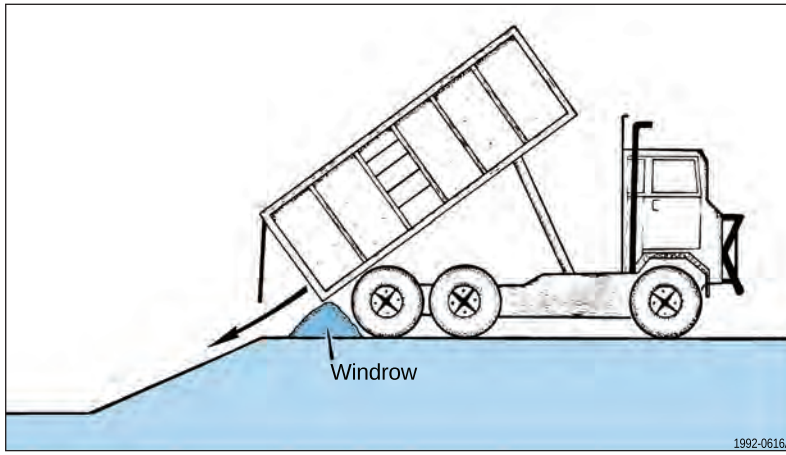


Fig. 12.3 *Truck with windrow.*

or heavy chain to conform to the general slope pattern of the area. In arid regions an irregular 'moon scape' surface may be created by dumping on the flat without levelling to trap seeds and encourage retention of the spasmodic and sparse rainfall.

In populated or environmentally sensitive areas it may be necessary to build the perimeter wall of the dump first and then backfill the centre. This allows rehabilitation of dump faces to occur early in the operation and reduces visual impact of the dumping operation from outside.

Factors to be considered in the design of dumps include:

- For a given volume the higher the dump the smaller the area required.
- Increasing the height increases the haulage cost but reduces the surface area needing reclamation; however it increases visual impact.

Slime Dams

Slime dams are needed to contain any unwanted fine material such as clay, slime or sand created by the quarrying.

- In dry screening operations fine particles may be generated by the movement of materials, traffic, quarry or crushing plants, or by blasting. The particles settle on stockpiles, benches, haulage roads and other areas from where they are washed by rain into drains or waterways.

- In a wet screening plant associated with the sand mining operation and occasionally with a quarry, unwanted fines may amount to 20% of the feed and provision has to be made to contain large quantities of slime which otherwise would almost certainly make its way into local water courses.

By forming a pond and so slowing water velocity, the particles are given a chance to settle out. In some cases it may take several days, even weeks, for this to happen. The size and type of particles and the quantity of water to be stored or held control the size of the dam. If topography does not allow sufficient area for a single large dam, then a series of small dams may be constructed.

Dams are best placed within a worked out area on a company's property or on a non-quarriable section. In the latter case it is topography which largely determines positioning of the dams. In either case a detailed investigation of the site should precede dam design.

State regulations may prevent interference with a water course by placing a dam across it, although special permission may be obtained if no other practical alternative is available. However, it must be remembered that building a dam across an existing drainage channel or water course can be fraught with danger in the event of heavy thunderstorms over a



Slime dam at a Golden Grove sand pit. (Photo 040459)

large catchment area. The design of a massive dam to cope with such a circumstance requires the services of an engineer qualified in this field.

A well constructed dam should have the following features:

- sufficient area of pondage to allow the particles to settle out
- a spillway designed to cope with peak rainfall events over the area draining into the dam and preferably located off the dam wall
- provision for periodic draining and cleaning out of accumulated silt.

Cleaning may be by sludge pump to an evaporation pond if water is always present in the dam, or by FEL if it can be dried out.

Among the factors to be considered in designing a dam are the weight bearing capacity of the subsurface formations and seepage control. The foundations of a dam wall must not only be built on suitable material, but the wall itself must be solidly built and it should have a relatively impervious core. The wall needs to be compacted after each short rise, with sufficient slope to ensure its ability to withstand water pressure from within the dam. Provision should be made for overflow and surface drainage, and the outside slope should be protected from erosion, preferably by promoting a cover of vegetation.

Where there is likely to be a depth of water on the inside of the face, care must be taken to provide for possible undercutting of the bank by wave action.

One common cause of failure in dams is breaching of the wall at the top by flood waters. Many dams in South Australia have failed in this way because they were operated with the water level too close to the top of the wall. There must be:

- sufficient height of wall above current filling operations
- a spillway to ensure that sudden rain storms and cloud bursts do not allow a dangerous buildup of water on the inside of the bank.

Spillways should be regularly maintained and kept free from blockages. Once a breach is made, the flow of water will soon destroy the best made wall.

It cannot be overstressed that the design and construction of a dam is a specialised engineering process and should not be attempted by unqualified practitioners.

Pumping and Drainage

There are two aspects of pumping and drainage which can be considered separately — prevention and cure. The first is to prevent water from entering the workings, the second is to remove it once it has entered. Water can flow into a quarry either from surface runoff or through the ground. Direct precipitation is rarely a major problem in South Australia except in large or deep cuts.

Surface Runoff

Surface water can be either permanent, as in a river, or ephemeral as in most Australian creeks. The treatment of both is much the same. Generally protection against the regional surface runoff is provided early in the life of the mine, and is a 'one-off' job.

It may be necessary to divert a creek completely so that it flows around the quarry instead of through it. In other cases it may be necessary only to restrict the spread of flooding by streamlining flow channels. Where the quarry is in a low lying area, diversion or retention dams may be necessary. At Leigh Creek coal mine, it was necessary to construct a retention dam to fully contain the flow of Leigh Creek with its catchment area of 100 km^2.

Once protection against regional flooding has been provided, it is necessary to protect against local flooding. This can include such measures as constructing shallow drains or bunds around the workings to divert water or direct it to storages. In an excavated pit the top of the ramp is always a danger point for floodwater ingress.

Groundwater

There are two ways of treating groundwater in hard rock quarries: allowing it come in and then pumping it out, or pumping it out through a series of de-watering wells. If the ground is hard and does not break down when wetted, and the stability of slopes is not seriously affected, it is generally cheaper to do nothing about groundwater beyond providing an effective in-pit drainage system to divert all the water to sumps.

However in coal mines, particularly lignite mines, it is usually necessary to dewater before mining commences. In soft rock, a dewatered slope may be stable at 40°; without dewatering it could be unsafe at 15°. There may be pressure aquifers present which, if not depressurised, will cause the floor to burst upwards.

The design of drainage systems and dewatering systems requires a good working knowledge of meteorology, hydrology, hydrogeology, pump technology and well construction, and usually requires the services of consultant engineers specialising in this field.

Removal of Inpit Water

Inpit drainage must be an integral part of any quarry design. Floors must be sloped (from about 1 in 100 to 1 in 200) towards a sump at the lowest point. Drains must be provided alongside haul roads, and culverts installed under roads. Sumps must be located where they are accessible for pump servicing.

Most quarries use diesel or electric powered, single or multiple stage centrifugal pumps for inpit pumping. Ideally, these pumps should deliver directly to the surface.

13. Operational Planning

General

Once a potential reserve of economically mineable mineral has been indicated by testing following exploration, planning of the whole operation with a view to the economic working of the deposit calls for consideration of:

- maximum use of the mineral resource
- preservation of the amenity of the area as far as possible
- safety of employees and the general public
- choice of an economic quarrying method and equipment
- safeguarding of adjoining properties
- use of the rehabilitated quarry site
- best rehabilitation techniques to achieve aftercare
- detailed layout of quarry, roads, plant, workshops, stores, offices and service areas and choice of a workforce with a view to creating an efficient operation.

In practice, as with all industry, a balance has to be struck between some of the above considerations.

One of the first aspects must be the safety of employees and the general public, and the safeguarding of the property. After consideration of these factors, the question of the amount of stone or ore to be extracted from a deposit must be determined. Where possible, for the benefit of the community, a mineral resource should be fully extracted. However, there may be limits imposed on the extractable tonnage by the quarrying method that has to be used. Then again the quarrying method itself may have to be modified because of the topography, the amenity of the area, or because of the nature and proximity of adjoining properties.

In practice it may be found that by taking a less-than-maximum tonnage by a modified method at a slightly greater cost, the operation might not unreasonably disturb the amenity of the area and might permit a good after-use for the excavation. Similarly, the stripping of overburden, choice of

equipment, siting of buildings, dumps, dams and so on, could be subjected in turn to such of the above principles as are applicable. The whole process of planning is a matter of considering various alternatives with a view to fixing a practical program of work. This should enable development of the State's mineral assets and, while permitting an efficient profitable operation, should satisfy to a reasonable extent the requirements which make an operation acceptable to the community.

Quarry Layout and Amenity

The essential first step in planning is to record on topographic maps all of the information on geology and vegetation that has been obtained during the exploration stage. Subsurface information, largely from drilling, is recorded on cross-sections together with geological interpolation and interpretation of the information. These plans would be on a scale of 1:1 000 or larger and the cross-sections at 20 m intervals or closer.

Transparent overlays are useful to show long term projections of the positions of benches as well as the position of roads, drills, loading equipment and other equipment for day-by-day operations.

Computers are being used increasingly in open cut planning for:

- creating two and three dimensional models
- assessing quantities of waste and mineral to be mined
- performing automatic design of ramps and roads
- scheduling equipment, such as the position and number of
- units required
- obtaining information on optimum equipment, blast designs and so on.

The size, shape, quality, position and attitude of the deposit, the physical nature of the stone (or ore) and of the rocks surrounding or covering it, and the topography of the area largely determine the quarrying method. The topography also influences the position of entrance and exit roads, the position of plant, workshops, housing, dumps, slime dams, stock piles and explosives magazines. It should be noted that, in order to keep haulage, conveying and pumping distances to a minimum, the use of adits, tunnels and cuts for drainage or access can be beneficial.

A quarry site invariably makes some initial alteration to the amenity of an area. Good public relations should be established right from the start of operations. It is an uphill battle to recover from a bad first impression, and the most important questions to answer in the beginning are: what are the possible nuisance effects of this quarry on the neighbours and how can these be eliminated or reduced to an acceptable minimum? Some answers to these questions are suggested in Chapter 3.

More will be said of dust and noise in Chapter 17. The obvious matters of concern to the public in any quarry operation are: trucks moving in or out of a quarry site; blasting noises and air concussion; the noise of drills and other machinery; dust arising from many different sources; pollution of water courses; quarry work started before normal traffic and general industry sets up a background noise, or work carried on after this noise has died down.

Particular note should be made of the prevailing wind and its effect on dust and noise. Water should be made available and used freely to allay dust. Good roads are needed in the quarry area — sealed where practicable. The speed of trucks should be kept low. Sound barriers should be arranged to deaden sounds in critical directions, either in the form of earth banks or stands of trees. Trees and shrubs also serve in restoring stripped or worked out areas, and in covering dumps and overburden.

The local government authority should be informed at an early stage about a quarry project. The best way to inform local people is to take them to the site initially, and to have them re-visit the site periodically. Full consultation with the community at an early stage can prevent many difficulties later.

Rate of Production

The initial rate of production will depend mainly on current demand for the stone, but planning must also take into account possible future demand, with increases in production by stages. The main items of equipment in the quarry and in the treatment plant should be capable of handling larger tonnages than the initial production, unless the quarry is merely being opened for a specific contract. However, expensive machinery must be fully used in productive work and not be required to move frequently or shut down unnecessarily. Leasing equipment as an alternative to outright purchase offers some advantages and flexibility. Areas should be left clear for possible future extensions of workings and plant, tailings dams, dumps and so on, to avoid costly reworking or resiting.

Stripping Overburden

Most quarries have overburden of one kind or another to strip before regular production begins. This stripping should not ordinarily be any more extensive than is required to maintain production, i.e. keeping ahead of extraction. However there might be circumstances in which, with a very limited effect on the environment, it could be more economic to clear as much overburden as possible in one operation. Indeed, it might be the main factor in making a project viable. In these circumstances the topsoil and vegetative cover should be stockpiled separately from the rest of the overburden for use in rehabilitation work.

Quarry Benches

The amount of stripping will vary from quarry to quarry, and it will vary also with the number of benches required to ensure continuous extraction. For regular production there must be enough separate benches, or faces on one bench, to enable drilling, blasting, loading and clean-up to be carried on without one task interfering with another. Access roads must be available when needed. Provision must be made in some instances to enable the selective quarrying necessary to cope with, among other things:

- special orders for a particular quality of stone
- variations in the character of the stone
- clayey materials in winter
- flooding of a lower bench.

In South Australia the height of faces in quarries is limited to a maximum of 20 m but not many quarry faces exceed 17 m and most are 13 m or less. The height of a face may be dictated by the thickness of the bed of stone being worked, or by planes of weakness in the bed. It may also be determined by the equipment (either drilling or loading) available for the job, or by environmental considerations.

The height is often selected for maximum digging efficiency but the total cost of drilling, blasting, loading, transportation to tip and cleanup should always be the prime consideration.

Higher benches are normally cheaper to work because of the lower overall cost of roads and the use of larger blasting patterns. Lower faces are usually needed to improve quality control. Low benches are sometimes required for harder stone or minerals. Bench clean-up and working areas needed on benches are factors also to be considered.

As a general rule for a rope shovel, face height should not exceed the height of the boom point because the shovel works close to the face and could be in danger from overhang. This factor is not so important where FELs are used because the muckpile is flatter, largely as a result of improved blasting practice and fragmentation and the loader is therefore further from the face.

Bench widths must provide sufficient space for:

- the loading machine and transport equipment to work in safety
- the pile of rock broken in blasting to rest on the bench floor and not spill onto the floor below.

Widths can vary from 8 to 40 m and more, depending on the equipment. A bench width of 16 m is common while 10 m is the minimum for efficient operation. If at all possible working one bench under another at the same time should be avoided.

Pit Slope

To ensure stability and safety of the final pit boundary, the angles of faces and the widths of berms left on the pit limits must be carefully designed. In many quarries final faces are pre-split or smooth-blasted at steeper angles than the working faces. Berms should be wide enough to allow a small dozer to clean up falls. The greater the distance between berms, the steeper can be the overall slope. In some open cuts, several benches are run together at the pit limits with berms installed at only every third or fourth bench. This enables the berms to be wider and more efficient at catching rocks. One 6 m berm every 20 m is far better than one 3 m berm every 10 m.

Factors which have a bearing on rock falls and slips, and the stability of rocks generally include (Fig. 13.1):

- type of rock
- strength of rock
- stratification and foliation
- mechanical fragmentation
- chemical defects
- relation between slope face and structural elements such as faults, joints and bedding time
- water in rock joints
- effect of vibrations
- slope height.

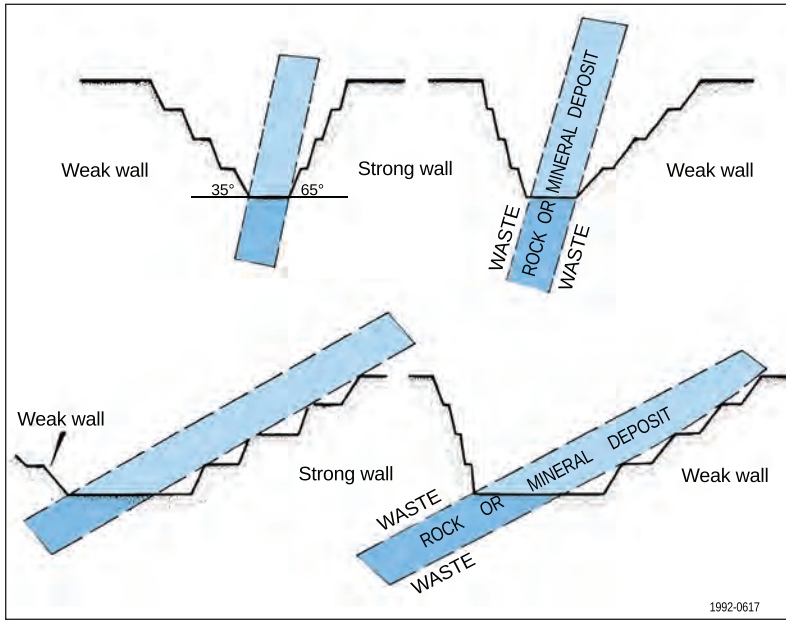


Fig. 13.1 *Pit slopes depend on ground conditions and orebody disposition.*

Some identifiable modes of slope failure are (Fig. 13.2):

- Rock fall
  - failure of the rock in tension
- Rock flow
  - pressure at base of a face causes failure by compression which progresses leading to a general breakdown of the rock mass particularly if it is jointed
- Rotational shear (circular failure)
  - occurs chiefly in weak rocks grading to sands
- Plane shear
  - failure occurs if the upper block is able to slide on the plane of weakness, usually a fault
- Wedge failure
  - two intersecting planes of weakness form a wedge shaped block which may be able to slip on one or both of the planes
- Toppling failure

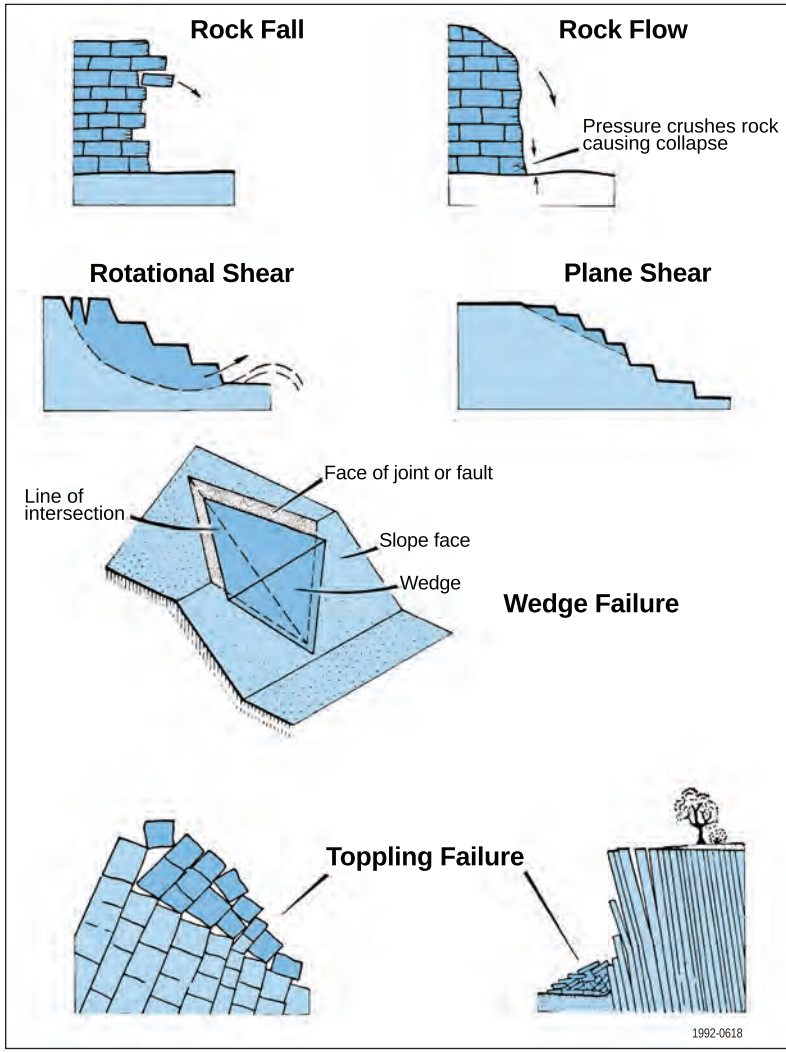


Fig. 13.2 *Modes of slope failure.*



Rockslide at Iron Monarch opencut causing displacement of benches. (Photo T013896)

Blasting or earth tremors may trigger failure in any of the above modes where appropriate conditions exist.

Typical overall pit slopes are given below:
Iron Knob ironstone quarry – 33-45°
Iron Duke ironstone quarry – 45°
Kanmantoo copper mine – 46-56°
Burra copper mine – 45°
Penrice Quarry – 65°
Stonyfell Quarry – 45-55°
Greenhill Quarry – 45-55°
Riverview Quarry – 48°
Linwood Quarry – 45-55°
Ardrossan Quarry – 46°
Rapid Bay Quarry – 45°
Sellicks Hill Quarry – 55°
Leigh Creek coal field – 38-55°.

In general, the flatter the overall pit slope, the greater the amount of overburden that has to be moved. Hence the need to have the pit slope as steep as possible consistent with safety, taking care (particularly with weak broken rock) to make allowances for the effect of weathering or water saturation on the rock. In some cases, particularly where the area of the lower benches is small, it may be feasible to increase the final slope on the lower levels. In addition, final land use may influence the angle of pit slopes left after mining. Nature will determine that, given time, the pit walls will eventually reach the natural angle of repose of the weathered rock.

14. Handling Plant

The main items of handling plant used to maintain the flow of material in a crushing and screening operation are belt conveyors, bucket elevators and pumps. Stockpiles and bins are essential to enable adequate storage and delivery of products to meet customer requirements.

Belt Conveyors

Belt conveyors are essentially endless belts running on idlers (which are spaced at intervals) and passing around pulleys or drums at either end. The idlers are usually troughed to carry the load but are flat on the return side of the system (Fig. 14.1).

The return side is often covered to prevent spillage from the loaded belt falling onto the return belt and building up on the head and tail drums and upper idlers. This could result in interference with the tracking and damage to the belt.

Belts are usually made of rubber reinforced with various fabrics such as cotton, nylon and polyester. Steel cords may be added for increased tensile strength. Using various rubber compounds, carcass types and numbers of plies, belt conveyors are available for many different service conditions such as acid or oil resistant environments, high temperature (to 800°C) and flame resistant applications, high wear, high lifts and high impact situations. Conveyor belting is available in most widths between 400 mm to 3 150 mm in increments of 50 mm or 100 mm. Choosing the most appropriate belt for an application is a complex matter and expert advice should be sought.

Conveyors used for mining are usually constructed with three-roll idlers and a trough angle of between 20 and 45° to suit the capacity, type and lump size of material transported and the belt construction. Four- and five-roll idlers are occasionally used, but they are initially more costly.



Belt conveyor in quarry crushing plant. (Photo 040460)

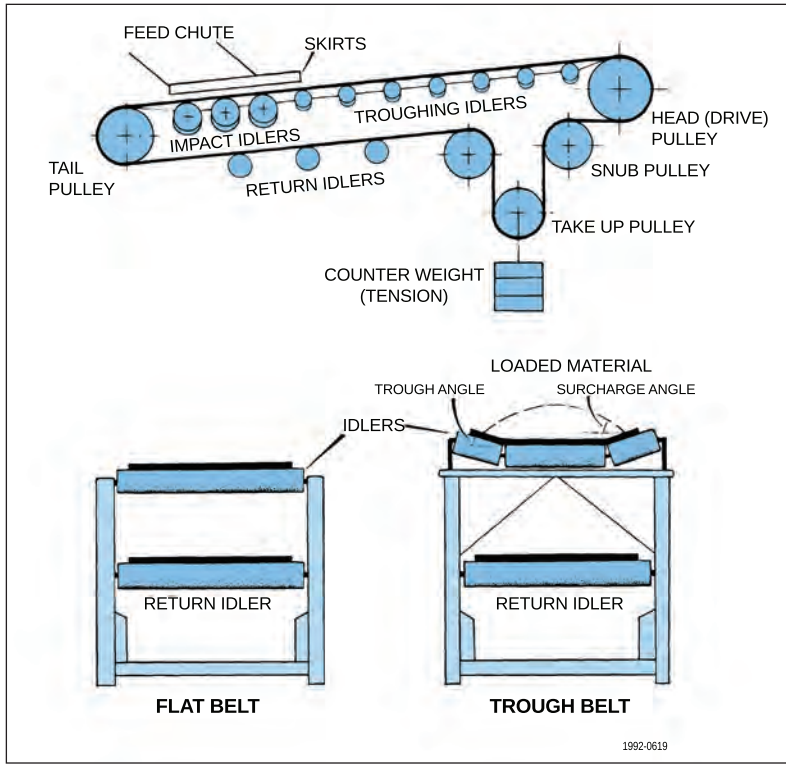


Fig. 14.1 *Belt conveyor installation.*

Table 14.1 *Maximum safe angle of inclination for troughed belt conveyors.*

Material	**Maximum angle of inclination** (°)
Dry sand	15
Damp sand	20-24
Gravel with sand	18-20
Screened gravel	15-18
Round shingle or pebbles	12-14
Limestone or clinker	18
Screened aggregate	16-18
Mixed crushed rock	18-20
Loose earth	20

Conveyors can run horizontally or be inclined, but the maximum angle of inclination is determined by the nature of the material to be transported, as shown in Table 14.1. For steeper inclines belts can be ribbed. Inclined belts are prevented from running back when power is cut off by means of a brake or, preferably, a ratchet holdback.

Belt conveyors have the following features:

- simple construction resulting in low maintenance requirement
- high efficiency, because of low friction losses and balanced loadings
- small power requirements and little labour
- conveying parts and running mechanism are completely separated
- the material to be moved is received directly onto the belt and is carried on it without further disturbance
- abrasive wear occurs only in the loading and discharge chutes, and where the material lands on and is accelerated by the belt
- clogging and jamming of material can be eliminated if chutes are correctly designed.

Width and Speed of Belts

After the quantity of material to be handled per hour has been determined, selection of the optimum combination of belt width and speed can be made.

Width is determined by the size of the largest lumps as indicated below, or by the formula for capacity when handling smaller lump size material.

Belt speed may be determined from the table of belt speeds given below or from the capacity formula. The speeds shown in the table should be considered as maxima for conservative design of short conveyors. For longer conveyors and extensive systems these speeds are often exceeded by up to 100% but chutes and loading points in particular and the whole structure in general must then be very carefully designed and installed.

Lump Size

Having arrived at belt width and speed as required by the quantity of material to be conveyed, the belt width should be examined to ensure that the largest lumps of material can be handled. A rough guide to belt width in this instance is:

- for uniform lump size: belt width = 4 x lump size
- for mixture of lump and fine sizes: belt width = 3 x largest lump size.

If a wider belt is required because of lump size, belt speed should be reduced so that the conveyor will be fully loaded when the required capacity is being handled. In most cases it is more economical to install a crusher early in the system, thus permitting installation of narrower, less expensive conveyors for the smaller material.

Belt Speed

While the speed of belt may vary between wide limits (0.2 m/s to 10 m/s), Table 14.2 is a reasonable guide to the maximum speed of small industrial conveyors.

Table 14.2 *Guide to belt speed (in m/s) for small industrial conveyors.*

Material	**Belt width** (mm)			
	Under 500	650-800	1000	1200 and over
Light, free flowing (grain, sand)	2.5	3.0	3.7	4.0
Moderately free flowing and fine sized (sand, gravel, fine stone, fine coal)	2.0	2.8	3.2	3.4
Moderately heavy with lumps (crushed ore, coarse stone, run-of-mine coal)	1.7	2.3	2.6	2.7
Heavy, sharp, abrasive (hard ore, hard coke, run-of-mine hard coal)	1.5	1.8	2.0	2.2

Determining Belt Capacity

The amount of material which a conveyor belt will carry depends on:

- belt width
- belt speed
- trough angle of the idlers
- angle of surcharge of the material
- bulk density of the material
- inclination of the conveyor.

To achieve maximum capacity, the belt must be loaded from well designed feeders or chutes and the skirting must be of sufficient length to ensure that the load is correctly placed. The capacity is found by the formula:

$$C = 3\,600\ SVP$$

where C = conveyor capacity (t/h)
S = material cross-sectional area (m^2)
V = belt speed (m/s)
P = material density (t/m^3)

For the volumetric capacity the following formula can be used:

$$K = 3\,600\,SV$$

where K = conveyor capacity (m^3/h)

Whilst the belt is physically capable of carrying a load indicated by this formula, it cannot be achieved unless the loading point is expertly designed, the loading area is level, and the material is loaded on to the belt at a speed about equal to the belt speed.

Table 14.3 provides a guide to conveyor belt capacity.

For other trough angles, the following multipliers can be used:

For a 0° trough use 0.40
20° trough use 0.84
35° trough use 1.06
45° trough use 1.15

Power

The drive may be by an electric motor or internal combustion engine, usually at the delivery end of the conveyor. The tensioning device is at the tail pulley if it has a screw adjustment. A gravity-type tension unit is placed at a convenient high point along the belt (Fig. 14.2).

The total power required to drive a belt conveyor is the sum of:

- power to move an unloaded belt over the idlers
- power to move the load horizontally
- power to lift (or lower) the load
- power to turn the pulleys
- power absorbed by trippers and belt cleaners
- power losses in the drive.

Table 14.3 *Conveyor capacities (t/h at a speed of 1 m/s) based on the above formulae.*

Belt width (mm)	**Cross-sectional area of load*** (m^2)	**Material density (loose)** (t/m^3)				
		0.8	1.2	1.6	2.0	2.8
450	0.019	55	82	110	137	191
500	0.023	66	100	133	165	232
600	0.036	104	156	208	260	364
650	0.040	115	173	230	288	400
750	0.058	167	250	334	418	585
800	0.064	184	275	368	460	644
900	0.086	248	370	500	620	870
1 000	0.103	300	440	590	740	1 040
1 050	0.120	347	528	690	863	1 210
1 200	0.152	447	656	876	1 090	1 530
1 300	0.200	577	862	1 150	1 440	2 020
1 400	0.210	604	907	1 209	1 510	2 110
1 500	0.250	720	1 100	1 440	1 800	2 520
1 600	0.280	805	1 210	1 610	2 010	2 820
1 800	0.370	1 060	1 600	2 130	2 650	3 750

*Based on a 30° trough angle and a 20° surcharge angle.

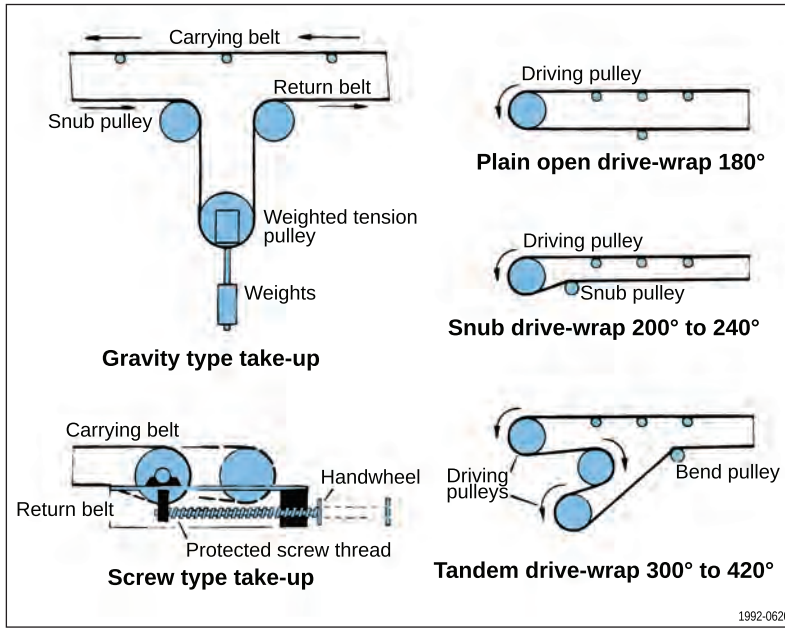


Fig. 14.2 *Belt configurations.*

Example. Determine the power required for a conveyor 250 m long to move 500 t/h on a 1 000 mm belt travelling at 3 m/s up a 15 degree incline (lift 65 m).

Power required at the belt:

Unloaded belt = 5.4 x 3 m/s	= 16.2 (from Table 14.4)
Material horizontal	= 10.2 (from Table 14.5)
Material elevation 6.9 + 27.2 + 54.4	= 88.5 (from Table 14.6)
Total power	= 114.9 kW
Effective tension in the belt	= 114.9 x 1000/3 = 38.3 kN

Loading and Unloading

The method of loading the belt is of particular importance as undue friction and impact at the loading point are the main causes of belt wear. The material should be delivered to the centre of the belt, if possible the fine material first, and the material should be moving at the same speed and in the same direction as the belt itself.

Table 14.4 *Power in kW to operate an unloaded conveyor per metre per second.*

Belt width (mm)	**Conveyor length** (m)							
	10	25	40	80	100	200	250	500
400	0.4	0.4	0.5	0.8	0.9	1.5	1.8	3.2
500	0.5	0.6	0.7	1.0	1.1	2.0	2.4	4.3
600	0.6	0.7	0.9	1.3	1.5	2.5	2.9	5.4
800	0.8	1.0	1.3	1.8	2.1	3.5	4.2	7.7
1 000	1.1	1.4	1.6	2.4	2.7	4.5	5.4	10.0
1 200	1.4	1.7	2.0	2.9	3.3	5.5	6.6	12.2
1 500	1.7	2.1	2.6	3.7	4.3	7.1	8.5	15.7
2 000	2.4	2.9	3.4	5.0	5.7	9.6	11.5	21.0

(Table supplied by Tecbelt Pacific)

Table 14.5 *Power in kW to convey material on a conveyor horizontally.*

Capacity (t/h)	**Conveyor length** (m)							
	10	25	40	80	100	200	250	500
40	0.2	0.2	0.2	0.4	0.4	0.7	0.8	1.5
100	0.4	0.5	0.6	0.9	1.0	1.7	2.0	3.7
160	0.7	0.8	1.0	1.4	1.6	2.7	3.2	5.9
200	0.8	1.0	1.2	1.8	2.0	3.3	4.1	7.5
250	1.0	1.3	1.5	2.2	2.6	4.1	5.1	9.7
400	1.6	2.0	2.4	3.5	4.1	6.8	8.2	15.0
500	2.0	2.6	3.1	4.4	5.1	8.3	10.2	18.7
800	3.3	4.1	4.9	7.1	8.2	13.6	16.3	29.9
1 000	4.1	5.1	6.1	8.8	10.2	16.5	20.4	37.4
1 250	5.1	6.4	7.7	11.1	12.8	21.4	25.7	47.1
1 600	6.6	8.1	9.8	14.1	16.3	27.2	32.6	59.8
2 000	8.2	10.2	12.2	17.7	20.4	33.0	40.8	74.8
2 500	10.2	12.8	15.3	22.0	25.5	42.5	51.0	93.5
3 200	13.1	16.0	19.3	27.3	32.1	53.5	64.2	117.7

(Table supplied by Tecbelt Pacific)

Table 14.6 *Power in kW to elevate material on an inclined conveyor.*

Capacity (t/h)	**Elevation** (m)			
	5	10	20	40
40	0.5	1.1	2.2	4.4
100	1.3	2.7	5.4	10.9
160	2.1	4.4	8.7	17.4
200	2.7	5.4	10.9	21.8
250	3.4	6.8	13.6	27.2
400	5.4	10.9	21.8	43.5
500	6.9	13.6	27.2	54.4
800	10.9	21.8	43.5	87.0
1 000	13.6	27.2	54.4	108.9
1 250	17.0	34.0	64.1	128.1
1 600	21.8	43.6	87.1	174.2
2 000	27.2	54.4	108.9	217.8
2 500	34.0	68.1	136.1	272.2
3 200	43.6	87.2	174.4	348.8

(Table supplied by Tecbelt Pacific)

Discharge is usually over the head pulley, but in some cases a tripper is used to direct the finished graded product into individual bins or to stockpiles or storage bins (Fig. 14.3).

Cleaning Conveyor Belts

A clean belt and clean pulleys and idlers lead to a longer belt life.

A diagonal cleaner bar can be placed on the return belt near the tail pulley to prevent sharp objects or stones, which may have fallen onto the return belt, from catching between belt and pulley and puncturing the belt or cutting it into strips.

Wet dirt and many other materials stick to belts and pulleys and will build up if permitted to do so. A scraper consisting of a rubber-on-steel blade or a serrated rubber roller or bristle brush may be placed just after the discharge point to clean the belt and direct scrapings into a chute to join the main stream.

In certain situations a water spray, with or without squeegees, may be used alone or in conjunction with a mechanical cleaner.

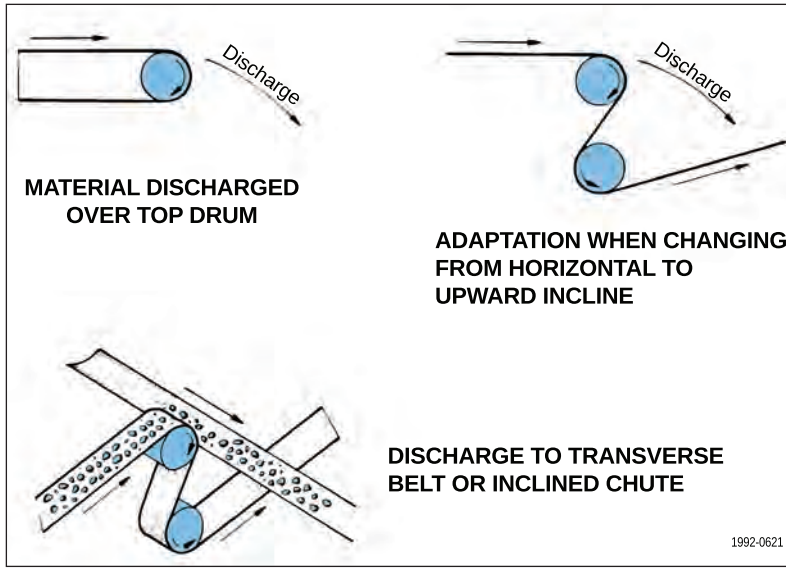


Fig. 14.3 *Discharging from belt.*

Special Belt Configurations

There is an application in appropriate circumstances for the Flexiwall Conveyor and the Pipe Conveyor, the former for steep lifts and the latter for curved paths (Figs 14.4 and 14.5).

Safety

The main safety requirements for conveyors are set out in Australian Standard AS 1755.

An important point in this standard is the requirement to provide guards for all pulleys and 'nip points' up to a height of 2.5 m above any floor, platform or walkway.

In accordance with State Government regulations conveyors are required to have a convenient means of stoppage at any point along the system. In practice this means that conveyors within 2.5 m from the floor or a normally accessible walkway must have a lanyard or pullwire emergency stop.

Many serious accidents have occurred during cleaning of return idlers while the belt is moving. If such work is required the belt must be stopped and 'tagged out' by placing a notice on the starter switch.

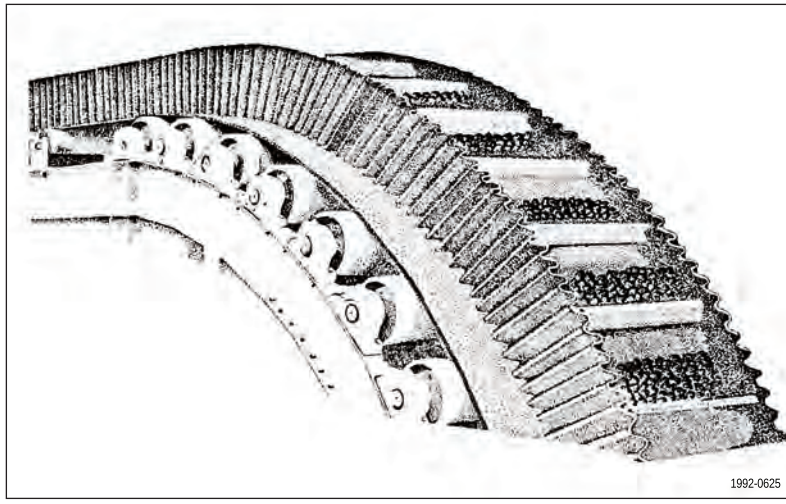


Fig. 14.4 *Flexiwall conveyor.*

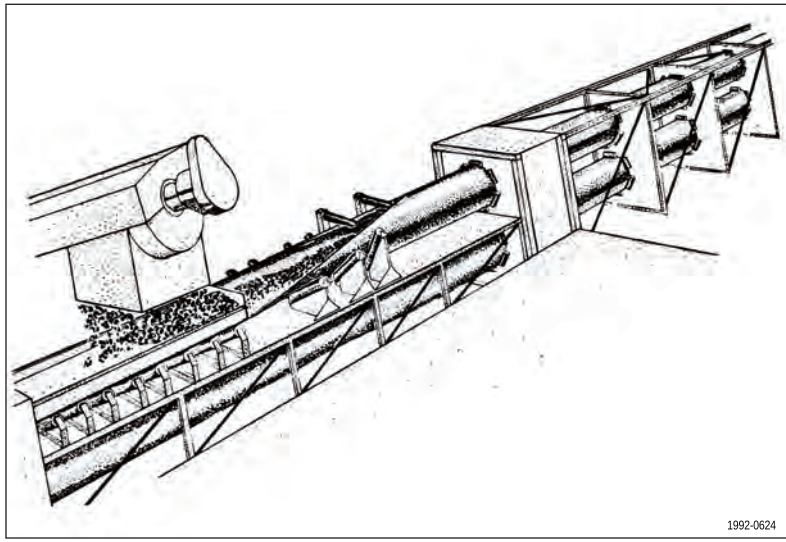


Fig. 14.5 *Pipe conveyor.*

Bucket Elevators

The commonest form of bucket elevator has mild-steel buckets rivetted to a belt (Fig. 14.6), although occasionally chain and bucket types are used. Bucket elevators are normally set at steep angles — around 70°. The load is picked up either by discharge from a chute or by a dredging action in a boot. Their best application is in a plant where space is restricted and the material is minus 50 mm size. Maintenance is high, particularly on coarse feed and unless totally enclosed (in which case inspection is difficult) they are a source of dust. The modern trend is away from bucket elevators, and towards belt conveyors. Plants can thus be spread out, and the suppression of dust is made much easier.

Stockpiles and Bins

Stockpiles

Stockpiles are reserves of material formed to cushion the effect of imperfectly matching consecutive stages of the quarrying operation from loading raw material to delivery of final product. They can be used to satisfy urgent customer demand or avoid plant stoppage when quarry or crushing plant equipment is held up for any reason.

Stockpiles generally involve double-handling: bins can avoid this.

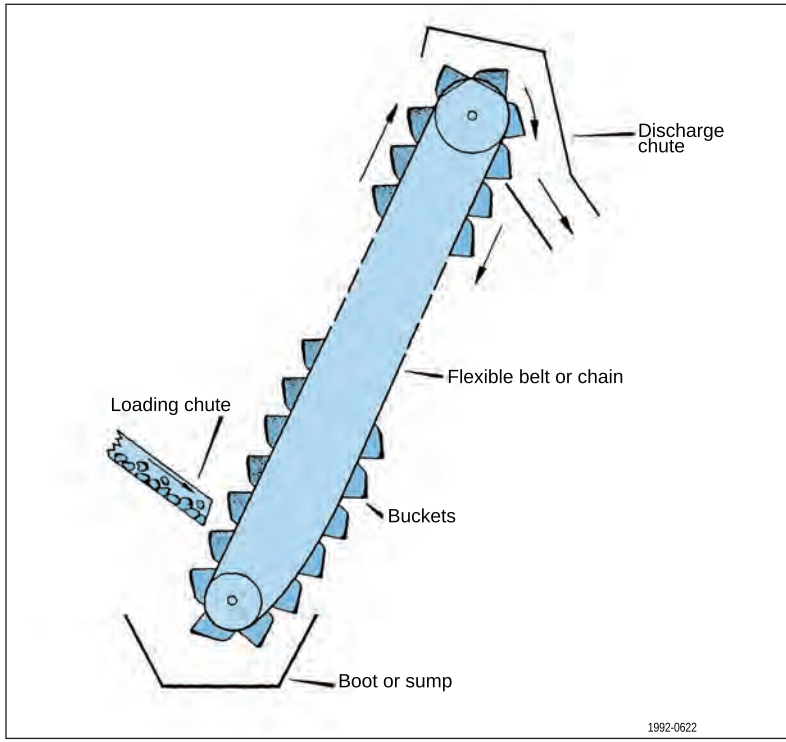


Fig. 14.6 *Bucket elevator.*

For final products it is advisable to stockpile different sizes separately and subsequently mix them as required unless there is an expected future requirement for a particular mix. Final stockpiles or bins are essential to cope with irregular demand.

The site for the stockpile should be selected to give easy access for dumping and reclaiming but generally should be clear of areas that may be used for future quarry or plant expansion except in particular circumstances.

Dumping can be by dumpers (directly over a disused bench or by ramp), conveyor belts or portable loaders. Reclaiming can be by a tunnel fitted with chutes and gates over which the stockpile is built, an FEL, grab or clamshell, dragline or bucket wheel excavator.



Final products bin with parabolic bottom. (Photo 040461)

To measure the volume of material in a stockpile the leading dimensions are taken, i.e. length and width of base and top and height in the case of a flat dump, or height and diameter or circumference of base for a conical dump. Heights can be obtained using a surveyor's theodolite and tape, a level and staff or, in the case of low dumps, using a pole pushed out horizontally from the top of the dump and a tape.

Formulae are then applied to calculate the volume of the whole dump or of parts.

Example. A rectangular flat dump is 50 x 30 m at base, and 30 x 10 m at top and 5 m high. Assume the base is flat and horizontal. The material weighs 1.8 t/m^3.

Divide into blocks (Fig. 14.7):

Volume	A	= 30 x 10 x 5 = 1 500 m^3
	B	= 2 x (½ x 30 x 10 x 5) = 1 500 m^3
	C	= 2 x (½ x 10 x 10 x 5) =500 m^3
	D	= 4 x (⅓ x 10 x 10 x 5) =667 m^3
Total volume		= 4167 m^3
Total weight		= 4 167 x 1.8 = 7 500 t

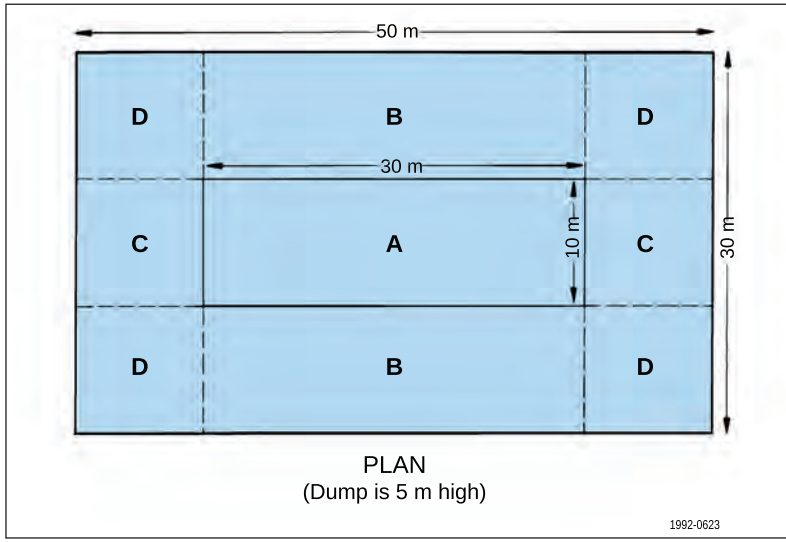


Fig. 14.7 *Volume of a rectangular stockpile.*

Bins

Bins are usually built so that trucks can load beneath them, chute doors being immediately above the trucks for a vertical flow of crushed material. Legs should be of steel, and should be widely spaced to give trucks plenty of room, particularly when it is necessary to back them into the loading position. If possible, drive-through should be arranged. Construction must be robust. It is usual to locate the final screens above the bins, the various size products being directed by chutes to their individual sections of the bin.

An alternative to conventional bins is the use of compartments separated by wood, steel, or concrete walls over a small tunnel in which a conveyor is installed. Trucks can then spot under the discharge of the conveyor, the belt is started and material from the desired compartment fed onto the conveyor through chutes in the tunnel.

In plants crushing aggregate for concrete or road use there is usually no flexibility between the main bin and chute feeding the primary crusher and the final stages of crushing. A continuous flow of material to the primary crusher from the primary bin is provided by means of a Ross chain feeder, roll grizzly, pan or apron feeder, or gate controlled chute (Figs 15.2 and 15.3). In the case of material requiring fine grinding for metallurgical treatment it is usual to provide a fine-storage bin, from which the material from the last stage of crushing is fed to the rod or ball mills.

Final product bins are usually built with the bottom inclined at 40° but can be flat bottomed which leaves a certain percentage of dead load. The slope of chutes to feed crushers should be 40° for material less than 76 mm gauge and 50° for material greater than 76 mm.

Pumps

Pumps of various kinds are needed in most plants to deal with waste and drainage water and to move sand pulps and sludge in washing plants. They are also required to pump water between screens and classifiers and to and from dams.

The type of pumps most commonly used for the low pressures (‘heads’) encountered in quarrying are centrifugal and diaphragm (positive displacement). The former is best suited for clear water and the latter for pulps although specially designed centrifugal pumps can cope with sludge. Diaphragm pumps are of much lower capacity and efficiency than centrifugal pumps and so are normally reserved for special purposes, e.g. thick sludges and small capacity applications.

15. Crushing and Screening

A crushing and screening plant is an integrated combination of receiving hoppers, feeding and scalping units, crushing machines, sizing apparatus (e.g. screens), conveyors and elevators, storage bins and control devices. The crushing plant should be designed to receive feed and to deliver a product or products according to consumer demands.

Definitions in Crushing Practice

Blocking	Obstruction of the feed opening by a stone which will not enter in any position.
Bridging	Obstruction of the feed opening by two or more pieces each of which could pass through.
Capacity	Quantity (tonnes or cubic metres) of material discharged by the crusher per hour.
Closed circuit operation	Type of operation where precise uniformity of product size is important; the product is screened as it is being discharged and the oversize material returned to the crusher; the product is discharged from the system only when it complies with size requirements.
Choke	Stoppage of the downward flow in the crushing chamber.
Choke feed	Feed in which the crushing chamber is as full as possible due to a continuous and sufficient head of material above the feed opening.
Discharge setting	Pre-determined distance between the crushing surfaces at discharge which regulates the product size.
Feed	Material that is to be crushed.
Feed opening	Determines maximum particle size of feed to fit into a machine.

Open circuit operations	Type of operation where uniformity of the product size is not important and the feed is run through the crusher once only.
Product size	Size of product after it has been crushed and screened.
Regulated feed	In contrast with choke feed, regulated feed occurs when the feed is throttled back to something below the full capacity of the crusher.
Reduction Ratio	There are two main methods of assessing this ratio: • Based on crusher dimensions. Ratio of the largest cube that will enter to the largest cube that can be discharged: $\frac{\text{Receiving opening (minimum dimension)}}{\text{Discharge opening (open setting)}}$ • Based on sieve sizing of feed and product: $\frac{\text{80\% passing size of feed}}{\text{80\% passing size of product}}$
Screenings	Quarry products that have been sorted through screens to meet customer requirements.

Crushing

Classification of Crushers (Fig. 15.1)

Coarse or primary breaking of hard rock is achieved using jaw and gyratory crushers and impactors. For soft material, hammer mills and single roll crushers can be used.

Intermediate or secondary crushing can be performed by gyratory crushers, cone crushers, impactors, hammer mills and roll crushers. Fine or tertiary crushers include short head cones, fine reduction gyratories, twin and triple roll crushers, hammer mills and vertical shaft impactors.

Jaw Crushers

Jaw crushers are reciprocating pressure breakers: they consist of two crushing surfaces called jaw plates, one moving and one fixed, set at a convergent angle downwards and housed in a rectangular frame. The frame

also supports bearings for the actuating shaft and contains the mechanism for transmitting the force from this shaft to the moving jaw.

There are two types of jaw crusher in use. The single toggle jaw crusher has the moving or swing jaw hung on an eccentric drive shaft with only one toggle at the bottom. The double toggle crusher has the swing jaw pivoted at the top and the jaw movement is caused by the pitman, which is hung from an eccentric on the drive shaft, and the oscillating action of the two toggles at the bottom. The difference in construction results in the single toggle jaw having up-and-down movement as well as backward-and-forward whereas the double toggle jaw swings only backward-and-forward.

The crusher size is determined by the feed opening (e.g. 900 x 600 mm). The discharge setting is measured at the bottom of the crushing chamber and is adjusted by varying the toggle size and/or shimming behind the toggle seat.

Jaw crushers produce a high proportion of stone the size of which is nearly that of the discharge setting with a low proportion of fines.

Gyratory Crushers

This crusher is made up of three assemblies: the top frame assembly, the main shaft assembly and the bottom frame assembly.

Table 15.1 *Typical jaw crusher capacities.*

Crusher size (mm)	**Min-Max setting** (mm)	**Capacity** (t/h)	**Drive** (kW)
500 x 360	40-75	17-45	20
760 x 500	50-115	40-100	55
1 060 x 760	100-175	140-300	110
1 370 x 1 150	160-225	450-650	185
1 800 x 1 500	175-300	630-1 050	300

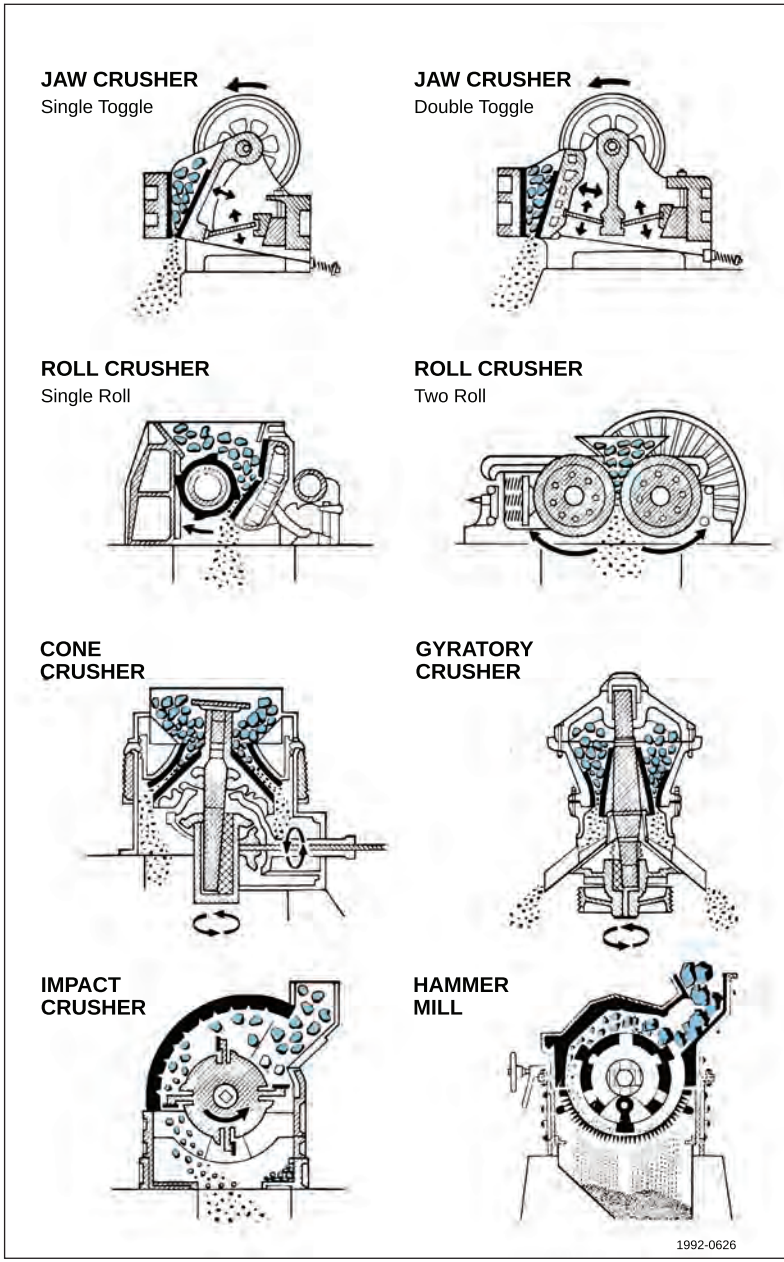


Fig. 15.1 *Crusher types.*

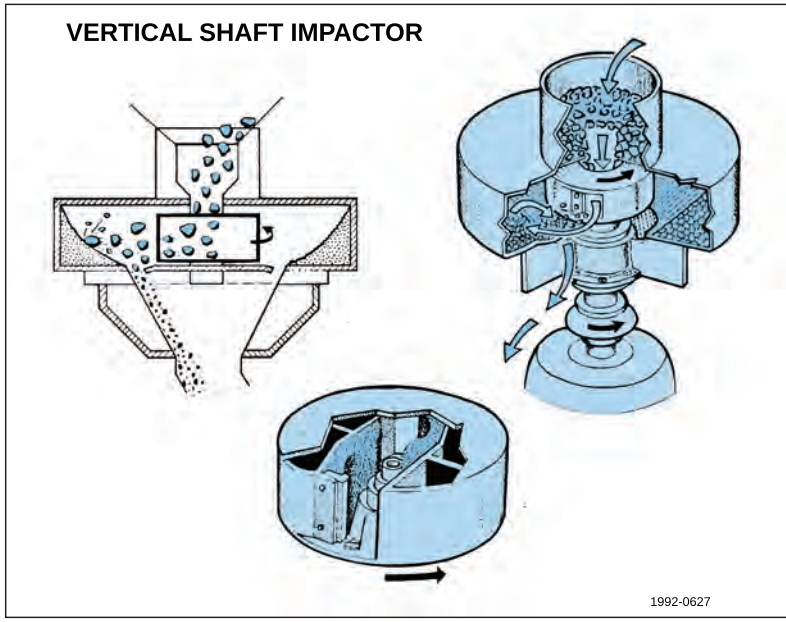


Fig. 15.1 *Crusher types (Contd.)*

The crushing chamber consists of the concave liner fixed to the top frame and the moving mantle liner fixed to the main shaft. In the bottom frame are the eccentric bearing and, in some models, the horizontal drive shaft.

The drive motor turns the eccentric through belting or it turns the horizontal shaft and rotation is transmitted through bevel gearing to the eccentric, which is mounted vertically in the bottom frame. The top of the main shaft pivots in a stationary bush in the top frame while the bottom is given a gyratory motion by the eccentric.

The crusher size is determined by the overall bottom diameter of the mantle liner (e.g. 900 mm, 1 200 mm), but for the large gyratory crushers used in primary applications the size of the feed opening is also specified; for example, a 1 500-2 500 mm gyratory crusher has a 1 500 mm feed opening and a 2 500 mm dimension over the mantle liner.

Gyratory crushers produce a coarse product of good shape but poorly shaped fines.

Table 15.2 *Typical gyratory crusher capacities.*

Crusher size (mm)	**Min-Max setting** (mm)	**Capacity** (t/h)	**Drive** (kW)
1 000	25-50	100-180	110
1 200	30-60	250-320	150
1 500	35-70	380-470	220
1 700	50-90	410-610	300

Cone Crushers

The cone crusher, like the gyratory crusher, consists of three main assemblies and is driven in the same way, but is designed for fine reduction and differs from the gyratory in three ways:

- the crushing chamber is much flatter and shorter in depth
- the main shaft movement is greater
- the head has no centralising bush at the top and is supported by the vertically mounted eccentric.

The size of the crusher is determined by the overall bottom diameter of the mantle liner. Feed openings vary within each size and crusher model to suit each desired application. The discharge setting is adjusted by moving the top frame assembly up or down.

Cone crushers perform best when choke fed; they produce products of good shape and a large amount of fines.

Table 15.3 *Typical cone crusher capacities.*

Crusher size (mm)	**Min-Max setting** (mm)	**Capacity** (t/h)	**Drive** (kW)
900	3-12	27-100	75
1 200	5-12	50-125	110
1 300	5-12	65-150	150
1 650	5-16	115-240	185
2 100	5-20	210-450	375

Single Roll Crushers

A single roll crusher consists essentially of a toothed roll and a stationary breaker plate. The input shaft drives the roll by gear and pinion. The breaker plate is hinged at the top on a support shaft. A cluster of compression springs holds the plate close to the roll. The whole assembly is housed in a rectangular frame.

Feed enters the machine from the top. Lumps are broken as they are struck by the heavy teeth projecting from the roll surface.

The crusher size is determined by the diameter of the roll and the width (e.g. 1 200 x 600 mm, 1 500 x 750 mm). Adjustment to the discharge setting is made by shimming behind the spring cluster.

Single roll crushers are suitable for non-abrasive products such as soft limestone, shale, coal and clays.

Twin and Triple Roll Crushers

Twin and triple roll crushers consist of two or three cylinders mounted on horizontal shafts which are driven in opposite directions, moving towards each other at the top. The triple roll crusher acts as two twin roll crushers in series, one roll being common with the other two being mounted one above the other and rotating in the same direction (opposite to the common roll). The cylinder liners are replaceable. The main frame carries the bearings for the drive shaft, the fixed roll bearings and the floating roll bearings. A cluster of compression springs holds the floating rolls against a stop, shims being used to alter the roll setting.

The fixed roll is driven through a pinion gear arrangement. Rotation of the floating roll or rolls is achieved from the fixed roll by means of a stargear or rubber tyre arrangement.

The crusher reduces rock by passing a ribbon of feed through the counter-rotating rolls.

Roll crusher sizes are determined by the diameter and width of the roll preceded by the number of rolls (e.g. 2-1 200 x 600 mm, 3-1 200 x 600 mm). Roll crushers are not suitable for abrasive materials; they can crush rock varying from hard to soft and they produce a product of good shape with few fines.

Hammer Mills

The hammer mill chamber consists of a thick steel plate body, lined with replaceable wear plates. Rows of hammers are suspended by pins between heavy steel disks spaced along and fixed to the drive shaft.

This assembly is called the rotor. The shaft is carried by heavy duty bearings on the main frame. The hammer mill is driven either by a motor directly coupled to the rotor shaft, or through a V-belt pulley arrangement.



Large crushing plant at Linwood Quarry. (Photo 040462)



Secondary HazeMag impactor with two tertiary cone crushers at Linwood Quarry. (Photo 040463)



Gyratory crusher with rock breaker, Penrice Quarry. (Photo 040464)



Crushing plant control room, Linwood Quarry. (Photo 040465)

The internal design of a hammer mill varies, particularly the placing and shapes of the breaker plates, the hammer design, the presence or absence of and the type of grating. The position of the feed inlet is important and varies with the application and the type of material to be processed. To compensate for wear and production changes, the breaker plates and/or the grates are adjusted towards or away from the rotor.

The mill size is determined by rotor diameter and width.

Hammer mills produce excellently shaped products throughout the full grade range.

Impactors

Impactors are similar to hammer mills but the rotor has fixed blow bars fitted to it instead of swinging hammers. These are either bolted to the rotor or slide into holders. The blow bars, being rigidly fixed to the heavy rotor, deliver maximum force because there is no 'give' in the impact. A lower speed can be used to give efficient breaking, resulting in less energy used, less machine wear and a lower fines percentage than in the case of swinging hammer mills.

To compensate for wear and production changes, blow bars are either turned or replaced or the impact plates are adjusted.

Impactor size is determined by the rotor diameter and width.

The particle shape throughout the full grade range is excellent.

Vertical Shaft Impactors

The vertical shaft impactor consists of a fast revolving impeller or rotor with vanes; this is fitted to the end of the vertical shaft supported on bearings in a drum-shaped crushing chamber. The rotor and the chamber walls have replaceable wear parts. The lower end of the vertical shaft is either coupled to an electric motor or driven by a V-pulley and belt arrangement, or it may be driven through a right angle gearbox with the above two options.

The feed material enters the rotor assembly from the top in the centre. High speed rotation with appropriately shaped vanes accelerates the particles outward, throwing them against stationary anvils or onto a stone bed lying along the chamber wall for impact crushing. The fragmented product then falls out of the discharge openings.

The degree of reduction and the final product size can be controlled by varying the rotor speed.

The vertical shaft impactor produces an excellently shaped product with a high fines content.

Other Crushers

Many other crushers have been developed for special purposes to deal with particular materials or produce particular products. Some are only slight modifications of the types mentioned already. The Omnicone crusher, for example, is similar in principle to the cone crusher but uses hydraulic cylinders instead of rotating the head to adjust the crusher setting. The MMD crusher consists of two contra-rotating rolls fitted with very large teeth; it combines higher capacity with low height requirement.

Two rather different crushers used for soft materials are the Bradford crusher (rather like a heavy trommel — see later under 'Screens') and the Stamler feeder-crusher which is similar to an impactor but the rotor with fixed blow bars is mounted over an apron feeder to minimise height requirements.

Selection of Crushers

Planning Factors

Normally the output of a quarry is governed by that of the crushing plant. Hence, in planning, it is first necessary to decide on the size, type, and number of crushers.

Sometimes, of course, the overall output is governed by the output of which the quarry is capable, and the nature of the material quarried may govern the choice of the crushing and other plant.

In selecting crushing equipment, the following factors will influence the choice:

- required ultimate output
- type of rock
- size of rock fed to the primary crusher from the quarry face
- size and shape of aggregate required.

The last factor influences both the type and number of crushers, since the reduction ratio for each stage must be kept within certain limits if good particle shape is required.

However, not all rock crushed is required as aggregate, and fine grinding may follow the last stage of crushing. An example is in cement manufacture and metallurgical processes where the aim is to achieve the maximum economic reduction ratio and particle shape is not important.

Having decided on the crushing plant, material must be fed to it from the quarry in such a way that the following requirements are met:

- a steady rate of feed is maintained
- dirt and unsuitable material are removed from the feed
- oversize rock is removed from the feed
- screened aggregate is cleared from the plant and appropriately handled according to requirements for each product.

The quality of the product will be governed by the purpose for which the stone is intended and the demands of the customers.

Reduction Ratio

The precise definition of reduction ratio varies but, in general terms, all definitions refer to the ratio between initial particle size before entering and final size after leaving a crusher.

Experience has shown that not only do high reduction ratios result in poor shape but they also adversely affect grading, so that to improve the quality of the product it may be necessary to have three or more crushing stages.

To maintain good particle-shape, individual reduction ratios should not be greater than 4 to 1. Thus, to produce good cubical screenings of 13 mm sizing, feed to the crusher should not exceed 52 mm.

It may also be found that the fines produced from the primary crusher are below the required quality, because its reduction ratio is too high.

Some types of crusher are more tolerant than others in the allowable reduction ratio for a good product. As a general guide with average material, the following table may be used:

Table 15.4 *Reduction ratio recommended for different types of crusher.*

Crusher Type	**Reduction Ratio**	
	Best	**Maximum**
Roll	2-3:1	4:1
Jaw	3-4:1	5:1
Gyratory	3-4:1	5:1
Impact	4:1	8:1

Primary Crushers

Primary crushers take over from blasting operations the work of size reduction of the product.

The choice of crushers is governed by several factors:

- characteristics of the material
- average daily, or hourly, capacity required
- product size or setting of crusher discharge
- quarry equipment, such as type and size of shovel dippers (buckets)
- drilling and blasting methods
- method of feeding the crusher.

Sometimes the last three are chosen after the crusher type and size have been selected.

Characteristics of Material. For thinly stratified flaky rock a gyratory crusher is suitable. If, in addition, the rock is non-abrasive or soft, a single roll crusher will do. For massive material, or for hard and tough rock, a jaw crusher is usually best.

Capacity. This must normally be greater than the overall crushing plant capacity as interruptions to feed are inevitable. This reserve capacity may be from 25 to 50%. Normally in a quarry there is little reserve between the quarry and the primary crusher, because of limitations on the size of a crusher feed bin and because run-of-quarry material is fed to the bin in surges. Dual primary crushers are sometimes installed, but normally only for high tonnages of, say, more than 800 t/h. For smaller outputs a crusher with dimensions necessary to handle the feed produced by the shovel will have enough output to meet total plant capacity. Where two smaller units can be used, a smaller primary setting is possible, and this may permit one less subsequent stage.

Product Size. Generally, setting for a smaller product will reduce capacity. Often a crusher with the greatest receiving opening for a certain capital outlay is chosen. This will pre-determine the product size. Such a crusher would have straight concaves or jaw plates which may not give the best overall economy. This is because in comparing similar sizes of, say, gyratories, the larger non-choking machine can be set to a smaller discharge opening and still give greater capacity and have slightly greater receiving opening than the smaller machine. The extra cost involved may be offset by savings in the cost of subsequent stages. It should be noted that wear of crusher plates precludes any setting from remaining constant. The margin to be allowed may vary from about 13 to 76 mm in larger machines.

Comparison of Gyratory and Jaw Crushers

Gyratory Crushers

- Large tonnage capacity per dollar invested as capital.
- Minimum of slabbing, good cube-shaped product.
- Receiving openings are so shaped as to easily receive slabby feed.
- Minimum of bridging due to large receiving openings.
- Can be fed from two sides because of double receiving openings.
- Easily adjusted to counter liner wear.
- Small flywheel effect gives low starting peaks and quick stopping after either overload or choking.
- Can be serviced by simple crawl-type hoist, whereas a jaw crusher normally requires a crane and two-way travel.
- High speeds of pinion shaft allow use of high speed motors and low ratio drives.
- Lubrication is simple and cheap.
- Lower cost for foundations than for jaw crushers.
- Safety guards less extensive than with jaw crushers.

Jaw Crushers

- Large receiving opening per dollar invested as capital.
- Shape of receiving opening suits blocky feed. Except in the largest sizes, jaw crushers are better suited to handling large blocks than are gyratory crushers.
- Greater range of adjustments to counter liner wear.
- Better for handling sticky or dirty feed, as there is no diaphragm below the crushing chamber.
- Easier maintenance and repairs.
- Quicker replacement of wearing plates.
- Can be easily strengthened for rough-rock duty. In normal designs they are more rugged than gyratory crushers.

Secondary and Reduction Crushers

Reduction crushers are usually regarded as those producing from 76 to 25 mm screen size material.

Scalping, or removal of material already of an appropriate size, between primary and secondary crushing is usually advisable.

The following three main factors govern the selection of the crusher.

Capacity of Secondary Stage. This can be less than that of the primary stage. For example, if a scalping screen between the primary and secondary crushers is used, the capacity requirement of the secondary crusher may be as low as 15% of the total feed to the primary crusher. If the discharge from the primary crusher is small enough for handling by a belt conveyor, a surge pile or bin can often be established between the primary and secondary stages. This will help maintain full output from the secondary crusher. Normally 10-15% loss of efficiency is allowed for stoppages and breaks in flow.

Size of Feed. The feed opening should be at least three times the setting of the primary crusher for jaw crusher secondaries, and two and a half times the setting of the primary crusher for gyratory secondaries.

Size of Product. A primary crusher is normally set at its minimum safe setting. This is usually possible owing to the excess capacity of most primary crushers. If the primary crusher is large, the setting will usually be too big to cause excessive fines from the first stage. If twin secondary crushers are used, a smaller setting can be adopted as the machines would each be smaller than one single unit. Less vibration due to wear occurs in the smaller secondary machines than in primary crushers.

Feeders

Any crusher operates most efficiently when fed uniformly at a rate near maximum capacity. This can be achieved in the case of a primary crusher if the quarry trucks tip into a bin large enough to smooth out the peaks in the flow of material resulting from the irregular arrival of the trucks.

However in most installations the trucks tip onto a scalping grizzly which doubles as a feeder or into a small bin or, rarely, directly into the crusher (particularly if it is a gyratory).

Feeders to primary crushers from bins are mainly of two types: apron (including pan) and chain (Fig. 15.2). Both types have infinitely variable speed control.

Feeders to secondary crushers and conveyors range from simple vertical doors with adjustable openings to rotating plate, reciprocating drum, vibratory or apron. The type depends on the size, nature and dampness of the feed.

The moving grizzly feeders will be dealt with later under ‘Grizzlies’.

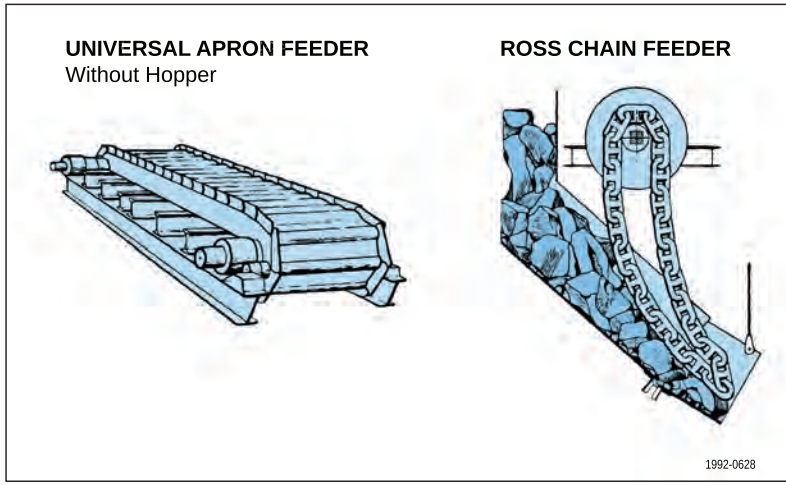


Fig. 15.2 *Apron and chain feeders.*

Screening

The purpose of screening is:

- To remove mud and small rubbish from feed to the primary crusher.
- To segregate between stages, as necessary, into
  - oversize material for recrushing in closed circuit crushing
  - undersize for either discarding or stockpiling
  - feed for the next crusher stage
  - material already fine enough to bypass the next stage of crushing
- To grade the final product into the required sizes.

A closed circuit crusher and screen unit is used when it is desired that the product from the unit should contain no material larger than a specified size. The discharge from the crusher is fed to the screen, the oversize from which is continuously returned to the crusher.

Washing is often allied to a screening operation to remove dirt and has the added benefit of settling dust.

Grizzlies

These are used to remove material which is too large for the primary crusher (or succeeding crushers) or to eliminate dirt and fines from run-of-quarry stone. They are of heavy construction.

Static grizzlies are normally improvised from bars or rails set longitudinally, without cross-bars. If used as a chute, a slope of 35-45° is used. Bars must be set with the narrow openings between them at the top to prevent material jamming. In some installations the oversize stones roll clear of the grizzly opening to an area where they can be broken by a pneumatic or hydraulic rock breaker.

There are three main types of **mechanically operated grizzlies**: the live-roll grizzly, the vibrating-bar grizzly, and the dropbar grizzly feeder (Fig. 15.3).

Screens

These are used to separate stone into gradings between specified limits or for scalping fines to prevent them passing through a crusher unnecessarily.

Rotary Screens (Trommels). Rotary screens or trommels are normally set on an inclined axis, and consist of one or more rotating cylinders with apertures of various sizes. Material is fed in at the upper end where the smallest apertures are positioned. For a diameter of a metre the peripheral speed should be about 45-60 m/min. Trommels are normally used for sizes of 51 mm and larger; screens cover the requirement below this. Trommels can be operated wet or dry.

Vibrating Screens. Vibrating screens consist essentially of a flat screen surface, usually stretched taut and normally inclined. The screen surfaces, single or up to four decks, are set as diaphragms in a rectangular frame having suitable sidewalls to confine the flow. The vibrating mechanisms are electrical (vibrating armature) or mechanical. Their impulses are applied directly to the screening surface or in most cases to the screen frame. The intensity of vibration must be sufficient to aid flow of the material over the obstructions formed by the uneven contour of the screen surface. If the inclination is too slight for a particle to roll, the vibratory impulses throw it clear of the surface and give a direction of travel. The impulse intensity must be sufficient to lift out particles caught in meshes.

The material used in making screen cloth is usually steel or steel reinforced rubber. Rubber balls may be used between decks to de-blind higher decks.

The selection of a vibrating screen is influenced by many factors. Variations in the crushing characteristics of rock, coal, ores and other materials are reflected in the screening technique.

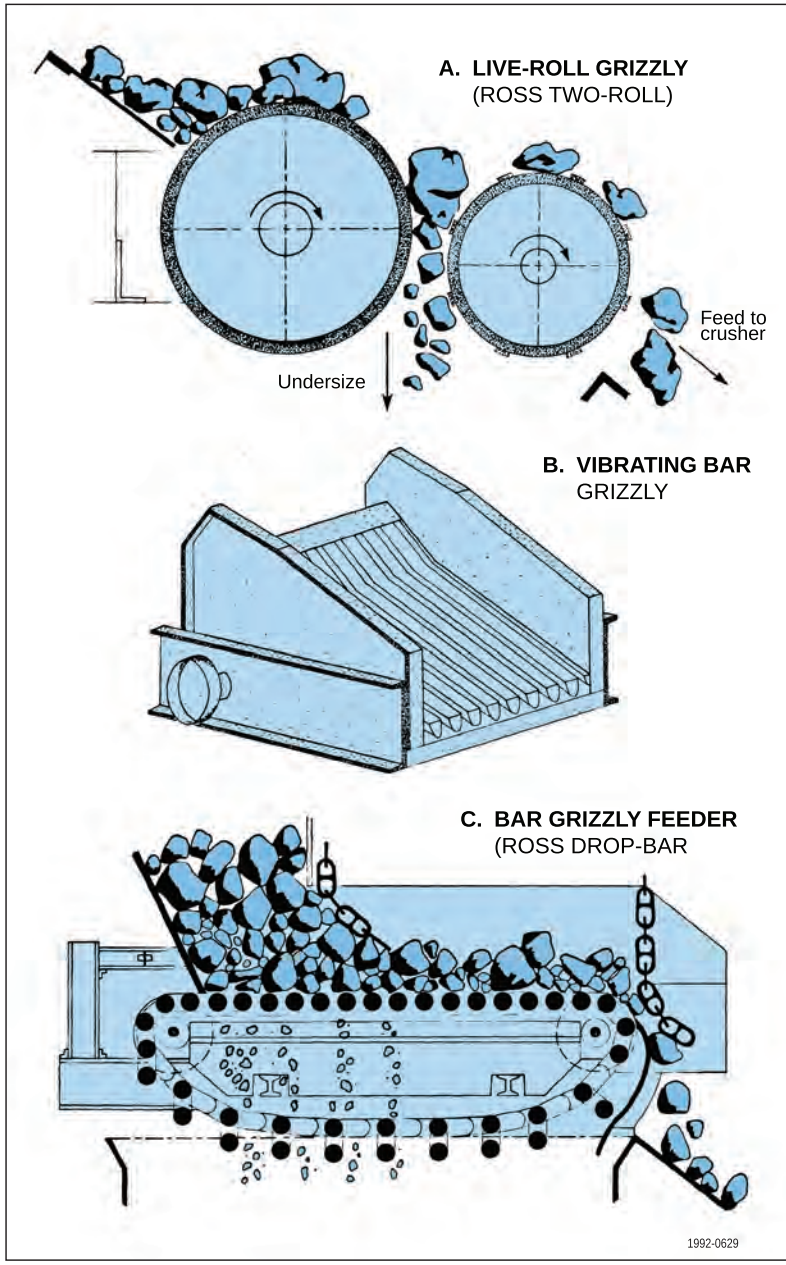


Fig. 15.3 *Three types of mechanically operated grizzly.*



Ross chain feeder, Penrice Quarry. (Photo 040466)



Screens in crushing plant, Stonyfell Quarry. (Photo 040467)



Trommel on sand washing plant at Golden Grove. (Photo 040468)

Selection of Vibrating Screens

The required surface area can be calculated using the formula:

$$A = \frac{R\ 1600}{CVHDB}$$

where A = Required surface area (m^2)

R = Rate of feed to the screen (t/h)

C = Factor determined from Figure 15.4A, based on the aperture of the screen wire.

V = Factor determined from Figure 15.4B, based on the percentage of oversize (larger than wire aperture) material in the feed.

H = Factor determined from Figure 15.4B, based on the percentage of half size (equal to or less than half the cloth aperture) material in the feed.

D = Factor determined from Figure 15.4D, deck area used on multi deck screens.

B = Bulk density of material (kg/m^3).

Example. Feed to a screen is 300 t/h of -85 mm limestone with a bulk density of 1 430 kg/m^3. Select a two deck vibrating screen, to be fitted with a 50 mm square aperture woven wire screen on the top deck and a 20 mm square aperture woven wire screen on the second deck. Each deck must be considered as a separate screen and percentages evaluated in terms of the feed to the deck under consideration.

Top Deck:

$$A = \frac{300 \times 1600}{76 \times 1.07 \times 0.68 \times 1.0 \times 1430} = 6.1\ m^2$$

using values:

R = 300 t/h feed rate to top deck

C = 76 for 50 mm aperture from Figure 15.4A

V = 1.07 from Figure 15.4B (interpolated value)

H = 0.68 from Figure 15.4B (interpolated value)

D = 1.00 from Figure 15.4D

B = 1430 kg/m^3

Second Deck:

$$A = \frac{189 \times 1600}{51 \times 1.58 \times 0.53 \times 0.90 \times 1\ 430} = 5.5\ m^2$$

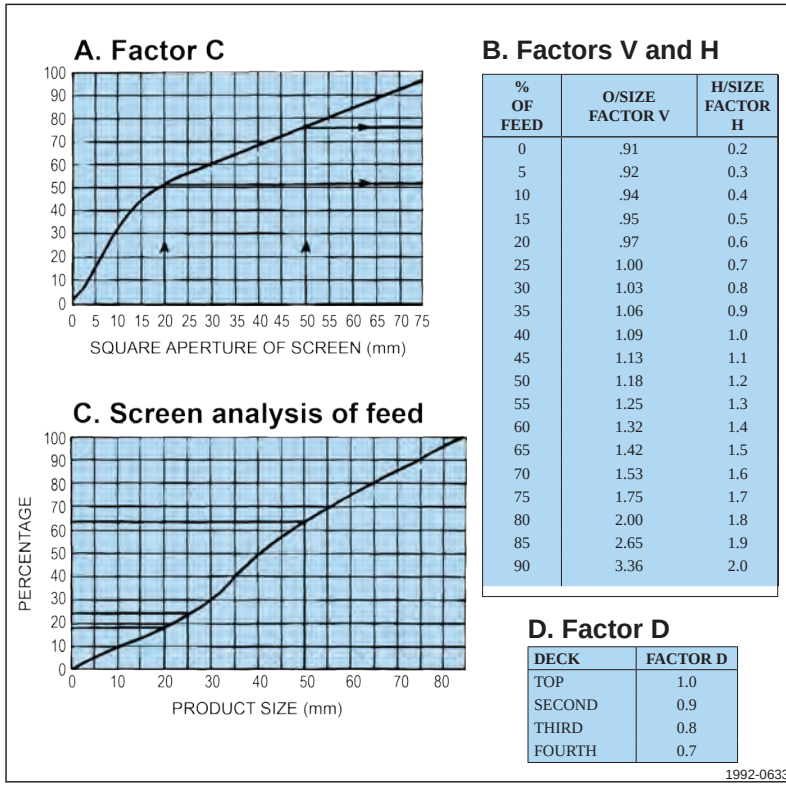


Fig. 15.4 *Charts for estimating screen size.*

using values:

R = 189 t/h feed rate to second deck (63% of original feed) from Figure 15.4C

C = 51 for 20 mm aperture from Figure 15.4A

V = 1.58 from Figure 15.4B (interpolated value)

H = 0.53 from Figure 15.4B (interpolated value)

D = 0.90 from Figure 15.4D

B = 1430 kg/m^3

In this example the top deck, with a net screening area of 6.1 m^2, controls the choice of screen size. The final choice is dependant on screen sizes currently available from manufacturers. For example, from catalogues it may be found that a 4.8 x 1.5 m screen with a net deck area of 6.5 m^2 is the nearest size with area greater than 6.1 m^2. Consideration should be given to using to a larger size to offset loss of open area if the screen cloth becomes clogged.

Installation and Operation

Installation

The following points are important in the installation of crushing and screening plants:

- Full advantage should be taken of gravity flow.
- The crushers should be mounted at the correct elevation so that feed can be controlled at a regular rate and removal of the crusher discharge is simplified.
- The foundations should be level, sufficiently rigid and of sufficient mass.
- Shafting and belt drives must be correctly aligned.
- The plant should be spread out to facilitate maintenance and clean-up access and to avoid concentrations of dust.
- Sufficient access should be provided for adjustment and servicing and for the removal and replacement of worn parts.
- Access is also required for clearing up along conveyors and other units of plant where spillage occurs.

Operating Instructions

The manufacturers normally supply detailed operating instructions, the principal points being:

- ensure that the crusher setting is correct
- run the crusher at the correct speed
- control the rate of feed
- watch for bridging of stones across the mouth
- tension drive belts correctly
- make frequent checks when in use to detect loose nuts or overheated bearings
- check lubrication.

Operation and Servicing

Machines used in quarrying and crushing are required to do heavy work over long periods, often with great fluctuations of loading and in bad conditions owing to dirt, moisture and the abrasive effect of grit, fine stone and sand particles. Although designed for the purpose — and robust in construction — they have working parts machined to fine tolerances and are susceptible to damage if not properly serviced and safeguarded.

Unless the machines are correctly aligned and installed on sufficiently rigid mountings, and unless precautions are taken to exclude dust and carry out systematic servicing, wear and tear may be excessive and break-downs will occur.

In most small plants, greasing should be done when the plant is idle. In large plants, automatic lubrication systems are normally used.

Matching the Crusher to the Quarry Plant

If blasting is the first stage of reduction, fragmentation obtained at the quarry face will influence the primary crusher selection. The loading machinery and the transport must be matched to it. The choice of secondary and subsequent stage crushers also follow from the choice of primary crusher.

Screening — whether for scalping, to work in closed circuit with the crushers or to sort out and size the final products — is an important part of the crushing process.

The movement of crushed materials through a crushing plant is mainly performed by belt conveyors.

Surge piles or storage bins are used to equalise the flow of material between stages. The capacity of these surge piles or bins depends on plant throughput. A large capacity surge storage between the primary section and the rest of the plant can have several functions. It can provide feed material for an extra shift, be a distributor to other crushing plants, or be a mixing device to average out the primary feed. Finally, bins are used to store the final products to be collected and transported for stockpiling or direct sales.

To control a crushing plant, interlocking automatic controls between different sections of the plant are essential for safe and economical operation. Using ammeters, temperature and pressure gauges, belt weighers, bin level indicators and closed-circuit television, crushing plant performance can be monitored every step of the way. The operation of the entire plant is usually controlled from an airconditioned control cabin adjacent to the primary crusher.

Dust Control

In many cases airborne dust made by the crushing plant can be successfully suppressed with the use of water sprays or wet screening — but avoid making the material sticky. Dust collection by suction fans is practised even though the collected dust may have no commercial value. The topic of dust incidence and control is dealt with in more detail in Chapter 17.

Washing

Washing is the name given to the process of removing clay from sand, or shale from coal. It can be a very simple or a very complex operation.

Ample supplies of water — up to 6 t of water per tonne of sand — are necessary for washing, but much of this water can be reclaimed after settling.

Washing is usually done at the screens to produce clean concrete aggregate or bituminous aggregate.

Organisation of Crushing and Screening

The aim in organisation is to achieve a continuous flow of material. The three overall stages to consider are:

Supply of Quarry Rock

Maintenance of a steady flow depends on sound quarry and transport practice, already discussed.

If necessary, provision should be made for the removal of dirt and fines or of oversize, by means of grizzlies before the primary crusher. If tramp iron is likely to be a problem, a magnetic device can be used for its removal or to stop the plant before damage is done.

Flow Through the Plant

Apart from operational controls to minimise mechanical breakdowns, a continuous flow can best be achieved by a well-planned use of gravity, elevators, conveyors, shaking screens and other suitable devices.

Redesign of minor points of obstruction is usually necessary after operational experience. Also, various expedients for alternative flow routes will allow partial production to proceed during the shutdown of one unit in a flow.

The use of stockpiles as discussed in Chapter 14 is an advantage.

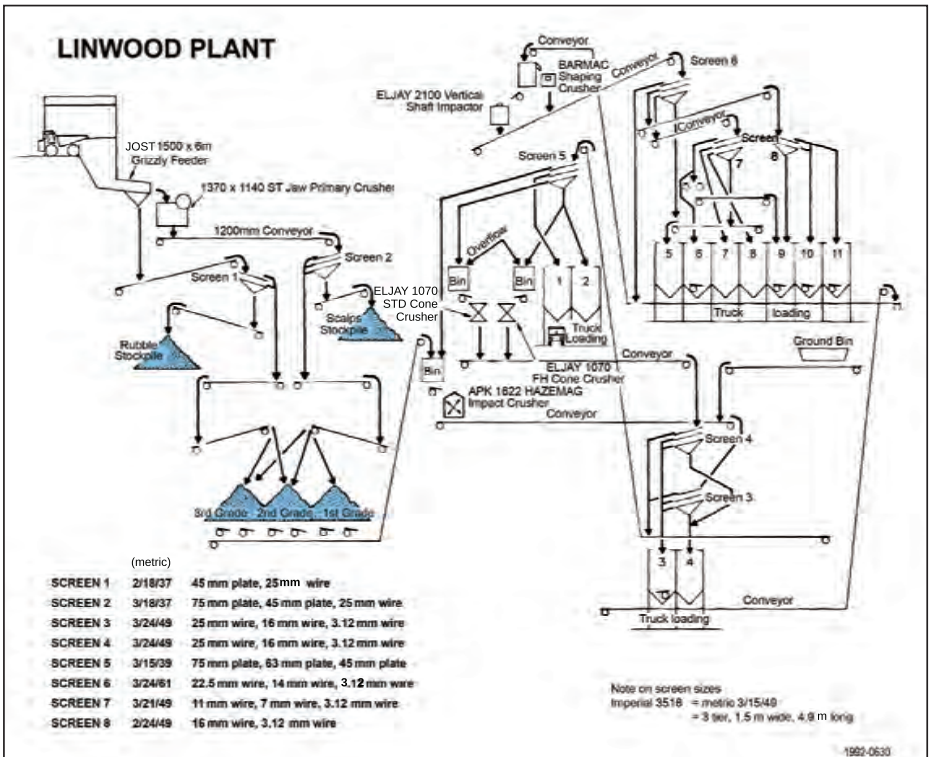
Crusher feed chutes and bins should be maintained at a reasonably full level, especially at change of shift.

Removal of product

Disposal of rejected material at each stage should be well planned, or build-ups will occur which may necessitate brief shutdowns to give time for clearance.

Examples of practice

Examples of typical plant flow sheets are given in Figures 15.5 to 15.7 and Table 15.5 gives details of the crushing and screening activities of some South Australian quarries.



(metric)

SCREEN 1	2/18/37	45 mm plate, 25mm wire
SCREEN 2	3/18/37	75 mm plate, 45 mm plate, 25 mm wire
SCREEN 3	3/24/49	25 mm wire, 16 mm wire, 3.12 mm wire
SCREEN 4	3/24/49	25 mm wire, 16 mm wire, 3.12 mm wire
SCREEN 5	3/15/39	75 mm plate, 63 mm plate, 45 mm plate
SCREEN 6	3/24/61	22.5 mm wire, 14 mm wire, 3.12 mm ware
SCREEN 7	3/21/49	11 mm wire, 7 mm wire, 3.12 mm wire
SCREEN 8	2/24/49	16 mm wire, 3.12 mm wire

Fig. 15.5 *Plant flow sheet, Linwood Quarry.*

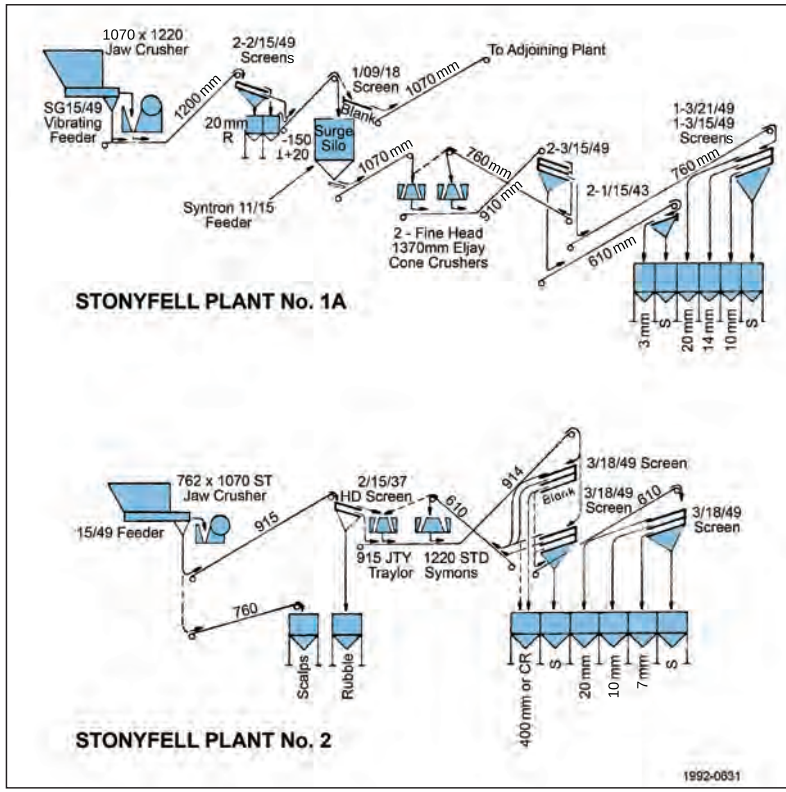


Fig. 15.6 *Plant flow sheets, Stonyfell Quarry and Salisbury Quarry.*

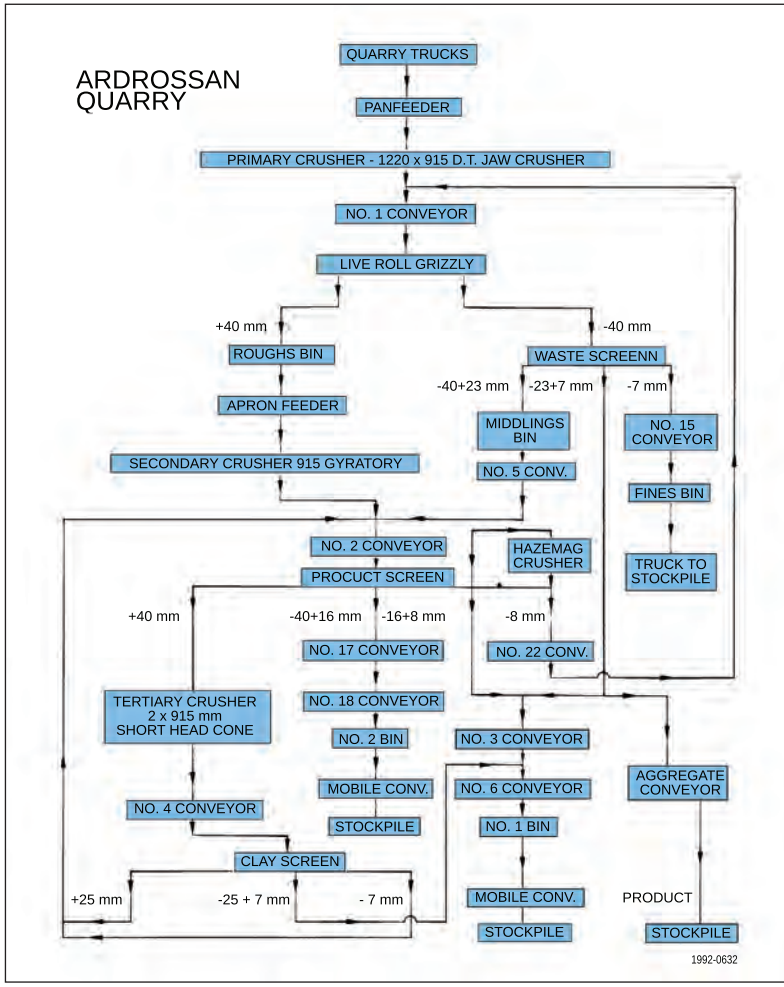


Fig. 15.7 *Plant flow sheet (alternative method of representation).*

Table 15.5 *Some South Australian crushing plants.*

	Ardrossan Quarry	**Iron Duke Quarry**	**Iron Knob Quarry**	**Klein Point Quarry**	**Leigh Creek coal open pit**
Size of shovel or FEL	2 x 1.9 m^3 shovels 1 x 2.5 m^3 shovel 2 x 4.5 m^3 loaders 1 x 3 m^3 loader	Cat 992C FEL	2 x Cat 992C FEL 1 x 5.4 m^3 Cat 988B	1 x 10m^3 Cat 992C 1 x 5.4 m^3 Cat 988B	Letourneau L800 9 x 9 m^3 FELs 1 x 3 m^3 loader Demag H24116m^3 hydraulic excavator
Primary crusher	1 220 x 915 DT jaw	2 133 x 1 524 mm Morgardshammer jaw	Jaques 2040 x 1 630 jaw	Hammermill	2 x rotary Bradford breakers (each 750 t/h)
Setting (mm)	100	250	127-165	30	250 mm
Secondary crusher	915 mm gyratory	2 x 1 676 mm Jaques gyratory	Jaques 1 350 gyratory	n.a.	2 x double ring rotary hammers
Setting (mm)	54	90	32	n.a.	40
Tertiary crusher	2 x 915 mm short head cone crusher	n.a.	n.a.	n.a.	n.a.
Setting (mm)	28	n.a.	n.a.	n.a.	—
Stone	Dolomite	Ironstone	Ironstone	Limestone	Coal
Products	22-38 mm 7-22 mm <7 mm	Crushed iron ore for steelworks	Crushed iron ore for steelworks	Shippable (-40 mm)	Coal for power station
Tonnes/8 hour day	1 200	Maximum capacity 12 000 t	5 285	7 000	8 000

Table 15.5 *Some South Australian crushing plants (Cont.).*

	Linwood Quarry	Mount Schank Quarry	Para Hills Quarry	Penrice Quarry	Rapid Bay Quarry
Size of shovel or FEL	Cat988b FEL 40 tonne Liebher excavator	5.4 m^3 988b FEL	Komatsu WA 600 loader	O&K RH 75 5.3 m^3 shovel WA 600FEL 5.3 m^3	2.7 m^3 RB100 shovel 4.6 m^3 FEL
Primary crusher	Jaques 1 371 x 1 143 mm ST jaw	1 066 x 762 mm ST jaw	Jaques 1 219 x 1 067 mm ST jaw	Gyrocrusher	1 524-1 219 mm Vickers DT jaw
Setting (mm)	200	150	150	250	150
Secondary crusher	APK 1622 Hazemag impactor	711 mm gyratory	330 x 914mm Allis Chalmers superior gyratory	Gyrocrusher	SBM impact
Setting (mm)	105 and 70	65	50	100	20
Tertiary crusher	2 x EL-JAY 1070 1-coarse and 1-fine	No. 35 gyracone	2 x 1 295 mm Symons short head cone crushers	n.a.	n.a.
Setting (mm)	16 and 10	21	12	n.a.	n.a.
Stone	Dolomite	Basalt	Quartzite	Limestone (Marble)	Limestone
Products	Aggregate: -65 +40 mm 20 mm l0 mm 7 mm Sand Crushed rock Rubble	Aggregate: 50 mm 20 mm 14 mm 7 mm 5 mm Sand Crushed rock	All road material Crushed rock and rubble Railway ballast Concrete Sand Quarry waste	150 mm + 50 mm -50 mm to fines: Screenings Rubble Scalps Sand	-20 mm rubble -40 mm rubble -20 + 10 mm gravel -10 mm quarry sand -150 + 40 mm Drainage rock
Tonnes/8 hour day	2 800	1 000	2 400	4 600	3 500

Table 15.5 *Some South Australian crushing plants (Cont.).*

	Riverview Quarry	Sellicks Hill Quarry	Stonyfell Quarry	White Rock Quarry
Size of shovel or FEL	2 x Cat 980	5.5 m^3 998B FEL 4.0 m^3 980C FEL	Cat 980C Cat 988B	5.0 m^3 FEL
Primary crusher	Jaques 1 219 x 915 mm DT jaw	1 067 x 762 mm ST jaw	Hadfield 219 x 1 067 mm DT jaw	Hadfield 915 x 610 mm DT jaw
Setting (mm)	150	150	150	150
Secondary crusher	EL-JAY	Kawasaki KC2513 gyratory	EL-JAY 1 372 mm cone	915 mm TX Traylor gyratory
Setting (mm)	38-51	30	35	30
Tertiary crusher	2 x EL-JAY 1 x BALMAC impactor	Jaques 536 impactor Kawasaki KFS 1000L gyratory	EL-JA Y 1 372 mm fine head cone	No. 50 Jaques cone
Setting (mm)	15-17	25 14	16	19
Stone	Dolomite	Dolomite	Quartzite	Quartzite
Products	-20 mm -14 mm -l0 mm -7 mm Sand	Concrete aggregate: 20 mm 14 mm 10 mm 7 mm Crushed rock: 40 mm 20 mm Screenings: 50 mm Rubble: 40 mm 20 mm	Crushed rock aggregate: 20 mm 10 mm 7 mm Sand Rubble Rock fill	+20 mm -20+ l0 mm -10+ 7 mm -7 + 3 mm Sand
Tonnes/8 hour day	1 440	2 500	1 800	2 778

16. Workshops and Stores

General Considerations

Most quarry operations require provision to be made for workshops and stores as well as a weighbridge, offices and staff rooms. The number, size and nature of these workshops and stores will depend on the scale of the operation and its particular needs. For example, the need will be greater if a long term operation is foreseen or the site is at a distance from a source of spare parts, engineering workshops, or specialist tradespersons who could deal with service and maintenance.

For a minor country operation a small concrete-floored, steelframed, color-bonded shed may suffice. In almost every quarry there must necessarily be a store to house the spare parts of machinery and general goods ordinarily required daily, plus a separate storage at an appropriate distance for lubricating and fuel oils as detailed later in 'Storage of Dangerous Substances'.

Some quarries would require more than one workshop as well as bays where routine repair jobs and maintenance work on plant and machinery could be carried out. It is only the larger open cut mines and quarries that would need to have fabricating workshops — particularly those operations with older type equipment for which proprietary spare parts are no longer obtainable.

In all enterprises of this nature production must be sustained. Assuming correct choice of equipment and a well built plant, preventive maintenance is the first consideration in regard to maintaining operations, followed by provisions to cover major breakdowns. Sufficient facilities should be available in any quarry for day-to-day maintenance. The workshops should be of adequate height and contain suitable lifting apparatus, both fixed and mobile, to handle heavy items.

The location, availability and cost of suitable outside engineering services must be balanced against losses that may arise from major breakdowns and the cost of providing major workshops on the job.

Preventive Maintenance

To keep plant and quarry equipment up to maximum capacity, planned preventive maintenance is the best policy to adopt on a mine or quarry. Preventive maintenance is the inspection, service, repair or replacement of parts or an item of equipment or plant, carried out to prevent a breakdown occurring at an unexpected or inconvenient time. This calls for regular, scheduled inspections of all the working machinery. These inspections may necessitate some overtime work but they can enable costly breakdowns or replacements to be anticipated.

Charts and cards or computer records of the inspections must be kept within the workshop environment and updated daily with the maintenance (labour and parts) undertaken.

Depending on the output of the plant, inspection intervals will vary:

- daily (power-shovel ropes, conveyors, idlers, chute wearing plates, pressure hoses)
- weekly (electrical wiring and the possibility of blowing out electric motors with compressed air, jaw plates, toggles and toggle seats, cones and mantles, cleaning of air and oil filters of diesel primary movers)
- quarterly (cone crusher sealing rings, crusher bearings etc.).

Oil analyses can give warning of wear and possible breakdown of gears and other moving parts.

The makers of such items as drills, shovels, trucks and crushers supply detailed instructions on their lubrication. These should be followed and the work done faithfully (more frequently if required) and the results recorded.

As well as the maintenance details, operating hours or days, distances, tonnages, wages, stores and replacement parts relevant to each unit should also be recorded. These figures combined with weekly or monthly totals for production items, distances drilled or travelled, tonnages, explosives used and so on, together with pertinent overheads, will form the basis for costing and indicate the efficiency of each section of the operation.

Facilities

Most quarries would have stores for:

- general items (tools, accessories etc.)
- spare parts and replacements
- fuel and lubricating oil (away from the other two)
- explosives (see Chapter 7).

Facilities may be required for the following work to be undertaken:

- Daily servicing, usually outside or in a drive-through service bay and carried out by drivers or semi-skilled service personnel. Service trucks and fuel trucks are often used.
- Routine (e.g. every 250 hours) service in workshop bays for mobile gear, service trucks etc.
- Repairs associated with major and minor breakdowns.
- Major overhauls.
- Changeout of modules (engines, transmissions etc.) at predetermined intervals before failure, based on historical records.

Possible requirements in workshops are for:

- servicing and repair of plant
- drill sharpening and repair of tools
- regular maintenance of mobile equipment
- construction of ancillary equipment or replacement parts.

The boiler shop should be separate or securely screened because of the danger of 'flash' to eyes.

Storage of Dangerous Substances

The storage of dangerous substances such as petrol, LPG, poisons, acids etc. must comply with the Dangerous Substances Act 1979, and may require a licence from the Department of Labour. In particular cases it will be necessary to comply with one or more of the following Acts:

Mining Act
Explosives Act
Occupational Health, Safety and Welfare Act
Water Resources Act.

Distillate, diesel and oils are not dangerous substances under the Dangerous Substances Act. It is recommended that any storage is kept in accordance with AS 1940, The Storage and Handling of Flammable and Combustible Liquids. Any tank used should comply with AS 1692, Tanks for Flammable and Combustible Liquids.

17. Occupational Health and Safety

Safety

An accident may be defined as an unforeseen interruption to the orderly progress of a process or action. All accidents should be investigated whether or not they result in injury. An accident can be very costly to the quarry owner due to damage or loss of operating time. However, if an injury is caused, the result is not only trauma suffered by injured persons and their associates but the cost to the owner will be many times the direct cost of providing maintenance and/or insurance cover for the people incapacitated.

Safety in any operation can best be achieved by a co-operative effort between management, employees and their associations to develop an atmosphere of safety consciousness throughout the workplace. 'Production with safety' should be the slogan by which all quarries operate.

To ensure its success the safety program must be initiated and supported by the chief executive. The best policy is to develop safety awareness within the workforce rather than to impose it. It then should be maintained by close supervision down the line. Adequate feedback of results is also essential to ensure the effectiveness of the program. Such results need to be made available both to the chief executive and to everyone within the organisation.

Prevention of accidents requires that all employees note any hazards or agencies that may possibly lead to accidents, reporting the observation and making sure that the necessary precautions or modifications are carried out.

Accident Recording

Whenever an employee suffers a work-related injury a record in accordance with AS 1885.1-1990, Workplace Injury and Disease Recording Standards, must be kept.

Not only can one monitor one's own safety performance but also make comparison with others. Caution is needed at times in making too close a comparison with others without taking into consideration all of

the circumstances. However, in regard to a company's own records, every accident (in some cases, incident or near accident) involving potential or actual injury to persons or damage to plant should be investigated, reported and corrective measures taken.

Legislative Requirements

South Australian legislation requires a written report to be sent within 24 hours to the Department of Labour where an accident occurs as the result of which any person is killed or suffers injury likely to cause their absence from work on three or more working days. The police must also be notified immediately upon the death of any person in a quarry.

The Occupational Health, Safety and Welfare Act provides for the joint participation of employers and employees in the development of safe working practices by means of:

- health and safety representatives, each representing workers in a designated workplace
- a health and safety committee, covering all operations in the quarry.

The employer must consult with the health and safety representatives of the employees on any proposed changes in the quarry. Changes to the policies, procedures, plant and substances used which might affect the health, safety and welfare of employees should always be subjected to close scrutiny.

The safety representative may inspect any part of the workplace, accompany an inspector during an inspection, issue default notices and direct work to cease if the risk is such that work should be discontinued immediately. The employer must also notify the safety representatives of an accident and provide means and assistance to representatives necessary to enable them to carry out their duties.

The safety committee should facilitate co-operation between an employer and the employees and assist in the resolution of issues relating to occupational health, safety and welfare that arise in the quarry.

Induction of New Employees

The best time to bring basic safety practices of the operation to the attention of all new employees is immediately on commencement of employment for work in a quarry. Employees should be given full instruction in the various types of work that they will be required to do.

Maintaining Safety Awareness

Safety policies, required by law in South Australia when more than five persons are employed, should be clearly stated, and be constantly displayed and reviewed as necessary.

Only by daily safety contact and constant repetition will people become and remain safety conscious. Parallel with this, on an individual or group basis, special recognition of good performance by the issuing of awards and prizes can be a positive factor in stimulating and maintaining interest in safety achievements elsewhere in the works. Achieving a good safety record depends largely on the development in all employees of continuing awareness of possible dangers.

Machinery

Guards must be provided to cover external moving parts of machinery in accordance with the Occupational Health, Safety and Welfare Act.

Any area where nip points occur, particularly on conveyor belts, pulleys and gears, must be effectively guarded.

Machinery must always be stopped, danger-tagged and switches 'locked-out' when guards are removed for oiling and greasing operations or any other purpose. Where a guard has been removed, the machine must not be restarted until the guard has been replaced.

Attendants in plants should keep the plant safe by good housekeeping, not only in the vicinity of moving machinery but also in walkways and aisles. All hoses, tools, bars, scraps and rubbish of any kind should be removed so as to minimise the danger of people tripping, stumbling or slipping on uneven surfaces.

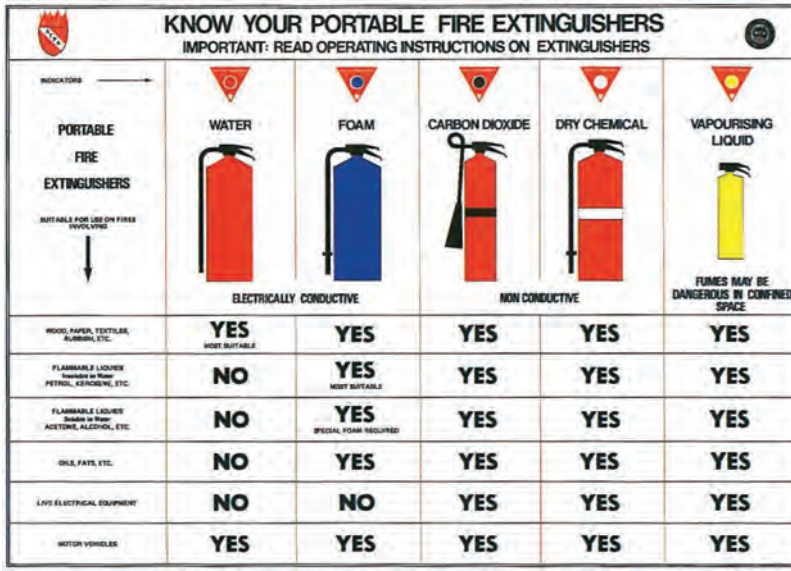
Close fitting clothes should be worn at all times to avoid being caught in machinery. Safety boots or shoes should be worn. Safety helmets are to be worn where danger exists from falling objects or head injuries could ocurr from bumping into objects. Safety glasses are essential for welding or working at grinding wheels . Gloves are necessary where hand injuries could occur. Respirators must be used in areas dangerous from gas or dust, and ear muffs or plugs where noise is a problem.

Fire Extinguishers

The positioning of fire extinguishers at strategic points through the works is essential whether or not there is a general overall fire service in a plant. For small buildings they should preferably be outside and not too close to equipment which may become a fire hazard.

Extinguishers are extremely useful in putting out small fires or in preventing them from spreading quickly. All personnel should know where they are located and how to use them. It is also important not to overlook knapsack sprays on bushfire hazard days.

There are many types of extinguishers including dry powder, foam, carbon dioxide and water. Each type has a specific application for which it should be used, for example: a wood, paper or fibre fire; an oil or liquid-fuel fire; or an electrical fire. While some extinguishers may be used in two or more of these instances, there are some which can be dangerous if used on fires for which they were not designed.



Fire extinguisher identification chart (National Safety Council of Australia).

Although the colour of the cylinder should relate to the type of fire for which it is used (see AS 1849-1976, Identification Colours for Portable Fire Extinguishers), this may not be the case with some of the older cylinders still in use. Consequently, it is safer to place the cylinders on walls or on boards with notices which clearly indicate how the equipment is to be used and on which fires it can be safely used. These cylinders must be regularly serviced by the makers or a local fire brigade. Practical training in the use of fire extinguishers should be given to all personnel. A used extinguisher should be placed on the floor and returned for refilling, never placed back on the board.

Employees should be aware that poisonous gases are often liberated as a result of burning a wide range of materials and also some fire extinguishers may produce toxic gases; safety breathing equipment should be available for use if required.

Electrical

Accidents arising out of the use of electricity continue to occur notwithstanding that the standards laid down in the SAA Wiring Rules for all installations are mandatory for mines and quarries. Electric power is such a common source of energy that it becomes at times too familiar both to tradespeople and users. Only qualified electricians are permitted by law to do installation or repair work.

Attention is drawn here particularly to the earthing of equipment and systems, the correct wiring of plugs and sockets, and to the proper use of danger tags and 'lock-outs' placed on machinery when any work is being done to the electrical part. Checking systems with a multimeter or megger is recommended — but only by qualified persons.

Care must be taken to prevent contact with live wires, particularly overhead cables, by mobile equipment such as cranes, drills, FELs and ladders or poles carried by persons.

General

The use of explosives and the working of quarry faces are covered by legislation (see Chapter 7). However there are a number of precautions, some of which are covered and some which are not, that are very often overlooked because people are in a hurry or because of negligence. Amongst the most important of these are:

- Checking and, if necessary, scaling of a quarry face.
- The necessity, when working close to and beneath a quarry face, of facing towards it.

- The need to watch the movement of loading equipment or trucks close by.
- Taking care when tipping trucks or working near the edge of a dump, a bench or roads.
- Repairing of any unsafe tools or equipment; the sharpening of blunt tools.
- Use of the right tool for the job.
- Use of the correct method when lifting any heavy object.

First Aid

In this area the Occupational Health, Safety and Welfare (Health and First Aid) Regulations must be complied with. This involves performance at least to the standard outlined in the Code of Practice for Occupational Health and First Aid in the Workplace.

The owner must ensure that adequate health and first aid facilities are available for any person who is injured or becomes ill while at work. This should be determined by consultation between employers and employees. Some points to be considered during consultation include specific hazards of quarrying or special conditions encountered (e.g. a remote worksite and/or the scattered workforce) and hence whether the following are required:

- A person trained in first aid present at all times; minimum standards are detailed in the code of practice.
- A first aid room.
- A motor vehicle to transport sick or injured persons to medical care.

For operations with 25 or less employees a basic first aid kit must be kept. If there are more than 25 employees an occupational first aid kit must be kept. The contents of these kits are given in Table 17.1.

Other modules should be added to the first aid kits if specific hazards exist. Examples are: eye, burns and remote area modules.

Injuries not Related to Accidents

Injuries/illnesses may be suffered by employees largely as a result of their employment but not directly as a result of identifiable accidents.

Examples are head or neck problems due to vibration, and hearing loss together with stress which may be related to work situations.

Table 17.1 *Contents of first aid kits.*

	Basic first aid kit	Occupational first aid kit
Gauze pieces, 75 x 75 mm, sterile packets containing 5 pieces	5 pkts	20 pkts
BPC wound dressings No. 15	1	2
Wound dressings sterile, non-adherent, small	3	12
Wound dressings sterile, non-adherent, large	1	3
Eye pads, sterile, individually wrapped	-	4
Conforming cotton bandages, 50 mm	3	6
Conforming cotton bandages, 100 mm	1	6
Triangular bandages (minimum width 90 mm)	2	6
Non-stretch adhesive tape, 25 mm x 2 m rolls (hypo-allergenic)	1	1
Adhesive dressing strips, independently wrapped, minimum quantity	50	100
Paracetamol tablets	—	240
Disposable wound cleaning swabs (1% Cetrimide BP)	10	25
Povidone-iodine 10% solution	15 ml	2 x 15 ml
Cotton-tipped applicators	-	50
Disposable latex gloves	5 pairs	10 pairs
Disposable eye wash (holding at least 30 ml)	—	5
Approved resuscitation face mask *	1	1
Scissors (sharp/blunt points)	1	1
Splinter forceps .	-	1
Splinter probe/remover	1	1
70% alcohol swabs (ethanol or methanol) for instrument disinfection	10	25
Safety pins	5	10
Recording book and pencil for recording injury and illness first aid	1	1
Leaflet *First aid instructions* (issued by the South Australian Occupational Health and Safety Commission)	1	1

* Persons using an approved resuscitation face mask should have received recent training in its use.

Vibration in quarries ranges from high speed oscillations (e.g. hand-held rock drills or tampers) through more or less continuous shaking (e.g. vehicles on rough roads) to single jolts (e.g. stepping or jumping down from high vehicles or plant). Alleviation of the problems in most cases is fairly obvious although sometimes difficult to put into practice. Some examples: using a firm rig for mounting vibrating equipment; better road construction; fitting vehicle cabin springing and installing subsidiary removable steps.

Noise

Noise is frequently defined as any unwanted sound. A vibrating source produces vibrations in the medium (air, water or elastic solid) in which it occurs. These vibrations are propagated as waves in the form of pressure variations. If these waves fall within the range of amplitude and frequency which excites the sense of hearing they are called 'sound waves'. Sound pressure level (SPL) is a measure of the pressure of a sound wave. The 'decibel' (dBA — measured on the A scale of the sound level meter) is the unit commonly used to rate sound levels. Because the human ear does not respond with equal sensitivity to all frequencies within the audible range, this A frequency weighting scale, which approximates the response of the human ear, is used on sound measuring equipment. The dBA symbol is commonly abbreviated to dB where the context is clear (that is, use in measuring noise levels).

The sound pressure level is defined as:

$$SPL = 20 \log_{10} (P/0.00002) \text{ (dB)}$$

where: P = pressure in Pascals (Pa)

0.00002 = reference level of 2×10^{-5} Pa

The decibel scale is logarithmic and so a difference of 20 dB from, say, 80 to 100 dB represents a ten-times increase in pressure. A consequence of this is that sound (or noise) levels are not added arithmetically; for example, if a 90 dB and a 100 dB noise occur together the resulting noise level is 102.4 dB and not 190 dB. The 102.4 dB is obtained by adding the relevant pressures: 0.63 + 2.00 = 2.63 Pa (which is the pressure for 102.4 dB).

Noise levels in decibels for some commonly heard sounds are given in Table 17.2.

Effects of Noise

Sound levels above 85 dB lead to fatigue or temporary loss of hearing, and if these are repeated frequently enough and for a long enough time, they will lead to some permanent hearing loss.

As an approximate guide one may say that if a noise prevents communication in a normal conversational voice, then that noise is approaching a level at which protection may be necessary.

Table 17.2 *Typical noise levels.*

Sound level (dB)	**Sound pressure** (Pa)	**Noise**
160	2 000	May break a plate glass window
120	20	Threshold of pain
100-120	2-20	Pneumatic drill
107	4.5	Shovel (diesel)
100	2.0	Timber saw
95-115	1.12-11.25	Screens
90-100	0.63-2	Compressor
90-110	0.63-6.3	Crusher
85-106	0.36-3.99	Bulldozer
80-95	0.20-1.12	PEL/Dump truck
65-75	0.04-0.11	City traffic
50	0.0062	Quiet office
0	0.00002	Threshold of hearing

The annoyance caused by noise is not solely related to the intensity of the sound; it may be influenced by familiarity and personal attitudes and by the state of the weather. It is largely an individual response which varies with persons and situations although, in general, loudness, pitch and intermittency or irregular occurrence are the main factors which cause annoyance.

Excessive noise can influence occupational accident rates by preventing the hearing of signals or by affecting an employee's movements, especially when accuracy is important. Noise demands attention and causes fatigue.

How to Deal with Noise

The problem of noise has been with the industry for a long time but until recently not much has been done about it. A standard for 'nuisance' noise is being drafted for assessing potential danger to hearing. In South Australia, hearing loss is a compensatable injury.

Hearing loss caused by noise cannot be recovered. Some hearing loss is due to ageing, but excessive noise exposure in early years can accelerate this process.

Other factors are the time interval between exposures to noise and the recovery time during or after the working day to enable the ear to revert to normal. A time limit should be set for exposure to different noise levels, together with an adequate recovery time.

AS 1269-1983 defines a level of 90 dB for 8 hours as a Daily Noise Dose of 1.0. Also giving the same dose of 1.0 are:

- 4 hours exposure at 93 dB
- 2 hours exposure at 96 dB
- 4 hours at 90 dB followed by one hour at 96 dB.

Instruments which can be worn as personal dosimeters are available for determining the Daily Noise Dose during a working shift.

Types of Noise

Most industrial noise is a combination of two or more of the following types though it is usually classified by its dominant characteristics:

- steady state noise (idling plant or equipment or constant speed plant, e.g. diesel engines)
- intermittent noise (noise of passing vehicles)
- impulse noises (gunfire, explosions)
- repeated impulse noise (riveting guns, pneumatic hammers, typewriters)
- broad band noise (jet engine exhausts, furnaces)
- narrow band noise (circular saws, fans).

Reducing Noise Exposure

Engineering noise control. The best form of protection is the reduction of noise to an acceptable level at the working place.

Noise control is an engineering problem and it requires the services of experts to reduce the noise output from a machine by redesigning it or installing acoustic covers and barriers.

Use of ear protection. Personal hearing protection should be used only where technical or economic considerations prevent the reduction of noise to below the levels prescribed. The function of the hearing protection device is to reduce the amount of noise reaching the inner ear of the wearer by covering the ear, covering the entrance to the ear, or blocking the ear canal with an ear plug.

The devices used by individuals must suit them personally and be appropriate to the working environment. There are many forms of ear protection available, but most fall into the following categories.

- disposable hearing protector, e.g. cotton wool pad (least effective)
- ear plug
- ear canal cap
- ear muff (most effective).

Reduction of exposure time. In some cases it is possible to have employees in sound-proof enclosures to perform certain tasks in order to shield them from a particular noise by means of barriers or baffles.

However, in many situations this cannot be done; where the noise approaches the set limit, an arrangement of tasks should be made for individuals so that they are not exposed to the noise for too long a period at any one time.

Part of a well designed hearing conservation program should involve education about the adverse effects of noise exposure away from the working place. Exposure to noises such as using a lawnmower, power tools, or listening to loud music can contribute significantly to a person's overall Daily Noise Dose.

It is becoming common practice to measure hearing loss of every employee on engagement and each year or so thereafter.

Nuisance Aspect of Noise

Noise as a nuisance can affect people at work as well as the general public. It can be a source of danger if it has the effect of distracting a worker at an unsafe moment. It can seriously affect work output as a result of influencing the mental state of the person exposed. However, it is from the point of view of nuisance to the general public that it is discussed in the following paragraphs.

AS 1055-1989 Parts 1 and 2 has been formulated to fix noise rating with respect to annoyance. This standard recommends methods for measuring and rating noises in urban residential, commercial and industrial areas with regard to their annoying qualities. It lays down how the measurements of the noise should be made and what corrections should be applied to the readings obtained. Measurements are also made at the same place at about the same time (but while the offending noise is absent) in order to obtain a 'background' reading. The Department of Environment and Planning has responsibility for recommending the allowable noise level for specified locations.

Remedies

In remedying a situation it is important to attack the noise at its source. In the case of engines, motors and other machines there is often a need to redesign them. Many manufacturers are now dealing successfully with the problem, as for example with compressors and some rock drills. For existing plant and machinery, baffles, vibration-dampening mountings, silenced exhausts, re-siting and enclosing of machines afford some dampening effect on the noise level. These should be used in conjunction with other noise dampening devices applicable to open workings such as stands of trees or earth banks.

Two common faults with truck operation within quarries are bumping at speed over bad roads with empty trucks, and revving engines (especially diesels) uphill or in order to accelerate quickly. Roads should be kept in good order and speed must be regulated to suit the environment. Engines should be properly muffled to deaden noise.

Noise from blasting often becomes excessive because of timing and bad design. By timing is meant the firing of a blast at a time of the day or under weather conditions that are likely to make the noise of the blast more apparent. Some blasts are excessively noisy because of defects or unseen fractures or cavities in the rock being blasted. However most noisy blasts are the result of bad blast design, faulty technique in initiation, wrong choice of explosive, or firing at the wrong time of day. One common fault leading to complaints is the use of detonating cord with uncovered lines on the surface.

Table 17.3 shows the source of noise from various pieces of equipment used in quarries together with some possible remedies to ameliorate the problem.

Dust

Health Hazards

All respirated dusts must be considered harmful in some degree. Even where there may be only slight danger to the lungs, there is very likely some adverse effect on the respiratory system, particularly to asthmatics or sufferers from hayfever.

There is no doubt that the hazards associated with dust are among the most serious of all those with which the quarry worker must contend. In South Australia, diseases resulting from excessive exposure to dust are compensatable.

Dust particles of size ranging from 0.001 to 0.1 mm (1 to 100 microns) pose a threat to health when they become airborne, reducing visibility, creating an uncomfortable environment (irritation of eyes, ears, nose, throat and skin) and possibly resulting in damage to the tissues of the lungs. Included amongst potentially harmful dusts are silica, asbestos, sugar cane fibre, carborundum, diatomite, talc and cotton dust — each of which can produce its own form of lung damage when dust control is inadequate.

Table 17.3 *Some noise sources and possible remedies.*

Machine	Source of noise	Possible remedies	Alternatives
Excavator, scraper	Engine	Fit more efficient silencer on exhaust Fit sound-proof cabin	Use electric motor
Rock drill		Fit more efficient silencer on exhaust Use the drill inside a portable acoustic enclosure or fit sound-proof cabin	Use more units thereby achieving desired result in a shorter time Use down hole hammer Use hydraulically operated machines
Compressor		'Dampen' metal casing Screen the compressor	
Pump	Engine pulsing	Enclose in acoustic screen (allow for engine cooling and exhaust)	
Trucks	Engine	Fit more efficient silencer Line bonnet of truck and driver's cabin with sound absorbent material	Build sound barriers Reduce speed limits Limit working hours
Crusher		Enclose or surround with barriers	Use remote control rooms built of sound absorbent material and, if possible, with double walls with an air space between
Chutes		Line with rubber or other sound absorbent material	
Screen	Deck screen Mountings	Enclose screen deck Install rubber vibration dampeners	



Driller wearing ear muffs and dust monitoring equipment. (Photo T024208)



Dust extraction unit on drilling rig. (Photo 040469)

The most common harmful dust in quarries is that which contains silica, and it would appear that the harmfulness increases with the increase in the percentage of silica in the dust. Also, the dust that is considered most harmful is that which is less than 5 microns in size, that is, particles smaller than 0.005 mm. As this dust is not visible to the naked eye, it does not follow that because dust cannot be seen there is no danger.

When dust-laden air is inhaled, most of the dust particles greater than 5 microns are caught in the mucus which coats the nose, sinuses, trachea and bronchi. They are moved up with the mucus towards the throat and either expectorated or swallowed.

Dust particles less than 5 microns can pass into the lungs. Some pass out again with the exhaled air but other particles are caught up in the lung tissue and may eventually result in fibrosis of the lungs. Individual susceptibility seems to play an important part in determining the degree of fibrosis produced. Other factors of importance are the age of the person when first exposed to the dust, the concentration of dust and the years of exposure.

Serious features of silicosis are that it predisposes a person to tuberculous infection, and that the silicosis may advance even after exposure to harmful dust concentrations has ceased.

There has been a remarkable drop in the incidence of silicosis in recent years in industries which employ stringent measures to suppress dust inhalation.

Dust Measurement

Concentrations of respirable dust can be expressed as follows:

- on a weight basis as milligrams per cubic metre (mg/m^3)
- on a numerical basis as particles per cubic centimetre or fibres per cubic centimetre when asbestos or similar fibrous dust are involved.

In the latter case, a microscope is used to assess the samples. In South Australia standards for respirable dust (less than 5 microns) are 5.0 mg/m^3 for non-toxic (nuisance) dust and 0.2 mg/m^3 for free silica (quartz) dust.

The main types of instruments used for measuring dust in the atmosphere are:

- impinger, including the konimeter and midget impinger
- thermal precipitator
- gravimetric instruments.

Currently in South Australia gravimetric instruments are used to monitor and measure the dustiness of working places. It should be the aim in all dust producing processes to reduce dust output to an acceptable limit and to use protective devices such as respirators only for emergencies.

Remedial Measures

Dust is created at every phase of a quarry operation. Sources of airborne dust include wind action, earth moving, vehicles and moving equipment, drilling, blasting, loading, dumping, feeding, crushing, screening, chutes, conveyors, transfer and discharge points, bins, stockpiles, dumps, loading points and road transport.

The methods by which dust in working places is suppressed or controlled are universal. They include the following:

- Wet drilling, water mist drilling and drilling with an exhaust system to remove and collect dust.
- Proper ventilation of working places, particularly dead ends (e.g. by supplying dust-free air to the face).
- Providing wheels or cutting compound of silicon carbide or aluminium oxide instead of sandstone (to reduce the amount of silica in the dust) and water sprays when using grinding wheels or sawing dimension stone.
- Use of wetting agents with water in selected processes.
- Use of clear water for sprays and mists.
- Use of total wet processes in crushing and screening plants.
- Good plant layout practices:
  - separating dusty from non-dusty operations
  - enclosing dusty machines and transfer points and extracting dust through an exhaust system
  - reducing the amount of fall of materials at transfer and discharge points with the discharge chute for fine material being sloped rather than perpendicular
  - using collapsible wind socks (elephant trunks) at the discharge point of stockpile conveyors
  - preventing chutes becoming empty
  - providing a dust-free operator's room under positive air pressure
  - employing good housekeeping methods by cleaning up spillage, paving the environs of the plant or keeping them oiled or damp
  - reducing the speed of all vehicles near the plant.
- Covering of dumps with vegetation as soon as practicable, but in the early stages spraying with chemicals to provide a protective coating.

- Paying attention to roads throughout the quarry area by:
  - providing paved surfaces where practicable
  - watering roads and tracks whether sealed or not
  - restricting vehicles to defined roads or tracks
  - restricting speed of vehicles
  - watering down of loads before leaving the quarry.
- Ensuring that any dust produced by blasting is blown away from neighbouring houses. Remember that too strong an explosive for the rock to be blasted will create excessive dust.

While the above considerations are mainly directed towards the elimination of health hazards to employees, it will be apparent that a dust suppression program, based on such of the above good practices as are appropriate, will also have the effect of reducing complaints from the community.

18. Administration, Management and Finance

The aim of a quarry organisation is to provide a basic commodity and in so doing make a profit. The operation of quarrying is the means of achieving this and success depends on co-ordination of sound methods both on the business and on the technical sides.

A quarry business may be owned by a private individual or one of several types of companies.

Company Structure and Procedures

Public Companies are those in which the public is able to participate in ownership.

Private Companies are usually smaller and must not have more than 50 members.

Partnerships are not regarded as companies under the Companies Act; however they must not have more than 20 members.

A Public Limited Liability Company consists of at least seven individuals who together subscribe the necessary capital to engage in a joint undertaking. Shares are allotted to each in the proportion in which they subscribe. The liability of each shareholder is normally limited to the face value of his or her shares.

Dividends are distributions of money from earned surpluses and are paid to each member in amount according to the number of shares held.

Control of large companies is usually exercised by a board of directors, which is a committee of shareholders. The policy and decisions of the board are executed by either a managing director, a chief executive officer or a general manager who controls operations, both business and technical.

Under the manager, various executive officers have control of different sections of the whole project. Each has a delegated authority to do one or more of the following:

- spend funds (capital or otherwise)
- control employees
- control engineering services
- carry out market research
- control quality or quantity of output
- determine operational procedures
- commit the company to contract obligations (usually sales).

Business management is usually given to an officer immediately subordinate to the general manager, e.g. secretary, commercial manager, accountant.

Technical management may be allocated to one officer but more often several are responsible, each with a particular sphere of responsibility, e.g. quarry manager, mill manager, chief engineer (mechanical and electrical), chief geologist, transport officer etc. (Fig. 18.1).

The annual report is based on the year's operation and trading results. It is prepared by the company secretary and presented to the annual general meeting of shareholders by the chairman of directors. It normally includes a balance sheet, a profit and loss account and the director's report.

The balance sheet summarises liabilities against assets and shows the company's financial position as at the end of the year.

The profit and loss account is a summary of the income and expenditure during the year. It reveals, as a balance, what surplus or deficit has resulted from the activities over the period. Capital expenditure does not appear in this account but enters it indirectly in the form of depreciation; this is the annual allowance on each capital item by which it is written off over an estimated life.

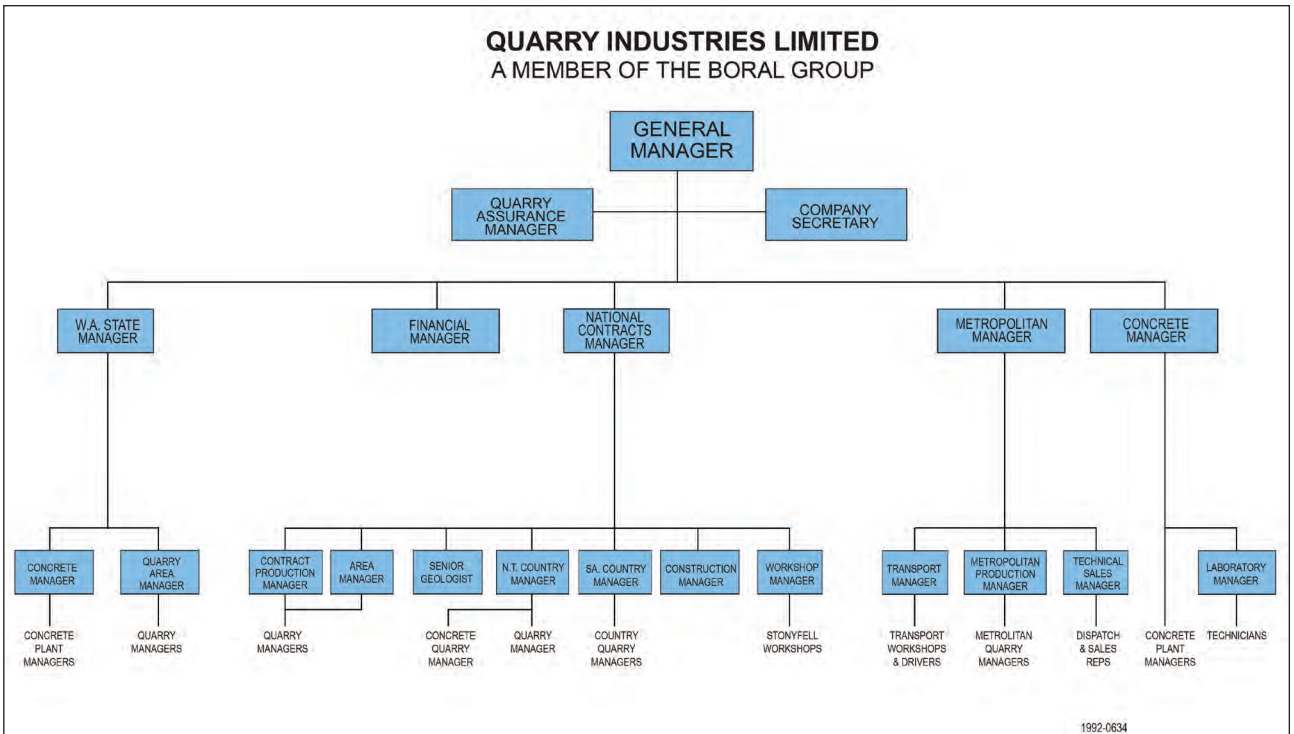


Fig. 18.1 *Organisation chart for a large company.*

Quarry Accountancy

Quarry accountancy is the continuous recording of transactions on a money basis. Some of its aims are to:

- Record all expenditure, e.g. wages, stores, power, rents, rates, taxes and receipts.
- Show clearly at any time the financial position of the business.
- Indicate what profit or loss has been made over any period and where these have been made.
- Record individual transactions with merchants or customers and so ensure prompt payment without duplication or other errors.
- Provide costing data both for the various phases and overall; this provides the basis for improving efficiency.

All transactions are recorded in individual accounts under suitable headings. There are two main classes of accounts:

- Capital accounts deal with assets and liabilities which are the main concern of the balance sheet.
- Revenue accounts deal with income and expenditure which ultimately form the material for the Profit and Loss Account.

Cost accountancy is a particular branch of accounting in which finance and technology are used to attain improved efficiency. Costing records are distinct from the financial records of the company, and are used by each department to regulate and control expenditure.

Whereas the normal accounting machinery will in time produce an accurate cost statement for, say, a monthly period, it is important to assess quickly — even if only approximately — how daily profit margins stand.

This can be done by calculating a 'unit cost' per employee. As a rule, the overall daily cost of running a quarry is approximately proportional to the number of employees. Knowing the unit cost per employee and the quarry production rate, a quarry manager can quickly estimate the cost per cubic metre of daily or weekly output.

Quarry costs

These are generally subdivided into overhead (or fixed) costs and operating (or variable) costs.

Overhead costs are those which do not vary immediately or directly with the output. For example:

- management and supervision
- office rent, office staff and supplies
- interest on loans or charges against capital investment
- insurances for fire, theft and liability
- ownership taxes on land and assets
- depreciation.

Operating costs are those directly related to output:

- wages of operators and maintenance personnel
- payroll taxes
- workers compensation insurance
- machinery hire
- fuel, oil, maintenance, repairs
- explosives
- other expendable stores
- royalties.

If these costs are subdivided into the various categories, for example drilling, blasting, loading, transport, crushing and so on, and combined with the relevant performance figures (distances, tonnages, explosives etc.) and the number of persons employed, the cost and efficiency of each section can be calculated. These figures (expressed as tonnes per shift, metres per machine shift etc.) are the basis for highlighting problem areas and making necessary changes to improve efficiency and reduce costs.

Administration

The three main concerns in administration are:

Labour

Payment of labour may be by:

- Day work, governed by industrial award rates.
- Contract work based on performance.
- Bonus work in which extra payment is received for performance beyond a set figure.

Labour turnover, if high, can cause low efficiency. It costs money to employ new staff and train them. Sometimes a bonus for length of service is paid to reduce losses of this kind.

Similarly, absenteeism can be reduced by a bonus for regular attendance.

Labour is usually the major item of expenditure in operating costs.

Stores

- These must be on hand to ensure continuous operations; however too great a stock adds to costs.
- The two main categories of stores are materials and machine parts.
- Stock on hand should be such that there is no chance of running short of a particular item.
- Often one month's consumption is taken as the minimum to be held for each line, but this can be greater in remote areas where replacement takes longer.
- Power, chiefly electric, is usually considered under 'stores', sometimes under a special heading 'services'.

Plant and Equipment

The main requirements for economy are:

- high utilisation (i.e. little idle time)
- sound lubrication and maintenance arrangements
- good quality of operators
- adequate workshop facilities.

The topics of workshops and maintenance are dealt with in Chapter 16.

Management

Management is the art of organising and directing human effort. Its prime objective is maximum efficiency with maximum safety and minimum interference with the environment. Management costs money but, if sound, will return the expense incurred many times over.

The first essential is to have a well-directed personnel organisation. This must be known to all and applied with a minimum of departure from the proper chain of command.

Attention to the comfort of employees, both physically and psychologically is important. Input from employees into the day to day operation of the quarry can be achieved through representation on consultative committees. Time studies of all operations provide data to reveal where idle or unproductive time occurs. Similarly, motion studies reveal where unproductive effort occurs. Such studies will also provide useful

data for estimating contract rates or output rates from various combinations of workers and machines. Consultation with employees is essential before beginning these studies.

Training

While it is true that not all people with technical qualifications are suitable as managers or supervisors, it is also true that those with managerial qualities need training and experience to bring them to their peak. The same applies to other categories of employees in the quarry, in the plant and in the office. All employees must be instructed in their duties and given appropriate experience to fit them for the work required.

When an engineer, a foreperson or a leading hand is assigned someone to train, there should be some executive in the organisation whose duty it is to keep watch on progress being made and to ensure that all along the line workers are receiving proper instruction and experience; in the case of key personnel the executive must ensure that reports are presented personally.

Whether general instruction classes for a particular job need to be held depends on the size of the operation and the organisation, but certainly where the operation is lacking in some aspect, classes must be considered. This requires not only lectures given in a classroom but also on-site training by skilled personnel.

Innovations

In any quarrying enterprise there is need for the development of new methods and techniques. While some innovations may be prompted by purely local conditions, including the environmental outlook, there are many ideas and changes continually being suggested, tried, modified, or adopted throughout the industry which must be considered for any operation in this highly competitive field.

Before a new technique can be introduced there must be experimentation to see whether the benefits or results obtained elsewhere are obtainable. Caution and proper planning of the test program are needed to ensure that normal operations are not interrupted and that sufficient work is done to bring the experiment to a satisfactory conclusion. When any major changes are made, they should be communicated to the statutory authorities concerned.

Project Evaluation

A manager is frequently faced with making a decision or a choice between two or more ways of achieving an objective or performing a particular task or project.

The most reliable technique used to assess the worthiness of investment projects is the Present Value (PV) method, also known as the Discounted Cash Flow (DCF) method.

This uses the idea that a sum of money placed in the bank at compound interest increases in amount year by year; and conversely that a certain sum of money in several years time has a current value which is less than the sum by the amount of interest added year by year.

The evaluation must take into account not only the estimates for revenue and direct costs but also factors such as inflation, discount and interest rates, depreciation, salvage values, injections of captial — all of which may vary from year to year.

Affecting the choice will also be the ease of operating the equipment concerned, noise created and other environmental and social considerations.

Computer software is available to assist in such analyses and makes it easier to compare options. Output of results can be presented in the form of charts, tables or statements with printouts being made to cover periods as found necessary. However, the basic information and estimates of future conditions must still be provided by the person undertaking the analysis.

19. Department of Mines and Energy

The role of the South Australian Department of Mines and Energy is to ensure that the mineral resources of South Australia are defined and managed in the best interests of the community. The Department provides advice on mineral deposits, their location, identification and development as well as on matters relating to environmental management.

Services Available To The Public

Mining Development

- Advice and guidance on mining and associated legislation and on consultation with Aboriginal communities.
- Advice on development of mineral deposits.
- Advice on environmental management of mining and quarrying operations.
- Advice on rehabilitation of mine and quarry sites.

Inspectors of Mines ensure that the provisions of the Mining Act and Regulations are carried out and assist the mining and quarrying industry with advice and discussions on problems encountered in working deposits of minerals or stone. It is always advisable to have discussions with the inspectors on such matters as pollution and the environment at a very early stage in the planning of a project.

Explosives, Occupational Health and Safety

Responsibility for these aspects of the mining and quarrying industry belongs to the Mining and Petroleum Branch and Dangerous Substances Branch of the Department of Labour.

Geological and Prospecting

- Advice on prospecting and geological matters.
- Identification of minerals and arranging for laboratory tests.
- Advice on the evaluation of commercial possibilities of mineral deposits.

- Specialised geophysical equipment, with operators, for hire.
- Storage and inspection of drill-core in the Department's core libraries.
- Crushing and assaying of ore parcels through operation of the State Government battery at Peterborough.

Groundwater

- Advice on well (bore) sites and the construction of wells.
- Testing of water samples for Total Dissolved Solids (salinity) and pH (acidity/alkalinity).
- Contract water-well drilling and well testing.

Publications

The Department publishes a wide range of material much of which is designed to assist the mine and quarry industry. Some examples:

- Mineral Industry Quarterly — a newsletter which provides current information and news items on all aspects of the South Australian mineral industry.
- The Annual Report — an illustrated summary of the Department's activities.
- Geological maps with Explanatory Notes.
- Digital geological information available from the SA_GEOLOGY database which can be purchased as a data tape or colour plot at various presentations and scales.
- The Mineral Exploration Index Series (MEIS) which provide locations of detailed geological, geophysical and geochemical surveys, drilling results and mineral occurrences, presented in map form.
- Mines and Energy Review — a collection of technical articles and reviews which contains South Australian mineral production statistics.
- Brochures and information sheets on a variety of topics related to mining, quarrying and the environment.
- Scientific bulletins and reports of investigations.
- Mineral exploration data packages which are compilations of technical information to assist the mineral industry.
- Abstracts of newly released open file unpublished reports from the Department's SAMREF database, published as the New Information Releases.
- List of South Australian mines and quarries.

A full list of publications is available from the Department.

Blasting Vibrations

- Departmental officers conduct tests in and near quarries, open cut mines and other excavations, measuring ground vibrations from blasting. Most of the tests are undertaken as a result of complaints from people residing near quarries.

Dust Sampling

- Environmental sampling of the dust content of air around quarry crushing plants in quarries and in metallurgical plants is carried out by officers of the Department.

Library

- A comprehensive service provides information on South Australia's geology, mineral commodities and related subjects. All material is accessible for public viewing and reading and copying facilities are available.

Research

The following services offered by various commercial organisations can be arranged through the Department:

- Mineralogical and petrological investigations.
- Inorganic and mineral assays and analyses.
- Metallurgical and ore-treatment investigations and plant design.
- Chemical research and development of chemical recovery processes.

Acknowledgements

This volume is an update of the 1983 edition and acknowledgement is given to its compiler Mr A.T. Armstrong and to those people who contributed to that work.

For the present edition the author wishes to record his appreciation of the assistance given to him in its compilation. It is not possible to list the names of all who supplied information, checked the draft and gave helpful advice and suggestions.

Mention must be made in particular of M.D. O'Brien for detailed examination and constructive comments on the whole text, to E.K. Hart and to officers of the Department of Mines and Energy and Department of Labour for critically reviewing the text and providing comments, most of which have been incorporated in the final production. Particular thanks also to Jon Selby for his painstaking editorial work.

I wish to acknowledge with thanks help and advice afforded by companies giving information, diagrams and photographs of equipment, explosives and other supplies.

Acknowledgement is made to the following explosives manufacturing and distributing companies for use of names of products with registered trademarks:

ICI Australia Pty Ltd and associated companies
Dyno Wesfarmers Ltd
ERT Explosives Australia Pty Ltd
Nitro Nobel AB of Sweden
Ensign-Bickford Co.

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The following journals are useful:

Mine and Quarry (UK), Quarry (UK), Quarry Management (UK), Resources (NSW, Australia).

Glossary

Adit	A passageway or opening driven, usually horizontally, into the side of a hill for the purpose of exploring or developing a mineral deposit.
Air blast	A sudden but transient increase in air pressure as the result of an overpressure compression wave travelling through the air and generated by a shock or explosion.
Alluvial deposit	Earth, sand, gravel or mineral transported by stream action and deposited where the velocity, and hence the carrying capacity, of the stream has decreased.
Angle of repose (Angle of rill)	The natural slope at which fragmented or loose material will settle when dumped in a pile.
Angle or inclined drilling	Refers to top-to-toe drilling in quarries. The angle usually given is that made by the hole with a vertical line through the collar.
Beethoven exploder	A capacitor-discharge electrical exploder for firing electric detonators.
Bench	The horizontal step or floor from which the material is quarried.
Berm	A ledge created at some level intermediate between the top and bottom of a quarry face to break the continuity of an otherwise long slope. May serve as a haulage road.

Blast	To loosen or move rock or dirt by means of explosives.
Brisance	The shattering effect of an explosive.
Bulling (chambering)	The enlargement of the bottom of a drilled hole by firing a small explosive charge in it.
Burden	The distance between an explosive charge and the free face of material to be blasted.
Chambering	See ‘Bulling’
Detonator	An explosive device used to initiate the primer or the main explosive charge. It is initiated itself either electrically, by a fuse or by other means.
Development	The work performed to make the valuable mineral in a deposit available for extraction.
Dimension stone	Naturally occurring rock material that can be used for paving or building construction, usually cut into blocks or slabs.
Dip	The angle at which a bed, stratum or vein is inclined from a horizontal plane.
Drop cut	The initial cut made in the floor of a quarry for the purpose of establishing a bench at a level below the floor.
Dump	A heap or stack of waste rock or other reject material.
Exploration	The work involved in gaining knowledge of the existence, size, shape, position and quality of a deposit.

Explosive	Substances which when detonated undergo a rapid chemical change, with production of a large volume of gas. In a confined space such as a drillhole, the high pressure developed shatters the surrounding rock.
Face	The vertical or near vertical surface exposed by excavation. A working place from which material is being excavated.
Fault	A fracture in the rock in which movement or displacement of one side relative to the other has taken place parallel to the fracture.
FEL	Front end loader.
Firing	The initiation of an explosive charge electrically, by igniting a fuse or by other means.
Floor	See 'Bench'.
Fuse	A flexible tube consisting of wrappings of different materials around a core of combustible material. It has a predetermined burning speed, and is used with a detonator to fire explosive charges. The fuse burning speed gives the delay between ignition and detonation.
Grab sample	A rough and random method of sampling to estimate the approximate value or quality of a material. It is not representative of the deposit.
Grade (of road)	The longitudinal rise or fall of a roadway relative to a horizontal plane. Expressed as a ratio (e.g. 1 in 20 would be a rise/fall of 1 m in every 20 m horizontally) or a percentage (e.g. 5% would be a rise/fall of 5 m per 100 m horizontally) or by degrees of the angle of inclination/declination. The term also applies to drains, benches etc.

Grade (of ore)	The percentage of valuable metal in the material (e.g. 5.0% copper). Also expressed as grams per tonne (e.g. 5 g/t gold).
Grizzly	A grating, usually of steel rails for stopping oversized rocks entering a crusher or bin; a coarse screen which may be fixed or vibrating.
Ground vibration	Particle movement in rock or soil as a result of energy radiating from an explosion in the form of wave motion.
Hanging wall	The wall or rock on the upper or top side of a vein, deposit or excavation.
Haul	The distance the quarried material or overburden must be moved, e.g. from the face to the treatment plant or from the quarry to a dump.
Hydraulicking	Excavating alluvial or other mineral deposits by means of a high pressure water jet.
Jackhammer	A hand-held percussion type rock drill, usually driven by compressed air.
Misfire	An explosive charge which has failed to explode, either wholly or in part, after initiation by any means.
Mullock	Waste rock of no known economic value.
Open pit, Open cut, Open cast mine	An open surface working from which stone or minerals are obtained.
Ore	The naturally occurring material from which a mineral or minerals of economic value can be extracted at a reasonable profit.
Outcrop	Surface exposure of rock or mineral.

Overburden	The surface soil or worthless material which has to be removed in order to mine the economic deposit beneath.
Pit limits	The vertical and lateral extent to which a deposit may be economically quarried.
Pit slope	The angle that the quarry wall from top to bottom makes with a horizontal plane. Where there are berms, it is the angle between the line joining the crests of the berms and a horizontal plane.
PPV	Peak particle velocity — measure of ground vibration intensity usually expressed in mm/s.
Primer	The cartridge or small explosive charge which is exploded by the detonator and which in turn sets off the main explosive charge.
Production	The work of quarrying and marketing the mineral.
Prospecting	The searching for stone and mineral deposits.
Quarry	See ‘Open pit’.
Screenings	See Chapter 15.
Slime	Very fine (less than 0.075 mm) reject material from washing sand or from mineral treatment.
Stemming	The inert material used in filling a drill hole, either between explosive charges or between the explosives column and the ground surface.
Stockpile	A heap or stack of saleable material held in stock for future use.

Strike	The horizontal course or bearing of a bed, stratum or vein, measured perpendicular to the dip.
Stripping	The removal of soil or overburden to expose the economic material beneath.
Sump	A hole or depression into which water is drained to facilitate its removal, generally by pumping.
Tailings	Reject material from mineral treatment.
Tamping	The compaction of stemming in a borehole.
Three-cone bit (tricone)	A roller bit having three cone-shaped cutters in the head of the bit.
Trommel	A rotary screen for sizing ore or aggregate.
Tunnel	A horizontal or nearly horizontal underground passage that is open to the atmosphere at both ends. By contrast an adit (q.v.) is open to the atmosphere at only one end.
Vein	A layer, seam or narrow irregular body of material different from surrounding formations.
VOD	Velocity of detonation — the speed of propagation of the shock wave through an explosive.

ISSN 0726-1519